THE STACKCALC LIBRARY • CURRICULUM SERIES — BOOK 3 OF 4

MIDDLE SCHOOL MATH LAB

Two-Method Student Practice Workbook with Physical Blocks, Real-World Stories & Four-Line Stack Verification

A self-contained Grades 6–8 practice workbook (10 Units, 45 Sections, 90 Practice A & B Worksheets, and 900 problems) for homeschooling families and classrooms. Teaches students to solve every concept by hand on paper in two independent ways, model quantities with our three physical block kits—Place Value Blocks, Fraction Strips, and Prime Factor Towers—and verify their paper work on the distraction-free StackCalc SC-6 calculator.

StackCalc

stackcalc.io

First Edition

BOOKS, BLOCKS & INSTRUMENTS IN THE STACKCALC LIBRARY

Illustrated Companion Trade Books (History of Mathematics & Stack Thinking)

- Volume I — From Blocks to the Stack: How Hands-On Blocks Make Numbers Click (Grades K–6 / StackCalc SC-6)
- Volume II — Thinking in the Stack: The Art of Calculating Without Parentheses (Grades 7–12, AP & Engineers / StackCalc SC-32)

K–12 Concept-First Curriculum & Student Workbooks (US Letter 8.5" × 11" Editions)

- Book 1 — Teaching with the Stack: Unified K–12 Master Teacher & Homeschool Guide, Lessons & Complete Answer Keys
- Book 2 — Hands-On Math Lab: K–12 Student Manipulative & Four-Line Stack Exploration Workbook
- Book 3 — Middle School Math Lab: Grades 6–8 Two-Method Student Practice Workbook (900 Problems)
- Book 4 — Stack Speed & Fluency: 15 Speed Lessons, Shortcut Playbooks & 60 Progressive Practice Tests (1,200 Problems)

Three Published Physical Manipulative Block Studios & Four-Line Stack Calculators (stackcalc.io)

- Place Value Blocks: 6×2 Whole & Decimal Trays, Digit Blocks (0–9) & Stackable Gold [+1] Carry Tokens
- Fraction Strips: 84.0 mm Proportional Fraction Towers (1 Whole to 1/12) & 12-Socket Comparator Lid
- Prime Factor Towers: Logarithmic Prime Blocks (2, 3, 5, 7 where equal products stand flush) & GCF/LCM Comparator
- StackCalc Calculators: SC-6 Elementary/Middle School Calculator & SC-32 Scientific Calculator (Distraction-Free Hardware)

COPYRIGHT, CURRICULUM ALIGNMENT & OPEN-EDUCATION LICENSE

MIDDLE SCHOOL MATH LAB: Two-Method Student Practice Workbook with Physical Blocks, Real-World Stories & Four-Line Stack Verification

Copyright © 2026 by StackCalc (https://stackcalc.io). Some rights reserved. Published by StackCalc Learning Lab, San Francisco, California. First Edition (US Letter 8.50 × 11.00 in / 215.9 × 279.4 mm).

Homeschool & Classroom Curriculum Compatibility:

Covers the 10 universal middle school mathematics domains (Grades 6–8). Works seamlessly alongside homeschool curricula (Saxon Math 7/6 & 8/7, Singapore Dimensions 6–8, Math Mammoth, RightStart, and AoPS Prealgebra) and state-mandated school textbooks (Big Ideas Math®, Illustrative Mathematics®, Eureka Math®, enVision®, Common Core CCSS & Texas TEKS).

Open-Education License (CC BY-NC-SA 4.0) & Perpetual Family/Classroom Reproduction Permission:

Homeschool families, classroom teachers, math coaches, interventionists, and schools are expressly permitted to photocopy or print unlimited copies of all 90 Practice A & B worksheets, the 1-Page Scoring Guide, and the 10 Chapter Packets for non-commercial educational instruction.

Regulatory & Physical Safety Notice (Reese's Law / P.L. 117-171 / 16 CFR Part 1263 & ASTM F963):

Physical calculator instruments that utilize coin or button cell batteries must maintain a captive, tool-secured battery compartment compliant with UL 4200A. 3D-printed manipulative blocks constitute small parts under 16 CFR Part 1501; adult supervision is required for children under 6 years of age.

Trademark Notice & Print Colophon:

StackCalc, SC-32, and SC-6 are trademarks of StackCalc. Big Ideas Math® is a trademark of Larson Texts, Inc., which does not sponsor or endorse this independent publication. Typeset in Computer Modern Unicode (CMU Serif), SpaceMono, and STIX Math in high-contrast monochrome ink on archival white paper.

Table of Contents

Section / Unit & Mathematical Topic	Sections & Standards	Hands-On Models & Focus	Pages
Preface: Why Two Paper Methods + Blocks + Stack Work	Students & Families	Paper First, 3 Block Kits & Quick-Start	**p. iv**
Master 10-Unit & 45-Section Progress Index	All 45 Sections	Daily Section & Score Check-Off Directory	**p. 1**
100-Point Two-Method Scoring & SC-6 Check Guide	Scoring Rules	+10 (1 Way) vs. +20 (2 Ways) & Stack Check	**p. 2**
Unit 01: Numerical Expressions & Factors	Sec 1.1–1.5 (10 Sheets)	Place Value & Prime Factor Blocks	**pp. 3–13**
Unit 02: Fractions & Decimals	Sec 2.1–2.5 (10 Sheets)	Fraction Strips & Place Value Blocks	**pp. 14–24**
Unit 03: Ratios, Rates & Percents	Sec 3.1–3.5 (10 Sheets)	Fraction Strips & Place Value Blocks	**pp. 25–35**
Unit 04: Algebraic Expressions & Properties	Sec 4.1–4.4 (8 Sheets)	Place Value & Prime Factor Blocks	**pp. 36–44**
Unit 05: Areas of Polygons & Elapsed Time	Sec 5.1–5.4 (8 Sheets)	Fraction Strips & Place Value Blocks	**pp. 45–53**
Unit 06: Integers, Rational Numbers & The Coordinate Plane	Sec 6.1–6.5 (10 Sheets)	Place Value Blocks & Fraction Strips	**pp. 54–64**
Unit 07: Equations & Inequalities	Sec 7.1–7.5 (10 Sheets)	Place Value Blocks & Fraction Strips	**pp. 65–75**
Unit 08: Surface Area & Volume of 3D Solids	Sec 8.1–8.4 (8 Sheets)	Place Value Blocks & Fraction Strips	**pp. 76–84**
Unit 09: Statistical Measures & Data Displays	Sec 9.1–9.4 (8 Sheets)	Place Value Blocks & Fraction Strips	**pp. 85–93**
Unit 10: Grade 7 Bridge — Signed Rationals, 2-Step Equations & Probability	Sec 10.1–10.4 (8 Sheets)	Place Value, Fraction & Prime Blocks	**pp. 94–102**
Appendix A: Middle School Math Formulas & Benchmarks	Units 1–10	Formulas, Conversions & Fraction Benchmarks	**p. 103**
Appendix B: Complete SC-6 Keycap & Two-Dot Guide	SC-6 Reference	4-Level Stack (T, Z, Y, X) & Exact Fractions	**p. 104**
Appendix C: Hands-On Manipulative & Visual Modeling Guide	All 10 Units	How to Model Textbook Concepts with the 3 Block Kits	**p. 105**
Appendix D: Mastery Certificate & Dot-Grid Scratchpad	Certificate & Grid	100-Point Mastery Certificate & Graph Workspace	**p. 106**
Appendix E: Compact Student Self-Check Answer Key	All 900 Problems	Final Answers for All 90 Practice A & B Sheets	**pp. 107–111**

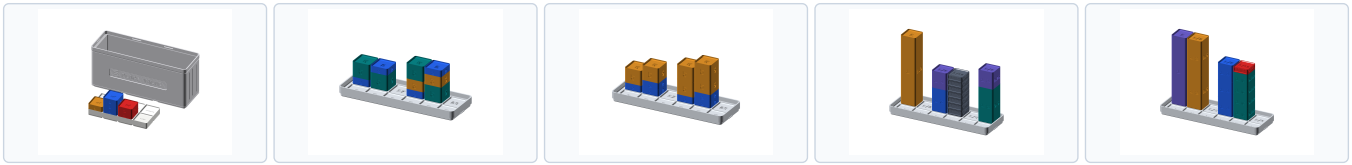
Preface: Why Two Paper Methods + Physical Blocks + Stack Work

1. PAPER & BLOCKS FIRST: WHY THIS WORKBOOK NEVER BECOMES A CALCULATOR CRUTCH

Middle school is where many students begin memorizing formulas ('flip and multiply,' 'move the x') without understanding why they work—and where standard [=] calculators often become a mental crutch. In this workbook, students NEVER use a calculator to skip paper-and-pencil math: they solve every problem by hand on paper first (often in two independent ways!), model quantities with physical blocks, and use the distraction-free SC-6 calculator strictly at the end to verify their work.

Why does the SC-6 have an [ENTER] key and a 4-line Stack screen (T, Z, Y, X) instead of an [=] sign? Because on paper, you write your first number on a top Waiting Line (Line Y), write your second number below it on your Workbench (Line X), and then apply the operation (+, −, ×, ÷). Try these two checks right now:

- **Try Basic Addition on the Stack ($8 + 5 = 13$):** 8 ENTER 5 +
- **Try Two-Dot Exact Fractions ($1/3 + 1/4 = 7/12$):** • 1 • 3 ENTER • 1 • 4 +



Place Value Regrouping | Prime Factor Blocks | Prime Factor Towers | Fraction Strips | Fraction Comparator

2. HOW EVERY SECTION WORKS: TWO PAPER WAYS (+10 / +20), HANDS-ON BLOCKS, THEN SELF-CHECK

- Step 1 — Solve by Hand in Your 1st Paper Way (+10 Points): Study the self-contained Worked Example and solve on paper using Method 1.
- Step 2 — Solve in a 2nd Independent Paper Way or Block Model (+20 Points Double!): Confirm your answer using Method 2 or our Place Value, Fraction, or Prime Factor blocks. Solving ANY 5 problems in 2 Ways earns a full 100 Points—twice the depth in half the tedium!
- Step 3 — Independent Self-Check on the SC-6 Stack: Verify your hand-written answer using the section's SC-6 Check Box and Appendix E (pp. 107–111).
- Works with Homeschool & School Curricula: Pairs effortlessly with Saxon 7/6 & 8/7, Singapore Dimensions 6–8, Math Mammoth, AoPS, or Big Ideas Math.

3. THE COMPLETE STACKCALC CURRICULUM LINEUP (100% SELF-CONTAINED)

BOOK 1 — TEACHING WITH THE STACK: UNIFIED K–12 MASTER TEACHER & HOMESCHOOL GUIDE

Organizes all K–12 concept domains and all 10 Middle School units into a single open-and-go reference with word-for-word scripts, target misconceptions, block remediations, and complete worked keys.

BOOK 2 & BOOK 3 — HANDS-ON MATH LAB (K–12) & MIDDLE SCHOOL MATH LAB (THIS VOLUME)

Self-contained lay-flat student workbooks pairing concrete manipulative models, real-world mathematical story openers, guided paper-and-pencil methods, and compact self-check answer keys.

BOOK 4 — STACK SPEED & FLUENCY: SPEED LESSONS, PLAYBOOKS & 60 PROGRESSIVE PRACTICE TESTS

15 speed lessons ('Slow Way vs. Champion Way' playbooks + 90 guided drills) and 60 progressive 20-problem practice tests (Tests A–D, 1,200 problems) with answer keys.

OPTIONAL COMPANION STORYBOOKS & TACTILE BLOCK STUDIOS

Explore From Blocks to the Stack (Grades K–6) and Thinking in the Stack (Grades 7–12), plus our Place Value, Fraction, and Prime Factor block kits at stackcalc.io.

4. FLEXIBLE HOMESCHOOL KITCHEN-TABLE & CLASSROOM PACING

- Homeschool Mastery Rhythm (No-Tears 5-Problem Option): Have your student solve 5 problems in 2 Ways (+20 pts each = 100 pts) and self-check on the SC-6.
- Daily Classroom Supplement (90 Days): Assign Practice A (Key Ideas) on Day 1 of a textbook section and Practice B (Modeling Real Life) on Day 2.
- 45-Day Summer Bridge or Pre-Algebra Readiness (9 Weeks): Complete 1 Section per day to build rock-solid arithmetic, fraction, and algebra foundations.

Middle School Math Lab: Two-Method Block & SC-6 Workbook

45 Core Sections × 2 Full Practice Sheets (Practice A: Key Ideas + Practice B: Real Life) • Works with Homeschool & School Curricula

1. SELF-CONTAINED PAPER MASTERY

Every worksheet teaches the core vocabulary, visual model, and two paper-and-pencil solution methods—compatible with Saxon, Singapore, AoPS, and Big Ideas Math.

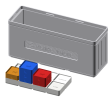
2. TWO WAYS = DOUBLE POINTS (+20)

Every Worked Example models TWO ways to solve on paper: Method 1 (+10 pts) and Method 2 (+20 pts DOUBLE), reinforced by tactile blocks and distraction-free SC-6 check.

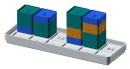
3. 5 IN 2 WAYS OR 10 IN 1 WAY

Students can earn 100 points by solving ANY 5 problems in 2 Ways (+20 each)—building twice the understanding without tedious drill—or all 10 problems in 1 Way (+10 each)!

HANDS-ON MANIPULATIVES REINFORCE TEXTBOOK CONCEPTS & TWO-METHOD MASTERY (+10 / +20 RUBRIC)



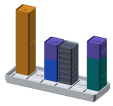
Place Value Regrouping



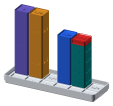
Prime Factor Blocks



Prime Factor Towers



Fraction Strips



Fraction Comparator

UNIT / TOPIC & CURRICULUM ALIGNMENT	SECTIONS & WORKSHEETS (PRACTICE A & B)	TWO-METHOD DOUBLE POINTS FOCUS
Unit 1 (p.3): Numerical Expressions & Factors Big Ideas Math Grade 6 — Chapter 1 • Place Value & Prime Factor Blocks	[ ] Sec 1.1 (pp.4-5): Whole Number Operations & Division with Remainders [ ] Sec 1.2 (pp.6-7): Powers and Exponents [ ] Sec 1.3 (pp.8-9): Order of Operations [ ] Sec 1.4 (pp.10-11): Prime Factorization [ ] Sec 1.5 (pp.12-13): Greatest Common Factor (GCF) & Least Common Multiple (LCM)	**Arrays vs Div Check; Trees vs Ladders** Model: Place Value [ ] Story + 10 Sheets (100 Probs)
Unit 2 (p.14): Fractions & Decimals Big Ideas Math Grade 6 — Chapter 2 • Fraction Strips & Place Value Blocks	[ ] Sec 2.1 (pp.15-16): Equivalent Fractions, Mixed Numbers & LCD Addition [ ] Sec 2.2 (pp.17-18): Multiplying Fractions and Mixed Numbers [ ] Sec 2.3 (pp.19-20): Dividing Fractions and Mixed Numbers [ ] Sec 2.4 (pp.21-22): Adding, Subtracting, and Multiplying Decimals [ ] Sec 2.5 (pp.23-24): Dividing Whole Numbers and Decimals	**LCD Strips vs Cross-Mult; Reciprocals** Model: Fraction Strips [ ] Story + 10 Sheets (100 Probs)
Unit 3 (p.25): Ratios, Rates & Percents Big Ideas Math Grade 6 — Chapters 3, 4 & 5 (Ratios, Rates & Percents) • Fraction Strips & Place Value Blocks	[ ] Sec 3.1 (pp.26-27): Ratios and Tape Diagrams [ ] Sec 3.2 (pp.28-29): Using Ratio Tables and Double Number Lines [ ] Sec 3.3 (pp.30-31): Rates, Unit Rates, and Comparing Rates [ ] Sec 3.4 (pp.32-33): Percents, Fractions, and Decimals [ ] Sec 3.5 (pp.34-35): Solving Percent Problems, Discounts & Conversions	**Tape Diagrams vs Ratio Tables & Rates** Model: Fraction Strips [ ] Story + 10 Sheets (100 Probs)
Unit 4 (p.36): Algebraic Expressions & Properties Big Ideas Math Grade 6 — Chapter 3 / Chapter 5 (Algebraic Expressions) • Place Value & Prime Factor Blocks	[ ] Sec 4.1 (pp.37-38): Writing and Evaluating Algebraic Expressions [ ] Sec 4.2 (pp.39-40): Simplifying Expressions by Combining Like Terms [ ] Sec 4.3 (pp.41-42): The Distributive Property [ ] Sec 4.4 (pp.43-44): Factoring Expressions Using the GCF	**Substitution, Like Terms & Area vs GCF** Model: Place Value [ ] Story + 8 Sheets (80 Probs)
Unit 5 (p.45): Areas of Polygons & Elapsed Time Big Ideas Math Grade 6 — Chapter 4 (Areas of Polygons) • Fraction Strips & Place Value Blocks	[ ] Sec 5.1 (pp.46-47): Areas of Parallelograms and Rectangles [ ] Sec 5.2 (pp.48-49): Areas of Triangles [ ] Sec 5.3 (pp.50-51): Areas of Trapezoids and Composite Figures [ ] Sec 5.4 (pp.52-53): Polygons in the Coordinate Plane & Elapsed Time	**Cut-and-Slide Grids vs Polygon Formulas** Model: Fraction Strips [ ] Story + 8 Sheets (80 Probs)
Unit 6 (p.54): Integers, Rational Numbers & The Coordinate Plane Big Ideas Math Grade 6 — Chapter 6 • Place Value Blocks & Fraction Strips	[ ] Sec 6.1 (pp.55-56): Positive and Negative Integers & Opposites [ ] Sec 6.2 (pp.57-58): Comparing and Ordering Integers & Rational Numbers [ ] Sec 6.3 (pp.59-60): Absolute Value and Magnitude [ ] Sec 6.4 (pp.61-62): The Four-Quadrant Coordinate Plane & Reflections [ ] Sec 6.5 (pp.63-64): Distances in the Coordinate Plane	**Number Lines vs Opposites & |x| Sums** Model: Place Value Blocks [ ] Story + 10 Sheets (100 Probs)
Unit 7 (p.65): Equations & Inequalities Big Ideas Math Grade 6 — Chapter 7 • Place Value Blocks & Fraction Strips	[ ] Sec 7.1 (pp.66-67): Writing Equations in One Variable & Testing Solutions [ ] Sec 7.2 (pp.68-69): Solving Equations Using Addition or Subtraction [ ] Sec 7.3 (pp.70-71): Solving Equations Using Multiplication or Division [ ] Sec 7.4 (pp.72-73): Writing and Graphing Inequalities [ ] Sec 7.5 (pp.74-75): Writing Equations in Two Variables (y = kx and y = x + b)	**Mental Inspection vs Inverse Operations** Model: Place Value Blocks [ ] Story + 10 Sheets (100 Probs)
Unit 8 (p.76): Surface Area & Volume of 3D Solids Big Ideas Math Grade 6 — Chapter 8 • Place Value Blocks & Fraction Strips	[ ] Sec 8.1 (pp.77-78): Three-Dimensional Figures, Nets & Rectangular Prisms [ ] Sec 8.2 (pp.79-80): Surface Areas of Triangular Prisms and Pyramids [ ] Sec 8.3 (pp.81-82): Volumes of Rectangular Prisms [ ] Sec 8.4 (pp.83-84): Circles, Cones, and the Pythagorean Theorem	**Unfolded 2D Nets vs Prism/Cone Formulas** Model: Place Value Blocks [ ] Story + 8 Sheets (80 Probs)
Unit 9 (p.85): Statistical Measures & Data Displays Big Ideas Math Grade 6 — Chapters 9 & 10 (Statistical Measures and Data Displays) • Place Value Blocks & Fraction Strips	[ ] Sec 9.1 (pp.86-87): Statistical Questions, Dot Plots, and the Mean [ ] Sec 9.2 (pp.88-89): Measures of Center: Median, Mode, and Range [ ] Sec 9.3 (pp.90-91): Quartiles, Interquartile Range (IQR) & Box Plots [ ] Sec 9.4 (pp.92-93): Mean Absolute Deviation (MAD)	**Fair-Share Leveling vs IQR & MAD Tables** Model: Place Value Blocks [ ] Story + 8 Sheets (80 Probs)
Unit 10 (p.94): Grade 7 Bridge — Signed Rationals, 2-Step Equations & Probability Big Ideas Math Grade 7 Bridge — Chapters 1–3 & Chapter 7 • Place Value, Fraction & Prime Blocks	[ ] Sec 10.1 (pp.95-96): Adding and Subtracting Rational Numbers [ ] Sec 10.2 (pp.97-98): Multiplying and Dividing Rational Numbers [ ] Sec 10.3 (pp.99-100): Solving Two-Step Equations (ax + b = c) [ ] Sec 10.4 (pp.101-102): Theoretical Probability, Combinations & Permutations	**Zero-Pair Chips vs Rules & Sample Space** Model: Place Value, Fraction [ ] Story + 8 Sheets (80 Probs)

Two-Method Mastery Scoring & SC-6 Work-Check Guide

How to Solve by Hand First, Earn +10 Single-Method or +20 Double Credit, and Verify Your Work on the SC-6

STEP 1: 1ST WAY (+10 PTS)

Way 1 [x] +10 → 1st Method = +10 PTS

Study the Worked Example first. Solve a problem by hand in the left column of your workspace using your 1st Textbook Way, and check off [x] +10 (1 Way) at the bottom of the box.

STEP 2: 2ND WAY (+20 PTS)

W1 W2 → 2 Methods = +20 PTS!

Solve the SAME problem a 2nd independent way in the right column (using the 2nd Textbook Way, a visual model, or one of the 3 block kits). When both ways agree, check off [x] +20 (2 Ways)!

STEP 3: CHECK ON SC-6

Y: 5 (Quotient)
X: 3 (Remainder)

2 3 ENTER 4 g ÷R

Never guess! Use the section's 'Check Your Work on the SC-6 Calculator' box to verify your hand-written answer—using Two-Dot fractions ([.] n [.] d), [≠] equality tests, and [f][LINE] mode.

STEP 4: 100-PT DAILY GOAL

100 / 100 MASTERY → 5×20 or 10×10 = 100 PTS!

Reach your 100-Point Daily Goal by solving any 5 problems in 2 Ways (5 × 20 = 100 pts), all 10 problems in 1 Way (10 × 10 = 100 pts), or any mix that totals 100 points on the page!

HOW TWO-METHOD (+10 / +20) WORKSHEET SCORING WORKS

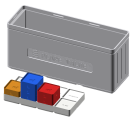
SAMPLE 10-PROBLEM WORKSHEET SCORE (#1–#10 = 100 PTS)

#1 +20	#2 +20	#3 +20	#4 10/20	#5 +20	100 GOAL
#6 10/20	#7 10/20	#8 10/20	#9 +20	#10 10/20	SC-6 CHECK

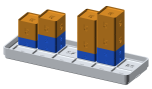
Problems #1–#10:
Check off +10 for solving in 1st Way; check off +20 for verifying in a 2nd Way.

100-Point Goal:
Reach 100 points by solving any 5 problems in 2 Ways OR all 10 in 1 Way.

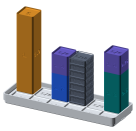
SC-6 Verification:
After solving on paper, use the SC-6 Work-Check box to confirm every answer.



1. Place Value Blocks



2. Prime Factor Towers



3. Fraction Strips

Solve by hand in 1 Way for +10 points, or verify with a 2nd Textbook Way for +20 Double Points!

3 FLEXIBLE PATHS TO THE 100-POINT DAILY GOAL

PATH A: DEEP MASTERY — ANY 5 PROBLEMS IN 2 WAYS = 100 PTS

5 Problems × 2 Ways (+20 pts each) = 100 Points

Best when a 10-problem page feels overwhelming! Pick any 5 problems (for example: Warm-Ups #1-#2, Core #3-#4, and Challenge #9), solve each using BOTH Textbook Ways, and verify on your SC-6. You do half as many problems, but understand them twice as deeply!

PATH B: FULL SHEET — ALL 10 PROBLEMS IN 1 WAY = 100 PTS

10 Problems × 1 Way (+10 pts each) = 100 Points

Best for building speed and accuracy across every problem type on the worksheet. Solve all 10 problems (#1-#10) using your favorite Textbook Way, check off [x] +10 on each card, and verify every answer on your SC-6.

PATH C: MIX & MATCH — ANY COMBINATION TOTALLING 100 PTS

Example: 3 Problems in 2 Ways (60 pts) + 4 Problems in 1 Way (40 pts) = 100 Points

Students and teachers can customize the assignment! Combine 2-Way (+20) and 1-Way (+10) solutions in any order to reach 100 points. Want an Honors Extension? Solve all 10 problems in 2 Ways for 200/100 Extra Credit!

MASTER SC-6 CALCULATOR WORK-CHECK CHEAT SHEET (UNITS 1–10: SOLVE BY HAND FIRST, THEN VERIFY!)

UNIT & TOPIC	SOLVE BY HAND FIRST (2 WAYS)	SC-6 WORK-CHECK KEYSTROKES	WHAT TO VERIFY ON THE SC-6 DISPLAY
Unit 1: Whole Numbers & Factors	Long Div / Tree vs. Array / Venn	2 3 ENTER 4 g ÷R	Line Y = Quotient, Line X = Remainder; [PRIME], [g][FACT], [GCF], [f][LCM], [x²], [x³]
Unit 2: Fractions & Decimals	LCD Strips & Reciprocals vs. Cross-Mult	• 2 • 3 ENTER • 1 • 4 +	HP-32 Two-Dot fraction entry ([.] n [.] d & w [.] n [.] d); [g][SIMP], [FDISP], [a b/c], [f][d/c], [PV], [$]
Unit 3: Ratios, Rates & Percents	Tape Diagrams vs. Ratio Tables & % Prop	1 2 ENTER 8 g a:b	[g][a:b] Ratio Mode (÷ [10F] table), [g][REP], [%] part of whole, [g][%CHG], [f][→cm]/[g][→in]
Unit 4: Algebraic Expressions	Area Model / Like Terms vs. GCF Factor	4 ENTER 5 × 7 +	Evaluate at x = 5 in both original & simplified forms and press [≠] (flashes NO when equal!)
Unit 5: Area & Elapsed Time	Cut-and-Slide Grid vs. Polygon Formulas	1 2 ENTER 8 f ■	[f][RECT] area/perim, [g]['TRI] 1/2bh, [Σ+]/[f][Σx] composite sum, [CLK] + [f][→HR]/[g][→HMS]
Unit 6: Integers & Coord Plane	Number Line Jumps vs. Absolute Value |x|	f ↔ 1 8 +/− 5 +/− f ≥	[f][LINE] Number Line jumps, [+/-] sign, [g][|x|] distance, [f][>]/[g][<] (flashes YES/NO!)
Unit 7: Equations & Inequalities	Balance Scale vs. Inverse Operations	4 1 ENTER 4 ENTER 9 × 5 + ≠	Equality test [≠] flashes NO when Left = Right (balanced!); [f][≥] & [g][≤] flash YES/NO
Unit 8: 3D Surface Area & Volume	Unfolded 2D Net vs. Prism/Cone Formulas	1 2 ENTER 8 ENTER 5 f ⬢	[f][PRISM] 3D volume, [g][CLΣ]+[Σ+]+[f][Σx] 2D net surface area, [f][CIRC], [g][HYP], [g][CONE]
Unit 9: Statistics & Distributions	Fair-Share Leveling / Peel vs. IQR & MAD	g CLΣ 1 8 Σ+ 2 4 Σ+ g x̄	Enter list with [Σ+] (or [FRQ]): check [g][x-bar], [f][MED], [g][MODE], [RNG], [f][1Q], [g][3Q], [IQR], [g][MAD]
Unit 10: Signed Ops & Probability	Zero-Pair Chips & Trees vs. Sign Rules	f ↔ 1 0 ENTER 4 +/− -	[f][LINE] signed hops, [+/-], [≠] 2-step equation check, [DICE]/[f][1..x], [x!], [f][nCr], [g][nPr]

45-SECTION 100-POINT MASTERY PROGRESS LOG (CHECK OFF EACH SECTION AS YOU EARN 100 POINTS ON PRACTICE A & B!)

Unit 1:	[ ] 1.1	[ ] 1.2	[ ] 1.3	[ ] 1.4	[ ] 1.5
Unit 2:	[ ] 2.1	[ ] 2.2	[ ] 2.3	[ ] 2.4	[ ] 2.5
Unit 3:	[ ] 3.1	[ ] 3.2	[ ] 3.3	[ ] 3.4	[ ] 3.5
Unit 4:	[ ] 4.1	[ ] 4.2	[ ] 4.3	[ ] 4.4	
Unit 5:	[ ] 5.1	[ ] 5.2	[ ] 5.3	[ ] 5.4	
Unit 6:	[ ] 6.1	[ ] 6.2	[ ] 6.3	[ ] 6.4	[ ] 6.5
Unit 7:	[ ] 7.1	[ ] 7.2	[ ] 7.3	[ ] 7.4	[ ] 7.5
Unit 8:	[ ] 8.1	[ ] 8.2	[ ] 8.3	[ ] 8.4	
Unit 9:	[ ] 9.1	[ ] 9.2	[ ] 9.3	[ ] 9.4	
Unit 10:	[ ] 10.1	[ ] 10.2	[ ] 10.3	[ ] 10.4	

Unit 1: Numerical Expressions & Factors — The Backcountry Trail Crew & Maker Cabin

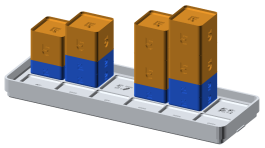
Whole Number Operations, Powers & Exponents, Order of Operations, Prime Factorization, GCF & LCM

Every summer in the Cascade Mountains, a group of sixth-grade volunteers helps the ranger station prepare backcountry campsites, wildlife observation blinds, and a small maker cabin. Maya signed up to organize the supply crates and battery banks, even though she secretly worried she would mess up the counts and leave a trail crew short on gear miles from the nearest road. Across the cabin, Leo tinkers with mountain bikes, solar signboards, and LEGO robotics prototypes, while Zoe hikes with the field biologists to watch Clark's nutcrackers hide thousands of pine seeds for the winter.

During their first week, guessing got all three of them into trouble. When Maya tried to pack 53 lithium battery cells into 6-cell drone packs, her phone calculator gave her a long decimal that didn't tell her how many real packs she could actually clip into a drone. When 78 campers showed up at the lake for 4-person canoes, Leo wasn't sure whether to round up or down so nobody got left on the dock. And when Sam and Leo added up the cost of a skateboard deck plus four wheels, they got two completely different totals from the exact same numbers.

In the bike and robotics shed, Leo noticed another puzzle: two gears with spiky teeth only mesh smoothly when their tooth counts share common building blocks, and two park shuttles leaving the trailhead on different schedules only meet up at the station when their loop times hit a shared multiple. Even Zoe's field notes on periodical cicadas that emerge every 12 or 15 years followed the exact same pattern.

None of them felt like 'math people' at the start of the summer. But by working out each supply puzzle by hand first and then checking their answers on their SC-6 calculators, they discovered that whole numbers have built-in patterns for leftovers, rapid doubling, grouping rules, and repeating cycles—turning messy real-world guesses into plans they could trust.



Real-World Mathematical Context & Two-Method Mastery • Hands-On Models: Place Value & Prime Factor Blocks

UNIT 1 LESSONS — LEARN TWO TEXTBOOK WAYS FIRST, THEN CHECK YOUR WORK ON THE SC-6 CALCULATOR

SEC 1.1 (pp.4-5) Whole Number Operations & Division with Remainders

Textbook Ways: Way 1: Standard Long Division | Way 2: Partial Quotients | Way 3: Equal-Groups Array Model

Learn by Hand: In Big Ideas Math, whole-number division splits a dividend into equal groups of the divisor: Dividend = (Quotient × Divisor) + Remainder, where the remainder is always less than the divisor. Modeling the dividend on the Place Value Tray physically separates the equal groups (Quotient) from the leftover ones (Remainder).

Check Your Work on the SC-6 Calculator:

2 3 ENTER 4 g ÷R

Type 23 [ENTER] 4 [g][÷R]. Read the Quotient (5) on line Y and the Remainder (3) on line X! Press [f][LASTX] [×] [+] to verify 5 × 4 + 3 = 23.

SEC 1.2 (pp.6-7) Powers and Exponents

Textbook Ways: Way 1: Multiply Step-by-Step | Way 2: Group Factors in Pairs | Way 3: Square & Cube Array Model

Learn by Hand: A power represents repeated multiplication of the same factor. In b^n, the base b is the repeated factor and the exponent n tells how many times the base is used as a factor. In the Prime Factor Towers, stacking n identical prime blocks models b^n directly (for example, three [2] blocks build $2^3 = 8$, and four [2] blocks build $4^2 = 2^4 = 16$).

Check Your Work on the SC-6 Calculator:

4 x^2 4 x^3

Press 4 [x^2] to verify $4^2 = 16$, and press 4 [x^3] to verify $4^3 = 64$.

SEC 1.3 (pp.8-9) Order of Operations

Textbook Ways: Way 1: PEMDAS Step-by-Step Funnel | Way 2: Distributive Property / Fraction-Bar Grouping

Learn by Hand: Big Ideas Order of Operations (PEMDAS): (1) Evaluate expressions inside grouping symbols (), [], or fraction bars first; (2) Evaluate powers; (3) Multiply and divide from left to right; (4) Add and subtract from left to right.

Check Your Work on the SC-6 Calculator:

8 ENTER 4 + 9 ENTER 3 - ×

Follow PEMDAS on paper first. To check (8 + 4) × (9 - 3) on your SC-6, solve the first parentheses with 8 [ENTER] 4 [+] (12 waits on Y), solve the second with 9 [ENTER] 3 [-] (6 on X), and press [×] to confirm 72!

SEC 1.4 (pp.10-11) Prime Factorization

Textbook Ways: Way 1: Factor Tree (Pair A) | Way 2: Factor Tree (Pair B) | Way 3: Division Ladder

Learn by Hand: In Big Ideas Math, a Factor Tree or Division Ladder splits a composite number until every branch ends at a prime number. In the Prime Factor Towers, block heights are logarithmic, so stacking [2]+[2]+[3]+[5] always reaches the exact 60 mark whether you split 60 as 6 × 10 or 4 × 15.

Check Your Work on the SC-6 Calculator:

6 0 g 人

Build your factor tree for 60 by hand until every branch ends at a prime. On your SC-6, test any branch with [PRIME] (flashes YES/NO) and type 60 [g][FACT] to verify the prime factorization $2^2 \times 3 \times 5$!

SEC 1.5 (pp.12-13) Greatest Common Factor (GCF) & Least Common Multiple (LCM)

Textbook Ways: Way 1: List Factors & Multiples | Way 2: Prime Factorization | Way 3: Prime Factor Venn Diagram

Learn by Hand: In Big Ideas Math, a Prime Factor Venn Diagram places shared prime factors in the center overlap and remaining primes in the outer circles: multiplying the center overlap gives the GCF, and multiplying all regions together gives the LCM! Placing Prime Factor Blocks into the Venn circles makes the shared overlap and union visible.

Check Your Work on the SC-6 Calculator:

2 0 ENTER 5 0 GCF 2 0 ENTER 5 0 f LCM

Press 20 [ENTER] 50 [GCF] to confirm GCF = 10, and 20 [ENTER] 50 [f][LCM] to confirm LCM = 100 (or [f][C.FAC] for shared primes).

HANDS-ON MODEL: PLACE VALUE BLOCKS

Section 1.1 — Practice A: Whole Number Operations & Division with Remainders

Key Vocabulary: dividend • divisor • quotient • remainder

Name: _____ Date: _____ Score: ____ / 100

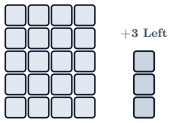
TEXTBOOK CONCEPT & HANDS-ON MODEL (PLACE VALUE BLOCKS)

Way 1: Long Division (23 ÷ 4)

```
   5  R 3
4 | 23
   -20      (5×4)
    3       (Rem < 4)
```

Check: (5 × 4) + 3 = 23

Equal-Groups Array: 5 Rows of 4 + 3



5 Full Groups (20) Rem = 3

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept: Divide Dividend ÷ Divisor to find the Quotient (full equal groups) and Remainder (leftover < Divisor).

Textbook Check: Always verify division with the Division Algorithm: (Quotient × Divisor) + Remainder = Dividend.

WATCH OUT: A remainder must ALWAYS be smaller than the divisor (Rem < Divisor). Check whether the problem asks for full groups, leftovers, or rounding up!

WORKED EXAMPLE — 3 TEXTBOOK WAYS (ANY 2 WAYS = +20 PTS)

Problem: Divide 23 ÷ 4 and check your answer using the Division Algorithm.

WAY 1: Standard Long Division
Divide 23 by 4: 4 goes into 23 five times (5 × 4 = 20). Subtract 23 - 20 = 3 → Quotient 5 R 3.

WAY 2: Partial Quotients (Subtract Multiples)
Split 23 into friendly multiples of 4: take 3 × 4 = 12 (leaves 11), then 2 × 4 = 8 (leaves 3) → (3 + 2) = 5 R 3.

WAY 3: Equal-Groups Array Model (With Blocks)
Arrange 23 tiles into rows of 4: you form 5 full rows of 4 (20 tiles) with 3 leftover tiles → 5 R 3. (Check: 5×4+3 = 23.)

Answer: 5 R 3 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 1.1 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[2] [3] [ENTER] [4] [g] [÷R]

Solve 23 ÷ 4 on paper first. Then check on your SC-6:
Type 23 [ENTER] 4 [g][÷R]. Read the Quotient (5) on line Y and the Remainder (3) on line X! Press [f][LASTX] [×] [+] to verify 5 × 4 + 3 = 23.

SECTION 1.1 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Divide 31 ÷ 6.**
Starter Step 1: Divide 31 ÷ 6 to find how many full groups of 6 fit inside 31, then check (Quotient × 6) + Remainder = 31.

Way 1 (Paper Work):

Q:___ R:___

Check: (___ × ___) + ___ = ___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Divide 47 ÷ 5.**
Starter Step 1: Find the largest multiple of 5 that is ≤ 47, then subtract it from 47 to find the remainder.

Way 1 (Paper Work):

Q:___ R:___

Check: (___ × ___) + ___ = ___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Divide 58 ÷ 7 and check your answer by multiplying.**

Way 1 (Paper Work):

Q:___ R:___

Check: (___ × ___) + ___ = ___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Divide 85 ÷ 9. How many full groups of 9 can be made, and how many are left over?**

Way 1 (Paper Work):

Q:___ R:___

Check: (___ × ___) + ___ = ___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Divide 146 ÷ 12.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Paracord Strap Cuts: Maya cuts 35 cm straps from a 250 cm roll of paracord. How many full lines can be cut, and how many centimeters of cord remain?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **When a number is divided by 8, the quotient is 14 and the remainder is 5. What is the number?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Error Analysis: Sam says 75 ÷ 8 = 8 R 11. Explain why a remainder of 11 is impossible and find the correct answer.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **LEGO Rover Wheels: Leo has a bin of 78 rover tires. Each rover requires 4 tires. How many complete 4-wheel rovers can be built, and how many tires are left in spares?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Wilderness Tent Expedition: 29 explorers are setting up 4-person tents for a storm. How many tents must they pitch so every explorer has shelter?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: PLACE VALUE BLOCKS

Section 1.1 — Practice B: Modeling Real Life (Whole Number Operations & Division with Remainders)

Textbook Ways: Standard Long Division | Partial Quotients | Equal-Groups Array Model Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — INTERPRETING THE REMAINDER (3 TEXTBOOK WAYS • ANY 2 = +20 PTS)

Problem: A robotics tournament has 135 students riding in vans that hold 8 students each. How many vans are needed so every student has a seat?

WAY 1: Standard Long Division
Divide 135 ÷ 8: 8 into 13 is 1 (13 - 8 = 5); bring down 5 → 8 into 55 is 6 (55 - 48 = 7) → 16 R 7.

WAY 2: Partial Quotients (10 Vans + 6 Vans)
Subtract 10 vans (10 × 8 = 80 students, leaving 55), then 6 vans (6 × 8 = 48, leaving 7) → 16 full vans + 7 students.

WAY 3: Interpret the Remainder in Context
16 vans hold 128 students, leaving 7 students waiting. Add 1 more van for the remaining 7 students → 17 vans total.

WATCH OUT: A remainder must ALWAYS be smaller than the divisor (Rem < Divisor). Check whether the problem asks for full groups, leftovers, or rounding up!

Answer: 17 vans (16 full vans + 1 van with 7 students)

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 1.1 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[1] [3] [5] [ENTER] [8] [g] [÷R] [f] [LASTx] [×] [+]

Work out 135 ÷ 8 by hand first, then press 135 [ENTER] 8 [g][÷R] on your SC-6. Line Y shows 16 full vans and line X shows 7 leftover students—confirming you must round up to 17 vans! Press [f][LASTX] [×] [+] to check 16 × 8 + 7 = 135.

At the ranger station, Maya stared at a bin of 53 lithium battery cells and felt her stomach sink. Each drone pack needed 6 cells, and when she typed 53 ÷ 6 into a phone calculator, it spat out 8.8333333—which made no sense at all when you are holding real metal batteries that cannot be cut into pieces. For a moment she worried she wasn't cut out to be supply quartermaster. Then she tried lining the batteries up in rows of 6 on the workbench and realized the messy decimal was hiding a simple whole-number remainder: 8 full rows of 6 use 48 cells, leaving 5 spare cells on the table. And when 78 campers arrived for 4-person canoes that afternoon, she saw that the same remainder meant 19 full canoes wouldn't be enough—they needed a 20th canoe so the last 2 campers wouldn't be left behind.

SECTION 1.1 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Drone Battery Packs: A robotics team has 53 lithium cells. Each drone battery pack requires 6 cells. How many full battery packs can the team build, and how many cells are left over?**

Starter Step 1: Divide 53 cells ÷ 6 cells per pack: the quotient is full packs and the remainder is leftover cells.

Way 1 (Paper Work):

Q:___ R:___

Check: (___ × ___) + ___ = ___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Arcade Prize Tickets: You win 94 tickets and trade them for mini-figures that cost 8 tickets each. How many mini-figures can you get, and how many tickets remain?**

Starter Step 1: Divide 94 tickets ÷ 8 tickets per figure: the quotient is mini-figures and the remainder is leftover tickets.

Way 1 (Paper Work):

Q:___ R:___

Check: (___ × ___) + ___ = ___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Camp Canoe Fleet: 78 campers are crossing a lake in canoes that hold 4 campers each. How many canoes are needed so every camper has a seat?**

Way 1 (Paper Work):

Q:___ R:___

Check: (___ × ___) + ___ = ___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Cookie Bakery Tray: A baker places 110 warm cookies onto cooling racks that hold 12 cookies per rack. How many racks are completely filled, and how many cookies go on the last rack?**

Way 1 (Paper Work):

Q:___ R:___

Check: (___ × ___) + ___ = ___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **3D Printer Spool: A filament spool has 450 meters of PLA. Each chess knight uses 16 meters. How many complete knights can be printed, and how much PLA is left over?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Nutcracker Seed Caches: A Clark's nutcracker hides 165 pine seeds in underground caches of 9 seeds each and remembers every cache months later! How many caches are full, and how many does it dig in all?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Mystery Marble Bags: When a bag of marbles is packed into pouches of 15, there are 12 full pouches and 7 marbles left over. How many marbles were in the bag to start?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Theme Park Coaster: A roller coaster train holds 24 riders. There are 200 people in line. How many full trains run, and how many riders are on the final train?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Pizza Party Slices: Each large pizza is cut into 8 slices. If 67 students each eat 1 slice, how many whole pizzas must the school order, and how many slices will be left over?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Apple Orchard Crates: An orchard harvests 325 apples and packs them into crates that hold 28 apples each. How many full crates are packed, and how many apples are left over?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: PRIME FACTOR TOWERS

Section 1.2 — Practice A: Powers and Exponents

Key Vocabulary: power • base • exponent • perfect square • perfect cube

Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & HANDS-ON MODEL (PRIME FACTOR TOWERS)

Power: Base × Base (Repeated Factor)

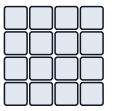
Exponent (2 factors)

$4^{2} = 4 \times 4 = 16$

Base

Watch Out: $4^2 = 16$ (NEVER $4 \times 2 = 8$!)

Prime & Square Model: 4^2 (16) vs. 4×2 (8)



$4^2 = 4 \times 4 = 16$



(Not a Square!)

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept: In b^n, the base b is the repeated factor and the exponent n counts how many times b is multiplied.
Textbook Watch-Out: 4^2 means $4 \times 4 = 16$ (a 4-by-4 perfect square), NEVER base × exponent ($4 \times 2 = 8$)!
WATCH OUT: 4^3 means $4 \times 4 \times 4 = 64$ — NEVER multiply the base by the exponent ($4 \times 3 = 12$)!

WORKED EXAMPLE — 3 TEXTBOOK WAYS (ANY 2 WAYS = +20 PTS)

Problem: Write 4^2 and 4^3 as repeated multiplication and find their values. Compare 4^2 to 4×2.

WAY 1: Multiply Step-by-Step Left to Right
Write the base 4 as a repeated factor: $4^2 = 4 \times 4 = 16$. For 4^3: $4 \times 4 = 16$, then $16 \times 4 = 64$.

WAY 2: Square Area & Cube Volume Model (With Blocks)
4^2 is the area of a 4×4 square (16 square units), and 4^3 is the volume of a $4 \times 4 \times 4$ cube (64 cubic units).

WAY 3: Decompose into Smaller Powers
Use the previous power: since $4^1 = 4$, $4^2 = 4 \times 4^1 = 16$, and $4^3 = 4 \times 4^2 = 4 \times 16 = 64$.

Answer: $4^2 = 16$, $4^3 = 64$ 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 1.2 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

4 [x^2] 4 [x^3]

Write out the repeated multiplication by hand first. Then check on your SC-6:
Press 4 [x^2] to verify $4^2 = 16$, and press 4 [x^3] to verify $4^3 = 64$.

SECTION 1.2 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Write 5^2 as repeated multiplication and find its value.**
Starter Step 1: Write the base 5 two times (5×5) and multiply (never multiply base × exponent!).

Way 1 (Paper Work):

Expand: ___ × ___ × ___

Value = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Evaluate 7^2 and compare its value to 7×2.**
Starter Step 1: Compute $7^2 = 7 \times 7$ first, then compute 7×2 so you can compare the square to the product.

Way 1 (Paper Work):

Expand: ___ × ___ × ___

Value = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Write 3^3 as repeated multiplication and find its value.**

Way 1 (Paper Work):

Expand: ___ × ___ × ___

Value = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Evaluate 5^3.**

Way 1 (Paper Work):

Expand: ___ × ___ × ___

Value = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Write 2^5 as repeated multiplication and find its value.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Evaluate 6^4.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Evaluate 10^4 and 1^4. How does the exponent relate to the number of zeros in a power of 10?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Error Analysis: Mia wrote $2^4 = 8$ because $2 \times 4 = 8$. Describe and correct her error.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Rope Pulley Multiplier: Each stage of a treehouse pulley doubles a crew's pulling force. With 5 stages, the multiplier is 2^5. If a crew pulls with 3 kN of force, what total force does the hoist lift?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Modular Storage Cubes: A camp gear station is assembled from cubic rooms measuring 2 m on each edge. What is the volume of 1 cubic room, and of 8 rooms?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: PRIME FACTOR TOWERS

Section 1.2 — Practice B: Modeling Real Life (Powers and Exponents)

Textbook Ways: Multiply Step-by-Step | Group Factors in Pairs | Square & Cube Array Model Name: _____ Date: _____ Score: _____ / 100

WORKED EXAMPLE — DOUBLING & TRIPLING GROWTH (3 TEXTBOOK WAYS • ANY 2 = +20 PTS)

Problem: A video clip is shared by 3 friends in Round 1, and the number of shares triples in each round. How many shares happen in Round 4?

WAY 1: Multiply Step-by-Step (Left to Right)
Write 3^4 as four factors of 3: $3 \times 3 = 9 \rightarrow 9 \times 3 = 27 \rightarrow 27 \times 3 = 81$ shares.

WAY 2: Group Factors in Pairs (Associative Property)
Pair the four factors of 3 into two squares: $(3 \times 3) \times (3 \times 3) = 9 \times 9 = 81$ shares.

WAY 3: Pattern of Powers Table
Track each round in a table: Round $1 = 3$, Round $2 = 9$, Round $3 = 27$, Round $4 = 81$ shares.

WATCH OUT: 4^3 means $4 \times 4 \times 4 = 64$ — NEVER multiply the base by the exponent ($4 \times 3 = 12$)!

Answer: 81 shares

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 1.2 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

3 ENTER 4 g y^x

Calculate 3^4 on paper first. To check any power b^n on your SC-6, type the base 3 [ENTER], type the exponent 4, and press [g][y^x] to confirm 81.

Leo was ready to give up when his biology lab sheet asked how many bacteria there would be after a single cell split in two every hour for 6 hours. He started drawing tiny branching lines on scratch paper, lost count around thirty, and muttered that he always messed up big numbers. Sitting next to him, Maya pointed out that every hour just multiplied the count by 2 again—just like his 8-by-8 LED matrix multiplied 8×8 for a square and his 6-by-6-by-6 storage crate multiplied $6 \times 6 \times 6$ for a cube. Once Leo saw that an exponent simply counts how many times a number multiplies by itself ($2^6 = 64$), a whole page of messy scribbles shrank into one step he could actually trust.

SECTION 1.2 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Pixel Art Canvas: A retro game sprite is drawn on an 8-by-8 square grid. Write the total number of pixels as a power and find its value.**

Starter Step 1: An 8-by-8 square grid has $8^2 = 8 \times 8$ pixels—multiply the side length by itself.

Way 1 (Paper Work):

Expand: ___ × ___ × ___

Value = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Cubic Storage Room: You dig a cubic room measuring 6 blocks long, 6 blocks wide, and 6 blocks tall. How many blocks do you mine in all?**

Starter Step 1: A 6-by-6-by-6 cube has volume $6^3 = 6 \times 6 \times 6$ blocks—multiply the first two 6s, then multiply by the third 6.

Way 1 (Paper Work):

Expand: ___ × ___ × ___

Value = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Cell Division Biology Lab: A single bacterium splits into 2 bacteria every hour. How many bacteria are there after 6 hours?**

Way 1 (Paper Work):

Expand: ___ × ___ × ___

Value = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Tournament Bracket: A tennis tournament has 4^4 players in the opening round. Write 4^4 as repeated multiplication and find how many players entered.**

Way 1 (Paper Work):

Expand: ___ × ___ × ___

Value = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Patio & Grill Pad: A patio has a 12 ft by 12 ft square main deck and a 5 ft by 5 ft square grill pad. What is the combined area of both squares?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Starling Flock Expansion: On call 1, 5 starlings alert 5 neighbors each. Each wave multiplies by 5. How many starlings are alerted on wave 4?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Perfect Square Garden: A square community garden has an area of 196 square meters. What is the side length of the garden?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Power Showdown: Which is greater, 2^5 or 5^2? By how much?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Binary Memory Kilobyte: One kilobyte in binary computer memory is 2^{10} bytes. How many bytes are in one kilobyte?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Sculpture Pedestal: A sculptor stacks a 3-inch cube on top of a 7-inch cube. What is the combined volume of the two cubes?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 1.3 — Practice A: Order of Operations

Key Vocabulary: numerical expression • evaluate • order of operations • grouping symbols

Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & VISUAL DIAGRAM

Textbook PEMDAS Funnel: (8 + 4) × (9 - 3)

(8 + 4) × (9 - 3)

= 12 × (9 - 3) [1st ( )]

= 12 × 6 [2nd ( )]

Way 2: 8(6) + 4(6) = 48 + 24

= 72

PEMDAS Priority Checklist

1. ( ) [ ] Grouping Symbols
2. b^n Exponents / Powers
3. × and ÷ Left → Right (Equal!)
4. + and - Left → Right (Equal!)

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept (PEMDAS): 1. Grouping Symbols ( ) [ ] 2. Exponents 3. Multiply & Divide (Left → Right) 4. Add & Subtract (Left → Right).

Textbook Watch-Out: Multiplication and division have EQUAL priority (go Left to Right); addition and subtraction also go Left to Right!

WATCH OUT: Multiply & Divide have EQUAL priority (go Left-to-Right!), and Add & Subtract have EQUAL priority (go Left-to-Right!).

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Evaluate (8 + 4) × (9 - 3).

WAY 1: Standard Order of Operations (PEMDAS Funnel)

Evaluate inside both sets of parentheses first: (8 + 4) = 12 and (9 - 3) = 6. Then multiply: 12 × 6 = 72.

WAY 2: Distributive Property Expansion

Simplify (9 - 3) = 6 first, then distribute 6 across (8 + 4): (8 × 6) + (4 × 6) = 48 + 24 = 72.

Answer: 72 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 1.3 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

8 ENTER 4 + 9 ENTER 3 - ×

Follow PEMDAS on paper first. To check (8 + 4) × (9 - 3) on your SC-6, solve the first parentheses with 8 [ENTER] 4 [+] (12 waits on Y), solve the second with 9 [ENTER] 3 [-] (6 on X), and press [×] to confirm 72!

SECTION 1.3 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 Evaluate $5 + 2 \times 3$.

Starter Step 1: By PEMDAS, multiply 2 × 3 FIRST before adding 5: 5 + (2 × 3).

Way 1 (Paper Work):

PEMDAS Funnel: ( ) → b^n → ×/÷ → +/-

= _____

= _____

= _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

2 Evaluate $(14 + 6) \div (7 - 2)$.

Starter Step 1: Evaluate both parentheses first—(14 + 6) and (7 - 2)—and then divide the first result by the second.

Way 1 (Paper Work):

PEMDAS Funnel: ( ) → b^n → ×/÷ → +/-

= _____

= _____

= _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

3 Evaluate $30 - 4 \times 6$.

Way 1 (Paper Work):

PEMDAS Funnel: ( ) → b^n → ×/÷ → +/-

= _____

= _____

= _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

4 Evaluate $(9 + 3) \times (15 - 8)$.

Way 1 (Paper Work):

PEMDAS Funnel: ( ) → b^n → ×/÷ → +/-

= _____

= _____

= _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

5 Evaluate $50 - (3 + 4)^2$.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

6 Evaluate $120 - 4 \times (3 + 2)^2$.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

7 Evaluate the fraction-bar expression $(48 - 4 \times 6) / (3 + 2 \times 4.5)$.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

8 Evaluate $3^3 + 15 \div 3 - 2 \times 4$.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

9 Claw-Bot Robotics Kit: A team orders 5 high-torque servos at $12 each and 4 bumper sensors at $8 each. What is the total cost of the parts?

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

10 3D-Printed Mini-Drone Fundraiser: A maker club sells 45 mini-drone frames at $2 each and subtracts $28 for PLA filament spools. What is the club's net profit?

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 1.3 — Practice B: Modeling Real Life (Order of Operations)

Textbook Ways: PEMDAS Step-by-Step Funnel | Distributive Property / Fraction-Bar Grouping Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — MULTI-STEP NUMERICAL EXPRESSION (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: Evaluate $40 - 3 \times (2 + 3)^2 \div 5$.

WAY 1: Standard PEMDAS Funnel (Left-to-Right Multiply/Divide)

1. Parentheses: $(2+3) = 5$. 2. Exponent: $5^2 = 25$. 3. Multiply then Divide L-to-R: $3 \times 25 = 75$, then $75 \div 5 = 15$. 4. Subtract: $40 - 15 = 25$.

WAY 2: Write Division as a Fraction Term

Rewrite $3 \times 5^2 \div 5$ as the fraction $(3 \times 25) / 5$. Divide $25 \div 5 = 5$ first, multiply $3 \times 5 = 15$, then subtract: $40 - 15 = 25$.

Watch-Out: Never subtract 40 - 3 first! Multiplication and division come before subtraction.

WATCH OUT: Multiply & Divide have EQUAL priority (go Left-to-Right!), and Add & Subtract have EQUAL priority (go Left-to-Right!).

Answer: 25

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 1.3 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

4 | 0 | ENTER | 3 | ENTER | 2 | ENTER | 3 | + | x^2 | × | 5 | ÷ | -

Work from the inside out on paper, then check on your SC-6 stack:
Type 40 [ENTER] 3 [ENTER] 2 [ENTER] 3 [+] [x^2] [×] 5 [÷] [-]—the 4-line stack holds each intermediate result without parentheses keys and confirms 25!

Sam and Leo both worked out the price of a $45 skateboard deck plus 4 wheels at $7 each, but Sam got $343 while Leo got $73. Sam was ready to crumple up his paper, convinced that math rules like PEMDAS were just made up to trick kids who aren't fast at arithmetic. Then Leo set the actual parts on the counter: one wooden deck and a plastic pack of four $7 wheels. Looking at the table, Sam suddenly saw what had happened—when he added $45 + 4$ first and multiplied by 7, he was pretending he bought 49 items at $7 each! Calculating the 4-wheel bundle ($4 \times \$7 = \$28$) before adding the $45 deck wasn't a random school rule after all; it matched how the parts really sit on the workbench.

SECTION 1.3 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Movie Night Snack Combo: Four friends each buy a $9 movie ticket and a $5 tub of popcorn. Write and evaluate an expression for the total cost.**

Starter Step 1: Group 1 friend's combo in parentheses: $4 \times (\$9 + \$5)$, or use the Distributive Property: $4(\$9) + 4(\$5)$.

Way 1 (Paper Work):

PEMDAS Funnel: ( ) → b^n → ×/÷ → +/-
= _____
= _____
= _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

2 **Skateboard Build: You buy a $45 skateboard deck and 4 wheels at $7 each. What is the total cost of the parts?**

Starter Step 1: Multiply the 4 wheels × $7 FIRST, then add the $45 deck: $45 + (4 × $7).

Way 1 (Paper Work):

PEMDAS Funnel: ( ) → b^n → ×/÷ → +/-
= _____
= _____
= _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

3 **Laser Tag Score: You start with 100 points, lose 8 points on each of 3 penalties, and then win a 25-point bonus. What is your final score?**

Way 1 (Paper Work):

PEMDAS Funnel: ( ) → b^n → ×/÷ → +/-
= _____
= _____
= _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

4 **Tile Border Courtyard: A 10 m by 10 m square courtyard has a 6 m by 6 m square fountain in the center. What is the area of the surrounding tile walkway?**

Way 1 (Paper Work):

PEMDAS Funnel: ( ) → b^n → ×/÷ → +/-
= _____
= _____
= _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

5 **Aquarium Ticket Group: Admission to an aquarium is $14 for adults and $9 for children. How much does admission cost for 2 adults and 5 children?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

6 **Raven Puzzle Boxes: A raven solves 6 puzzle boxes. Each box awards 3^2 points for pulling a pin plus 5 points for sliding a latch, minus a 14-point time penalty total. Find the raven's final score.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

7 **Craft Fair Booth: You make 72 bracelets, keep 12 for friends, and pack the rest equally into 6 display boxes. How many bracelets go into each box?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

8 **Evaluate $48 \div 6 \times 4 + 10$. Explain why division is performed before multiplication in this expression.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

9 **Bakery Muffin Boxes: A baker makes 5 trays of 12 blueberry muffins and 3 trays of 8 banana muffins, then packs all of them equally into 7 gift baskets. How many muffins are in each basket?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

10 **Evaluate $(4^3 - 14) \div (2 + 3) + 8^2$.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

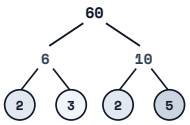
CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

Section 1.4 — Practice A: Prime Factorization

Key Vocabulary: factor pair • prime number • composite number • prime factorization • factor tree Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & HANDS-ON MODEL (PRIME FACTOR TOWERS)

Way 1: Factor Tree



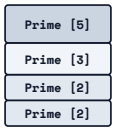
Circle prime leaves!

Way 2: Division Ladder

2 | 60
2 | 30
3 | 15
5 (Prime!)

60 = $2^2 \times 3 \times 5$

Way 3: Prime Blocks



Tower = 2·2·3·5 = 60

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Method 1 (Factor Tree): Branch a composite number into any factor pair until every leaf at the bottom is prime.

Textbook Method 2 (Division Ladder): Divide repeatedly by the smallest prime factors (2, 3, 5, ...) until the quotient is prime.

WATCH OUT: Keep branching until EVERY bottom leaf is PRIME (do not stop at composite numbers like 4, 6, 8, or 9; and remember 1 is NOT prime).

WORKED EXAMPLE — 3 TEXTBOOK WAYS (ANY 2 WAYS = +20 PTS)

Problem: Write the prime factorization of 60 using exponents.

WAY 1: Factor Tree Starting with 6 × 10

Branch 60 → 6 × 10. Split 6 → 2 × 3 and 10 → 2 × 5. Circle the prime leaves: $2 \times 2 \times 3 \times 5 = 2^2 \times 3 \times 5$.

WAY 2: Factor Tree Starting with 4 × 15

Branch 60 → 4 × 15. Split 4 → 2 × 2 and 15 → 3 × 5. Same prime leaves (and same prime blocks): $2^2 \times 3 \times 5$.

WAY 3: Division Ladder (Repeated Prime Division)

Divide by smallest primes step-by-step: 60 ÷ 2 = 30 → 30 ÷ 2 = 15 → 15 ÷ 3 = 5 (prime) → $2^2 \times 3 \times 5$.

Answer: 60 = $2^2 \times 3 \times 5$ 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 1.4 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

6 0 g [FACT]

Build your factor tree for 60 by hand until every branch ends at a prime. On your SC-6, test any branch with [PRIME] (flashes YES/NO) and type 60 [g][FACT] to verify the prime factorization $2^2 \times 3 \times 5$!

SECTION 1.4 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 Write the prime factorization of 12 using exponents.

Starter Step 1: Branch 12 into two factors (such as 2 × 6 or 3 × 4), keep branching until all leaves are prime, and write with exponents.

Way 1 (Paper Work):

Factor Tree

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

2 Write the prime factorization of 36 using exponents.

Starter Step 1: Branch 36 into two factors (such as 6 × 6), then branch each factor until only prime numbers remain.

Way 1 (Paper Work):

Factor Tree

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

3 Determine whether 29 and 91 are prime or composite. If composite, write the prime factorization.

Way 1 (Paper Work):

Factor Tree

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

4 Write the prime factorization of 45.

Way 1 (Paper Work):

Factor Tree

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

5 Two Factor Trees: Write the prime factorization of 72 using two different factor trees (starting with 8 × 9 and 6 × 12).

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

6 Write the prime factorization of 84.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

7 Find the prime factorization of 210.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

8 Write the prime factorization of 504 using exponents.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

9 Arabian Nights Mystery Number: Factor 1,001 into its three prime factors.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

10 Base-60 Ancient Astronomer Number: There are 3,600 seconds in 1 hour. Write the prime factorization of 3,600 using exponents.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

HANDS-ON MODEL: PRIME FACTOR TOWERS

Section 1.4 — Practice B: Modeling Real Life (Prime Factorization)

Textbook Ways: Factor Tree (Pair A) | Factor Tree (Pair B) | Division Ladder Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — MARCHING BAND FORMATIONS (3 TEXTBOOK WAYS • ANY 2 = +20 PTS)

Problem: A marching band has 120 musicians. Write the prime factorization of 120 and find two different rectangular row formations.

WAY 1: Factor Tree (12 × 10)
Split $120 = 12 \times 10 \rightarrow (4 \times 3) \times (2 \times 5) \rightarrow (2 \times 2 \times 3) \times (2 \times 5) = 2^3 \times 3 \times 5$.

WAY 2: Division Ladder by Smallest Primes
$120 \div 2 = 60 \rightarrow 60 \div 2 = 30 \rightarrow 30 \div 2 = 15 \rightarrow 15 \div 3 = 5 \rightarrow 2^3 \times 3 \times 5$.

WAY 3: Regroup Prime Blocks into Factor Pairs
Group [2][2][2][3][5] in two ways: $(2^3) \times (3 \times 5) = 8$ rows of 15, or $(2 \times 5) \times (2^2 \times 3) = 10$ rows of 12.

WATCH OUT: Keep branching until EVERY bottom leaf is PRIME (do not stop at composite numbers like 4, 6, 8, or 9; and remember 1 is NOT prime).

Answer: $2^3 \times 3 \times 5$ (e.g., 8 × 15 or 10 × 12)

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 1.4 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

2 [x^3] 3 × 5 ×

To check the whole number built from $2^3 \times 3 \times 5$ on your SC-6, press 2 [x^3] 3 [×] 5 [×] to confirm 120 (then press [g][FACT] to see its prime factors or [f][C.FAC] to compare two numbers).

When Maya tried to arrange 48 small solar cells into a neat rectangle on the cabin roof, she kept losing track of which row widths worked evenly and which ones left awkward gaps, while a spare batch of 37 cells wouldn't form any rectangle at all except one long, skinny line. She doubted she could ever find every possible layout without guessing all afternoon. Then Leo showed her how to branch 48 down a factor tree until only prime numbers ($2 \times 2 \times 2 \times 2 \times 3$) remained at the tips. Even when Maya started her tree with 6×8 and Leo started with 4×12, both trees ended at the exact same prime building blocks—giving her a foolproof way to see every factor pair inside a number.

SECTION 1.4 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

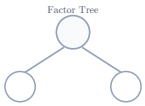
100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Solar Panel Grid: A satellite wing has 48 solar cells. List all factor pairs of 48 and write its prime factorization using exponents.**

Starter Step 1: Test divisors 1, 2, 3, 4, and 6 to list all factor pairs of 48, then build a factor tree to find its prime factorization.

Way 1 (Paper Work):

Factor Tree



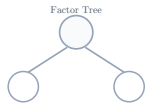
WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Drumline Formation: A drumline has 54 musicians. Write the prime factorization of 54.**

Starter Step 1: Divide 54 by the smallest prime (2), then keep branching the remaining factor into prime numbers.

Way 1 (Paper Work):

Factor Tree

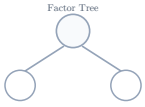


WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Prime Jersey Check: Three players wear jersey numbers 37, 51, and 61. Which jersey number is composite, and what is its prime factorization?**

Way 1 (Paper Work):

Factor Tree

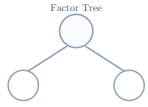


CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Mineral Display Case: A museum case holds 80 mineral specimens. Write the prime factorization of 80.**

Way 1 (Paper Work):

Factor Tree



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Mystery Composite Number: What whole number has the prime factorization $2 \times 3^2 \times 7$?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Century Mark: Write the prime factorization of 100 using exponents.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Songbird Brain Cells: Crows and parrots pack 140 million neurons into a single gram of brain tissue—denser than primates! Write the prime factorization of 140.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Gross of Pencils: A gross is 144 pencils (12 dozen). Write the prime factorization of 144 using exponents.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Two Factor Trees: Draw two different factor trees for 90—one starting with 9×10 and one starting with 6×15. How do their prime leaves compare?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Primorial Number: What is the smallest whole number that has the first four prime numbers as factors?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: PRIME FACTOR TOWERS

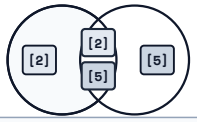
Section 1.5 — Practice A: Greatest Common Factor (GCF) & Least Common Multiple (LCM)

Key Vocabulary: common factor • greatest common factor (GCF) • common multiple • least common multiple (LCM) • Venn diagram

Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & HANDS-ON MODEL (PRIME FACTOR TOWERS)

Prime Factor Venn Diagram with Blocks (20 = 2·2·5 and 50 = 2·5·5)



Overlap = GCF: 2 × 5 = 10 | All Blocks = LCM: 2 × 2 × 5 × 5 = 100

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Method 1 (Lists): List all factor pairs to find the Greatest Common Factor (GCF); list multiples to find the Least Common Multiple (LCM).

Textbook Method 2 (Prime Venn Diagram): Overlap primes = GCF; union of all regions (highest power of each prime) = LCM.

WATCH OUT: GCF is a factor that divides IN (smaller or equal, Venn overlap); LCM is a multiple they build UP to (larger or equal, all Venn blocks).

WORKED EXAMPLE — 3 TEXTBOOK WAYS (ANY 2 WAYS = +20 PTS)

Problem: Find the greatest common factor (GCF) and least common multiple (LCM) of 20 and 50.

WAY 1: List Factors (for GCF) & Multiples (for LCM)
Factors of 20: 1, 2, 4, 5, 10, 20; of 50: 1, 2, 5, 10, 25, 50 → GCF = 10. Multiples: 20, 40, 60, 80, 100 & 50, 100 → LCM = 100.

WAY 2: Prime Factorization Powers
$20 = 2^2 \times 5$ and $50 = 2 \times 5^2$. Shared prime factors = $2 \times 5 = 10$ (GCF). Highest power of each prime = $2^2 \times 5^2 = 100$ (LCM).

WAY 3: Prime Factor Venn Diagram (With Blocks)
Left circle only: [2]. Center overlap: [2] and [5] → GCF = 2×5 = 10. Right circle only: [5]. All regions: 2×10×5 = 100 (LCM).

Answer: GCF = 10, LCM = 100 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 1.5 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

2 0 ENTER 5 0 GCF 2 0 ENTER 5 0 f LCM

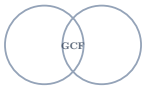
Find the GCF and LCM of 20 and 50 on paper first. Then check on your SC-6:
Press 20 [ENTER] 50 [GCF] to confirm GCF = 10, and 20 [ENTER] 50 [f][LCM] to confirm LCM = 100 (or [f][C.FAC] for shared primes).

SECTION 1.5 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Find the GCF and LCM of 12 and 18.**
Starter Step 1: Factor 12 and 18 into prime numbers; multiply the shared overlap primes for GCF, and all Venn primes for LCM.

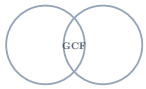
Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Find the GCF and LCM of 8 and 12.**
Starter Step 1: List all factors of 8 and 12 to find the GCF, then count up by multiples of 12 to find the LCM.

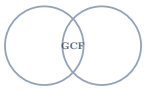
Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Find the GCF and LCM of 15 and 25.**

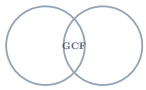
Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Find the GCF and LCM of 14 and 35.**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Trail Crew Snack Packs: Maya has 48 granola bars and 72 fruit leather strips to pack into identical trail-snack pouches with none left over. What is the greatest number of identical pouches she can make?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Field Trip Snack Bags: You have 84 cheddar crackers and 140 pretzel sticks to pack into identical snack bags for a hike with none left over. What is the greatest number of identical snack bags you can make?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Find the GCF and LCM of 45 and 75.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Park Shuttle Cycle: The North Trail shuttle completes a loop every 15 minutes and the South Trail shuttle every 20 minutes. If they leave the station together, in how many minutes will they meet at the station again?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Municipal Traffic Signals: Along an avenue, the North-South green light cycles every 45 seconds and the pedestrian walk signal cycles every 60 seconds. If both trigger together now, how many seconds pass before they trigger together again?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Robotics Part Bags: Leo has 36 red LEGO Technic axles and 48 blue connector pins to pack into identical team kit bags with none left over. What is the greatest number of kit bags he can make, and how many of each part go into each bag?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: PRIME FACTOR TOWERS

Section 1.5 — Practice B: Modeling Real Life (Greatest Common Factor (GCF) & Least Common Multiple (LCM))

Textbook Ways: List Factors & Multiples | Prime Factorization | Prime Factor Venn Diagram Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — IDENTICAL SNACK PACKS (3 TEXTBOOK WAYS • ANY 2 = +20 PTS)

Problem: You have 18 pretzel bags and 24 fruit bars to make identical snack packs with none left over. What is the greatest number of packs you can make?

WAY 1: List All Factors of 18 and 24
Factors of 18: 1, 2, 3, 6, 9, 18. Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24. Greatest common factor = 6 packs.

WAY 2: Prime Factor Venn Diagram
$18 = 2 \times 3 \times 3$ and $24 = 2 \times 2 \times 2 \times 3$. Center overlap has [2] and [3] → GCF $= 2 \times 3 = 6$ packs (left has 3 pretzels; right has 4 bars).

WAY 3: Common Prime Division Ladder
Divide (18, 24) by 2 → (9, 12); divide by 3 → (3, 4). Product of side divisors $= 2 \times 3 = 6$ packs (with 3 and 4 inside).

WATCH OUT: GCF is a factor that divides IN (smaller or equal, Venn overlap); LCM is a multiple they build UP to (larger or equal, all Venn blocks).

Answer: 6 packs (3 pretzels and 4 fruit bars per pack)

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 1.5 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

1 8 ENTER 2 4 GCF 1 8 ENTER 2 4 f LCM

After solving by hand, press 18 [ENTER] 24 [GCF] on your SC-6 to verify 6 identical snack packs (and press 18 [ENTER] 24 [f][LCM] to confirm 72 when checking repeating cycles).

Leo used to dread word problems with GCF and LCM because the two three-letter names sounded almost identical, and he always guessed the wrong one on quizzes. While helping pack 28 bandages and 42 antiseptic wipes into identical trail first-aid kits, he paused and asked himself what the physical objects were actually doing. Splitting supplies into smaller equal bags with zero leftovers meant dividing IN to find the biggest shared building block—the Greatest Common Factor (14 kits). Later, watching two camp shuttles on 18-minute and 24-minute loops and two meshing 14-tooth and 21-tooth LEGO gears, he realized repeating cycles build UP into the future until they meet at the Least Common Multiple.

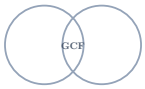
SECTION 1.5 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **First-Aid Kits: A school nurse has 28 bandages and 42 antiseptic wipes to pack into identical first-aid kits with none left over. What is the greatest number of kits she can make?**

Starter Step 1: 'Greatest number of identical kits' means GCF—find the Greatest Common Factor of 28 bandages and 42 wipes.

Way 1 (Paper Work):

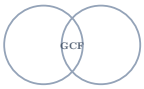


WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Periodical Insect Cycles: Insect Species A emerges every 12 years and Species B peaks every 15 years. If both peak together this year, in how many years will they next peak together?**

Starter Step 1: 'Next peak together' means LCM—find the Least Common Multiple of the 12-year and 15-year cycles.

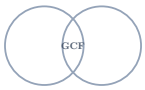
Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Cookout Packages: Hot dogs come in packs of 8 and buns come in packs of 12. What is the least number of hot dogs and buns you can buy to have an equal number of each?**

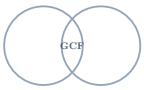
Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Choir Riser Rows: The 6th-grade choir has 32 altos and 48 sopranos. Each row must have the same number of singers and only one voice part. What is the greatest number of singers that can be placed in each row?**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Bus Station Loop: Route A returns to the station every 18 minutes and Route B returns every 24 minutes. Both leave at 9:00 AM. At what time will both buses next be at the station together?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Square Quilt Patches: You have a sheet of fabric 36 inches wide by 60 inches long and want to cut it into the largest possible equal squares with no fabric left over. What is the side length of each square?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Meshing LEGO Gears: As a 14-tooth gear and a 21-tooth gear turn together, their teeth click 1-for-1. (a) What is the largest equal tooth-group size for both gears? (b) How many teeth click past before two marked teeth meet again?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Triple Track Laps: Three go-karts complete a lap in 4, 6, and 10 minutes. If they start together, after how many minutes will all three cross the start line together again?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Balloon Bouquets: You have 45 gold balloons and 75 silver balloons to tie into identical bouquets using all the balloons. What is the greatest number of bouquets you can make, and how many gold balloons go in each?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **GCF & LCM Product Pattern: Find the GCF and LCM of 15 and 20. Compare the product GCF × LCM to the product 15 × 20. What do you notice?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

Unit 2: Fractions & Decimals — The Kitchen Bakery, the Avian Lab & the 3D Print Shop

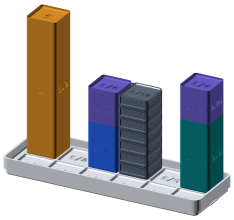
Equivalent Fractions, Multiplying & Dividing Fractions, and All Four Decimal Operations

If you spend an afternoon helping Elena in her family's bakery kitchen, volunteering with Zoe at the bird-rehab aviary, or watching Marcus run the school's desktop 3D printer, you quickly run into an annoying truth: real measurements almost never land on whole numbers. Elena's measuring scoops are stamped in halves, thirds, quarters, and eighths, and she used to dread fractions because every rule felt backward—why did a bigger number on the bottom mean a smaller scoop?

Next door at the wildlife center, Zoe watches New Caledonian crows bend hooked twigs to pull grubs out of logs and drop pebbles into water tubes, raising the water level by tiny fractions of a centimeter with each splash until a floating treat rises within reach. To know how many pebbles a crow needs, Zoe has to figure out how many 3/8-centimeter splashes fit inside a 4 1/2-centimeter rise.

Meanwhile, in the maker room, Marcus deals with decimals every time he prints a part: the 3D printer's 'slicer' software chops a 14.4-millimeter bracket into paper-thin horizontal layers, each just 0.20 millimeters thick, and melts corn-based plastic one layer at a time. Out on the track, the stopwatch times 100-meter sprints to the hundredth of a second, where 0.55 seconds is the gap between first and second place.

At first, Elena, Zoe, and Marcus all second-guessed themselves whenever fractions or decimals showed up. Across these lessons, they learn how to trade unequal fraction slices for matching pieces, why taking a 'part of a part' shrinks a fraction, how dividing fractions counts how many scoops fit inside a whole, and how to verify every fraction and decimal step on the SC-6 calculator.



Real-World Mathematical Context & Two-Method Mastery • Hands-On Models: Fraction Strips & Place Value Blocks

UNIT 2 LESSONS — LEARN TWO TEXTBOOK WAYS FIRST, THEN CHECK YOUR WORK ON THE SC-6 CALCULATOR

SEC 2.1 (pp.15-16) Equivalent Fractions, Mixed Numbers & LCD Addition

Textbook Ways: Way 1: Divide Once by GCF / Use LCD | Way 2: Step-by-Step Common Factors | Way 3: Fraction Strips Model

Learn by Hand: In Big Ideas Math, divide the numerator and denominator by their GCF to write a fraction in simplest form, and rewrite unlike fractions using their Least Common Denominator (LCD) to add or subtract. With Fraction Strips, equivalent fractions match in length, and trading 1/3 + 1/4 for twelfths shows 4/12 + 3/12 = 7/12.

Check Your Work on the SC-6 Calculator:

[.] [1] [2] [.] [1] [8] [ENTER] [g] [SIMP] [g] [SIMP]

Type [.] 12 [.] 18 [ENTER], then press [g][SIMP] to step through common factors to 2/3!

SEC 2.2 (pp.17-18) Multiplying Fractions and Mixed Numbers

Textbook Ways: Way 1: Multiply Straight Across | Way 2: Cross-Cancel Common Factors First | Way 3: Area Model / Distributive Split

Learn by Hand: In Big Ideas Math, multiply fractions by multiplying the numerators and multiplying the denominators: (a/b) × (c/d) = (ac)/(bd). For mixed numbers, write each mixed number as an improper fraction first, and divide out common factors before multiplying. Partitioning a 3/4 Fraction Strip in half visually shows why 1/2 of 3/4 is 3/8.

Check Your Work on the SC-6 Calculator:

[.] [1] [.] [2] [ENTER] [.] [3] [.] [4] [×]

Type [.] 1 [.] 2 [ENTER] [.] 3 [.] 4 [×] to verify 3/8 in simplest form.

SEC 2.3 (pp.19-20) Dividing Fractions and Mixed Numbers

Textbook Ways: Way 1: Multiply by the Reciprocal | Way 2: Common-Denominator Division | Way 3: Fraction Strip / Number-Line Model

Learn by Hand: In Big Ideas Math, you can divide fractions in two ways: (1) Multiply by the Reciprocal of the divisor: (a/b) ÷ (c/d) = (a/b) × (d/c), or (2) Rewrite both fractions with a Common Denominator and divide their numerators! Placing 1/8 Fraction Strips alongside a 3/4 strip shows why six 1/8s fit inside 3/4.

Check Your Work on the SC-6 Calculator:

[.] [3] [.] [4] [ENTER] [.] [1] [.] [8] [f] [1/x] [×]

Type [.] 3 [.] 4 [ENTER] [.] 1 [.] 8 [÷] (or press [f][1/x] [×] to flip 1/8 to 8/1 and multiply) to confirm 6!

SEC 2.4 (pp.21-22) Adding, Subtracting, and Multiplying Decimals

Textbook Ways: Way 1: Vertical Place-Value Algorithm | Way 2: Convert to Fractions/Hundredths | Way 3: Place-Value Split / Area Model

Learn by Hand: To add or subtract decimals, line up the decimal points vertically so ones, tenths, and hundredths align, and annex trailing zeros as needed. To multiply decimals, multiply as whole numbers and count the total number of decimal places in the factors—always checking with a whole-number front-end estimate.

Check Your Work on the SC-6 Calculator:

[1] [4] [.] [6] [ENTER] [3] [.] [8] [5] [+] [f] [LAST x] [-]

Line up the decimal points and add 14.60 + 3.85 by hand first. On your SC-6, check any place-value digit with [PV] (e.g., 14.6 [ENTER] 0.1 [PV]), then press 14.6 [ENTER] 3.85 [+] to confirm 18.45!

SEC 2.5 (pp.23-24) Dividing Whole Numbers and Decimals

Textbook Ways: Way 1: Power-of-10 Decimal Shift & Long Division | Way 2: Equivalent Fraction Scaling | Way 3: Reciprocal Fraction Division

Learn by Hand: When dividing by a decimal (such as 6.75 ÷ 0.25), multiply both the divisor and the dividend by the same power of 10 (shift both decimal points the same number of places to the right) so the divisor becomes a whole number (675 ÷ 25 = 27).

Check Your Work on the SC-6 Calculator:

[2] [3] [ENTER] [4] [÷] [FDISP] [f] [d/c]

Divide 23 ÷ 4 by hand first. On your SC-6, press 23 [ENTER] 4 [÷] and tap [FDISP], [a b/c], and [f][d/c] to cycle through 5.75, 5 3/4, and 23/4 (or use [f][0.3-rep] for repeating decimals)!

HANDS-ON MODEL: FRACTION STRIPS

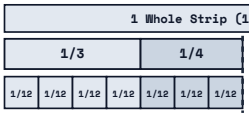
Section 2.1 — Practice A: Equivalent Fractions, Mixed Numbers & LCD Addition

Key Vocabulary: equivalent fractions • simplest form • least common denominator (LCD) • mixed number • improper fraction

Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & HANDS-ON MODEL (FRACTION STRIPS)

Fraction Strips: Trade 1/3 + 1/4 for LCD Twelfths (7/12)



4/12 + 3/12 = 7/12 (Exact Length Match!)

Simplify:
12/18 = 2/3

Mixed Number:
2 1/3 = 7/3
(3×2 + 1)/3

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept: Divide numerator and denominator by their GCF to simplify; use LCD = LCM of denominators to add or subtract fractions.

Textbook Mixed Numbers: Multiply Whole × Denominator + Numerator for improper fractions (a b/c = (ac+b)/c); divide to convert back.

WATCH OUT: Never add denominators directly (1/3 + 1/4 ≠ 2/7)! Rewrite with the Least Common Denominator (LCD) first, then add only the numerators.

WORKED EXAMPLE — 3 TEXTBOOK WAYS (ANY 2 WAYS = +20 PTS)

Problem: Write 12/18 in simplest form.

WAY 1: Divide by the GCF in One Step
Find GCF(12, 18) = 6. Divide numerator and denominator by 6: (12 ÷ 6) / (18 ÷ 6) = 2/3.

WAY 2: Divide Step-by-Step by Small Prime Factors
Divide top and bottom of 12/18 by 2 first → 6/9. Then divide top and bottom by 3 → 2/3.

WAY 3: Prime Factor Cancellation / Fraction Strips
Write 12/18 = (2 × 2 × 3) / (2 × 3 × 3). Cancel the shared (2 × 3) → 2/3.

Answer: 2/3 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 2.1 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

• 1 2 • 1 8 ENTER g SIMP g SIMP

Simplify 12/18 by hand first. On the SC-6 (like the HP-32), enter fractions with the Two-Dot [.] syntax ([.] n [.] d): Type [.] 12 [.] 18 [ENTER], then press [g][SIMP] to step through common factors to 2/3!

SECTION 2.1 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Write 8/12 in simplest form.**
Starter Step 1: Find the Greatest Common Factor (GCF) of numerator 8 and denominator 12, then divide both top and bottom by that GCF.

Way 1 (Paper Work):

Divide Top & Bottom by GCF:

(___ ÷ ___) / (___ ÷ ___) = ___/___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Championship Pizza Party: A team eats 48 slices from trays cut into 18ths (48/18). Write 48/18 in simplest form and as a mixed number.**
Starter Step 1: Simplify 48/18 by dividing top & bottom by their GCF, then divide numerator ÷ denominator for a mixed number.

Way 1 (Paper Work):

Whole × Denom + Num:

Improper = ___/___ Mixed = ___ ___/___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Write the mixed number 2 3/5 as an improper fraction and as a decimal.**

Way 1 (Paper Work):

Whole × Denom + Num:

Improper = ___/___ Mixed = ___ ___/___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Pizza Slice Inventory: After a festival, 23 sixth-slices (23/6) remain. How many whole pizzas and leftover sixth-slices can be boxed?**

Way 1 (Paper Work):

LCD = _____

___/___ ± ___/___ = ___/___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Add 1/2 + 1/3. Write the sum in simplest form.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Camp Lemonade Blend: Elena mixes 3/4 gallon of lemonade with 2/5 gallon of berry tea in a cooler. How many gallons of drink are in the cooler?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Lookout Balcony Rail: A carpenter welds a 1 1/2-foot bracket to a 2 1/3-foot rail. What is the combined length?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Sawing a Diagonal Cross-Brace: A timber frame carpenter trims 2 3/4 feet from a 5 1/3-foot cedar beam. How long is the remaining beam?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Climbing Wall Progress: Zoe climbs 1/2 of a rock wall in Minute 1, 1/4 in Minute 2, and 1/8 in Minute 3. What total fraction of the wall has she climbed, and what fraction remains?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Smoothie Pitcher Mix: Elena blends 3 1/2 pints of strawberry puree with 1 3/4 pints of peach juice. How many pints of smoothie does this make?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: FRACTION STRIPS

Section 2.1 — Practice B: Modeling Real Life (Equivalent Fractions, Mixed Numbers & LCD Addition)

Textbook Ways: Divide Once by GCF / Use LCD | Step-by-Step Common Factors | Fraction Strips Model Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — COMBINING TRAIL LENGTHS (3 TEXTBOOK WAYS • ANY 2 = +20 PTS)

Problem: A hiker walks a 1 2/3-mile ridge trail and a 1 1/4-mile creek trail. How many miles does the hiker walk in all?

WAY 1: Add Whole Numbers & LCD Fractions Separately
LCD(3, 4) = 12. Rename: 1 8/12 + 1 3/12. Add wholes (1 + 1 = 2) and twelfths (8/12 + 3/12 = 11/12) → 2 11/12 miles.

WAY 2: Convert Both to Improper Fractions First
1 2/3 = 5/3 = 20/12 and 1 1/4 = 5/4 = 15/12. Add: 20/12 + 15/12 = 35/12 = 2 11/12 miles.

WAY 3: Fraction Strips Trade (Twelfths)
Combine 2 whole bars, trade two 1/3 bars for eight 1/12 bars, and trade one 1/4 bar for three 1/12 bars → 2 11/12 miles.

WATCH OUT: Never add denominators directly (1/3 + 1/4 ≠ 2/7)! Rewrite with the Least Common Denominator (LCD) first, then add only the numerators.

Answer: 2 11/12 miles (35/12 miles)

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 2.1 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

`1` `•` `2` `•` `3` `ENTER` `1` `•` `1` `•` `4` `+`

Add 1 2/3 + 1 1/4 on paper first. On your SC-6, enter mixed numbers as w [.] n [.] d:
Type 1 [.] 2 [.] 3 [ENTER] 1 [.] 1 [.] 4 [+] to confirm 2 11/12 (press [f][d/c] for 35/12 or [a b/c] for mixed view).

In the bakery kitchen, Elena poured 3/8 cup of blueberries and 1/4 cup of strawberries into a blender and froze when she tried to write down the total. Part of her wanted to just add across the top and bottom to get 4/12, even though she had a nagging feeling that wasn't right—how could adding berries together give her less than she started with? Picking up the clear glass measuring cup, she looked closely at the etched lines and noticed that every 1/4-cup mark was the exact same height as two 1/8-cup marks. Once she traded 1/4 for 2/8 so all the scoops were the same size (eighths), adding 3/8 + 2/8 = 5/8 cup finally made complete sense.

SECTION 2.1 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Smoothie Recipe: You blend 3/8 cup of blueberries and 1/4 cup of strawberries. How many cups of berries did you use in all?**

Starter Step 1: Rewrite 1/4 as an equivalent fraction with denominator 8 so both fractions share LCD = 8, then add the numerators.

Way 1 (Paper Work):

LCD = _____

___/___ ± ___/___ = ___/___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **3D Print Progress: A print is 18/24 complete. Write the completed fraction in simplest form, and find what fraction of the print remains.**

Starter Step 1: Simplify 18/24 by dividing top & bottom by their GCF, then subtract from 1 whole to find the fraction remaining.

Way 1 (Paper Work):

LCD = _____

___/___ ± ___/___ = ___/___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Skateboard Hardware: A bolt is 1 5/8 inches long. Write 1 5/8 as an improper fraction and as a decimal.**

Way 1 (Paper Work):

Whole × Denom + Num:

Improper = ___/___ Mixed = ___ ___/___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Crow Twig Tools: A New Caledonian crow bends a hooked twig 5/6 foot long to fish a grub from a log, while a second crow uses a 3/8-foot twig. How much longer is the first crow's tool?**

Way 1 (Paper Work):

LCD = _____

___/___ ± ___/___ = ___/___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Pizza Leftovers: After a party, there are 2 1/3 cheese pizzas and 1 1/2 pepperoni pizzas left. How many pizzas are left in all?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Carpentry Trim: You cut 1 3/4 feet off a 4 1/2-foot board. How many feet of board are left?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Free-Throw Practice: Maya made 15 out of 25 shots. Write the fraction of shots she made in simplest form and as a decimal.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Science Beaker Mix: A beaker holds 2/5 liter of water. You pour in 3/10 liter more, then pour out 1/2 liter. How much water is in the beaker now?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Ribbon Spool: You need 2 2/3 yards of blue ribbon and 1 3/4 yards of gold ribbon. If your spool holds 5 yards, how much ribbon do you use and how much is left?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Triathlon Training: You swim 3/8 mile, bike 2 1/2 miles, and run 1 3/4 miles. What is the total distance of your workout?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: FRACTION STRIPS

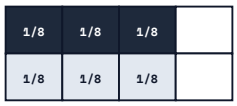
Section 2.2 — Practice A: Multiplying Fractions and Mixed Numbers

Key Vocabulary: product of fractions • area model • cross-canceling (dividing out common factors) • improper fraction

Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & HANDS-ON MODEL (FRACTION STRIPS)

Way 1: Area Model (1/2 of 3/4 = 3/8)



3/4 Shaded Columns → 3/8 Overlap

Way 2 & 3: Multiply or Cross-Cancel

Multiply Across: (1×3) / (2×4) = 3/8

Mixed Numbers: 2 2/3 × 1 1/2 = 8/3 × 3/2

Cross-Cancel Common Factors First:

(8÷2)/(3÷3) × (3÷3)/(2÷2) = 4/1 × 1/1 = 4

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept: Multiply numerators and multiply denominators: (a/b) × (c/d) = (a·c)/(b·d). Write mixed numbers as improper fractions first.

Textbook Shortcut (Cross-Canceling): Divide out common factors between any numerator and denominator BEFORE multiplying.

WATCH OUT: For mixed numbers, ALWAYS convert to improper fractions (a b/c = (ac+b)/c) BEFORE multiplying!

WORKED EXAMPLE — 3 TEXTBOOK WAYS (ANY 2 WAYS = +20 PTS)

Problem: Find 1/2 × 3/4.

WAY 1: Multiply Numerators & Multiply Denominators

Multiply numerators across (1 × 3 = 3) and denominators across (2 × 4 = 8) → (1 × 3) / (2 × 4) = 3/8.

WAY 2: 2D Area Model Grid (Shaded Overlap)

Draw a 4-column by 2-row grid (8 eighths). Shade 3 columns (3/4) and 1 row (1/2); the overlap is 3 of 8 boxes = 3/8.

WAY 3: Partition Fraction Strips

Trade three 1/4 strips for six 1/8 strips, then take half of those six strips → three 1/8 strips = 3/8.

Answer: 3/8

1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 2.2 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[•] [1] [•] [2] [ENTER] [•] [3] [•] [4] [×]

Multiply numerators and denominators (or cross-cancel) by hand first. Then check on your SC-6 using Two-Dot fraction entry: Type [.] 1 [.] 2 [ENTER] [.] 3 [.] 4 [×] to verify 3/8 in simplest form.

SECTION 2.2 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Find 1/2 × 2/3. Write the product in simplest form.**

Starter Step 1: Cross-cancel the common factor 2 on the diagonal (or multiply numerators and denominators and simplify).

Way 1 (Paper Work):

Setup: (___/___) × (___/___)

= (___ × ___) / (___ × ___) = ___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Multiply 2/3 × 3/4. Divide out common factors before multiplying.**

Starter Step 1: Cross-cancel the common factors on the diagonals BEFORE multiplying the numerators and denominators.

Way 1 (Paper Work):

Setup: (___/___) × (___/___)

= (___ × ___) / (___ × ___) = ___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Multiply 3/5 × 5/9.**

Way 1 (Paper Work):

Setup: (___/___) × (___/___)

= (___ × ___) / (___ × ___) = ___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Cranberry Trail Mix: A recipe for 16 cups of trail mix is 3/8 dried cranberries. How many cups of cranberries are in the batch?**

Way 1 (Paper Work):

Setup: (___/___) × (___/___)

= (___ × ___) / (___ × ___) = ___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Multiply 1 3/4 × 2 2/7.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Multiply 2 1/4 × 1 1/3.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **High-Energy Trail Cookies: One batch of trail cookies uses 2 1/4 cups of rolled oats. How many cups of oats are needed to bake 4 batches?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Without calculating, is 5 × 2/3 greater than 5 or less than 5? Explain why, then find the product.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Light-Rail Blueprint Scale: On an transit map, 1 inch represents 3 1/2 miles. If a proposed rail line measures 4 1/4 inches on the map, how many miles long is the actual line?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Robot Speed Gears: When a small gear turns a bigger gear, the spin slows down. In a LEGO robot, Gear 1 spins at 1/2 the motor's speed, Gear 2 at 2/3 of Gear 1, and Gear 3 at 3/4 of Gear 2. What fraction of the input speed comes out of Gear 3?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

Section 2.2 — Practice B: Modeling Real Life (Multiplying Fractions and Mixed Numbers)

Textbook Ways: Multiply Straight Across | Cross-Cancel Common Factors First | Area Model / Distributive Split Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — SCALING A RECIPE (3 TEXTBOOK WAYS • ANY 2 = +20 PTS)

Problem: A batch of granola uses 1 1/2 cups of oats. You make 2 2/3 batches. How many cups of oats do you need?

WAY 1: Improper Fractions & Multiply Straight Across
Write 1 1/2 = 3/2 and 2 2/3 = 8/3. Multiply across: (3 × 8) / (2 × 3) = 24 / 6 = 4 cups.

WAY 2: Divide Out (Cross-Cancel) Common Factors First
In (3/2) × (8/3), divide both 3s by 3 (→ 1) and divide 8 and 2 by 2 (→ 4 and 1): (1/1) × (4/1) = 4 cups.

WAY 3: Distributive Property (2 Batches + 2/3 Batch)
2 batches use 2 × 1 1/2 = 3 cups. Two-thirds of a batch uses (2/3) × (3/2) = 1 cup. Total = 3 + 1 = 4 cups.

WATCH OUT: For mixed numbers, ALWAYS convert to improper fractions (a b/c = (ac+b)/c) BEFORE multiplying!

Answer: 4 cups

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 2.2 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

`1` `•` `1` `•` `2` `ENTER` `2` `•` `2` `•` `3` `×`

Convert mixed numbers to improper fractions and multiply on paper first. Then check on your SC-6:
Type 1 [.] 1 [.] 2 [ENTER] 2 [.] 2 [.] 3 [×] to confirm 4 cups.

Elena always assumed multiplying had to make a number get bigger, so when her phone battery was at 4/5 charge and a game used 1/2 of that charge, she was sure she needed to divide—and got a bigger number that made no sense. Discouraged, she sketched a bar split into fifths, shaded 4 of them, and sliced that shaded part in half. Seeing only 2/5 of the whole bar left, she realized that taking a 'fraction of a fraction' shrinks the piece because you are taking part of a part. Better yet, when scaling a 1 3/4-cup pancake recipe by 2 1/2 batches, dividing out matching factors on the diagonal kept the numbers small enough to multiply without getting lost.

SECTION 2.2 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Garden Herb Plot: Two-thirds of a garden is planted with vegetables, and 3/5 of the vegetables are tomatoes. What fraction of the whole garden is planted with tomatoes?**

Starter Step 1: '3/5 of 2/3' means multiply (2/3) × (3/5)—look for a common factor on the diagonal to cross-cancel first!

Way 1 (Paper Work):

Setup: (___/___) × (___/___)

= (___ × ___) / (___ × ___) = ___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Phone Battery Drain: Your phone battery is at 4/5 charge. Playing a video game uses 1/2 of that charge. What fraction of a full battery did the game use?**

Starter Step 1: '1/2 of 4/5' means multiply (4/5) × (1/2)—cross-cancel the even numbers on the diagonal before multiplying.

Way 1 (Paper Work):

Setup: (___/___) × (___/___)

= (___ × ___) / (___ × ___) = ___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Pancake Recipe Doubler: A pancake recipe calls for 1 3/4 cups of flour. You make 2 1/2 batches for a team breakfast. How many cups of flour do you need?**

Way 1 (Paper Work):

Improper: (___/___) × (___/___)

= (___ × ___) / (___ × ___) = ___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Laptop Sticker Area: A rectangular laptop sticker measures 2 1/4 inches wide by 1 1/3 inches tall. What is its area?**

Way 1 (Paper Work):

Setup: (___/___) × (___/___)

= (___ × ___) / (___ × ___) = ___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Trail Hike Distance: You hike at a pace of 3 1/5 miles per hour for 1 1/4 hours. How many miles do you hike?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Art Canvas Frame: A rectangular painting is 3/4 yard wide and 8/9 yard tall. What is the area of the painting in square yards?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Rain Barrel Water: A 24-gallon rain barrel is 5/8 full. You use 2/3 of the water in the barrel for the garden. How many gallons of water did you use?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Fractional Cube Volume: What is the volume of a cube whose edges are each 1/2 inch long?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Skate Ramp Plywood: A rectangular plywood sheet measures 3 1/2 feet by 2 2/7 feet. What is the area of the plywood sheet?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Sleeping in Flight: Migrating frigatebirds sleep 1/3 of each 24-hour day, and 1/4 of that sleep happens while flying with one brain hemisphere awake! What fraction of the day (and how many hours) do they sleep in flight?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: FRACTION STRIPS

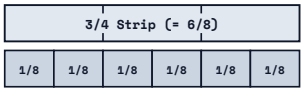
Section 2.3 — Practice A: Dividing Fractions and Mixed Numbers

Key Vocabulary: reciprocals • multiplicative inverses • common-denominator division • measurement division

Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & HANDS-ON MODEL (FRACTION STRIPS)

Way 1: Fraction Strips (How many 1/8s in 3/4?)



Six 1/8-strips fit inside 3/4!

Way 3 (LCD): 6/8 ÷ 1/8 = 6 ÷ 1 = 6

Way 2: Multiply by the Reciprocal

3/4 ÷ 1/8

Keep Change Flip!

= 3/4 × 8/1 = 24/4 = 6

Reciprocals: 1/8 × 8/1 = 1

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Method 1 (Multiply by the Reciprocal): Keep the first fraction, change ÷ to ×, and flip the divisor: (a/b) ÷ (c/d) = (a/b) × (d/c).

Textbook Method 2 (Common Denominators): Rewrite both fractions with the LCD and divide their numerators: (a/d) ÷ (b/d) = a ÷ b.

WATCH OUT: Keep the FIRST fraction as-is and flip ONLY the SECOND fraction (the divisor): a/b ÷ c/d = a/b × d/c.

WORKED EXAMPLE — 3 TEXTBOOK WAYS (ANY 2 WAYS = +20 PTS)

Problem: Find 3/4 ÷ 1/8.

WAY 1: Multiply by the Reciprocal

Keep 3/4, change ÷ to ×, and flip 1/8 to its reciprocal 8/1: (3/4) × (8/1) = 24/4 = 6.

WAY 2: Common Denominator Method

Rewrite both fractions with LCD = 8: 6/8 ÷ 1/8. Divide the numerators: 6 eighths ÷ 1 eighth = 6 ÷ 1 = 6.

WAY 3: Fraction Strips / Number-Line Jumps (With Blocks)

Mark 0 to 3/4 (6/8) on a number line or fraction tray and count jumps of 1/8: it takes 6 jumps of 1/8 to reach 3/4.

Answer: 6 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 2.3 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[•] [3] [•] [4] [ENTER] [•] [1] [•] [8] [f] [1/x] [×]

Multiply by the reciprocal on paper first. On your SC-6, check with Two-Dot fractions:
Type [.] 3 [.] 4 [ENTER] [.] 1 [.] 8 [÷] (or press [f][1/x] [×] to flip 1/8 to 8/1 and multiply) to confirm 6!

SECTION 2.3 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Divide 1/2 ÷ 1/4.**

Starter Step 1: Keep the first fraction (1/2), change ÷ to ×, and flip 1/4 to its reciprocal: (1/2) × (4/1).

Way 1 (Paper Work):

Keep × Flip (Reciprocal)

(___/___) × (___/___) = ___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Divide 2/3 ÷ 1/6.**

Starter Step 1: Keep the first fraction (2/3), change ÷ to ×, and flip 1/6 to its reciprocal: (2/3) × (6/1).

Way 1 (Paper Work):

Keep × Flip (Reciprocal)

(___/___) × (___/___) = ___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Divide 5/6 ÷ 5/12. Write the quotient in simplest form.**

Way 1 (Paper Work):

Keep × Flip (Reciprocal)

(___/___) × (___/___) = ___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Divide 6 ÷ 3/4.**

Way 1 (Paper Work):

Keep × Flip (Reciprocal)

(___/___) × (___/___) = ___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Divide 3/4 ÷ 2/5. Write the answer as a mixed number.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Quarter-Batch Polymer Clay: A sculpture recipe uses 3 1/2 cups of base clay. If you divide the recipe equally into 4 mini-batches, how much clay is in each mini-batch?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Ceramic Bike Frame Polish: A mechanic has 2/3 liter of polish. Each bike frame requires 1/4 liter. How many frames (including fraction of a frame) can be coated?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Divide 5 1/4 ÷ 1 3/4 and check your answer by multiplying.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Oak Crate Slats: A carpenter saws an 8-foot (96-inch) oak board into crate slats that are each 18 1/2 inches long. How many full slats can be sawed from the board?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Smoothie Pitcher Servings: A 3 3/4-cup pitcher of berry smoothie is poured into 5/8-cup glasses. How many glasses does it fill?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: FRACTION STRIPS

Section 2.3 — Practice B: Modeling Real Life (Dividing Fractions and Mixed Numbers)

Textbook Ways: Multiply by the Reciprocal | Common-Denominator Division | Fraction Strip / Number-Line Model Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — CUTTING WOOD PLANKS (3 TEXTBOOK WAYS • ANY 2 = +20 PTS)

Problem: A carpenter cuts a 6 3/4-foot cedar board into shelves that are each 1 1/8 feet long. How many shelves can be cut?

WAY 1: Improper Fractions & Multiply by the Reciprocal
Write 6 3/4 = 27/4 and 1 1/8 = 9/8. Multiply by the reciprocal: (27/4) × (8/9) = (3/1) × (2/1) = 6 shelves.

WAY 2: Common Denominator (Eighths)
Rewrite both in eighths: 6 3/4 = 54/8 and 1 1/8 = 9/8. Divide the numerators: 54 eighths ÷ 9 eighths = 54 ÷ 9 = 6 shelves.

WAY 3: Repeated Subtraction / Grouping
Two shelves use 2 × 1 1/8 = 2 1/4 ft; four shelves use 4 1/2 ft; six shelves use 6 3/4 ft → 6 shelves.

WATCH OUT: Keep the FIRST fraction as-is and flip ONLY the SECOND fraction (the divisor): a/b ÷ c/d = a/b × d/c.

Answer: 6 shelves

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 2.3 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:



Rewrite 6 3/4 ÷ 1 1/8 as improper fractions and multiply by the reciprocal by hand. Then check on your SC-6: Type 6 [.] 3 [.] 4 [ENTER] 1 [.] 1 [.] 8 [÷] to verify 6 shelves.

Whenever a teacher said 'keep, change, flip' for dividing fractions, Zoe followed the steps like a robot but secretly hated that she had no idea why flipping the second fraction worked. At the bird-rehab center, she watched a crow drop pebbles into a clear tube where each pebble pushed the water up by 3/8 centimeter until the water rose 4 1/2 centimeters. She asked herself a simpler question first: how many 3/8-cm splashes does it take to raise the water 1 full centimeter? Since 1 centimeter is 8/8, it takes 8/3 pebbles per centimeter—which is the flipped fraction (the reciprocal)! Multiplying 4 1/2 centimeters by 8/3 pebbles per centimeter gave 12 pebbles, and for the first time the flip rule wasn't magic anymore.

SECTION 2.3 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Pizza Slice Sharing: How many 1/8-size slices are in 3/4 of a pizza?**
Starter Step 1: Set up (3/4) ÷ (1/8), then multiply by the reciprocal of the slice size.

Way 1 (Paper Work):

Keep × Flip (Reciprocal)
(___/___) × (___/___) = ___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Crow Water Puzzle: To reach a floating grub in a lab tube, a crow drops pebbles that each raise the water level by 3/8 cm. How many pebbles must the crow drop to raise the water by 4 1/2 cm?**
Starter Step 1: Convert 4 1/2 to an improper fraction FIRST, then multiply by the reciprocal of 3/8.

Way 1 (Paper Work):

Keep × Flip (Reciprocal)
(___/___) × (___/___) = ___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Relay Race Legs: A 3 3/4-mile charity relay is split into equal legs of 5/8 mile each. How many runners are on each relay team?**

Way 1 (Paper Work):

Keep × Flip (Reciprocal)
(___/___) × (___/___) = ___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **LED Strip Cutting: You have a 5 1/4-meter LED light strip and cut it into 3/4-meter segments for desk shelves. How many segments can you cut?**

Way 1 (Paper Work):

Keep × Flip (Reciprocal)
(___/___) × (___/___) = ___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **3D Print Nozzle Layers: A 6-millimeter bracket is printed in layers that are 3/20 millimeter thick. How many layers tall is the bracket?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Garden Stepping Stones: A 7 1/2-foot garden path has stepping stones spaced every 1 1/4 feet. How many 1 1/4-foot sections are along the path?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Solar Panel Width: A rectangular solar panel has an area of 4 1/2 square meters and a length of 2 1/4 meters. What is its width?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Juice Pitcher Servings: A pitcher holds 2 2/3 quarts of lemonade. How many 1/3-quart cups can be filled from the pitcher?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Reciprocal Check: Write the reciprocal of 2 2/5, and show what happens when you multiply 2 2/5 by its reciprocal.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Lumber Yard Cuts: You have a 10 1/2-foot board. You cut off a 1 1/2-foot piece first, then cut the remaining board into 3/4-foot pieces. How many 3/4-foot pieces do you get?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 2.4 — Practice A: Adding, Subtracting, and Multiplying Decimals

Key Vocabulary: tenths • hundredths • thousandths • place-value alignment • trailing zeros • estimation Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & VISUAL DIAGRAM

Add/Subtract: Line Up Decimal Points

```
  14.60  ← Annex 0
+  3.85
-------
  18.45
```

Estimate: 15 + 4 = 19 (Close!)

Multiply: Count Total Decimal Places

```
  4.5    (1 decimal place)
× 1.80   (2 decimal places)
---------
= 8.100  (1 + 2 = 3 places)
```

= $8.10 (Check: 45×18 = 810)

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Addition/Subtraction: Line up the decimal points vertically and annex trailing zeros so both numbers have the same place values.

Textbook Multiplication: Multiply as whole numbers, then count the total decimal places in both factors to place the decimal point.

WATCH OUT: Line up decimal points for + and - (annex trailing 0s), but for × count the TOTAL decimal places in both factors!

WORKED EXAMPLE — 3 TEXTBOOK WAYS (ANY 2 WAYS = +20 PTS)

Problem: Find 14.6 + 3.85 and check by subtracting.

WAY 1: Line Up Decimal Points Vertically
Annex a zero to write 14.6 as 14.60. Add hundredths (0+5=5), tenths (6+8=14, carry 1), and ones (14+3+1=18) → 18.45.

WAY 2: Decompose by Place Value (Wholes + Hundredths)
Add whole numbers (14 + 3 = 17) and decimals (0.60 + 0.85 = 1.45) separately, then combine: 17 + 1.45 = 18.45.

WAY 3: Add as Mixed Numbers (Hundredths)
14 60/100 + 3 85/100 = 17 145/100 = 18 45/100 = 18.45.

Answer: 18.45 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 2.4 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

1 4 • 6 ENTER 3 • 8 5 + f LAST*x* -

Line up the decimal points and add 14.60 + 3.85 by hand first. On your SC-6, check any place-value digit with [PV] (e.g., 14.6 [ENTER] 0.1 [PV]), then press 14.6 [ENTER] 3.85 [+] to confirm 18.45!

SECTION 2.4 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **3D Printer Calibration Shims: Marcus stacks a 345 mm base plate, a 0.6 mm spacer, and a 0.07 mm feeler shim. What is the combined thickness in millimeters?**

Starter Step 1: Line up the decimal points vertically (345.00 + 0.60 + 0.07) and add column by column from right to left.

Way 1 (Paper Work):

Line Up Decimals:

.
± .
.

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Find 12.8 + 4.75 and check by subtracting.**

Starter Step 1: Annex a trailing zero (12.80 + 4.75), add vertically, and check by subtracting 4.75 from your sum.

Way 1 (Paper Work):

Line Up Decimals:

.
± .
.

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Find 20 - 8.65.**

Way 1 (Paper Work):

Line Up Decimals:

.
± .
.

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Multiply 4.5 × 0.6.**

Way 1 (Paper Work):

Multiply & Count Dec Places:

_____ × _____ = _____ (___ places)

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Evaluate 25.4 - 9.86 + 3.2.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Multiply 3.6 × 2.25 and check that your answer is reasonable using a whole-number estimate.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Multiply 0.15 × 0.4. Write the product as a decimal and as a simplified fraction.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Bioplastic Funnel Mass: Every cubic centimeter of PLA bioplastic weighs 1.24 grams. If Marcus's 3D-printed funnel has a volume of 28.86 cm^3, what is its mass in grams (rounded to the nearest hundredth)?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **You pay with a $20 bill for an $8.75 burger and a $4.60 smoothie. How much change should you receive?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Greenhouse Strut Tubing: A camp greenhouse uses 30 struts of 1.20 m, 40 struts of 1.35 m, and 50 struts of 1.42 m. How many meters of aluminum tubing are required in all?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 2.4 — Practice B: Modeling Real Life (Adding, Subtracting, and Multiplying Decimals)

Textbook Ways: Vertical Place-Value Algorithm | Convert to Fractions/Hundredths | Place-Value Split / Area Model Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — BUYING BULK TRAIL MIX (3 TEXTBOOK WAYS • ANY 2 = +20 PTS)

Problem: You buy 3.6 pounds of trail mix at $2.25 per pound. What is the total cost of the trail mix?

WAY 1: Multiply Whole Numbers & Count Decimal Places
Multiply 225 × 36 = 8,100. Count decimal places (1 place in 3.6 + 2 places in 2.25 = 3 places) → 8.100 = $8.10.

WAY 2: Distributive Split (3 lb + 0.6 lb)
3 lb × $2.25 = $6.75, and 0.6 lb × $2.25 = $1.35. Add the partial products: $6.75 + $1.35 = $8.10.

WAY 3: Multiply as Fractions / Mixed Numbers
3.6 = 18/5 and 2.25 = 9/4. Multiply: (18/5) × (9/4) = 162/20 = 81/10 = $8.10.

WATCH OUT: Line up decimal points for + and - (annex trailing 0s), but for × count the TOTAL decimal places in both factors!

Answer: $8.10

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 2.4 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

3 • 6 ENTER 2 • 2 5 × FDISP

Multiply 3.6 × 2.25 on paper and count decimal places first. Then check on your SC-6:
Press 3.6 [ENTER] 2.25 [×] (and press [$] for Currency Mode or [FDISP] for fraction/decimal view) to confirm $8.10.

Marcus used to panic when decimal problems had uneven lengths, like adding a 345-mm base plate, a 0.6-mm spacer, and a 0.07-mm feeler shim, because he wasn't sure whether to line up the right-hand digits or the decimal points. And when multiplying 3.25 pounds of apples by $2.40 per pound, the decimal alignment rules seemed to change completely! After accidentally getting $780 for a bag of apples, he realized he could use common-sense estimation as a safety net: when adding or subtracting, lining up the decimal points keeps tenths with tenths; when multiplying, about 3 pounds at $2.40 HAS to be near $7.80, and counting the total decimal places lands the point in that exact spot.

SECTION 2.4 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **3D Filament Order: You buy a spool of Matte Navy PLA for $18.75 and a spool of Silk Gold PLA for $22.49. What is the total cost?**

Starter Step 1: Line up the decimal points vertically ($18.75 + $22.49) and add hundredths first.

Way 1 (Paper Work):

Line Up Decimals:

. ± . .

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Change at the Book Fair: You buy a graphic novel for $13.65 and pay with a $20 bill. How much change should you receive?**

Starter Step 1: Write $20 as $20.00, line up the decimal points ($20.00 - $13.65), and regroup across the zeros.

Way 1 (Paper Work):

Line Up Decimals:

. ± . .

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Sprint Stopwatch: In the 100-meter dash, Maya ran 13.42 seconds and Leo ran 12.87 seconds. How much faster was Leo's time?**

Way 1 (Paper Work):

Line Up Decimals:

. ± . .

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Bulk Honeycrisp Apples: Apples cost $2.40 per pound. How much does a 3.25-pound bag of apples cost?**

Way 1 (Paper Work):

Multiply & Count Dec Places:

_____ × _____ = _____ (___ places)

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Microchip Dimensions: A silicon processor die measures 1.4 cm by 0.85 cm. What is its area in square centimeters?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Road Trip Fuel: Your family car takes 11.5 gallons of gas at $3.60 per gallon. How much does the fill-up cost?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Peregrine Falcon Thermal Climb: A falcon rises 24.8 meters on a thermal updraft, drops 6.35 meters, and climbs another 11.9 meters. What is its net altitude gain?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Skate Shop Receipt: You buy 2 sets of bearings at $14.50 each and grip tape for $6.75. How much do you spend in all?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Workshop Drawer Clearance: Four 42.0 mm storage boxes sit side-by-side in a drawer that is 170.5 mm wide. How much clearance width remains?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Snack Bar Profit: You make 25 smoothies for $1.80 each and sell all 25 for $3.20 each. What is your total profit?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

Section 2.5 — Practice A: Dividing Whole Numbers and Decimals

Key Vocabulary: terminating decimal • power-of-10 shift • equivalent division expression • mixed number Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & VISUAL DIAGRAM

Way 1: Annex Zeros (23 ÷ 4 = 5.75)

```
   5.75
4 ) 23.00
```

Way 2 (Mixed #): 23/4 = 5 3/4

Way 3 (÷100): 575/100 = 5.75

Decimal Divisor: Shift Both Points Right!

6.75 ÷ 0.25 (×100 both)

Shift 2 places right → 675 ÷ 25

675 ÷ 25 = 27

Check: 27 × 0.25 = 6.75

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Whole-Number Division: Annex a decimal point and zeros (23.00 ÷ 4) to continue dividing until the remainder is 0.

Textbook Decimal Divisor Rule: Multiply divisor and dividend by the same power of 10 (shift both decimal points right) to make the divisor a whole number.

WATCH OUT: When dividing by a decimal, shift the decimal point the SAME number of places right in BOTH the divisor and the dividend!

WORKED EXAMPLE — 3 TEXTBOOK WAYS (ANY 2 WAYS = +20 PTS)

Problem: Divide 23 ÷ 4. Write the quotient as a remainder, a mixed number, and an exact decimal.

WAY 1: Annex a Decimal Point & Zeros (Long Division)

Write 23 as 23.00 and divide by 4: 23 ÷ 4 = 5 (3 left) → 30 tenths ÷ 4 = 7 tenths (2 left) → 20 hundredths ÷ 4 = 5 → 5.75.

WAY 2: Write as a Mixed Number & Scale to Hundredths

23 ÷ 4 = 23/4 = 5 3/4. Scale 3/4 by ×25: (3×25)/(4×25) = 75/100 = 0.75 → 5.75.

WAY 3: Decompose Dividend into Friendly Parts

Split 23 = 20 + 2 + 1: (20 ÷ 4) + (2 ÷ 4) + (1 ÷ 4) = 5 + 0.50 + 0.25 = 5.75.

Answer: 5 R 3 = 5 3/4 = 5.75 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 2.5 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

2 3 ENTER 4 ÷ FDISP f d/c

Divide 23 ÷ 4 by hand first. On your SC-6, press 23 [ENTER] 4 [÷] and tap [FDISP], [a b/c], and [f][d/c] to cycle through 5.75, 5 3/4, and 23/4 (or use [f][0.3-rep] for repeating decimals)!

SECTION 2.5 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Write 39 ÷ 8 as (a) a quotient with remainder, (b) a mixed number, and (c) a decimal.**

Starter Step 1: Divide 39 ÷ 8 for Quotient & Remainder, write Remainder/8 for the mixed number, and annex .000 for the decimal.

Way 1 (Paper Work):

Shift Decimal Equally →

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Divide 18.4 ÷ 8.**

Starter Step 1: Bring the decimal point straight up into the quotient above 18.4, then divide 18.4 ÷ 8.

Way 1 (Paper Work):

Shift Decimal Equally →

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **How many quarters ($0.25) are in $11.25?**

Way 1 (Paper Work):

Shift Decimal Equally →

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Divide 7.2 ÷ 0.9 and check by multiplying.**

Way 1 (Paper Work):

Shift Decimal Equally →

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Divide 15.6 ÷ 0.12.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Divide 9.45 ÷ 1.5. Write the answer as a decimal and as a mixed number in simplest form.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Team Material Budget Split: Six engineering students split a $124.80 hardware store bill equally. How much does each student pay?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Error Analysis: Leo says 4.8 ÷ 0.2 = 0.24. Use estimation or multiplication to explain and correct his error.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Track Race Laps: A runner completes a 4.5-mile race on a track where each lap is 0.75 mile. How many laps is the race?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Sports Drink Case: A 6-pack of sports drinks costs $7.50. What is the price of 1 bottle, and how many bottles can you buy with $20.00?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 2.5 — Practice B: Modeling Real Life (Dividing Whole Numbers and Decimals)

Textbook Ways: Power-of-10 Decimal Shift & Long Division | Equivalent Fraction Scaling | Reciprocal Fraction Division

Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — ARCADE TOKEN MACHINE (3 TEXTBOOK WAYS • ANY 2 = +20 PTS)

Problem: An arcade token machine takes quarters ($0.25). A player inserts $6.75. How many quarters did the player insert?

WAY 1: Multiply Divisor & Dividend by 100 (Decimal Shift)
Move both decimal points 2 places right (× 100) so the divisor is a whole number: 6.75 ÷ 0.25 = 675 ÷ 25 = 27 quarters.

WAY 2: Convert to Fractions & Multiply by Reciprocal
$6.75 = 6 3/4 = 27/4 and $0.25 = 1/4. Divide: (27/4) ÷ (1/4) = (27/4) × (4/1) = 27 quarters.

WAY 3: Count Quarters per Dollar
Each $1.00 is 4 quarters: $6.00 = 6 × 4 = 24 quarters, plus $0.75 = 3 quarters → 24 + 3 = 27 quarters.

WATCH OUT: When dividing by a decimal, shift the decimal point the SAME number of places right in BOTH the divisor and the dividend!

Answer: 27 quarters

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 2.5 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[6] [•] [7] [5] [ENTER] [0] [•] [2] [5] [÷] [f] [LASTx] [×]

Shift the decimal point 2 places right (675 ÷ 25) and divide on paper first. Then check on your SC-6:
Press 6.75 [ENTER] 0.25 [÷] to confirm 27 quarters, and [f][LASTX] [×] to verify 27 × 0.25 = 6.75.

Looking at the 3D-printer screen, Marcus needed to find how many 0.20-millimeter plastic layers it would take to build a 14.4-millimeter tall bracket, and how many 1.75-millimeter quarters were in a 28-millimeter stack. Long division was hard enough with whole numbers; dividing by 1.75 made him want to close his notebook. Then he thought about changing millimeters into hundredths of a millimeter: multiplying both 28 and 1.75 by 100 slid both decimal points two places right, turning 28 ÷ 1.75 into 2,800 ÷ 175. Because both numbers scaled by the same factor, the answer (16 quarters) stayed identical—and the scary decimal divisor disappeared.

SECTION 2.5 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Four-Way Representation: Express 35 ÷ 8 as (a) a quotient with remainder, (b) a mixed number, (c) an improper fraction, and (d) an exact decimal.**

Starter Step 1: Divide 35 ÷ 8 to find the whole-number quotient and remainder, then express the remainder as eighths and as a decimal.

Way 1 (Paper Work):

Shift Decimal Equally →

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Quarter Stack Height: A stack of quarters is 28.0 mm tall. If each quarter is 1.75 mm thick, how many quarters are in the stack?**

Starter Step 1: Shift the decimal point 2 places right in BOTH numbers so the divisor 1.75 becomes the whole number 175.

Way 1 (Paper Work):

Shift Decimal Equally →

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Splitting the Pizza Bill: Four friends split a $34.60 pizza order equally. How much does each friend pay?**

Way 1 (Paper Work):

Shift Decimal Equally →

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **3D Print Layer Slicer: A 14.4 mm tall model is sliced into layers that are each 0.20 mm thick. How many layers are in the print?**

Way 1 (Paper Work):

Shift Decimal Equally →

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Hummingbird Nectar: A ruby-throated hummingbird drinks 5.25 milliliters of nectar by visiting flowers that hold 0.35 mL each. How many flowers did it visit?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Track Lap Pace: A runner completes 4.5 miles in 0.75 hours. What is her average speed in miles per hour?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Garden Fence Posts: A 22.5-meter garden wall is divided into equal sections that are 1.25 meters long. How many sections are there?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Lemonade Stand Cups: A cooler holds 13.5 liters of lemonade. How many 0.45-liter cups can be filled?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Movie Ticket Group: A group of friends spent $87.50 on movie tickets that cost $12.50 each. How many tickets did they buy?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Evaluate 10.5 ÷ 1.75 in two ways: by dividing decimals and by converting both numbers to mixed numbers.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

Unit 3: Ratios, Rates & Percents — Bicycle Sprockets, Supermarket Aisles & Field Surveys

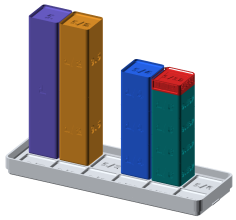
Ratios & Tape Diagrams, Ratio Tables, Unit Rates, Percents, Discounts & Unit Conversions

Devon always wondered why pedaling his mountain bike uphill felt impossible in high gear, while shifting to low gear made his feet spin fast and easy. Flipping his bike upside down in the driveway, he counted 48 spiky teeth on the front pedal sprocket and 16 teeth on the back wheel sprocket. Because each link of the metal chain grabs one tooth at a time, pulling 48 front teeth forward forces the 16-tooth back gear to spin around three full times! One turn of Devon's feet spins the rear tire three times—trading harder pedal push for triple the speed.

That same day, Priya was at the grocery store buying granola bars for the robotics club and felt stuck in front of the shelf. Box A had 6 bars for $4.50, while Box B had 10 bars for $7.00. At first she thought the $4.50 box had to be the better deal simply because $4.50 is smaller than $7.00, until she wondered if she was paying more per bar without realizing it.

Over at the wildlife center, Zoe had a similar comparison problem: how do you fairly compare 17 out of 20 gigabytes used on a field tablet, 9 out of 25 penalty kicks made on the soccer field, and 21 out of 24 counting questions answered by an African grey parrot when every total is different?

Whether Devon is comparing bike gears, Priya is stretching the club snack budget, or Zoe is scaling scores to a 0-to-100 percent meter, all three of them learn how ratios, unit rates, and percents let you compare almost anything fairly—and how to check every ratio, unit rate, and percent change on the SC-6.



Real-World Mathematical Context & Two-Method Mastery • Hands-On Models: Fraction Strips & Place Value Blocks

UNIT 3 LESSONS — LEARN TWO TEXTBOOK WAYS FIRST, THEN CHECK YOUR WORK ON THE SC-6 CALCULATOR

SEC 3.1 (pp.26-27) Ratios and Tape Diagrams

Textbook Ways: Way 1: Tape Diagram (Equal Boxes) | Way 2: Equivalent Ratio Fraction | Way 3: Divide by GCF

Learn by Hand: A ratio compares two quantities (a to b, a:b, or a/b). In a Tape Diagram, each equal box (or equal-length Fraction Strip in) represents the same unit amount. For a 3:2 ratio, 3 equal strips + 2 equal strips = 5 total strips, making the Part-to-Whole ratios 3:5 and 2:5.

Check Your Work on the SC-6 Calculator:

[1] [2] [ENTER] [8] [g] [a:b]

8 on paper first. Then check on your SC-6 using Ratio Mode: press 12 [ENTER] 8 [g][a:b] to simplify directly to 3:2 (and tap [10F] to view the interactive Ratio Table)!

SEC 3.2 (pp.28-29) Using Ratio Tables and Double Number Lines

Textbook Ways: Way 1: Multiplicative Scale Factor | Way 2: Double Number Line / Repeated Addition | Way 3: Scale Down Then Up

Learn by Hand: In Big Ideas Math, you can find equivalent ratios in a Ratio Table or on a Double Number Line by (1) adding the base ratio quantities repeatedly, (2) multiplying or dividing both quantities by the same scale factor, or (3) scaling down to a smaller equivalent ratio and then scaling up.

Check Your Work on the SC-6 Calculator:

[8] [g] [REP] [+] [+] [+] [+]

B] and press [10F] to step through the Ratio Table (or repeat steps with [g][REP]) to verify 15 : 40!

SEC 3.3 (pp.30-31) Rates, Unit Rates, and Comparing Rates

Textbook Ways: Way 1: Divide to Find the Unit Rate (1 Unit) | Way 2: Common Multiple Comparison | Way 3: Double Number Line

Learn by Hand: A rate compares two quantities measured in different units (such as miles and hours). A unit rate gives the amount per 1 unit of the second quantity (divide numerator ÷ denominator). Comparing unit prices shows which package is the better buy.

Check Your Work on the SC-6 Calculator:

[3] [1] [5] [ENTER] [5] [÷]

Press 315 [ENTER] 5 [÷] to confirm 63 words per minute.

SEC 3.4 (pp.32-33) Percents, Fractions, and Decimals

Textbook Ways: Way 1: Equivalent Fraction over 100 | Way 2: Divide Numerator ÷ Denominator | Way 3: Percent Bar / Benchmark Model

Learn by Hand: 'Percent' (%) means 'per hundred' or 'out of 100.' Any fraction, decimal, or percent can be converted into the other two forms: scale the denominator to 100 (or divide numerator ÷ denominator and multiply by 100%). To find p% of a whole w, multiply (p/100) × w.

Check Your Work on the SC-6 Calculator:

[•] [7] [•] [2] [0] [ENTER] [FDISP] [1] [0] [0] [×]

Type [.] 7 [.] 20 [ENTER] [FDISP] to see 0.35, and press [f][→%] (or 100 [×]) to confirm 35%.

SEC 3.5 (pp.34-35) Solving Percent Problems, Discounts & Conversions

Textbook Ways: Way 1: Find Discount & Subtract | Way 2: Multiply by Remaining % | Way 3: Percent Proportion / Ratio Bar

Learn by Hand: In Big Ideas Math, you can find a discounted sale price in two ways: (1) find the discount amount and subtract it from the original price, or (2) subtract the discount percent from 100% first and multiply! For percent change, divide (Change ÷ Original Amount) × 100%. For unit conversions, multiply by a conversion factor so the old units cancel.

Check Your Work on the SC-6 Calculator:

[6] [0] [ENTER] [2] [5] [%] [-]

Type 60 [ENTER] 25 [%] (which computes 25% of 60 = $15 while keeping 60 on Y!) and press [-] to get the $45 sale price.

HANDS-ON MODEL: FRACTION STRIPS

Section 3.1 — Practice A: Ratios and Tape Diagrams

Key Vocabulary: ratio • part-to-part ratio • part-to-whole ratio • equivalent ratios • tape diagram Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & HANDS-ON MODEL (FRACTION STRIPS)

Ratio Tape Diagram: 12 Builders & 8 Programmers (5 Total Boxes = 20 Members)

Builders (3): [4] [4] [4] = 12 Builders (3 × 4)

Program (2): [4] [4] = 8 Programmers (2 × 4)

Part-to-Part = 12:8 = 3:2 | Part-to-Whole = 12:20 = 3:5 (Each box = 4 members)

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept: A ratio compares two quantities as a to b, a:b, or a/b. Distinguish Part-to-Part (3:2) from Part-to-Whole (3:5).

Textbook Visual Model (Tape Diagram): Draw equal-sized boxes for each ratio part so every box represents the same unit amount.

WATCH OUT: Order matters in ratios! Check whether the question asks for Part-to-Part (3:2) or Part-to-Whole (3:5).

WORKED EXAMPLE — 3 TEXTBOOK WAYS (ANY 2 WAYS = +20 PTS)

Problem: A robotics team has 12 builders and 8 programmers. Write the simplified Part-to-Part and Part-to-Whole ratios.

WAY 1: Divide Both Terms by Their GCF

Part-to-Part = 12:8. Divide both terms by GCF(12, 8) = 4 → 3:2. Total = 12+8 = 20 → Part-to-Whole = 12:20 = 3:5.

WAY 2: Tape Diagram Model (With Blocks)

Place 4 members in each tile box: 12 is 3 boxes and 8 is 2 boxes (5 boxes total) → Part-to-Part = 3:2; Part-to-Whole = 3:5.

WAY 3: Equivalent Ratio Table

Scale down in steps: (12, 8, 20) ÷ 2 → (6, 4, 10) ÷ 2 → (3, 2, 5).

Answer: Part-to-Part = 3:2; Part-to-Whole = 3:5

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 3.1 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[1] [2] [ENTER] [8] [g] [a:b]

Simplify the ratio 12:

8 on paper first. Then check on your SC-6 using Ratio Mode: press 12 [ENTER] 8 [g][a:b] to simplify directly to 3:2 (and tap [10F] to view the interactive Ratio Table)!

SECTION 3.1 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Write the ratio 12 : 18 in simplest form.**

Starter Step 1: Find the Greatest Common Factor (GCF) of 12 and 18, then divide both terms of the ratio by that GCF.

Way 1 (Paper Work):

Divide Both Terms by GCF:

GCF = ___ → ___ : ___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Write the ratio 24 : 16 in simplest form.**

Starter Step 1: Find the Greatest Common Factor (GCF) of 24 and 16, then divide both terms of the ratio by that GCF.

Way 1 (Paper Work):

Divide Both Terms by GCF:

GCF = ___ → ___ : ___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **A pouch holds 4 red marbles and 6 blue marbles. Find the simplified ratio of red marbles to total marbles.**

Way 1 (Paper Work):

Tape Model:

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **In a 3:4 tape diagram, each box represents 6 items. Find the two parts and the total.**

Way 1 (Paper Work):

Tape Model:

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Living-Wall Hydroponic Reservoir: Nutrient solution mixes 2 parts plant concentrate with 15 parts water. How many liters of concentrate and water are in a 34-liter tank?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **The ratio of cats to dogs at an animal shelter is 4:5. If there are 36 animals in all, how many are dogs?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **The ratio of sixth graders to seventh graders in a chess club is 3:5. If there are 15 sixth graders, how many seventh graders are in the club?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **The ratio of acoustic guitars to electric guitars in a music shop is 2:5. There are 18 more electric guitars than acoustic guitars. How many electric guitars are there?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Robotics Wheel Split: Two robotics squads split 84 wheels in a 4:3 ratio. How many wheels does each squad receive?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **The ratio of gold coins to silver coins in a chest is 7:4. If there are 21 more gold coins than silver coins, how many coins are in the chest in all?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: FRACTION STRIPS

Section 3.1 — Practice B: Modeling Real Life (Ratios and Tape Diagrams)

Textbook Ways: Tape Diagram (Equal Boxes) | Equivalent Ratio Fraction | Divide by GCF Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — TEAM WIN-LOSS RATIO (3 TEXTBOOK WAYS • ANY 2 = +20 PTS)

Problem: A robotics team's ratio of wins to losses is 5:3. If the team played 32 matches in all, how many matches did they win?

WAY 1: Tape Diagram (Count All Boxes)
Draw 5 Win boxes and 3 Loss boxes = 8 equal boxes total. Each box = 32 ÷ 8 = 4 matches. Wins = 5 × 4 = 20 matches.

WAY 2: Scale the Part-to-Whole Ratio
Wins to Total is 5 to (5 + 3) = 5/8. Since 8 × 4 = 32, multiply numerator by 4: (5 × 4)/(8 × 4) = 20/32 → 20 wins.

WAY 3: Repeated Addition Ratio Table
(Wins, Losses, Total): (5, 3, 8) → (10, 6, 16) → (15, 9, 24) → (20, 12, 32) → 20 wins.

WATCH OUT: Order matters in ratios! Check whether the question asks for Part-to-Part (3:2) or Part-to-Whole (3:5).

Answer: 20 wins (and 12 losses)

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 3.1 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

3 2 ENTER 5 ENTER 3 + ÷ 5 ×

Draw the 5 + 3 = 8-box tape diagram on paper first. Then check on your SC-6:
Press 32 [ENTER] 5 [ENTER] 3 [+] [÷] (4 games per box) and 5 [×] to confirm 20 wins.

When a smoothie recipe called for strawberries and bananas in a 4:3 ratio using 28 pieces of fruit in all, Devon initially tried dividing 28 by 4 and 28 by 3—and when 28 ÷ 3 didn't even come out whole, he thought the problem was broken. Then he remembered how he had simplified his bike's 48-tooth front sprocket and 16-tooth rear sprocket into equal groups, and sketched 4 strawberry boxes and 3 banana boxes side by side. Looking at the sketch, he saw his mistake right away: there were 4 + 3 = 7 equal boxes sharing the 28 fruits! Dividing 28 ÷ 7 put 4 fruits in every box, making 16 strawberries and 12 bananas without any guesswork.

SECTION 3.1 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Robotics Club Roles: A club has 15 coders and 10 builders. Write the simplified ratio of coders to builders, and the simplified ratio of coders to total club members.**

Starter Step 1: Part-to-Part compares coders to builders (15:10); Part-to-Whole compares coders to total club members (15 + 10).

Way 1 (Paper Work):

Tape Model:

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Smoothie Berry Mix: The ratio of strawberries to bananas in a smoothie is 4:3. If there are 28 pieces of fruit in all, how many are strawberries and how many are bananas?**

Starter Step 1: Add the ratio parts (4 + 3) to find the total equal boxes in the tape diagram, then divide 28 fruit by that total.

Way 1 (Paper Work):

Tape Model:

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Paint Color Mixing: Forest green paint uses 5 parts blue to 2 parts yellow. If you use 15 cups of blue paint, how many cups of yellow paint do you need?**

Way 1 (Paper Work):

Tape Model:

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Scrub-Jay Seed Caches: When a western scrub-jay sees a rival jay watching, it re-hides its food in a 7:5 ratio of secret trays to open trays. Out of 60 seeds, how many are hidden in secret trays vs. open trays?**

Way 1 (Paper Work):

Tape Model:

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Treasure Chest Coins: The ratio of gold coins to silver coins in a chest is 7:3. There are 24 more gold coins than silver coins. How many gold coins and how many silver coins are in the chest?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **How Bike Gears Work: A bike's 48-tooth front pedal gear pulls a chain to spin a 16-tooth back wheel gear (every 16 front teeth spin the back wheel 1 full turn). What is the simplified ratio of front teeth to back teeth?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **School Band Instruments: A band has 16 flutes, 12 clarinets, and 8 trumpets. What is the simplified ratio of flutes to total instruments?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Screen Aspect Ratio: A monitor is 1,920 pixels wide by 1,080 pixels tall. Write the simplified width-to-height aspect ratio.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Trail Mix Ratio: A trail mix uses almonds, cashews, and raisins in a 3 : 2 : 5 ratio. How many grams of each ingredient are in a 500-gram bag?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Chess Tournament Record: You won 24 games and lost 8 games. What is the simplified ratio of wins to total games played?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 3.2 — Practice A: Using Ratio Tables and Double Number Lines

Key Vocabulary: ratio table • equivalent ratios • double number line • scaling (multiplying or dividing) Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & VISUAL DIAGRAM

Way 1 & 2: Ratio Table (×5 Scale)

Time	3	6	9	15
Dist	8	16	24	40

3 × 5 = 15 s and 8 × 5 = 40 m

Way 3: Double Number Line



HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Method 1 (Ratio Table — Addition): Add the base ratio quantities repeatedly to build a table of equivalent ratios.

Textbook Method 2 (Scale Factor — Multiplication/Division): Multiply or divide both quantities in a ratio by the same nonzero number.

WATCH OUT: Never add the same number to both rows of a ratio table — always MULTIPLY or DIVIDE both quantities by the same scale factor!

WORKED EXAMPLE — 3 TEXTBOOK WAYS (ANY 2 WAYS = +20 PTS)

Problem: A drone flies 8 meters every 3 seconds. How far does it fly in 15 seconds?

WAY 1: Multiply Both Quantities by the Scale Factor
To scale Time from 3 s to 15 s, multiply by 5 (3 × 5 = 15). Multiply Distance by 5: 8 m × 5 = 40 m.

WAY 2: Repeated Addition in a Ratio Table
Add (3 s, 8 m) step-by-step: (3, 8) → (6, 16) → (9, 24) → (12, 32) → (15, 40) → 40 m.

WAY 3: Double Number Line Jumps
Mark jumps of +3 s on the top line (0, 3, 6, 9, 12, 15) and matching jumps of +8 m on the bottom line (0, 8, 16, 24, 32, 40).

Answer: 40 meters 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 3.2 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

8 g REP + + + +

Complete the ratio table columns by hand first. On your SC-6, enter 3 [ENTER] 8 |g||a: B] and press [10F] to step through the Ratio Table (or repeat steps with [g][REP]) to verify 15 : 40!

SECTION 3.2 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Write the first 4 equivalent ratios of 4 : 9.**
Starter Step 1: Multiply both 4 and 9 by 1, 2, 3, and 4 to build the first four equivalent columns of the ratio table.

Way 1 (Paper Work):

Ratio Table:

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **In a ratio table where A : B = 4 : 9, find B when A = 28.**
Starter Step 1: Find what 4 is multiplied by to reach 28, then multiply 9 by that exact same scale factor.

Way 1 (Paper Work):

Ratio Table:

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **In a ratio table where X : Y = 5 : 12, find X when Y = 60.**

Way 1 (Paper Work):

Ratio Table:

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **A laser printer prints 15 pages every 2 minutes. How many pages does it print in 8 minutes?**

Way 1 (Paper Work):

Ratio Table:

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Hot cocoa uses 3 tablespoons of cocoa for every 8 ounces of milk. How many ounces of milk are needed for 18 tablespoons of cocoa?**

Way 1 (Paper Work):

Ratio Table:

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **8 studio markers cost $12. Find the cost of 10 markers.**

Way 1 (Paper Work):

Ratio Table:

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **A runner's heart beats 21 times in 15 seconds. At this rate, how many times does it beat in 1 minute (60 seconds)?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Error Analysis: To find a ratio equivalent to 3:5, Sam added 4 to both terms and wrote 7:9. Explain why 3:5 and 7:9 are not equivalent.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Smoothie Bar: 9 berry smoothies require 12 cups of berries. How many cups of berries are needed to blend 15 smoothies?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Electric Scooter Range: An e-scooter travels 14 miles on 35% of its battery. At this rate, how many miles can it travel on a 100% full charge?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 3.2 — Practice B: Modeling Real Life (Using Ratio Tables and Double Number Lines)

Textbook Ways: Multiplicative Scale Factor | Double Number Line / Repeated Addition | Scale Down Then Up Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — SKATE PARK PASSES (3 TEXTBOOK WAYS • ANY 2 = +20 PTS)

Problem: Six skate park passes cost $45. At this rate, how much do 10 skate park passes cost?

WAY 1: Scale Down Then Scale Up in a Ratio Table
Divide (6 passes, $45) by 3 to get (2 passes, $15). Then multiply (2 passes, $15) by 5 to get (10 passes, $75).

WAY 2: Add Columns in the Ratio Table (6 Passes + 4 Passes)
Since 2 passes cost $15, 4 passes cost $30. Add 6 passes ($45) + 4 passes ($30) = 10 passes for $75.

WAY 3: Unit Rate (Cost per 1 Pass)
Divide $45 ÷ 6 = $7.50 per pass. Multiply by 10 passes: 10 × $7.50 = $75.

WATCH OUT: Never add the same number to both rows of a ratio table — always MULTIPLY or DIVIDE both quantities by the same scale factor!

Answer: $75

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 3.2 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:
4 5 ENTER 3 ÷ 5 ×

Scale the ratio table on paper first (divide by 3 to find 1 bag, then multiply by 5). Check on your SC-6: Press 45 [ENTER] 3 [÷] 5 [×] to confirm $75.

Priya had to bake 10 pizza crusts for the robotics club using a recipe written for 4 crusts (6 cups of flour for 4 crusts), and she got stuck because 4 doesn't multiply evenly into 10. For a minute she thought she would have to deal with messy fractions in every column. Then she drew a quick ratio table and realized nobody forces you to jump from 4 to 10 in one step: she could divide both numbers by 2 first to find the amount for 2 crusts (3 cups of flour), and then multiply by 5 to reach 10 crusts (15 cups of flour). Stepping down to a smaller friendly column first made scaling any recipe feel effortless.

SECTION 3.2 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **3D Printing Speed: A printer extrudes 15 grams of filament every 20 minutes. At this rate, how many grams does it extrude in 60 minutes?**

Starter Step 1: Find how many 20-minute blocks fit in 60 minutes, then multiply 15 grams by that same scale factor.

Way 1 (Paper Work):

Ratio Table:

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Aquarium Salt Mix: A marine tank needs 7 scoops of sea salt for every 2 gallons of water. How many scoops are needed for a 16-gallon tank?**

Starter Step 1: Find what 2 gallons is multiplied by to reach 16 gallons, then multiply 7 scoops by that same scale factor.

Way 1 (Paper Work):

Ratio Table:

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **School Store Notebooks: If 8 notebooks cost $12, how much do 10 notebooks cost at the same rate?**

Way 1 (Paper Work):

Ratio Table:

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Dove Heartbeat Rate: A resting dove's heart beats 28 times in 15 seconds. At this rate, how many times does its heart beat in 1 minute?**

Way 1 (Paper Work):

Ratio Table:

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Maple Syrup Sap: It takes 40 gallons of tree sap to boil down 1 gallon of pure maple syrup. How many gallons of sap are needed for 6.5 gallons of syrup?**

Way 1 (Paper Work):

Ratio Table:

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Moon Weight Ratio: An object weighing 60 pounds on Earth weighs 10 pounds on the Moon. How much would a 150-pound astronaut weigh on the Moon?**

Way 1 (Paper Work):

Ratio Table:

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Pizza Dough Flour: Six cups of flour make 4 pizza crusts. How many cups of flour are needed to make 10 pizza crusts?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Solar Charger: A portable solar panel charges 18% of a battery every 15 minutes. At this rate, how many minutes does it take to charge 90% of the battery?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Model Train Scale: A scale model uses 2 inches to represent 15 feet of a real train. If a real boxcar is 60 feet long, how many inches long is the model?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Lemonade Pitcher: Nine lemons make 6 cups of lemonade. How many lemons do you need to make 20 cups of lemonade?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 3.3 — Practice A: Rates, Unit Rates, and Comparing Rates

Key Vocabulary: rate • unit rate • unit cost (unit price) • equivalent rates

Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & VISUAL DIAGRAM

Unit Rate: Denominator of 1 Unit

$$\frac{315 \text{ words}}{5 \text{ minutes}} \xrightarrow{\div 5} \frac{63 \text{ words}}{1 \text{ minute}}$$

Unit Rate = 63 words/min

In 8 min: 63 × 8 = 504 words

Comparing Rates (Better Buy)

Pack A: $5.80 ÷ 4 = $1.45 / battery

(12 batteries = 3 × $5.80 = $17.40)

Pack B: $8.40 ÷ 6 = $1.40 / battery

Pack B is cheaper! ($16.80 for 12)

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept: A rate compares two quantities with different units. A unit rate has a denominator of 1 unit (divide numerator ÷ denominator).

Textbook Comparison Strategy: To find the 'Better Buy' or faster speed, compare the unit rates (cost per 1 item) or scale to a common multiple.

WATCH OUT: For unit price ('per item' or 'per ounce'), divide Money ÷ Quantity. The smaller unit cost is the better buy!

WORKED EXAMPLE — 3 TEXTBOOK WAYS (ANY 2 WAYS = +20 PTS)

Problem: You type 315 words in 5 minutes. Find your unit rate in words per minute.

WAY 1: Divide Numerator and Denominator by 5

Write 315 words / 5 min. Divide top and bottom by 5: (315 ÷ 5) / (5 ÷ 5) = 63 words per 1 minute.

WAY 2: Partial Quotients / Place-Value Split

Split 315 = 300 + 15: (300 ÷ 5) + (15 ÷ 5) = 60 + 3 = 63 words per minute.

WAY 3: Double Number Line Partition

Divide the 0-to-315 segment into 5 equal 1-minute intervals: each interval is 63 words.

Answer: 63 words per minute 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 3.3 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

3 1 5 ENTER 5 ÷

Divide 315 words by 5 minutes on paper first to find the unit rate per 1 minute. Then check on your SC-6: Press 315 [ENTER] 5 [÷] to confirm 63 words per minute.

SECTION 3.3 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **A car travels 195 miles in 3 hours. Find the average speed in miles per hour.**

Starter Step 1: Divide total distance by total time (195 miles ÷ 3 hours) to find the unit rate for 1 hour.

Way 1 (Paper Work):

Unit Rate = Total ÷ Units

_____ ÷ _____ = _____ per 1

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **You earn $36 for 4 hours of tutoring. What is your hourly pay rate?**

Starter Step 1: Divide total pay by total hours ($36 ÷ 4 hours) to find the unit pay rate for 1 hour.

Way 1 (Paper Work):

Unit Rate = Total ÷ Units

_____ ÷ _____ = _____ per 1

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **A 6-pound bag of apples costs $7.80. What is the price per pound?**

Way 1 (Paper Work):

Unit Rate = Total ÷ Units

_____ ÷ _____ = _____ per 1

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **A 3D printer extrudes 90 grams of filament in 6 hours. How many grams does it extrude per hour?**

Way 1 (Paper Work):

Unit Rate = Total ÷ Units

_____ ÷ _____ = _____ per 1

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Craft Foil Comparison: Supplier A sells an 18-oz roll of copper craft foil for $3.24. Supplier B sells a 24-oz roll for $4.08. Find both unit prices—which supplier is cheaper per ounce?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **A 6-pack of gel pens costs $5.10, and a 10-pack costs $8.20. Which pack has the lower cost per pen?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **A train travels 210 miles in 3.5 hours. At this constant speed, how far will it travel in 5 hours?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Pump A fills 45 gallons in 5 minutes. Pump B fills 64 gallons in 8 minutes. Which pump has the faster flow rate?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **3D Printer Feed Speed: Marcus's 3D printer pushes 45 mm of filament into the nozzle in 15 seconds. At this rate, how many millimeters of filament does it push in 1 minute (60 seconds)?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Car A drives 168 miles on 6 gallons of gas. Car B drives 232 miles on 8 gallons. Which car gets better fuel mileage, and by how many miles per gallon?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 3.3 — Practice B: Modeling Real Life (Rates, Unit Rates, and Comparing Rates)

Textbook Ways: Divide to Find the Unit Rate (1 Unit) | Common Multiple Comparison | Double Number Line Name: _____ Date: _____ Score: _____ / 100

WORKED EXAMPLE — COMPARING UNIT PRICES (3 TEXTBOOK WAYS • ANY 2 = +20 PTS)

Problem: Pack A has 8 AA batteries for $11.60. Pack B has 12 AA batteries for $16.80. Which pack is the better buy?

WAY 1: Compare Unit Prices (Cost per 1 Battery)
Pack A: $11.60 ÷ 8 = $1.45 per battery. Pack B: $16.80 ÷ 12 = $1.40 per battery. Pack B is cheaper ($1.40 < $1.45).

WAY 2: Scale Both Packs to LCM(8, 12) = 24 Batteries
For 24 batteries: 3 of Pack A cost 3 × $11.60 = $34.80; 2 of Pack B cost 2 × $16.80 = $33.60. Pack B saves $1.20.

WAY 3: Compare Cost of 4 Batteries (GCF = 4)
Divide Pack A by 2 → 4 batteries cost $5.80. Divide Pack B by 3 → 4 batteries cost $5.60. Pack B is the better buy.

WATCH OUT: For unit price ('per item' or 'per ounce'), divide Money ÷ Quantity. The smaller unit cost is the better buy!

Answer: Pack B ($1.40 per battery vs. $1.45 per battery)

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 3.3 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[1] [1] [•] [6] [0] [ENTER] [8] [÷] [1] [6] [•] [8] [0] [ENTER] [1] [2] [÷]

Find both unit prices on paper first. Then check on your SC-6:
Press 11.60 [ENTER] 8 [÷] ($1.45 sits on Y) and 16.80 [ENTER] 12 [÷] ($1.40 sits on X), then press [g][<] (flashes YES!) to confirm $1.40 < $1.45.

Standing in the snack aisle, Priya held a 6-bar box of granola bars priced at $4.50 and a 10-bar box priced at $7.00. Her first instinct was to grab the $4.50 box because the price tag looked smaller, while Devon guessed the 10-bar box because bulk boxes are usually cheaper—neither of them was actually sure. Instead of arguing, Priya divided the cost by the number of bars to find the price of 1 single bar: $4.50 ÷ 6 = $0.75 per bar versus $7.00 ÷ 10 = $0.70 per bar. Stripping both packages down to a 'per 1' unit rate settled the question in seconds.

SECTION 3.3 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **High-Speed Train: A bullet train travels 450 miles in 3 hours. What is its average speed in miles per hour?**
Starter Step 1: Divide 450 miles ÷ 3 hours so the denominator becomes 1 hour.

Way 1 (Paper Work):

Unit Rate = Total ÷ Units
_____ ÷ _____ = _____ per 1

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Best-Buy Granola Bars: Box A has 6 bars for $4.50. Box B has 10 bars for $7.00. Find the unit price per bar for each box. Which box is the better buy?**
Starter Step 1: Divide Total Cost ÷ Number of Bars for each box ($4.50 ÷ 6 and $7.00 ÷ 10) to compare the unit price per bar.

Way 1 (Paper Work):

Unit Rate = Total ÷ Units
_____ ÷ _____ = _____ per 1

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **3D Filament Comparison: Spool A is 500 grams for $11.00. Spool B is 1,000 grams for $19.50. Find the price per 100 grams for each spool. Which is the better buy?**

Way 1 (Paper Work):

Unit Rate = Total ÷ Units
_____ ÷ _____ = _____ per 1

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Swimming Pool Pump: A pump fills 180 gallons in 12 minutes. What is its unit rate in gallons per minute, and how many gallons does it pump in 25 minutes?**

Way 1 (Paper Work):

Unit Rate = Total ÷ Units
_____ ÷ _____ = _____ per 1

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Woodpecker Tapping Speed: Woodpecker A taps a tree 220 times in 4 minutes to locate larvae. Woodpecker B taps 285 times in 5 minutes. Find each bird's unit rate in taps per minute. Which bird taps faster?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Fuel Economy Test: Car A drives 336 miles on 12 gallons of gas. Car B drives 405 miles on 15 gallons. Which car gets more miles per gallon?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Dog Walking Job: You earn $54 for walking 6 dogs. At this rate, how many dogs must you walk to earn $99?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Juice Box Bulk Pack: An 8-pack of juice boxes costs $3.60. A 12-pack costs $5.04. Find the unit price per juice box for each pack. Which is cheaper per box?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Snowfall Rate: During a snowstorm, 14 inches of snow fall in 3.5 hours. What is the snowfall rate in inches per hour?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Rover Speed: A planetary rover drives 27 meters in 6 seconds. What is its speed in meters per second, and how far does it travel in 20 seconds?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

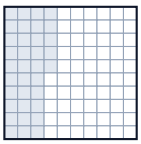
Section 3.4 — Practice A: Percents, Fractions, and Decimals

Key Vocabulary: percent ('per hundred') • hundredths grid • benchmark percent • percent proportion

Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & VISUAL DIAGRAM

10×10 Grid: 7/20 = 35/100 = 35%



35 of 100 squares

= 0.35

= 35%

3 Ways: Fraction = Decimal = %

Way 1 (Scale to 100):
7/20 × 5/5 = 35/100 = 35%

Way 2 (Divide):
7 ÷ 20 = 0.35 → 35%

Way 3 (% of Whole):
35% of 80 = 0.35 × 80 = 28

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept: Percent (%) means 'per hundred' (out of 100). Write an equivalent fraction with denominator 100, or divide numerator ÷ denominator.

Textbook Finding a Percent of a Number: Convert the percent to a decimal or fraction and multiply by the whole (p% of w = (p/100) × w).

WATCH OUT: 'Percent' means per 100: move the decimal 2 places RIGHT for % (0.06 = 6%, NOT 60%), and 2 places LEFT for a decimal.

WORKED EXAMPLE — 3 TEXTBOOK WAYS (ANY 2 WAYS = +20 PTS)

Problem: Write 7/20 as a decimal and as a percent.

WAY 1: Equivalent Fraction with Denominator 100
Since 20 × 5 = 100, multiply top and bottom of 7/20 by 5: (7 × 5)/(20 × 5) = 35/100 = 0.35 = 35%.

WAY 2: Divide Numerator by Denominator
Divide 7 ÷ 20 = 0.35. Multiply by 100% (move decimal 2 places right) → 35%.

WAY 3: Benchmark 1/20 = 5% Bar Model
On a percent bar, 1/20 of 100% is 5%. Seven twentieths is 7 × 5% = 35%.

Answer: 0.35 and 35%

1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 3.4 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[•] [7] [•] [2] [0] [ENTER] [FDISP] [1] [0] [0] [×]

Convert 7/20 to a decimal and percent on paper first. Then check on your SC-6 using Two-Dot fraction entry: Type [.] 7 [.] 20 [ENTER] [FDISP] to see 0.35, and press [f][→%] (or 100 [×]) to confirm 35%.

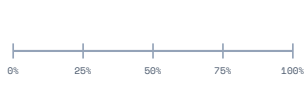
SECTION 3.4 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 Write 3/5 as a decimal and as a percent.

Starter Step 1: Scale the denominator of 3/5 to 100 by multiplying top and bottom by 20, then write as a decimal and percent.

Way 1 (Paper Work):

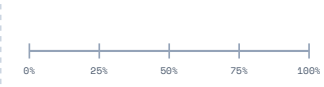


WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

2 Write 9/25 as a decimal and as a percent.

Starter Step 1: Scale the denominator of 9/25 to 100 by multiplying top and bottom by 4, then write as a decimal and percent.

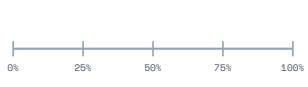
Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

3 Write 45% as a decimal and as a fraction in simplest form.

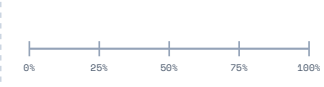
Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

4 Write 0.125 as a percent and as a fraction in simplest form.

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

5 Harvest Co-Op Banquet: A community hall seats 450 guests. If 60% of the seats are reserved for farm families, how many seats are reserved?

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

6 Bakery Dough Hydration: Elena uses a 65% hydration recipe (water weight is 65% of flour weight). For 5,400 grams of flour, how many grams of water are needed?

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

7 Grading Golden Honey Jars: An inspector grades 42 out of 50 jars of wildflower honey as Grade A Amber. What percent of the jars earned Grade A?

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

8 Write 150% as a decimal and as a mixed number in simplest form.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

9 Order 5/8, 60%, and 0.61 from least to greatest.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

10 Phone Battery Charge: A phone battery is at 35% of its 4,000 mAh capacity. How many mAh of charge remain?

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 3.4 — Practice B: Modeling Real Life (Percents, Fractions, and Decimals)

Textbook Ways: Equivalent Fraction over 100 | Divide Numerator ÷ Denominator | Percent Bar / Benchmark Model Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — FREE-THROW PERCENTAGE (3 TEXTBOOK WAYS • ANY 2 = +20 PTS)

Problem: A basketball player makes 18 out of 24 free throws. What percent of the free throws did the player make?

WAY 1: Simplify by GCF First, Then Scale to 100
Divide 18 and 24 by GCF = 6: 18/24 = 3/4. Scale 3/4 by ×25: 75/100 = 75%.

WAY 2: Divide Part ÷ Whole (18 ÷ 24)
Divide 18 ÷ 24 = 0.75. Write the decimal as a percent: 0.75 = 75%.

WAY 3: Percent Proportion (Part / Whole = p / 100)
Write 18 / 24 = p / 100 → 24p = 1,800 → p = 1,800 ÷ 24 = 75%.

WATCH OUT: 'Percent' means per 100: move the decimal 2 places RIGHT for % (0.06 = 6%, NOT 60%), and 2 places LEFT for a decimal.

Answer: 75% (3/4 = 0.75)

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 3.4 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

1 8 ENTER 2 4 ÷ FDISP

Simplify 18/24 and scale to 100 on paper first. Then check on your SC-6:
Press 18 [ENTER] 24 [÷] [FDISP] to toggle between 3/4 and 0.75 (and [f][→%] for 75%).

Zoe was trying to compare three different results—17 out of 20 gigabytes used on a tablet, 9 out of 25 soccer penalty kicks made, and 21 out of 24 counting trials answered by Alex the African grey parrot—and felt overwhelmed because every fraction had a different denominator. Finding a common denominator for 20, 25, and 24 looked miserable. Then she remembered that 'percent' simply means 'out of 100.' By turning each score into a percent on the same 0-to-100 scale (85%, 36%, and 87.5%), she could compare all three at a single glance.

SECTION 3.4 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Tablet Storage: You have used 17 out of 20 gigabytes on your tablet. Write the fraction of storage used as a decimal and as a percent.**

Starter Step 1: Scale the denominator of 17/20 to 100 by multiplying top and bottom by 5 to find the decimal and percent.

Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Soccer Goal Rate: A striker scores on 9 out of 25 penalty kicks. Write her scoring rate as a decimal and as a percent.**

Starter Step 1: Scale 9/25 to an equivalent fraction out of 100 (or divide 9 ÷ 25) to find the decimal and percent.

Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Quiz Mastery Score: You earn 40% of the points on Part A (worth 50 points) and 80% of the points on Part B (worth 50 points). How many total points did you score?**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **School Survey: In a poll of 160 middle schoolers, 65% voted for a robotics elective. How many students voted for robotics?**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Battery Charge Remaining: A laptop shows 15% battery remaining. Write 15% as a decimal and as a fraction in simplest form.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Science Club Membership: Eighteen students in the science club are 6th graders. If 6th graders make up 40% of the club, how many total students are in the science club?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Mental Math Tip: What is 15% of $60?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Alex the Parrot's Counting Trials: In Dr. Irene Pepperberg's lab, Alex the African grey parrot answered 21 out of 24 counting questions correctly! Write his accuracy as a simplified fraction, a decimal, and a percent.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Fundraiser Goal: A class has raised $360, which is 75% of their goal. What is their total fundraiser goal?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Order 3/5, 58%, and 0.62 from least to greatest.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 3.5 — Practice A: Solving Percent Problems, Discounts & Conversions

Key Vocabulary: discount • sale price • markup • sales tax • percent of change • conversion factor

Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & VISUAL DIAGRAM

Percent Discount Bar Model ($60 Hoodie at 25% Off) & Unit Conversion

Sale Price: 75% of $60 = $45	-25% ($15)

Way 1: $60 - $15 = $45 | Way 2: 0.75 × $60 = $45

% Change = (Change ÷ Original) × 100%

Watch Out: Always divide by the ORIGINAL amount!

Unit Conversion

Cancel old units:

32 in × (2.54 cm / 1 in)

= 81.28 cm

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Discounts & Markups: Method 1: Find the discount/tax and subtract or add. Method 2: Multiply by the remaining/total percent (e.g., 75% or 107.5%).

Textbook Percent of Change & Conversions: Percent Change = (Amount of Change ÷ Original Amount) × 100%; multiply by unit conversion factors so old units cancel.

WATCH OUT: For percent change, ALWAYS divide the change by the ORIGINAL starting amount — never the new ending amount!

WORKED EXAMPLE — 3 TEXTBOOK WAYS (ANY 2 WAYS = +20 PTS)

Problem: A $60 video game is on sale for 25% off. Find the discount and the sale price.

WAY 1: Find the 25% Discount First, Then Subtract

Discount = 25% of $60 = 0.25 × $60 = $15. Subtract from original price: $60 - $15 = $45.

WAY 2: Multiply by the Remaining Percent (75%)

Taking 25% off leaves 100% - 25% = 75% of the price: 0.75 × $60 = (3/4) × $60 = $45.

WAY 3: Benchmark Quarters Bar Model

Split $60 into four 25% quarters ($15 each). Remove 1 quarter ($15); 3 quarters ($45) remain.

Answer: Discount = $15; Sale Price = $45

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 3.5 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[6] [0] [ENTER] [2] [5] [%] [-]

Calculate the 25% discount on paper first. Then check on your SC-6:

Type 60 [ENTER] 25 [%] (which computes 25% of 60 = $15 while keeping 60 on Y!) and press [-] to get the $45 sale price.

SECTION 3.5 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Wool Work Coat: An $85 wool coat is on sale for 20% off. Find the discount amount and the sale price.**

Starter Step 1: Multiply $85 × 0.20 to find the 20% discount amount first, then subtract that discount from $85.

Way 1 (Paper Work):

Part = % × Whole: ___ × ___ = ___

Orig: _____ ± %Amt: _____ = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Garden Cart Checkout: A steel garden cart costs $120.00 before an 8.5% sales tax. Find the sales tax and the final price.**

Starter Step 1: Multiply $120 × 0.085 to find the sales tax amount first, then add the tax to $120.00.

Way 1 (Paper Work):

Part = % × Whole: ___ × ___ = ___

Orig: _____ ± %Amt: _____ = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Seedling Height Surge: A pine seedling grows from 15 mm to 75 mm. What is the percent of increase in its height?**

Way 1 (Paper Work):

Part = % × Whole: ___ × ___ = ___

Orig: _____ ± %Amt: _____ = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Find the percent of decrease from 80 down to 60.**

Way 1 (Paper Work):

Part = % × Whole: ___ × ___ = ___

Orig: _____ ± %Amt: _____ = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Use the conversion factor 1 in = 2.54 cm to convert 10 inches to centimeters.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Use the conversion factor 1 in = 2.54 cm to convert 50.8 centimeters to inches.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Workshop Tool Backpack: A $56 canvas tool backpack is on sale for 30% off. What is the sale price?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **18 is 25% of what whole number?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Stacking Two Discounts on a Solar Battery: A $200 solar battery is marked 20% off, and an extra 10% clearance coupon is applied to the reduced sale price. What is the final price?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **A 32-inch skateboard deck weighs 5 pounds. Convert its length to centimeters (1 in = 2.54 cm) and its mass to kilograms (1 lb ≈ 0.4536 kg, rounded to the nearest hundredth).**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 3.5 — Practice B: Modeling Real Life (Solving Percent Problems, Discounts & Conversions)

Textbook Ways: Find Discount & Subtract | Multiply by Remaining % | Percent Proportion / Ratio Bar Name: _____ Date: _____ Score: _____ / 100

WORKED EXAMPLE — PERCENT INCREASE & UNIT CONVERSION (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: A robotics club grows from 250 to 325 members. Find the percent of increase, and convert a 32-inch robot arm to centimeters (1 in = 2.54 cm).

WAY 1: Amount of Change ÷ Original Amount

Change = 325 - 250 = 75. Divide by original: 75 ÷ 250 = 0.30 = 30% increase. Conversion: 32 in × (2.54 cm / 1 in) = 81.28 cm.

WAY 2: Scale Ratio to 100 (or New ÷ Original)

75/250 = 15/50 = 30/100 = 30% (or 325 ÷ 250 = 130% → +30%). For 32 in: 30(2.54) + 2(2.54) = 76.2 + 5.08 = 81.28 cm.

Check: 250 + 0.30(250) = 250 + 75 = 325 members.

WATCH OUT: For percent change, ALWAYS divide the change by the ORIGINAL starting amount — never the new ending amount!

Answer: 30% increase; 81.28 cm

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 3.5 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

2 5 0 ENTER 3 2 5 g %CHG 3 2 f ▶cm

Find the percent change and unit conversion on paper first. On your SC-6, type 250 [ENTER] 325 [g][%CHG] to verify +30%, and 32 [f][→cm] to convert 32 inches to 81.28 cm!

When a $100 bike helmet bundle went on sale for 20% off and Priya had an extra coupon for '10% off the sale price,' she confidently added 20% + 10% = 30% off and expected to pay $70. When the register rang up $72 instead, she thought the store had overcharged her! Writing out the two steps on paper, she saw why: the first 20% off dropped the $100 price to $80, and the extra 10% coupon came off that new $80 sale price (10% of $80 is $8, leaving $72). She realized you always have to ask what starting 'whole' a percent is being taken from.

SECTION 3.5 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Gaming Headset Deal: A $75 wireless headset is on sale for 20% off. What is the discount amount, and what is the sale price?**

Starter Step 1: Multiply $75 × 0.20 to find the 20% discount amount, then subtract it from $75 (or multiply $75 × 0.80).

Way 1 (Paper Work):

Part = % × Whole: ___ × ___ = ___

Orig: _____ ± %Amt: _____ = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Team Dinner Tip: A team dinner costs $64.00 before tip. Your family leaves a 15% tip. What is the tip amount, and what is the total bill?**

Starter Step 1: Find 15% of $64.00 (find 10% + 5%, or multiply $64 × 0.15), then add the tip to $64.00.

Way 1 (Paper Work):

Part = % × Whole: ___ × ___ = ___

Orig: _____ ± %Amt: _____ = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Sales Tax Checkout: You buy a $48.00 build plate plus 7.5% sales tax. Find the tax amount and the total cost.**

Way 1 (Paper Work):

Part = % × Whole: ___ × ___ = ___

Orig: _____ ± %Amt: _____ = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Wildlife Sanctuary Bald Eagle Recovery: A sanctuary's nesting eagle count rose from 40 pairs last year to 52 pairs this year. What is the percent of increase?**

Way 1 (Paper Work):

Part = % × Whole: ___ × ___ = ___

Orig: _____ ± %Amt: _____ = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Winter Jacket Clearance: A jacket was marked down from $90 to $63. What is the percent of decrease?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Race Distance Conversion: How many kilometers are in a 10-mile race? (Use 1 mi ≈ 1.609 km.)**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Luggage Weight: Your suitcase weighs 20 kilograms. How many pounds does it weigh? (Use 1 kg ≈ 2.205 lb.)**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Water Jug Capacity: A sports jug holds 1 gallon of water. How many liters is that (1 gal ≈ 3.79 L), and how many fluid ounces (1 gal = 128 fl oz)?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Successive Discounts: A $100 bike is on sale for 20% off, and you have an additional 10%-off coupon that applies to the sale price. What is the final price, and why is it not $70?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Temperature Lab: Room temperature in a science lab is 20°C. What is this temperature in degrees Fahrenheit? (Use F = 1.8C + 32.)**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

Unit 4: Algebraic Expressions & Properties — Coding an Arcade Game & Running the School Craft Booth

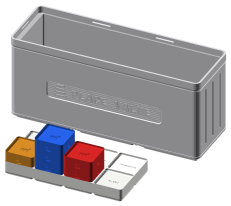
Evaluating Algebraic Expressions, Combining Like Terms, Distributive Property & Factoring with GCF

When Noah tried to code his first arcade game for the school STEM showcase, he almost deleted the whole project in frustration. To track a player's carnival tickets, he started writing one line of code for 1 ride, another line for 2 rides, and another for 3 rides. By twenty lines, his screen was a tangled mess. And when letters like x and g showed up in his math homework that same week, he complained that letters belonged in English class, not math.

Then his older sister showed him that a letter in code—and in algebra—isn't a mystery to be afraid of: it's just a labeled box that holds whatever number changes from turn to turn. Writing one short rule like 50 - 4r let the computer plug in any number of rides r and find the remaining tickets in a split second.

Across the gym at the craft booth, Maya was having the same trouble keeping track of custom soccer uniform bundles—where every player needed both a jersey and $7 shorts—and packing leftover stickers into identical gift bags without coming up short.

As Noah, Maya, and Zoe work through these lessons, their nervousness around letters in math fades away. They see how plugging numbers into a formula, sorting matching items into neat piles, and bundling or unbundling group orders makes their work shorter—and how substituting a test number on the SC-6 proves whether two algebraic expressions are truly equivalent.



Real-World Mathematical Context & Two-Method Mastery • Hands-On Models: Place Value & Prime Factor Blocks

UNIT 4 LESSONS — LEARN TWO TEXTBOOK WAYS FIRST, THEN CHECK YOUR WORK ON THE SC-6 CALCULATOR

SEC 4.1 (pp.37-38) Writing and Evaluating Algebraic Expressions

Textbook Ways: Way 1: Substitute in Parentheses & Follow PEMDAS | Way 2: Equal-Groups Bar / Place Value Model

Learn by Hand: An algebraic expression contains numbers, variables (letters representing unknown quantities), and operations. In an expression like 4x + 7, 4x and 7 are terms, 4 is the coefficient of x, and 7 is a constant. To evaluate an algebraic expression, substitute the given number for each variable and follow the Order of Operations (PEMDAS).

Check Your Work on the SC-6 Calculator:

[4] [ENTER] [5] [×] [7] [+]

Press 4 [ENTER] 5 [×] 7 [+] to confirm 27.

SEC 4.2 (pp.39-40) Simplifying Expressions by Combining Like Terms

Textbook Ways: Way 1: Reorder & Add Coefficients (Properties) | Way 2: Sort by Variable Power Columns

Learn by Hand: Terms that have the exact same variable raised to the exact same exponent are called Like Terms (such as 4x and 2x, or $3x^2$ and $2x^2$). Constant terms (such as 5 and 3) are also like terms. To simplify an expression, use the Commutative and Associative Properties to group like terms together and add or subtract their coefficients.

Check Your Work on the SC-6 Calculator:

[4] [ENTER] [2] [+] [5] [ENTER] [3] [+]

Group the x-terms (4+2=6) and constants (5+3=8) on paper first. On your SC-6, test x = 2 in both forms (20 [ENTER] 6 [ENTER] 2 [×] 8 [+]) and press [≠]—it flashes NO (not unequal, proving both match!).

SEC 4.3 (pp.41-42) The Distributive Property

Textbook Ways: Way 1: Multiply Each Term Inside $a(b + c) = ab + ac$ | Way 2: Split-Rectangle Area Model | Way 3: Repeated Addition of Rows

Learn by Hand: To multiply a sum or difference by a number, multiply each term inside the parentheses by the number outside the parentheses: a(b + c) = ab + ac and a(b - c) = ab - ac. Geometrically, a rectangle of height 3 and split width (x + 4) has two partial areas: 3(x) + 3(4) = 3x + 12.

Check Your Work on the SC-6 Calculator:

[6] [ENTER] [1] [0] [×] [6] [ENTER] [4] [×] [+]

Distribute 6(10 + 4) on paper first. On your SC-6, evaluate 10 [ENTER] 4 [+] 6 [×] (84 on Y) and 6 [ENTER] 10 [×] 6 [ENTER] 4 [×] [+] (84 on X), then press [≠]—it flashes NO to prove both forms are equal!

SEC 4.4 (pp.43-44) Factoring Expressions Using the GCF

Textbook Ways: Way 1: Factor Out the GCF Using Prime Factors | Way 2: Equal-Rows Area Model / Division Ladder

Learn by Hand: Factoring an expression means writing a sum or difference as a product—the exact reverse of the Distributive Property! To factor an expression completely, find the Greatest Common Factor (GCF) of its terms, write the GCF outside the parentheses, and divide each term by the GCF to find the terms inside.

Check Your Work on the SC-6 Calculator:

[1] [8] [ENTER] [2] [4] [GCF]

Press 18 [ENTER] 24 [GCF] to confirm GCF = 6, giving 6(3x + 4), and test x = 2 in both forms with [≠] (flashes NO!).

HANDS-ON MODEL: PLACE VALUE BLOCKS

Section 4.1 — Practice A: Writing and Evaluating Algebraic Expressions

Key Vocabulary: algebraic expression • variable • term • coefficient • constant • evaluate • substitute

Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & HANDS-ON MODEL (PLACE VALUE BLOCKS)

Parts of an Expression: 4x + 7

4x + 7

Coefficient = 4
Variable = x
Constant = 7

2 Terms: 4x and 7

Substitute x = 5 into 4x + 7



4(5) + 7 = 20 + 7 = 27

Multiply 4(5) BEFORE adding 7 (PEMDAS)!

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept: To evaluate an algebraic expression, substitute the given value for each variable in parentheses and follow PEMDAS.

Textbook Vocabulary: Identify terms (parts added), coefficients (number multiplied by a variable), and constants (numbers alone).

WATCH OUT: When substituting x into 4x + 7, 4x means 4 × x (multiply 4 × 5 = 20 first; do NOT write '45')!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Evaluate 4x + 7 when x = 5.

WAY 1: Substitute with Parentheses & Follow PEMDAS

Replace x with (5) in 4x + 7: 4(5) + 7. Multiply first (4 × 5 = 20), then add 7: 20 + 7 = 27.

WAY 2: Equal-Groups Bar / Place Value Model (With Blocks)

Write 4x + 7 as four x-groups plus 7: (5 + 5 + 5 + 5) + 7 = 20 + 7 = 27.

Answer: 27

1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 4.1 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[4] [ENTER] [5] [×] [7] [+]

Substitute x = 5 into 4x + 7 on paper first. Then check on your SC-6:
Press 4 [ENTER] 5 [×] 7 [+] to confirm 27.

SECTION 4.1 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Evaluate 6x - 9 when x = 7.**

Starter Step 1: Substitute (7) in place of x in 6x - 9, multiply 6 × 7 first, and then subtract 9.

Way 1 (Paper Work):

Substitute in ( ):

____(___) ± ____ = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Evaluate 5n + 14 when n = 8.**

Starter Step 1: Substitute (8) in place of n in 5n + 14, multiply 5 × 8 first, and then add 14.

Way 1 (Paper Work):

Substitute in ( ):

____(___) ± ____ = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Evaluate 40 - 3m when m = 9.**

Way 1 (Paper Work):

Substitute in ( ):

____(___) ± ____ = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Evaluate x^2 + 11 when x = 6.**

Way 1 (Paper Work):

Substitute in ( ):

____(___) ± ____ = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Evaluate $2y^2$ - 5 when y = 4.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Evaluate 7a + 3b when a = 4 and b = 9.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Evaluate (x + y) ÷ 4 when x = 19 and y = 13.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Evaluate 5k - 2.4 when k = 3.5.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Skate Park Rental: A skate park charges an $8 helmet fee plus $4.50 per hour. Write an algebraic expression for the total cost of h hours, and find the total cost for 4 hours.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Cube Display Case: The surface area of a cube with edge length s is given by the expression $6s^2$. What is the surface area of a cube with edge length s = 5 cm?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: PLACE VALUE BLOCKS

Section 4.1 — Practice B: Modeling Real Life (Writing and Evaluating Algebraic Expressions)

Textbook Ways: Substitute in Parentheses & Follow PEMDAS | Equal-Groups Bar / Place Value Model Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — TWO-VARIABLE SCORE FORMULA (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: In a game, each boss trophy (a) is worth $3a^2$ points and each gem (b) is worth 4b points. What is your total score if you collect 5 trophies and 6 gems?

WAY 1: Substitute into $3a^2 + 4b$ & Follow PEMDAS

Write $3(5)^2 + 4(6)$. Evaluate exponent first ($5^2 = 25$), multiply ($3 \times 25 = 75$ and $4 \times 6 = 24$), then add: $75 + 24 = 99$ points.

WAY 2: Find Each Item's Sub-Score Separately

1 trophy = $3(5\times5) = 75$ pts. 6 gems = $6 \times 4 = 24$ pts. Combine both sub-scores: $75 + 24 = 99$ points.

Watch-Out: Evaluate $5^2 = 25$ before multiplying by 3 (never multiply 3×5 first!).

WATCH OUT: When substituting x into 4x + 7, 4x means 4 × x (multiply 4 × 5 = 20 first; do NOT write '45')!

Answer: 99 points

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 4.1 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[5] [x^2] [3] [×] [4] [ENTER] [6] [×] [+]

Substitute t = 5 and b = 4 into $3t^2 + 6b$ on paper first. Then check on your SC-6:
Press 5 [x^2] 3 [×] (75 waits on Y), then 4 [ENTER] 6 [×] (24 on X), and [+] to confirm 99.

Noah used to freeze up whenever letters showed up inside math expressions like $6 + 5g$ or $16t^2$, convinced that algebra was way over his head. While coding the checkout screen for a bowling game ($6 shoe rental plus $5 per game g), he realized that the letter g wasn't a number he had to solve for—it was just an empty input box waiting for the player to type how many games they bowled! Once he drew parentheses around the slot—$6 + 5(\ )$—and dropped the number 4 inside, it turned right back into ordinary arithmetic he already knew how to do.

SECTION 4.1 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Bowling Alley: Bowling costs $6 for shoe rental plus $5 per game g. Write an expression for the total cost and find the cost of playing 4 games.**

Starter Step 1: Multiply the $5 per-game cost by g games and add the $6 shoe rental, then substitute g = 4.

Way 1 (Paper Work):

Substitute in ( ):

____(___) ± ____ = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Carnival Wristband: You start with 50 tickets and spend 4 tickets on each ride r. Write an expression for your remaining tickets and find how many are left after 9 rides.**

Starter Step 1: Start with 50 tickets and subtract 4 tickets per ride r, then substitute r = 9 rides.

Way 1 (Paper Work):

Substitute in ( ):

____(___) ± ____ = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Square Tile Patio Cost: Paving a square patio of side length s feet costs $8s^2 + 25$ dollars. What is the cost when s = 5 feet?**

Way 1 (Paper Work):

Substitute in ( ):

____(___) ± ____ = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Kea Parrot Puzzle Box: In a New Zealand wildlife lab, a kea parrot earns $10w + 3d$ reward points for unlocking w wire latches and d sliding doors. Find its score when w = 7 and d = 4.**

Way 1 (Paper Work):

Substitute in ( ):

____(___) ± ____ = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Falling Object Physics: A ball dropped from a tower falls $16t^2$ feet in t seconds. How far does it fall in 3 seconds?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Cube Surface Area: The surface area of a cube with edge length s is $6s^2$. What is the surface area when s = 4 inches?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Snack Stand Order: Popcorn costs $7 and drinks cost $4. Write an expression for p popcorns and d drinks, and find the total cost for 3 popcorns and 5 drinks.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Evaluate $1/2h(a + b)$ when h = 6, a = 11, and b = 15.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **3D Print Batch Time: Printing k keycaps takes $12k + 15$ minutes (including 15 minutes for bed warm-up). How many minutes does a batch of 8 keycaps take?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Evaluate $2x^3 - 3y^2$ when x = 4 and y = 5.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

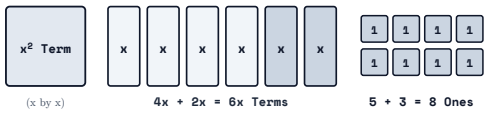
CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

Section 4.2 — Practice A: Simplifying Expressions by Combining Like Terms

Key Vocabulary: like terms • simplest form • Commutative Property • Associative Property • equivalent expressions Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & HANDS-ON MODEL (PLACE VALUE BLOCKS)

Combining Like Terms by Variable Power: $4x + 5 + 2x + 3 = 6x + 8$



Watch Out: 6x and 8 are unlike terms → NEVER combine into 14x!

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept: Like terms have the exact same variable raised to the exact same power. Add or subtract their coefficients; keep the variable part unchanged.

Textbook Equivalence Check: Substitute a test value (such as $x = 2$ or $x = 10$) into both the original and simplified expressions to verify they match.

WATCH OUT: Combine only terms with the EXACT same variable and exponent — never combine $6x + 8$ into $14x$!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Simplify the expression $4x + 5 + 2x + 3$.

WAY 1: Commutative & Associative Grouping

Reorder and group like terms: $(4x + 2x) + (5 + 3) = (4 + 2)x + 8 = 6x + 8$.

WAY 2: Sort by Place-Value / Variable Columns

Place x-terms in Column 1 ($4x + 2x = 6x$) and constants in Column 2 ($+5 + 3 = +8$) → $6x + 8$.

Answer: $6x + 8$ 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 4.2 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

4 ENTER 2 + 5 ENTER 3 +

Group the x-terms (4+2=6) and constants (5+3=8) on paper first. On your SC-6, test x = 2 in both forms (20 [ENTER] 6 [ENTER] 2 [×] 8 [+]) and press [≠]—it flashes NO (not unequal, proving both match!).

SECTION 4.2 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Simplify the expression: $7x + 4 + 2x + 9$.**

Starter Step 1: Group the like x-terms ($7x + 2x$) together and the constant numbers ($4 + 9$) together, then add.

Way 1 (Paper Work):

Var Terms	Constants

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Simplify the expression: $8x + 11 - 3x - 4$.**

Starter Step 1: Group the x-terms ($8x - 3x$) and constant terms ($11 - 4$), keeping the + or - sign in front of each term.

Way 1 (Paper Work):

Var Terms	Constants

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Simplify the expression: $9y - y + 6$.**

Way 1 (Paper Work):

Var Terms	Constants

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Simplify the expression: $12a + 5 - 4a + 8$.**

Way 1 (Paper Work):

Var Terms	Constants

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Simplify the expression: $4x^2 + 3x + 2x^2 + 5x + 7$.**

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Simplify the expression: $6n^2 + 9n - 2n^2 - 4n + 10$.**

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Simplify the expression: $3.5x + 4.2 + 1.8x - 1.5$.**

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Error Analysis: Leo simplified $5x + 3$ to $8x$. Explain why $5x$ and 3 cannot be combined, and prove they are not equivalent by substituting $x = 10$ into both expressions.**

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Garden Border: A rectangular garden has two sides of length $(2x + 3)$ meters and two sides of length $(x + 5)$ meters. Write a simplified algebraic expression for its perimeter, and find the perimeter when $x = 10$.**

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Pennant Frame: A triangular frame has side lengths of $(3x + 2)$ cm, $(2x + 7)$ cm, and $(4x - 1)$ cm. Write a simplified expression for its perimeter, and find the perimeter when $x = 5$.**

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: PLACE VALUE BLOCKS

Section 4.2 — Practice B: Modeling Real Life (Simplifying Expressions by Combining Like Terms)

Textbook Ways: Reorder & Add Coefficients (Properties) | Sort by Variable Power Columns Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — COMBINING x^2, x, AND CONSTANT TERMS (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: Simplify the expression $3x^2 + 5x + 2x^2 - x + 6$.

WAY 1: Group Terms with the Same Variable & Exponent

Rewrite -x as -1x: $(3x^2 + 2x^2) + (5x - 1x) + 6 = (3 + 2)x^2 + (5 - 1)x + 6 = 5x^2 + 4x + 6$.

WAY 2: Sort into x^2, x, and Constant Columns

Combine $3 + 2 = 5$ in the x^2-column ($5x^2$), $5 - 1 = 4$ in the x-column ($4x$), and 6 in the ones column ($+6$) → $5x^2 + 4x + 6$.

Check with x = 2: Original = 12 + 10 + 8 - 2 + 6 = 34; Simplified = 5(4) + 4(2) + 6 = 34.

WATCH OUT: Combine only terms with the EXACT same variable and exponent — never combine $6x + 8$ into $14x$!

Answer: $5x^2 + 4x + 6$

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 4.2 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

3 ENTER 2 + 5 ENTER 1 -

Combine the x^2-terms, x-terms, and constants on paper first. Check each coefficient sum on your SC-6 (3 [ENTER] 2 [+] = $5x^2$ and 5 [ENTER] 1 [-] = 4x), then test x = 2 in both expressions with [≠] (flashes NO!).

Looking at the expression $6x + 9 - 2x + 5$, Noah was tempted to mash everything together into 18x so there would only be one piece left. To check if that made sense, he set 6 sealed pouches of x tokens and 9 loose tokens on his desk, took away 2 pouches, and added 5 loose tokens. Immediately he laughed at his first guess: you can't dump loose single tokens into the pouch count! Sorting the pouches together ($6x - 2x = 4x$) and the loose tokens together ($9 + 5 = 14$) showed him why only 'like terms' can be combined into $4x + 14$.

SECTION 4.2 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Triangle Perimeter: A triangle has side lengths $(2x + 3)$, $(3x + 1)$, and $(4x + 5)$. Write and simplify an expression for its perimeter.**

Starter Step 1: Add all three side expressions together by grouping the x-terms ($2x + 3x + 4x$) and the constants ($3 + 1 + 5$).

Way 1 (Paper Work):

Var Terms	Constants

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

2 **Rectangle Perimeter: A garden is $(3x + 4)$ feet long and $(x + 2)$ feet wide. Write and simplify an expression for its perimeter.**

Starter Step 1: Perimeter adds all 4 sides—double the length $2(3x + 4)$ and double the width $2(x + 2)$, then combine like terms.

Way 1 (Paper Work):

Var Terms	Constants

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

3 **Arcade Token Pouches: You have 6 pouches of x tokens plus 9 loose tokens, give away 2 pouches, and win 5 more loose tokens. Write and simplify an expression for your total tokens.**

Way 1 (Paper Work):

Var Terms	Constants

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

4 **Simplify $4x^2 + 7x + 3 + 2x^2 - 3x + 8$.**

Way 1 (Paper Work):

Var Terms	Constants

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

5 **Bake Sale Revenue: On Saturday you earn $(8c + 12)$ dollars and on Sunday you earn $(5c + 19)$ dollars. Write a simplified expression for your weekend total.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

6 **Simplify $9x + 14 - 2x + 5$, then state the coefficient of x and the constant term.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

7 **Simplify $(3/8)n + 4 + (1/8)n + 9$.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

8 **Simplify $2.4y + 5.8 + 1.6y - 2.3$.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

9 **Simplify $10m - 4m + 15 - 6$, then check your work by substituting $m = 5$ into both the original and simplified expressions.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

10 **Simplify $5x^2 + 6x + 10 - 2x^2 - 6x - 3$.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

HANDS-ON MODEL: PLACE VALUE BLOCKS

Section 4.3 — Practice A: The Distributive Property

Key Vocabulary: Distributive Property • area model • partial products • equivalent expressions

Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & HANDS-ON MODEL (PLACE VALUE BLOCKS)

Area Model: $6(10 + 4) = 60 + 24$

6	10	+ 4
	6 × 10 = 60	6×4=24

Total Area: 6(14) = 60 + 24 = 84

Algebraic Distribution: $a(b + c) = ab + ac$

3 (2x + 5)
= 3(2x) + 3(5)
= 6x + 15

Multiply BOTH terms inside ( ) by 3!

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept (Distributive Property): To multiply a sum or difference by a number, multiply EACH term inside parentheses: $a(b + c) = ab + ac$.

Textbook Area Model: A rectangle of height a and split width (b + c) has total area a(b + c), which equals the sum of the two sub-areas ab + ac.

WATCH OUT: Multiply the outside factor by EVERY term inside the parentheses: $3(2x + 5) = 6x + 15$ (do not forget 3 × 5)!

WORKED EXAMPLE — 3 TEXTBOOK WAYS (ANY 2 WAYS = +20 PTS)

Problem: Evaluate $6(10 + 4)$ using the Order of Operations and using the Distributive Property.

WAY 1: Use the Distributive Property $a(b + c) = ab + ac$
Multiply 6 by each term inside the parentheses: $6(10 + 4) = 6(10) + 6(4) = 60 + 24 = 84$.

WAY 2: Use the Order of Operations (Combine Width First)
Add inside the parentheses first: $6(10 + 4) = 6(14) = 84$.

WAY 3: Split-Rectangle Area Model (Partial Products)
A 6-by-14 rectangle splits into a 6×10 section (60) and a 6×4 section (24): 60 + 24 = 84.

Answer: 84 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 4.3 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

6 ENTER 1 0 × 6 ENTER 4 × +

Distribute 6(10 + 4) on paper first. On your SC-6, evaluate 10 [ENTER] 4 [+] 6 [×] (84 on Y) and 6 [ENTER] 10 [×] 6 [ENTER] 4 [×] [+] (84 on X), then press [≠]—it flashes NO to prove both forms are equal!

SECTION 4.3 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 Use the Distributive Property to find the two partial products and total of $7(20 + 3)$.

Starter Step 1: Multiply the outside factor 7 by EACH term inside parentheses—(7 × 20) + (7 × 3)—and add the products.

Way 1 (Paper Work):

--- Term 1 | Term 2

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

2 Use the Distributive Property to simplify $4(x + 5)$.

Starter Step 1: Multiply the outside factor 4 by BOTH terms inside the parentheses: 4(x) + 4(5).

Way 1 (Paper Work):

--- Term 1 | Term 2

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

3 Use the Distributive Property to simplify $6(3x + 4)$.

Way 1 (Paper Work):

--- Term 1 | Term 2

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

4 Use the Distributive Property to simplify $8(2y - 5)$.

Way 1 (Paper Work):

--- Term 1 | Term 2

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

5 Evaluate 9×43 mentally using the Distributive Property by splitting 43 into $(40 + 3)$.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

6 Simplify the expression: $3(4x + 2) + 5x$.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

7 Simplify the expression: $4(3n + 5) - 7$.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

8 Error Analysis: Mia wrote $5(x + 6) = 5x + 6$. Explain her error and write the correct simplified expression.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

9 Simplify $5(2x + 3) + 4x$, and verify your simplified expression by evaluating both expressions when $x = 2$.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

10 Team Lunch Order: Six teammates each order a sandwich costing x dollars and a smoothie costing $3.50. Write a factored expression and an expanded expression for the total cost, then find the total if each sandwich costs $6.50.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

HANDS-ON MODEL: PLACE VALUE BLOCKS

Section 4.3 — Practice B: Modeling Real Life (The Distributive Property)

Textbook Ways: Multiply Each Term Inside a(b + c) = ab + ac | Split-Rectangle Area Model | Repeated Addition of Rows

Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — SIMPLIFYING A COMBO ORDER EXPRESSION (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: Five friends each buy 2 tacos (x dollars each) and a $3 drink, and the group orders 4 extra tacos: 5(2x + 3) + 4x. Simplify the expression.

WAY 1: Distribute First, Then Combine Like Terms

Distribute 5 across (2x + 3): 5(2x) + 5(3) + 4x = 10x + 15 + 4x. Combine x-terms: (10x + 4x) + 15 = 14x + 15.

WAY 2: Equal-Group Rows (5 Groups of 2x + 3)

Five groups of (2x + 3) give 10x + 15. Add the 4 extra tacos (4x) → 14x + 15.

Check at x = $2: 5(2·2 + 3) + 4(2) = 35 + 8 = $43, and 14(2) + 15 = $43.

WATCH OUT: Multiply the outside factor by EVERY term inside the parentheses: 3(2x + 5) = 6x + 15 (do not forget 3 × 5)!

Answer: 14x + 15

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 4.3 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

5 ENTER 2 × 4 + 5 ENTER 3 ×

Distribute 5(2x + 3) + 4x = 14x + 15 on paper first. Check the coefficients on your SC-6 (5 [ENTER] 2 [×] 4 [+] = 14; 5 [ENTER] 3 [×] = 15) and test x = 2 in both forms with [≠] (flashes NO!).

When Maya wrote 4(x + 7) for four soccer players each ordering an $x jersey and $7 shorts, she accidentally simplified it to 4x + 7—and couldn't figure out why her budget came up $21 short. Laying out four separate order cards on the craft booth table, she saw her mistake right away: all four players got the $x jersey AND all four players got the $7 shorts (4x + 28), whereas 4x + 7 only bought shorts for one kid! Better yet, she realized the same Distributive Property let her multiply 8 × $53 in her head without scratch paper by splitting $53 into 8($50 + $3) = $400 + $24 = $424.

SECTION 4.3 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Mental Math Ticket Split: Use the Distributive Property to find the cost of 8 tickets at $53 each by splitting 53 into 50 + 3.**

Starter Step 1: Write 8(50 + 3), then multiply (8 × 50) + (8 × 3) and add the two partial products.

Way 1 (Paper Work):

Term 1 | Term 2

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Price Tag Trick: Use the Distributive Property to find the cost of 6 shirts at $9.90 each by writing $9.90 as ($10 - $0.10).**

Starter Step 1: Write 6($10 - $0.10), then multiply (6 × $10) - (6 × $0.10) and subtract.

Way 1 (Paper Work):

Term 1 | Term 2

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Split Rectangle Garden: A rectangular garden is 7 feet wide. Its length is split into an x-foot herb bed and an 8-foot flower bed. Write the total area in factored form and expanded form.**

Way 1 (Paper Work):

Term 1 | Term 2

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Soccer Uniform Order: Nine players each buy a jersey for 2x dollars and shorts for $7. Write and simplify an expression for the team's total cost.**

Way 1 (Paper Work):

Term 1 | Term 2

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Coupon Discount: Four friends each buy a ticket priced at t dollars using a $3-off coupon. Write and expand an expression for the total paid, and find the total when t = $15.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Simplify 3(4x + 5) + 2x.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Simplify 4(2x + 3) + 5(x + 2).**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Simplify (1/2)(6x + 10).**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Rectangle Perimeter: A rectangle has length (2x + 5) and width (x + 3). Write and simplify an expression for its perimeter.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Evaluate 12(10 + 5) in two ways: using the Order of Operations and using the Distributive Property.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: PRIME FACTOR TOWERS

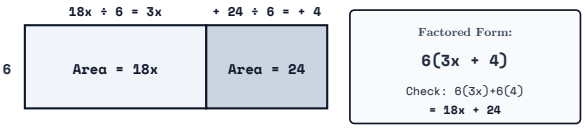
Section 4.4 — Practice A: Factoring Expressions Using the GCF

Key Vocabulary: factoring an expression • greatest common factor (GCF) • factored form • expanded form

Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & HANDS-ON MODEL (PRIME FACTOR TOWERS)

Factoring = Undoing the Distributive Property Using the GCF: 18x + 24 = 6(3x + 4)



Factored Form:

6(3x + 4)

Check: 6(3x)+6(4) = 18x + 24

6 Equal Rows of (3x + 4) Tiles | GCF(3, 4) = 1 → Factored Completely!

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept (Factoring out the GCF): Find the GCF of all terms, write the GCF outside parentheses, and divide each term by the GCF: ab + ac = a(b + c).

Textbook Verification: Distribute your factored expression back out and check that the inside terms have no common factors left (GCF = 1).

WATCH OUT: Factor out the GREATEST common factor (GCF) so the terms left inside parentheses share no common factor > 1.

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Factor 18x + 24 completely using the GCF.

WAY 1: Find GCF & Divide Each Term

GCF(18, 24) = 6. Rewrite each term as a product with 6: 18x + 24 = 6(3x) + 6(4) = 6(3x + 4).

WAY 2: Equal-Rows Area Model / Division Ladder (With Blocks)

Arrange 18x + 24 into 6 equal rows of (3x + 4), or use a ladder: divide (18x + 24) by 2 → 3 → GCF = 6, leaving (3x + 4).

Answer: 6(3x + 4) 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 4.4 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

1 8 ENTER 2 4 GCF

Find the GCF of 18 and 24 and factor on paper first. Check on your SC-6:

Press 18 [ENTER] 24 [GCF] to confirm GCF = 6, giving 6(3x + 4), and test x = 2 in both forms with [≠] (flashes NO!).

SECTION 4.4 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Factor the numerical expression 12 + 20 using the GCF.**

Starter Step 1: Find the Greatest Common Factor (GCF) of 12 and 20, write it outside parentheses, and divide both terms by that GCF.

Way 1 (Paper Work):

GCF = _____

Factored: ____(_____ ± _____)

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Factor 8x + 12 completely using the GCF.**

Starter Step 1: Find the GCF of 8x and 12, write it outside the parentheses, and divide both terms by that GCF.

Way 1 (Paper Work):

GCF = _____

Factored: ____(_____ ± _____)

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Factor 15y - 25 completely using the GCF.**

Way 1 (Paper Work):

GCF = _____

Factored: ____(_____ ± _____)

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Factor 24x + 36 completely using the GCF.**

Way 1 (Paper Work):

GCF = _____

Factored: ____(_____ ± _____)

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Factor 32n - 48 completely using the GCF.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Factor 42a + 14 completely using the GCF.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Factor 36m - 60 completely using the GCF.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Error Analysis: Sam factored 24x + 36 as 6(4x + 6). Explain why his expression is not factored completely, and write the completely factored form.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Club Banner Width: A rectangular banner has an area of (28x + 42) square inches. Its height in inches is the greatest whole number that divides evenly into both 28 and 42. What are the height and width of the banner?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Square Garden Perimeter: A square garden has a total perimeter of (36x + 48) feet. What is the length of one side of the garden?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: PRIME FACTOR TOWERS

Section 4.4 — Practice B: Modeling Real Life (Factoring Expressions Using the GCF)

Textbook Ways: Factor Out the GCF Using Prime Factors | Equal-Rows Area Model / Division Ladder Name: _____ Date: _____ Score: _____ / 100

WORKED EXAMPLE — FINDING GARDEN DIMENSIONS BY FACTORING (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: A rectangular garden has an area of (45x - 30) square feet. Factor the expression using the GCF to find possible dimensions of the garden.

WAY 1: Prime Factorization GCF Method

$45 = 3^2 \times 5$ and $30 = 2 \times 3 \times 5 \rightarrow$ GCF $= 3 \times 5 = 15$. Divide each term by 15: $45x - 30 = 15(3x - 2)$.

WAY 2: Step-by-Step Division Ladder

Divide (45x - 30) by $5 \rightarrow 5(9x - 6)$; then factor out $3 \rightarrow 5 \cdot 3(3x - 2) = 15(3x - 2)$ (Width = 15 ft, Length = 3x - 2 ft).

Check by distributing: 15(3x) - 15(2) = 45x - 30.

WATCH OUT: Factor out the GREATEST common factor (GCF) so the terms left inside parentheses share no common factor > 1.

Answer: 15(3x - 2)

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 4.4 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

4 5 ENTER 3 0 GCF 4 5 ↑↓ ÷

Factor 45x - 30 by hand first. Then check on your SC-6:
Press 45 [ENTER] 30 [GCF] to confirm GCF = 15, divide $45 \div 15 = 3$ and $30 \div 15 = 2$ for 15(3x - 2), and test x = 2 with [≠] (flashes NO!).

At the end of the craft fair, Maya had (16x + 24) stickers to pack into the greatest possible number of identical gift bags, and the word 'factor the expression' on her worksheet sounded intimidating. Then she noticed that packing identical bags was just the uniform-bundle problem from the day before running in reverse! Instead of multiplying the number of bags into the parentheses, she pulled out the Greatest Common Factor of 16 and 24 (which is 8) to write 8(2x + 3)—showing 8 identical gift bags with (2x + 3) stickers inside each bag.

SECTION 4.4 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Identical Gift Bags: You have (16x + 24) stickers to divide equally into the greatest possible number of identical gift bags. Factor the expression to find the number of bags and the contents of each bag.**

Starter Step 1: Find the GCF of 16 and 24 to get the greatest number of bags, then divide 16x and 24 by that GCF.

Way 1 (Paper Work):

GCF = _____

Factored: ____(_____ ± _____)

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Rectangular Solar Array: A solar panel array has area (21x + 35) square meters and width 7 meters. What is the length of the array?**

Starter Step 1: Factor out the width 7 from (21x + 35) by dividing both 21x and 35 by 7 to find the length.

Way 1 (Paper Work):

GCF = _____

Factored: ____(_____ ± _____)

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Square Perimeter: A square has a perimeter of (20x + 36) inches. What is the length of one side of the square?**

Way 1 (Paper Work):

GCF = _____

Factored: ____(_____ ± _____)

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Equilateral Triangle Side: An equilateral triangle has a perimeter of (18y - 27) centimeters. What is the length of one side?**

Way 1 (Paper Work):

GCF = _____

Factored: ____(_____ ± _____)

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Factor 40x + 64 completely using the GCF.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Club Uniform Kits: Every player on a team receives an identical kit of shirts and hats, and the team's total cost is (36s + 54) dollars. Write the expression in factored form to find the greatest possible number of players and the cost of one kit.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Factor 12x + 18y + 30 completely using the GCF.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Classroom Desk Rows: A classroom has (35d - 14) square feet of desk space arranged in 7 equal rows. Write an expression for the desk space in each row.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Combine like terms in 10x + 18 + 14x + 18, and then factor the result completely.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Factor 27x - 18 completely, and check by evaluating both expressions when x = 2.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

Unit 5: Areas of Polygons & Elapsed Time — Building Skate Ramps, Measuring Sails & Navigating by Grid

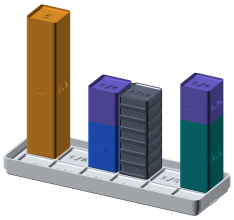
Areas of Parallelograms, Triangles, Trapezoids, Composite Figures, Coordinate Polygons & Base-60 Time

When Liam set out to build a wooden skateboard launch ramp and a pair of cornhole boards in his garage, he hit a wall right away at the hardware store. Plywood and grip tape are sold by the square foot, yet almost none of the pieces in his sketch were plain rectangles. The side panels of his ramp were right triangles, the gym vaulting box he wanted to copy had trapezoid sides, and the cornhole board had a square hole cut right out of the middle.

Down at the marina, Sora felt just as unsure when she had to order replacement sailcloth for a triangular mainsail and measure diagonal boat-trailer spaces in the lot. Whenever slanted sides and strange angles appeared, both Liam and Sora worried they would mix up which numbers to multiply and waste expensive materials.

Even plotting a rectangular field on a map grid or adding flight times across the 60-minute clock mark tripped them up when they tried to count squares one by one or add minutes like ordinary base-10 decimals.

Everything changed when they took a pair of scissors to scrap cardboard and saw that every parallelogram, triangle, and trapezoid is secretly built from ordinary rectangles. Once they learned to slice and slide shapes by hand—and check their areas and elapsed times using the geometry and clock keys on the SC-6—they never had to guess a formula again.



Real-World Mathematical Context & Two-Method Mastery • Hands-On Models: Fraction Strips & Place Value Blocks

UNIT 5 LESSONS — LEARN TWO TEXTBOOK WAYS FIRST, THEN CHECK YOUR WORK ON THE SC-6 CALCULATOR

SEC 5.1 (pp.46-47) Areas of Parallelograms and Rectangles

Textbook Ways: Way 1: Area Formula A = bh | Way 2: Cut-and-Slide Rectangle Decomposition

Learn by Hand: If you cut a right triangle from one end of a parallelogram and translate (slide) it to the other end, the parallelogram forms a rectangle with the exact same base b and perpendicular height h! Therefore, the area of any parallelogram is A = bh, where h is the perpendicular vertical height (never the slanted side length).

Check Your Work on the SC-6 Calculator:

[1] [2] [ENTER] [8] [f] [■]

Press 12 [ENTER] 8 [f][RECT] to confirm Area = 96 cm^2 on X (and Perimeter = 40 cm on Y).

SEC 5.2 (pp.48-49) Areas of Triangles

Textbook Ways: Way 1: Formula A = (b × h) ÷ 2 | Way 2: Take Half of Base or Height First (1/2b × h)

Learn by Hand: Any triangle can be duplicated and rotated to form a parallelogram with the same base b and height h. Because the two congruent triangles make up one bh parallelogram, the area of a single triangle is exactly half the area of the parallelogram: A = 1/2bh = (b × h) ÷ 2.

Check Your Work on the SC-6 Calculator:

[1] [0] [ENTER] [6] [g] [▲]

Press 10 [ENTER] 6 [g][TRI] (which automatically computes 1/2 × b × h) to confirm 30 cm^2!

SEC 5.3 (pp.50-51) Areas of Trapezoids and Composite Figures

Textbook Ways: Way 1: Decompose & Add Sub-Areas | Way 2: Trapezoid Formula A = 1/2h(b_1 + b_2) | Way 3: Enclosing Box Minus Cutouts

Learn by Hand: A trapezoid has two parallel bases, b_1 and b_2, and a perpendicular height h. Its area is half the product of the height and the sum of the bases: A = 1/2h(b_1 + b_2). To find the area of a composite figure, decompose it into non-overlapping rectangles, triangles, and trapezoids (or subtract cutout areas from an enclosing rectangle).

Check Your Work on the SC-6 Calculator:

[g] [CLΣ] [1] [2] [ENTER] [8] [f] [■] [Σ+] [1] [2] [ENTER] [5] [g] [▲] [Σ+] [f] [Σx]

Split the shape into a rectangle and triangle on paper first. On your SC-6, clear stats with [g][CLΣ], add the rectangle (12 [ENTER] 8 [f][RECT] [Σ+]) and triangle (12 [ENTER] 5 [g][TRI] [Σ+]), and press [f][Σx] to confirm 126 ft^2!

SEC 5.4 (pp.52-53) Polygons in the Coordinate Plane & Elapsed Time

Textbook Ways: Way 1: Subtract Coordinates / Regroup 60 min = 1 hr | Way 2: Count Grid Squares / Time-Line Jumps

Learn by Hand: To find the length of a horizontal segment in Quadrant I, subtract the x-coordinates; for a vertical segment, subtract the y-coordinates. When calculating elapsed time, remember that hours and minutes use Base-60 (60 minutes = 1 hour), so regroup 60 minutes whenever minutes exceed 59 or when borrowing 1 hour.

Check Your Work on the SC-6 Calculator:

[9] [ENTER] [2] [-] [8] [ENTER] [3] [-] [f] [■]

Press 9 [ENTER] 2 [-] (7) and 8 [ENTER] 3 [-] (5), then [f][RECT] to see Area = 35 on X and Perimeter = 24 on Y!

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 5.1 — Practice A: Areas of Parallelograms and Rectangles

Key Vocabulary: polygon • parallelogram • base (b) • perpendicular height (h) • slant length • perimeter Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & VISUAL DIAGRAM

Parallelogram Cut-and-Slide Proof: Area = Base b × Perpendicular Height h



HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept: Area of a rectangle = lw; Area of a parallelogram = bh, where height h is PERPENDICULAR (90°) to the base b.

Textbook Visual Proof: Cut the right triangle from one end of a parallelogram and slide it to the other end to form a b × h rectangle.

WATCH OUT: Always multiply Base b × Perpendicular Height h (at 90°) — ignore the slanted side length when finding area!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find the area of a parallelogram with base b = 12 cm, perpendicular height h = 8 cm, and slanted side 10 cm.

WAY 1: Use the Parallelogram Area Formula A = bh

Multiply base b = 12 cm by perpendicular height h = 8 cm: A = 12 × 8 = 96 cm² (ignore the 10 cm slant side for area!).

WAY 2: Cut-and-Slide into a Rectangle

Cut the right triangle off one end and slide it to the other end to form a 12 cm by 8 cm rectangle → Area = 96 cm².

Answer: 96 cm² 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 5.1 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

1 2 ENTER 8 f ■

Multiply base × perpendicular height (12 × 8) on paper first. Then check on your SC-6: Press 12 [ENTER] 8 [f][RECT] to confirm Area = 96 cm² on X (and Perimeter = 40 cm on Y).

SECTION 5.1 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Find the area of a parallelogram with base b = 15 m, perpendicular height h = 9 m, and slanted side 11 m.**

Starter Step 1: Multiply base b = 15 m by the PERPENDICULAR height h = 9 m (ignore the 11 m slanted side!).

Way 1 (Paper Work):

Base b = ___ Height h = ___

Area A = b × h = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Parallelogram Flower Bed: A botanical garden bed is shaped like a parallelogram with a base of 8 meters and a perpendicular height of 4.5 meters. What is the area of the flower bed?**

Starter Step 1: Multiply base b = 8 m by the perpendicular height h = 4.5 m: A = b × h.

Way 1 (Paper Work):

Base b = ___ Height h = ___

Area A = b × h = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Find the area of a parallelogram with base b = 8.5 m and perpendicular height h = 6.4 m.**

Way 1 (Paper Work):

Base b = ___ Height h = ___

Area A = b × h = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Find the area of a square courtyard tile with side length 13 cm.**

Way 1 (Paper Work):

Base b = ___ Height h = ___

Area A = b × h = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **A parallelogram has a base of 6 1/2 inches and a perpendicular height of 4 inches. What is its area?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **A parallelogram has an area of 108 cm² and a base of 12 cm. What is its perpendicular height?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **A parallelogram has an area of 135 ft² and a perpendicular height of 7.5 ft. What is its base?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Error Analysis: A parallelogram has base b = 10 cm, perpendicular height h = 6 cm, and slanted side s = 7.5 cm. Sam calculated the area as 75 cm². Explain his error and find the correct area.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Angled Parking Spaces: Each angled parking space is a parallelogram with a base of 9 feet and a perpendicular length of 18.5 feet. How many square feet of asphalt are needed to pave 4 adjacent spaces?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Patio Planter Cutout: A rectangular stone patio measuring 15 feet by 12 feet has a parallelogram-shaped herb planter (base 4 feet, perpendicular height 3 feet) built in the center. How many square feet of stone paving surround the planter?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 5.1 — Practice B: Modeling Real Life (Areas of Parallelograms and Rectangles)

Textbook Ways: Area Formula A = bh | Cut-and-Slide Rectangle Decomposition Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — FINDING A MISSING DIMENSION (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: A parallelogram-shaped team banner has an area of 168 square inches and a base of 14 inches. What is the vertical height of the banner?

WAY 1: Substitute into A = bh & Solve Using Division

Write 168 = 14h. Divide both sides by 14: h = 168 ÷ 14 = 12 inches.

WAY 2: Partial Quotients / Area Decomposition

Split 168 in^2 into 140 in^2 + 28 in^2: (140 ÷ 14) + (28 ÷ 14) = 10 in + 2 in = 12 inches.

Check: b × h = 14 in × 12 in = 168 in^2.

WATCH OUT: Always multiply Base b × Perpendicular Height h (at 90°) — ignore the slanted side length when finding area!

Answer: 12 inches

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 5.1 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

1 6 8 ENTER 1 4 ÷ f LASTx f ■

Solve 168 ÷ 14 by hand first to find the missing height. Then check on your SC-6:
Press 168 [ENTER] 14 [÷] to confirm 12 inches, and |f|[LASTX] |f|[RECT] to verify 12 × 14 = 168 in^2.

When Liam measured a slanted parallelogram parking space in the marina lot, his tape measure showed a 10-foot base, an 18-foot straight-up perpendicular height, and a 20-foot slanted side stripe. Having three numbers confused him, and he multiplied 10 × 20 first because those were the outer edges. Unsure if 200 square feet was right, he sketched the parallelogram on grid paper, snipped off the slanted right triangle on one end, and slid it to the other end. It locked into a 10-by-18 rectangle (180 square feet)! That one cut showed him why area always uses perpendicular height, while the 20-foot slanted edge is only used when walking the perimeter.

SECTION 5.1 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Diagonal Parking Space: A parallelogram parking space has a base of 10 feet, a perpendicular height of 18 feet, and a slanted side of 20 feet. What is the area of the parking space?**

Starter Step 1: Multiply base b = 10 ft by the perpendicular height h = 18 ft (ignore the 20 ft slanted side!).

Way 1 (Paper Work):

Base b = ___ Height h = ___

Area A = b × h = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Parking Space Stripes: What is the perimeter around all four outer edges of the parking space in Problem 1?**

Starter Step 1: Perimeter measures the outer boundary—add the two 10 ft bases and the two 20 ft slanted sides.

Way 1 (Paper Work):

Outer Sides: s_1=___ s_2=___

Perimeter P = 2(s_1 + s_2) = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Staircase Railing Glass: A parallelogram glass panel has a base of 4.5 feet and a vertical height of 3.2 feet. What is its area?**

Way 1 (Paper Work):

Base b = ___ Height h = ___

Area A = b × h = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Garden Patio Stone: A parallelogram patio stone has an area of 135 square inches and a perpendicular height of 9 inches. What is its base length?**

Way 1 (Paper Work):

Base b = ___ Height h = ___

Area A = b × h = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Quilt Pattern Rhombus: A quilt uses 8 identical parallelogram patches, each with base 6 cm and perpendicular height 4.5 cm. What is the total area of all 8 patches?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Basketball Key Paint: The rectangular free-throw lane is 19 feet long by 12 feet wide. How many square feet of paint are needed to cover it?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Lawn Area: A rectangular yard is 65 feet by 40 feet. A storage shed covers a 10 ft by 8 ft corner. How many square feet of grass remain?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Find the area of a parallelogram with base 5 1/2 inches and perpendicular height 4 inches.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Rectangle A and Parallelogram B both have a base of 16 cm and a perpendicular height of 9 cm. How do their areas compare, and why?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Banner Cost: A parallelogram banner has a base of 8 feet and a height of 3.5 feet. If vinyl printing costs $4.00 per square foot, how much does the banner cost?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

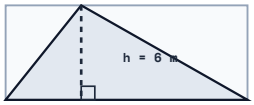
HANDS-ON MODEL: FRACTION STRIPS

Section 5.2 — Practice A: Areas of Triangles

Key Vocabulary: triangle • congruent triangles • perpendicular base & height • Area Formula A = 1/2bh Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & HANDS-ON MODEL (FRACTION STRIPS)

Triangle Area Proof: Every Triangle Is 1/2 of a b × h Rectangle (A = 1/2bh)



Rectangle = 10 × 6 = 60 m²

Triangle = 1/2(10)(6) = 30 m²

Missing Height Formula:

h = 2A ÷ b (Double Area first!)

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept: Two congruent triangles always form a parallelogram of area bh, so every triangle has area A = 1/2bh = (b × h) ÷ 2.

Textbook Mental-Math Tip: You can either multiply b × h first and divide by 2, OR take half of the even dimension first and then multiply.

WATCH OUT: Don't forget to divide by 2 for triangles (A = 1/2bh)! And when solving for a missing height, double the area first (h = 2A ÷ b).

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find the area of a triangle with base b = 10 cm and perpendicular height h = 6 cm.

WAY 1: Multiply Base × Height, Then Divide by 2

Find the enclosing rectangle area (10 × 6 = 60 cm²), then divide by 2: 60 ÷ 2 = 30 cm².

WAY 2: Take Half of an Even Dimension First

Take 1/2 of base 10 first (1/2 × 10 = 5), then multiply by height 6: 5 × 6 = 30 cm² (or 10 × 1/2(6) = 10 × 3 = 30 cm²).

Answer: 30 cm² 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 5.2 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

1 0 ENTER 6 g ▲

Calculate 1/2 × 10 × 6 on paper first. Then check on your SC-6 using the Triangle Area key:
Press 10 [ENTER] 6 [g][TRI] (which automatically computes 1/2 × b × h) to confirm 30 cm²!

SECTION 5.2 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Triangular Park Pavilion Pennant: A nylon celebration pennant for the park pavilion has a base of 25 cm and a height of 60 cm. What is the area of the pennant?**

Starter Step 1: Use Triangle Area A = 1/2bh: multiply 1/2 × 25 cm × 60 cm (remember to divide by 2!).

Way 1 (Paper Work):

Base b = ___ Height h = ___

Area A = 1/2 × b × h = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **A right triangle has legs of 9 feet and 4 feet, and a slanted hypotenuse of 9.85 feet. What is its area?**

Starter Step 1: Use the two perpendicular legs (9 ft and 4 ft) as base and height in A = 1/2bh (ignore the 9.85 ft slanted side!).

Way 1 (Paper Work):

Base b = ___ Height h = ___

Area A = 1/2 × b × h = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Find the area of a triangle with base b = 16 cm and height h = 11 cm.**

Way 1 (Paper Work):

Base b = ___ Height h = ___

Area A = 1/2 × b × h = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Find the area of a triangle with base b = 12.5 m and height h = 8.4 m.**

Way 1 (Paper Work):

Base b = ___ Height h = ___

Area A = 1/2 × b × h = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Find the area of a triangle with base b = 4 1/2 inches and height h = 6 inches.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **A triangle has an area of 54 cm² and a base of 12 cm. What is its perpendicular height?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **A triangle has an area of 84 m² and a perpendicular height of 14 m. What is its base?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Error Analysis: Leo found the area of a triangle with base 8 cm and height 5 cm as 40 cm². What did he forget to do, and what is the correct area?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Sailboat Rigging: A sailboat has a triangular mainsail (base 12 ft, height 25 ft) and a triangular jib sail (base 8 ft, height 15 ft). What is the combined area of both sails?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Diamond Kite Area: Sora's kite is made of two identical triangles that share a 40 cm center crossbar (base). Each triangle has a height of 35 cm (70 cm from top tip to bottom tip). What is the total area of the kite?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: FRACTION STRIPS

Section 5.2 — Practice B: Modeling Real Life (Areas of Triangles)

Textbook Ways: Formula A = (b × h) ÷ 2 | Take Half of Base or Height First (1/2b × h) Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — FINDING MISSING HEIGHT OF A TRIANGLE (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: A triangular sports pennant has an area of 60 square inches and a base of 15 inches. What is the height of the pennant?

WAY 1: Substitute into A = 1/2bh & Solve

Write 60 = 1/2(15)h → 60 = 7.5h. Divide both sides by 7.5: h = 60 ÷ 7.5 = 8 inches.

WAY 2: Double the Triangle into a Parallelogram First

Two identical pennants form a parallelogram of area 2 × 60 = 120 in^2. Divide by base 15: h = 120 ÷ 15 = 8 inches.

Check: A = 1/2 × 15 × 8 = 15 × 4 = 60 in^2.

WATCH OUT: Don't forget to divide by 2 for triangles (A = 1/2bh)! And when solving for a missing height, double the area first (h = 2A ÷ b).

Answer: 8 inches

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 5.2 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

6 0 ENTER 2 × 1 5 ÷

Double the area and divide by the base on paper first. Then check on your SC-6: Press 60 [ENTER] 2 [×] 15 [÷] to confirm h = 8 in. (and test 15 [ENTER] 8 [g][TRI] = 60!).

Sora always forgot to divide by 2 when finding the area of a triangle, especially when she was in a hurry measuring a 12-foot by 25-foot sailboat mainsail. While helping Liam cut two triangular side panels for his skate ramp out of a single rectangular sheet of plywood, she watched him draw one diagonal line from corner to corner across the rectangle. Seeing the two identical right triangles nested inside the rectangle made it impossible to forget: two matching triangles always make one rectangle or parallelogram of area b × h, so a single triangle is always half (1/2bh).

SECTION 5.2 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Sailboat Mainsail: A triangular sail has a base of 12 feet and a height of 25 feet. What is the area of the sail?**

Starter Step 1: Use Triangle Area A = 1/2bh with base b = 12 ft and height h = 25 ft.

Way 1 (Paper Work):

Base b = ___ Height h = ___

Area A = 1/2 × b × h = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Skateboard Launch Ramp: Each triangular side panel of a launch ramp has a base of 8 feet and a height of 3.5 feet. What is the area of one side panel, and what is the combined area of both side panels?**

Starter Step 1: Find the area of 1 triangular side panel using A = 1/2bh, then multiply by 2 for both panels.

Way 1 (Paper Work):

Base b = ___ Height h = ___

Area A = 1/2 × b × h = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Hang Glider Wing: A triangular hang glider wing has a wingspan (base) of 30 feet and a center chord (height) of 9.4 feet. What is its wing area?**

Way 1 (Paper Work):

Base b = ___ Height h = ___

Area A = 1/2 × b × h = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Yield Sign Base: A triangular road sign has an area of 450 square inches and a height of 25 inches. What is the base of the sign?**

Way 1 (Paper Work):

Base b = ___ Height h = ___

Area A = 1/2 × b × h = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Tent Door Height: A triangular tent door has an area of 18 square feet and a base of 6 feet. What is the height of the tent door?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Diagonal Sandwich Cut: A square slice of bread measures 4.5 inches on each side and is cut diagonally into 2 congruent triangles. What is the area of one triangle?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Corner Flower Bed: A right-triangular flower bed in the corner of a yard has perpendicular sides of 9 feet and 12 feet, and a 15-foot slanted front edge. What is its area?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Falcon Stoop Wing Geometry: During a high-speed dive, a peregrine falcon sweeps its wings into two right triangles, each with base 18 cm and length 35 cm. What is their combined wing area?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Find the area of a triangle with base 6 1/2 inches and height 4 inches.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Skylight Panes: A skylight is built from 4 congruent triangular glass panes, each with base 5 feet and height 6.4 feet. What is the total glass area?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

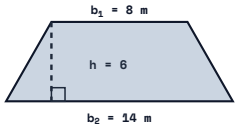
Section 5.3 — Practice A: Areas of Trapezoids and Composite Figures

Key Vocabulary: trapezoid • parallel bases (b_1, b_2) • height (h) • composite figure • decompose

Name: _____ Date: _____ Score: ____ / 100

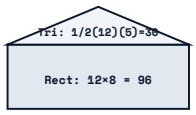
TEXTBOOK CONCEPT & VISUAL DIAGRAM

Trapezoid: $A = 1/2h(b_1 + b_2)$



$A = 1/2(6)(8+14) = 3(22) = 66\ m^2$

Composite: Decompose or Subtract



Total = 96 + 30 = 126 ft²

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Trapezoid Formula: $A = 1/2h(b_1 + b_2)$ — add the two parallel bases, multiply by the perpendicular height h, and divide by 2 (average of bases × h).

Textbook Composite Figures: Either DECOMPOSE (split into non-overlapping rectangles/triangles and add) or SUBTRACT (enclosing box minus cutouts).

WATCH OUT: In a trapezoid, add the two parallel bases ($b_1 + b_2$) FIRST before multiplying by 1/2h!

WORKED EXAMPLE — 3 TEXTBOOK WAYS (ANY 2 WAYS = +20 PTS)

Problem: Find the total area of a 12 ft by 8 ft rectangular shed wall topped by a roof gable triangle (base 12 ft, height 5 ft).

WAY 1: Decompose into a Rectangle + Triangle

Split the house wall horizontally: bottom 12 × 8 rectangle = 96 ft²; top roof triangle = 1/2 × 12 × 5 = 30 ft². Sum: 96 + 30 = 126 ft².

WAY 2: Enclosing Rectangle Minus Corner Triangles

Full 12 × 13 enclosing box = 156 ft². Subtract the two unshaded top right triangles (2 × 1/2 × 6 × 5 = 30 ft²): 156 - 30 = 126 ft².

WAY 3: Split Vertically into Two Congruent Trapezoids

Left and right halves are vertical trapezoids with parallel sides 8 ft and 13 ft and width 6 ft: 2 × 1/2(6)(8 + 13) = 6 × 21 = 126 ft².

Answer: 126 ft² 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 5.3 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[g] [CLΣ] 1 2 [ENTER] 8 [f] [■] [Σ+] 1 2 [ENTER] 5 [g] [▲] [Σ+] [f] [Σx]

Split the shape into a rectangle and triangle on paper first. On your SC-6, clear stats with [g][CLΣ], add the rectangle (12 [ENTER] 8 [f][RECT] [Σ+]) and triangle (12 [ENTER] 5 [g][TRI] [Σ+]), and press [f][Σx] to confirm 126 ft²!

SECTION 5.3 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Find the area of a trapezoid with parallel bases $b_1 = 10$ cm and $b_2 = 14$ cm, and height $h = 6$ cm.**

Starter Step 1: Add the two parallel bases (10 + 14) inside parentheses first, then multiply by 1/2 and by the height 6.

Way 1 (Paper Work):

$b_1 + b_2 =$ ___ + ___ = ___

$A = 1/2 \times h \times (b_1+b_2) =$ _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Find the area of a trapezoid with parallel bases $b_1 = 18$ ft and $b_2 = 12$ ft, and height $h = 9$ ft.**

Starter Step 1: Add the two parallel bases (18 + 12) inside parentheses first, then multiply by 1/2 and by the height 9.

Way 1 (Paper Work):

$b_1 + b_2 =$ ___ + ___ = ___

$A = 1/2 \times h \times (b_1+b_2) =$ _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **An L-shaped floor plan consists of a 10 ft by 6 ft section and an 8 ft by 4 ft section. What is the total floor area?**

Way 1 (Paper Work):

Part 1 Area = _____

Part 2 Area = _____ → Sum = ___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **A directional sign is made of a 15 in. by 6 in. rectangle attached to a triangular arrowhead with base 10 in. and height 6 in. What is the total area of the sign?**

Way 1 (Paper Work):

Part 1 Area = _____

Part 2 Area = _____ → Sum = ___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Find the area of a trapezoid with parallel bases $b_1 = 8.5$ m and $b_2 = 5.5$ m, and height $h = 4.2$ m.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Picture Frame Mat: A 10 in. by 8 in. rectangular mat board has a 6 in. by 4 in. rectangular photo window cut out of the center. What is the remaining area of the mat border?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **A trapezoid has an area of 90 cm² and parallel bases of 11 cm and 7 cm. What is its perpendicular height h?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Mini-Golf Fairway: A mini-golf hole is made of a 12 ft by 5 ft rectangular fairway connected to a trapezoid green (bases 5 ft and 9 ft, height 4 ft). How much turf is needed to cover the hole?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Two Ways to Measure a Trapezoid: Find the area of a right trapezoid with bases 10 cm and 16 cm and height 8 cm both by using the trapezoid formula and by splitting it into a 10×8 rectangle and a 6×8 right triangle.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Backyard Landscaping: A 20 ft by 15 ft rectangular yard contains a triangular flowerbed (base 8 ft, height 6 ft) and a 5 ft by 5 ft square storage shed. How many square feet of grass lawn remain?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 5.3 — Practice B: Modeling Real Life (Areas of Trapezoids and Composite Figures)

Textbook Ways: Decompose & Add Sub-Areas | Trapezoid Formula $A = 1/2h(b_1 + b_2)$ | Enclosing Box Minus Cutouts Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — TRAPEZOID CLASSROOM DESK (3 TEXTBOOK WAYS • ANY 2 = +20 PTS)

Problem: A trapezoid classroom desk has parallel bases of 48 inches and 36 inches, and a depth (height) of 24 inches. What is the area of the desktop?

WAY 1: Trapezoid Area Formula $A = 1/2h(b_1 + b_2)$
Add the parallel bases (48 + 36 = 84 in) and multiply by 1/2 of height (1/2 × 24 = 12 in): 12 × 84 = 1,008 in².

WAY 2: Decompose into a Rectangle + Side Triangles
Center rectangle = 36 × 24 = 864 in². Extra base width (48 - 36 = 12 in) forms triangles of area 1/2 × 12 × 24 = 144 in². Total: 864 + 144 = 1,008 in².

WAY 3: Average the Two Bases × Height
Average of bases = (48 + 36) ÷ 2 = 42 in. Multiply by height 24 in: 42 × 24 = 1,008 in².

WATCH OUT: In a trapezoid, add the two parallel bases ($b_1 + b_2$) FIRST before multiplying by 1/2h!

Answer: 1,008 square inches

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 5.3 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

4 8 ENTER 3 6 + 2 4 g ▲

Add the parallel bases (48 + 36) and multiply by 1/2 × 24 on paper first. On your SC-6, press 48 [ENTER] 36 [+] 24 [g][TRI] (which takes 1/2 × ($b_1 + b_2$) × h!) to confirm 1,008 in².

The trapezoid area formula—$A = 1/2h(b_1 + b_2)$—looked so crowded with subscripts and parentheses that Liam was convinced he would never remember it on a test. Looking at the trapezoid side of a gym vaulting box (16 inches wide at the top and 28 inches wide at the bottom), he wondered what would happen if he just found the halfway average of the top and bottom widths: halfway between 16 and 28 is 22 inches. Multiplying that 22-inch average width by the 20-inch height gave 440 square inches—and he realized that 'average the two parallel bases, then multiply by the height' is the exact same formula in plain English!

SECTION 5.3 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Gym Vaulting Box: The trapezoid side of a gym vaulting box has a top base of 16 inches, a bottom base of 28 inches, and a height of 20 inches. What is its area?**

Starter Step 1: Add the top and bottom parallel bases (16 + 28) first, then multiply by 1/2 and by the perpendicular height 20.

Way 1 (Paper Work):

$b_1 + b_2$ = ___ + ___ = ___

A = 1/2 × h × (b_1+b_2) = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Popcorn Tub Panels: Each of the 4 sides of a popcorn tub is a trapezoid with bases of 4 inches and 6 inches and a height of 8 inches. What is the total area of all 4 sides?**

Starter Step 1: Find the area of 1 trapezoid side using $A = 1/2h(b_1 + b_2)$, then multiply that area by 4 sides.

Way 1 (Paper Work):

$b_1 + b_2$ = ___ + ___ = ___

A = 1/2 × h × (b_1+b_2) = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Mini-Golf Putting Green: An L-shaped putting green is formed by a 14 ft by 6 ft section and a 10 ft by 5 ft section. What is the total turf area?**

Way 1 (Paper Work):

Part 1 Area = _____

Part 2 Area = _____ → Sum = ___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Model Rocket Fins: A model rocket has 3 identical trapezoid fins. Each fin has parallel edges of 5 cm and 9 cm and a height of 6 cm. What is the total area of all 3 fins?**

Way 1 (Paper Work):

$b_1 + b_2$ = ___ + ___ = ___

A = 1/2 × h × (b_1+b_2) = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Directional Arrow Sign: An arrow sign is made of a 15 in by 6 in rectangle attached to a triangle with base 12 in and height 8 in. What is the total area of the sign?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Cornhole Board Cutout: A rectangular cornhole board measures 48 inches by 24 inches, with a 6 in by 6 in square opening cut out. What is the area of the remaining wood surface?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Trapezoid Patio Height: A trapezoid patio has parallel bases of 10 feet and 16 feet and an area of 104 square feet. What is the perpendicular distance (height) between the bases?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Barn Gable Wall: A barn wall is a 20 ft by 14 ft rectangle topped by a triangular roof gable with base 20 ft and height 6 ft. What is the total area of the wall?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Swimming Pool Deck Border: A 20 m by 10 m rectangular pool is surrounded by a concrete deck whose outer boundary measures 26 m by 16 m. What is the area of the concrete deck?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Hexagon Tile: A regular hexagon tile is made of 2 identical trapezoids, each with bases of 4 cm and 8 cm and height 3.5 cm. What is the area of the hexagon?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

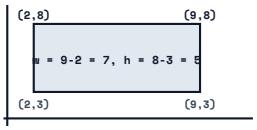
TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 5.4 — Practice A: Polygons in the Coordinate Plane & Elapsed Time

Key Vocabulary: coordinate grid • vertices • horizontal side • vertical side • elapsed time • base-60 regrouping Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & VISUAL DIAGRAM

Polygon on Coordinate Grid



P = 2(7+5) = 24 | Area = 35

Elapsed Time (Base-60: 1 hr = 60 min)



8:40 → 9:00 (+20m)
9:00 → 12:30 (+3h 30m)
Total = 3 hr 50 min

15 min = 0.25 hr | 30 min = 0.5 hr

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Coordinate Polygons: Subtract x-coordinates for horizontal side lengths and y-coordinates for vertical side lengths, then find Perimeter and Area.
Textbook Elapsed Time: Regroup 60 minutes = 1 hour when adding/subtracting times, or jump to the nearest hour on a time number line.
WATCH OUT: Wait! 1 hour = 60 minutes (NOT 100): when borrowing 1 hour in elapsed time, add 60 minutes to the minutes column.

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: A rectangle has vertices (2, 3), (9, 3), (9, 8), and (2, 8). Find its width, height, perimeter, and area.

WAY 1: Subtract Coordinates to Find Side Lengths

For A(2,3), B(9,3), C(9,8), D(2,8): Width = 9 - 2 = 7; Height = 8 - 3 = 5. Perimeter = 2(7+5) = 24; Area = 7 × 5 = 35.

WAY 2: Count Grid Intervals & Decompose

Count 7 grid units right from x=2 to x=9 and 5 grid units up from y=3 to y=8 → 5 rows of 7 squares = 35 square units.

Answer: Width = 7, Height = 5, Perimeter = 24, Area = 35

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 5.4 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

9 [ENTER] 2 [-] 8 [ENTER] 3 [-] [f] [■]

Subtract coordinates to find the width and height on paper first. Check on your SC-6:
Press 9 [ENTER] 2 [-] (7) and 8 [ENTER] 3 [-] (5), then [f][RECT] to see Area = 35 on X and Perimeter = 24 on Y!

SECTION 5.4 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Find the length of the horizontal segment connecting (3, 4) and (11, 4) in the coordinate plane.**

Starter Step 1: Since both points share the same y-coordinate (4), subtract their x-coordinates (11 - 3) to find the horizontal length.

Way 1 (Paper Work):

0

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

2 **A rectangle has vertices at (1, 2), (10, 2), (10, 8), and (1, 8). What is its area?**

Starter Step 1: Subtract the x-coordinates (10 - 1) for the base and subtract the y-coordinates (8 - 2) for the height, then multiply.

Way 1 (Paper Work):

0

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

3 **Aquarium Filter Cycle: The main bio-filter pump starts its morning cycle at 7:10 AM and runs for 3 hours and 48 minutes. At what time does the cycle finish?**

Way 1 (Paper Work):

0

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

4 **Convert 2 hours 30 minutes into decimal hours, and convert 4 hours 45 minutes into decimal hours.**

Way 1 (Paper Work):

0

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

5 **Science Museum Fountain Shift: The courtyard water fountain turns on at 8:15 AM and shuts down for maintenance at 3:05 PM. For how many hours and minutes did the fountain run?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

6 **Convert 1.75 hours into hours and minutes.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

7 **A train departs at 8:50 AM and arrives at 11:25 AM. How long was the train ride in hours and minutes?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

8 **A right triangle in the coordinate plane has vertices at (2, 1), (10, 1), and (2, 7). What is the area of the triangle?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

9 **Saturday Pet-Sitting: You pet-sit a neighbor's dogs from 9:15 AM until 2:45 PM on Saturday at a rate of $18.50 per hour. How many hours did you work, and how much did you earn?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

10 **Coordinate Park Blueprint: A rectangular park has vertices (2, 2), (14, 2), (14, 11), and (2, 11), where each grid unit represents 10 yards. What are the perimeter (in yards) and area (in square yards) of the park?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 5.4 — Practice B: Modeling Real Life (Polygons in the Coordinate Plane & Elapsed Time)

Textbook Ways: Subtract Coordinates / Regroup 60 min = 1 hr | Count Grid Squares / Time-Line Jumps Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — ELAPSED TIME & DECIMAL HOURS (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: A 3D print starts at 9:40 AM and takes 2 hours 50 minutes. At what time does the print finish?

WAY 1: Add Hours & Minutes, Then Regroup 60 min = 1 hr

9 hr 40 min + 2 hr 50 min = 11 hr 90 min. Regroup 90 min = 1 hr 30 min → 12 hr 30 min = 12:30 PM.

WAY 2: Jump to Benchmark Hours on a Time Line

From 9:40 AM, add 20 min to reach 10:00 AM (2 hr 30 min left to go). Add 2 hr 30 min → 12:30 PM.

Check: From 9:40 AM to 12:40 PM is 3 hours; 10 minutes earlier is 12:30 PM (2 hr 50 min).

WATCH OUT: Wait! 1 hour = 60 minutes (NOT 100): when borrowing 1 hour in elapsed time, add 60 minutes to the minutes column.

Answer: 12:30 PM

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 5.4 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[CLK] 9 • 4 0 [ENTER] 2 • 5 0 + 3 • 1 5 [f] [▸HR]

Add the hours and minutes on paper first (rolling 60 minutes into 1 hour). Check on your SC-6 using [CLK] Planetary Clock Mode or convert H.MM values with [f][→HR] and [g][→HMS] to confirm 12:30 PM!

When Sora plotted a rectangular park with corners at (1, 2) and (11, 8) on a map grid, she started counting little grid squares one by one and kept losing her place. And when she tried to add a 3-hour 40-minute flight to an 8:35 AM takeoff using regular base-10 column addition, she got a weird time like '11:75 AM.' Taking a breath, she realized she could subtract coordinates (11 - 1 = 10 wide and 8 - 2 = 6 tall) instead of counting squares—and on a clock, every 60 minutes bundles into 1 full hour, turning 75 minutes into 1 hour and 15 minutes (12:15 PM).

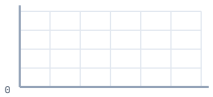
SECTION 5.4 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Coordinate Soccer Field: A field on a coordinate map has corners at (1, 2), (11, 2), (11, 8), and (1, 8). Find its perimeter and area.**

Starter Step 1: Subtract the x-coordinates (11 - 1) for the width and the y-coordinates (8 - 2) for the height.

Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Coordinate Right Triangle: A triangle has vertices at (2, 1), (10, 1), and (2, 7). Find its area.**

Starter Step 1: Subtract x-coordinates (10 - 2) for the horizontal leg and y-coordinates (7 - 1) for the vertical leg, then use A = 1/2bh.

Way 1 (Paper Work):

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Flight Duration: A flight departs at 8:35 AM and lands 3 hours 40 minutes later. At what time does the flight land?**

Way 1 (Paper Work):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

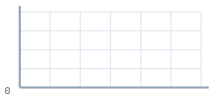
4 **Science Lab Timer: A chemistry experiment starts at 10:45 AM and ends at 2:20 PM. How many hours and minutes did the experiment run?**

Way 1 (Paper Work):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Marathon Split Sum: A runner completes the first half of a marathon in 1 hour 52 minutes and the second half in 1 hour 58 minutes. What is her total time?**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Lifeguard Shift Pay: You work a lifeguard shift from 1:15 PM to 4:45 PM at $14 per hour. How much do you earn for the shift?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Battery Endurance Test: A laptop battery test lasts 2.75 hours. How many hours and minutes is that?**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **3D Print Queue: Print A takes 1 hr 25 min, Print B takes 2 hr 15 min, and Print C takes 50 min. What is the total print time in hours and minutes?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Missing Vertex: Three corners of a rectangle are at (3, 4), (9, 4), and (9, 10). What are the coordinates of the fourth corner, and what is the area of the rectangle?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Train Travel Distance: A train travels for 2 hours 15 minutes at an average speed of 80 miles per hour. How many miles does it travel?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

Unit 6: Integers, Rational Numbers & The Coordinate Plane — Deep-Sea Rovers, Plunging Seabirds & Four-Quadrant Radar

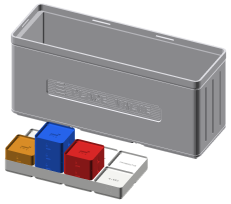
Positive & Negative Integers, Comparing Rational Numbers, Absolute Value, and the 4-Quadrant Coordinate Plane

For Kai, who helps pilot a tethered underwater camera rover with the harbor oceanography club, zero isn't nothing—it's the glittering surface of the sea. High above the boat, a northern gannet circles at +32 meters in the air before folding its wings and plunging straight through the waves down to -14 meters underwater. Meanwhile, in the high Andes mountains, Zoe reads field logs about tiny hummingbirds that survive freezing nights by dropping their body temperature by 28 degrees.

At first, working below zero made Kai and Mia doubt their own common sense. On a negative number line, everything seemed backward: why was -18 degrees colder than -11 degrees, yet -11 was the greater number? And when Kai winched his rover from +25 feet on the boat deck down to -140 feet in the bay, he accidentally subtracted the numbers and came up short on cable.

Mia ran into the same sign traps when plotting landmarks across all four quadrants of the harbor radar screen, mixing up which way to move first and when to add versus subtract distances between coordinates.

By picturing a vertical thermometer and an underwater depth line, Kai, Zoe, and Mia untangle the difference between where a number sits (its signed value) and how far you travel from zero (its absolute value)—and use the sign-change `[+/-]` and absolute-value `[|x|]` keys on the SC-6 to verify every step.



Real-World Mathematical Context & Two-Method Mastery • Hands-On Models: Place Value Blocks & Fraction Strips

UNIT 6 LESSONS — LEARN TWO TEXTBOOK WAYS FIRST, THEN CHECK YOUR WORK ON THE SC-6 CALCULATOR

SEC 6.1 (pp.55-56) Positive and Negative Integers & Opposites

Textbook Ways: Way 1: Equal Distance on Opposite Sides of 0 (Number Line) | Way 2: Additive Inverse & Zero Pairs

Learn by Hand: Integers are the set of whole numbers and their opposites (... -3, -2, -1, 0, 1, 2, 3 ...). Positive numbers are greater than 0 and represent gains, deposits, or elevations above sea level; negative numbers are less than 0 and represent losses, withdrawals, or depths below sea level. Two numbers that are the same distance from 0 on opposite sides are called opposites, and their sum is always 0 (a Zero Pair).

Check Your Work on the SC-6 Calculator:

f ↔ 1 5 +/- ENTER 1 5 +

Plot -15 and +15 on a number line first. On your SC-6, open [f][LINE] Number Line Mode to see signed jumps across 0, or press 15 [+/-] [ENTER] 15 [+] to confirm that opposites add to 0!

SEC 6.2 (pp.57-58) Comparing and Ordering Integers & Rational Numbers

Textbook Ways: Way 1: Convert to Decimals & Plot on Number Line | Way 2: Rewrite with Least Common Denominator (LCD)

Learn by Hand: On a horizontal number line, numbers increase from left to right and decrease from right to left. For negative numbers, a number closer to 0 is further to the right and therefore GREATER (so -5 > -18 because -5°F is warmer than -18°F). To compare negative fractions and decimals, convert them to the same form or common denominator first.

Check Your Work on the SC-6 Calculator:

1 8 +/- ENTER 5 +/- f >

Type 18 [+/-] [ENTER] 5 [+/-] [f][>]—it compares X (-5) > Y (-18) and flashes YES to confirm -5 > -18!

SEC 6.3 (pp.59-60) Absolute Value and Magnitude

Textbook Ways: Way 1: Measure Distance from 0 on a Number Line | Way 2: Evaluate |x| Grouping Bars Before Outside Signs

Learn by Hand: The absolute value of a number is its distance from 0 on a number line, written as |x|. Because distance is never negative, the absolute value of any non-zero number is always positive: |-6| = 6 and |6| = 6. Be careful not to confuse Opposite (which flips + and -) with Absolute Value (which measures positive distance from 0).

Check Your Work on the SC-6 Calculator:

1 4 +/- g |x|

Find the distance from 0 on paper first. On your SC-6, view the distance on [f][LINE] Number Line Mode or press 14 [+/-] [g][|x|] to confirm |-14| = 14.

SEC 6.4 (pp.61-62) The Four-Quadrant Coordinate Plane & Reflections

Textbook Ways: Way 1: Plot & Count Distance to the Mirror Axis | Way 2: Coordinate Reflection Sign Rules

Learn by Hand: The horizontal x-axis and vertical y-axis intersect at the origin (0, 0) and divide the coordinate plane into four quadrants: QI (+, +), QII (-, +), QIII (-, -), and QIV (+, -). To plot an ordered pair (x, y), move horizontally along the x-axis first, then vertically along the y-axis. Reflecting a point across the x-axis changes the sign of y; reflecting across the y-axis changes the sign of x.

Check Your Work on the SC-6 Calculator:

6 +/- ENTER 5

Check the signs of (x, y) on paper first to name each quadrant. On your SC-6, load (-6, 5) with 6 [+/-] [ENTER] 5 so Y holds x = -6 and X holds y = 5.

SEC 6.5 (pp.63-64) Distances in the Coordinate Plane

Textbook Ways: Way 1: Add/Subtract Absolute Values to the Axis | Way 2: Absolute Difference of Coordinates $|x_1 - x_2|$

Learn by Hand: To find the distance between two points that share the same x-coordinate or the same y-coordinate: if the points lie in DIFFERENT quadrants (across an axis), add the absolute values of their differing coordinates; if they lie in the SAME quadrant, subtract the absolute values of their differing coordinates. Equivalently, distance is always |a - b|.

Check Your Work on the SC-6 Calculator:

6 +/- ENTER 9 - g |x|

Press 6 [+/-] [ENTER] 9 [-] [g][|x|] to confirm 15 units!

HANDS-ON MODEL: PLACE VALUE BLOCKS

Section 6.1 — Practice A: Positive and Negative Integers & Opposites

Key Vocabulary: integer • positive number • negative number • opposites • additive inverse • zero pair Name: _____ Date: _____ Score: ____ / 100

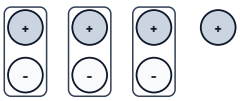
TEXTBOOK CONCEPT & HANDS-ON MODEL (PLACE VALUE BLOCKS)

Opposites on a Number Line (-15 and +15)



Additive Inverses: -15 + 15 = 0

Zero-Pair Cancellation: (+1)+(-1)=0



3 Zero Pairs cancel → +1 Left!

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept: Positive and negative integers represent opposite directions from 0. Opposites (a and -a) are the same distance from 0 on opposite sides.

Textbook Additive Inverse Property: The sum of any number and its opposite is always 0 (-15 + 15 = 0), and -(-a) = a.

WATCH OUT: Zero (0) is its own opposite and is neither positive nor negative; -(-x) means the opposite of -x, which is +x!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find the opposite of -15 and the value of -(-15).

WAY 1: Plot Equal Distance Across 0 on a Number Line

-15 is 15 units to the left of 0. Its opposite is 15 units to the right of 0, which is +15.

WAY 2: Additive Inverse / Zero-Pair Rule

The opposite of -15 is -(-15) = +15, because 15 negatives and 15 positives form 15 Zero Pairs: -15 + 15 = 0.

Answer: Opposite of -15 is +15; -15 + 15 = 0

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 6.1 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

f [↔] 1 5 [+/-] ENTER 1 5 +

Plot -15 and +15 on a number line first. On your SC-6, open [f][LINE] Number Line Mode to see signed jumps across 0, or press 15 [+/-] [ENTER] 15 [+] to confirm that opposites add to 0!

SECTION 6.1 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Write the integer that represents '25 feet below sea level' and state its opposite.**

Starter Step 1: Write 'below sea level' as a negative integer (-); its opposite is the same distance on the positive side of 0 (+).

Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

2 **Find the opposite of -42, and then evaluate -(-42).**

Starter Step 1: The opposite of a negative number is positive, and -(-42) means 'the opposite of -42'.

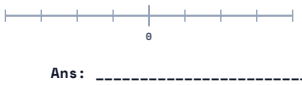
Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

3 **A research submarine is at an elevation of -18 meters and rises 18 meters. What is its new elevation?**

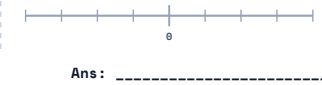
Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

4 **You combine 6 positive (+1) units and 6 negative (-1) units. What is the net value?**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

5 **A desert canyon hiker starts at -120 feet relative to sea level and climbs 120 feet up the canyon wall. What is her elevation now?**

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

6 **Which integer is its own opposite? Explain why using a number line.**

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

7 **A freezer thermometer starts at 0°F, drops 12°F, and then rises 19°F. What is the new temperature?**

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

8 **In golf, -3 means 3 strokes under par. If a golfer shoots -3 on Saturday and +3 on Sunday, what is her combined score relative to par?**

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

9 **An elevator starts at the Ground Lobby (Floor 0), goes up 14 floors, goes down 20 floors to the underground garage, and then goes up 9 floors. On which floor does it stop?**

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

10 **Four-Season Robotics Club Treasury: Over four seasons, the robotics club treasury records +$14,500 in fall grants, -$3,200 in winter parts, +$8,900 in spring sponsorships, and -$1,400 in travel fees. What is the club's net treasury balance?**

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

HANDS-ON MODEL: PLACE VALUE BLOCKS

Section 6.1 — Practice B: Modeling Real Life (Positive and Negative Integers & Opposites)

Textbook Ways: Equal Distance on Opposite Sides of 0 (Number Line) | Additive Inverse & Zero Pairs Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — ELEVATION & BANK LEDGER CHANGES (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: You start with a $0 balance, deposit $35, spend $50, and deposit $20. What is your final balance?

WAY 1: Step-by-Step Jumps on a Number Line

Start at 0 → move right to +35 → move 50 left (passing 0 to -15) → move 20 right (-15 + 20) → lands at +$5.

WAY 2: Group All Positives & All Negatives (Zero Pairs)

Combine positive deposits (+35 + 20 = +55) and pair with -50: 50 zero pairs cancel, leaving +$5.

Check: 35 + 20 - 50 = 55 - 50 = +$5.

WATCH OUT: Zero (0) is its own opposite and is neither positive nor negative; -(-x) means the opposite of -x, which is +x!

Answer: +$5

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 6.1 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

f [↔] 3 5 ENTER 5 0 [+/-] + 2 0 +

Track the deposits and withdrawals on paper first. Then check on your SC-6 (in [f][LINE] mode or standard stack): Press 35 [ENTER] 50 [+/-] [+] 20 [+] to confirm a final balance of +$5.

At first, Kai felt silly writing negative signs in front of numbers—how could you have less than nothing? Then he spent Saturday on the oceanography club boat watching the depth finder. With the water's surface marked as 0 feet, a gull hovering 15 feet in the air was at +15, while the camera rover 15 feet underwater was at -15. He realized that negative numbers aren't about holding 'less than nothing' in your hand; they simply tell you which direction you moved from a starting baseline like sea level or 0°C, and every number has a mirror-image opposite the same distance from 0.

SECTION 6.1 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Submarine Dive & Climb: A submarine starts at sea level (0 ft), dives 120 feet, and then rises 120 feet. What is its final elevation, and how are -120 and +120 related?**

Starter Step 1: Diving 120 ft is -120 and rising 120 ft is +120; combine these opposite quantities to find the final elevation.

Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Death Valley Elevation: Badwater Basin in Death Valley is 282 feet below sea level. Write this elevation as an integer and find its opposite.**

Starter Step 1: Write 'below sea level' with a negative sign (-), then flip the sign to find its opposite.

Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Football Drive: A football team gains 12 yards, loses 8 yards, and then gains 15 yards. What is their net yardage change?**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Hummingbird Overnight Sleep: To survive freezing mountain nights, an Andean hummingbird enters deep sleep (torpor) and drops its body temperature by 28°C. Write this temperature change as an integer and state its opposite.**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Opposite Quantities: You owe $18 to your brother and earn $25 walking a neighbor's dog. After paying your brother back, how much money do you have left?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Golf Tournament Score: In golf, par is 0. A player shoots 4 under par (-4) on Friday, 2 over par (+2) on Saturday, and 3 under par (-3) on Sunday. What is her 3-day score?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **If a scuba diver's elevation is x = -35 feet, what is -x, and what is -(-x)?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Gift Card Balance: You have a $25 gift card, spend $25 on a game, and then receive a $10 store credit. What is your balance?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Canyon Trail Hiker: A hiker starts in a canyon at -40 meters, climbs 95 meters, and then descends 20 meters. What is her final elevation?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Zero-Sum Puzzle: Three integers have a sum of 0. Two of the integers are -14 and -9. What is the third integer?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

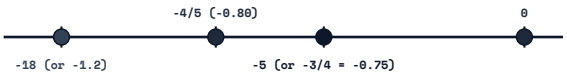
TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 6.2 — Practice A: Comparing and Ordering Integers & Rational Numbers

Key Vocabulary: rational number • inequality (<, >) • number line order (left = less, right = greater) Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & VISUAL DIAGRAM

Comparing Negative Integers (-5 > -18) & Rationals (-0.75 > -0.80): Right Is Greater!



-5 > -18 (since -5 is right of -18) | -3/4 (-0.75) > -4/5 (-0.80) (Closer to 0 = Greater!)

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept: On a horizontal number line, numbers increase from Left to Right (a number to the right is always greater).

Textbook Rational Ordering: To compare negative fractions and decimals, convert to a common form (all decimals or LCD fractions) and plot on a number line.

WATCH OUT: With negative numbers, a larger digit means FURTHER LEFT (smaller/colder): -18 < -5, and -0.80 < -0.75!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Compare -5 and -18 using > or <.

WAY 1: Horizontal Number Line Position

Plot -5 (5 units left of 0) and -18 (18 units left of 0). Since -5 is to the right of -18, -5 > -18.

WAY 2: Vertical Number Line (Thermometer / Elevation)

On a vertical scale, -5 is higher (closer to 0) than -18, so -5 > -18 (and -18 < -5).

Answer: -5 > -18 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 6.2 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

1 8 +/− ENTER 5 +/− f >

Plot -5 and -18 on a number line first. Then check on your SC-6 (in [f][LINE] mode or comparison test): Type 18 [+/-] [ENTER] 5 [+/-] [f][>]—it compares X (-5) > Y (-18) and flashes YES to confirm -5 > -18!

SECTION 6.2 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Which temperature is warmer (greater): -12°C or -5°C? Write an inequality comparing them.**

Starter Step 1: Plot both temperatures on a number line—whichever number sits further to the RIGHT is greater (>).

Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Compare -9 and -2 using > or <.**

Starter Step 1: On a horizontal number line, numbers further to the LEFT are lesser (<) and numbers further to the RIGHT are greater (>).

Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Compare -15 and 0 using > or <.**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Order the integers from least to greatest: 4, -7, 0, -2, -10.**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Compare -3/4 and -0.6 using > or <.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Compare -2 1/2 and -2.3 using > or <.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Compare -5/8 and -7/8 using > or <.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Error Analysis: Leo says -4.5 > -4.1 because 4.5 > 4.1. Explain why -4.1 is actually greater using a number line.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Order the rational numbers -1.5, 1 1/4, -1 3/4, and -0.5 from least to greatest.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Three marine biologists are diving at elevations of -14.8 m, -14 1/2 m, and -15.2 m. Which diver is the deepest (least elevation), and which is the shallowest (greatest elevation)?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 6.2 — Practice B: Modeling Real Life (Comparing and Ordering Integers & Rational Numbers)

Textbook Ways: Convert to Decimals & Plot on Number Line | Rewrite with Least Common Denominator (LCD) Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — COMPARING DIVING DEPTHS (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: Diver A is at -4/5 meter and Diver B is at -0.75 meter. Which diver is closer to the surface (at a greater elevation)?

WAY 1: Convert Both to Decimals & Plot on a Number Line

Write -4/5 = -0.80. On a number line, -0.75 is to the right of -0.80 (closer to 0), so -0.75 > -4/5 (Diver B).

WAY 2: Rewrite Both as Fractions with LCD = 20

-4/5 = -16/20 and -0.75 = -3/4 = -15/20. Since -15 > -16, -15/20 > -16/20, so Diver B (-0.75 m) is higher.

Watch-Out: For negative numbers, a smaller distance from 0 means a GREATER value!

WATCH OUT: With negative numbers, a larger digit means FURTHER LEFT (smaller/colder): -18 < -5, and -0.80 < -0.75!

Answer: -0.75 > -4/5 (Diver B is closer to the surface)

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 6.2 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

0 • 7 5 +/− ENTER • 4 • 5 +/− FDISP g <

Convert -4/5 to -0.80 and compare with -0.75 on paper first. On your SC-6, load Y with 0.75 [+/-] [ENTER], enter X as [.] 4 [.] 5 [+/-] [FDISP] (-0.8), and press [g][<]—it flashes YES to confirm -4/5 < -0.75!

Zoe got tripped up comparing a puffin diving to -45.8 meters and a cormorant diving to -45.25 meters. Because 45.8 is bigger than 45.25 on the positive side of zero, she initially wrote that -45.8 > -45.25—and felt frustrated when it was marked wrong. Sketching a vertical depth line down into the water cleared up her confusion: going from 0 down through -45.25 to -45.80 means -45.80 is lower (deeper and colder), while -45.25 is higher (closer to the surface, so it is the greater number). Adding a trailing zero (-45.80 vs. -45.25) made comparing negative decimals way easier.

SECTION 6.2 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Arctic Weather Station: Monday's temperature was -18°C and Tuesday's was -11°C. Write an inequality comparing the two temperatures. Which day was warmer?**

Starter Step 1: Compare -18°C and -11°C on a thermometer—the temperature closer to 0°C is warmer (greater).

Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

2 **Seabird Dive Depths: A puffin dives to an elevation of -45.8 meters and a cormorant dives to -45.25 meters. Which bird dives deeper, and which elevation is greater?**

Starter Step 1: Compare the tenths place of -45.80 and -45.25: the more negative number is deeper, while the number closer to 0 is greater.

Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

3 **Compare -2/3 and -5/6 using < or >.**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

4 **Golf Leaderboard: Four golfers have scores of -6, +2, -9, and -1. Order the scores from least to greatest.**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

5 **Freezing Points: Water freezes at 0°C, olive oil at -6°C, mercury at -39°C, and rubbing alcohol at -89°C. Order these freezing points from coldest to warmest.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

6 **Reef & Sea Turtle: A coral reef is at -3 1/4 meters and a sea turtle is at -3.1 meters. Which one is closer to the water's surface?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

7 **Stock Market Change: Stock A changed by -1.8% and Stock B changed by -1.08%. Which stock had the greater (less negative) percent change?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

8 **Name one fraction and one decimal that lie between -4 and -3 on a number line.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

9 **Order -7/4, -1.6, 3/2, and -2 from least to greatest.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

10 **If a < b and both a and b are negative numbers, which number is farther from 0 on the number line? Give an example.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 6.3 — Practice A: Absolute Value and Magnitude

Key Vocabulary: absolute value (|x|) • magnitude • distance from zero • opposite vs. absolute value Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & VISUAL DIAGRAM

Absolute Value |x| = Distance from 0 on the Number Line



Both -14 and +14 are 14 units from 0!

Inside vs. Outside '-' Sign:

|-14| = +14

|14| = +14

-|-14| = -14

(Evaluate | | bars first!)

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept: The absolute value |x| is the distance between x and 0 on a number line. Since distance is never negative, |x| ≥ 0.

Textbook Order of Operations with |x|: Absolute value bars act as grouping symbols—evaluate inside the bars FIRST, then apply any outside '-' sign.

WATCH OUT: Absolute value |x| is always ≥ 0, but a negative sign OUTSIDE the bars (-|-14| = -14) stays negative!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Evaluate |-14|, |14|, and -|-14|.

WAY 1: Count Distance from 0 on a Number Line

-14 is 14 units from 0 (|-14| = 14) and +14 is 14 units from 0 (|14| = 14).

WAY 2: Evaluate Inside Bars First, Outside Sign Second

In -|-14|, first evaluate |-14| = 14 inside the bars, then apply the outside '-' sign → -14.

Answer: |-14| = 14, |14| = 14, -|-14| = -14

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 6.3 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

1 4 +/− g |x|

Find the distance from 0 on paper first. On your SC-6, view the distance on [f][LINE] Number Line Mode or press 14 [+/-] [g][|x|] to confirm |-14| = 14.

SECTION 6.3 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 Evaluate |-125|.

Starter Step 1: Absolute value |-125| measures the distance from -125 to 0 on the number line, so distance is always positive.

Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

2 Evaluate |-18| + |12|.

Starter Step 1: Find the positive distance inside each absolute value bar first (|-18| and |12|), then add the two distances.

Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

3 Evaluate |-4.8| + |-3.2|.

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

4 Evaluate |-35| - |-19|.

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

5 Compare |-12| and |9| using > or <.

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

6 Evaluate -|-24|.

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

7 Evaluate 50 - |-18|.

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

8 Order the values from least to greatest: |-8|, -6, |-3|, 0, -|-9|.

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

9 A scuba diver is at -32 feet and a seagull is flying at +25 feet relative to sea level (0 ft). Which is closer to sea level, and by how many feet?

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

10 Two bank accounts have balances of -$45 and -$28. Which account has the greater debt (magnitude), and which account has the greater balance?

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 6.3 — Practice B: Modeling Real Life (Absolute Value and Magnitude)

Textbook Ways: Measure Distance from 0 on a Number Line | Evaluate |x| Grouping Bars Before Outside Signs Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — TOTAL VERTICAL TRAVEL ACROSS SEA LEVEL (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: A northern gannet spots a fish from +32 meters in the air and plunges to -14 meters underwater. What is the bird's total vertical travel distance, and what is 50 - |-18|?

WAY 1: Add Absolute Value Distances to Sea Level (0)

Distance from +32 m to 0 is |32| = 32 m; distance from 0 to -14 m is |-14| = 14 m. Total plunge = 32 + 14 = 46 m.

WAY 2: Evaluate |x| Bars as Grouping Symbols

In 50 - |-18|, evaluate |-18| = 18 first, then subtract: 50 - 18 = 32 (never change - |-18| into + 18!).

Check: Distance from 0 is always positive or zero.

WATCH OUT: Absolute value |x| is always ≥ 0, but a negative sign OUTSIDE the bars (-|-14| = -14) stays negative!

Answer: 46 m plunge; and 50 - |-18| = 32

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 6.3 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

5 0 ENTER 1 8 +/- g |x| -

Add the distances to and past sea level on paper first. Check on your SC-6:
For 50 - |-18|, press 50 [ENTER] 18 [+/-] [g][|x|] [-] to confirm 32.

When a northern gannet plunged from +32 meters in the air down to -14 meters underwater, Kai subtracted 32 - 14 and said the bird traveled 18 meters. Looking at the photo on the screen, though, he knew that couldn't be right—the bird fell 32 meters just to hit the waves, plus another 14 meters underwater! He realized that physical distance on a tape measure or winch cable is always positive. Using absolute value to add the distance to zero plus the distance past zero (|32| + |-14| = 32 + 14 = 46 meters) kept the negative sign from tricking him.

SECTION 6.3 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Plunging Gannet Dive: A northern gannet spots a fish from +32 meters in the air and plunges to -14 meters underwater. What is the total vertical distance of its dive?**

Starter Step 1: Find the air distance |+32| and the underwater distance |-14|, then add both positive distances together.

Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

2 **Account Balances: Account A has a balance of -$65 and Account B has a balance of -$40. Which account has the greater balance, and which account has the greater debt?**

Starter Step 1: Compare positions on a number line for greater balance, then compare absolute values (|-65| vs. |-40|) for greater debt.

Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

3 **Drone Flight Distance: A drone flies 18 meters west (-18) from its launch pad, hovers, and then flies 14 meters east (+14) from the launch pad. What is the sum of the two distances from the pad?**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

4 **Evaluate |-35| and -|-35|. Explain why the two answers have opposite signs.**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

5 **Archery Bullseye Error: Three arrows land at -2.4 cm, +1.8 cm, and -1.5 cm from the center of a target (0). Which arrow landed closest to the bullseye?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

6 **Evaluate |-42| ÷ |-6| + |-11|.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

7 **Submarine Camera Cable: A research vessel winches a camera from +25 feet on the deck down to -140 feet in the ocean. How many feet of cable are let out?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

8 **Find all numbers on the number line whose absolute value is 19.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

9 **Order |-8|, -6, |5|, and -|-9| from least to greatest.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

10 **Evaluate |-3 1/2| + |2 1/4| - |-1 1/8|.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

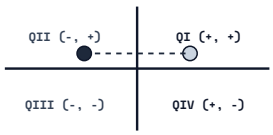
Section 6.4 — Practice A: The Four-Quadrant Coordinate Plane & Reflections

Key Vocabulary: coordinate plane • origin (0, 0) • x-axis • y-axis • quadrants (I, II, III, IV) • ordered pair (x, y) • reflection

Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & VISUAL DIAGRAM

Four-Quadrant Coordinate Plane



Axis Reflection Rules

Reflect Across x-Axis:
Keep x, flip y: (x, y) → (x, -y)

Reflect Across y-Axis:
Flip x, keep y: (x, y) → (-x, y)

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept: For an ordered pair (x, y), start at the origin (0, 0), move horizontally along the x-axis first, then vertically parallel to the y-axis.

Textbook Reflection Rules: Across the x-axis → keep x, flip y: (x, y) → (x, -y). Across the y-axis → flip x, keep y: (x, y) → (-x, y).

WATCH OUT: In (x, y), move horizontally (Left/Right) for x FIRST, then vertically (Up/Down) for y. Reflecting across the x-axis flips y!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Identify the quadrant for A(-6, 5), B(4, -7), and C(-3, -8).

WAY 1: Move Horizontally (x) Then Vertically (y)

A(-6, 5): left 6, up 5 → Quad II. B(4, -7): right 4, down 7 → Quad IV. C(-3, -8): left 3, down 8 → Quad III.

WAY 2: Quadrant Sign Chart

QI = (+, +), QII = (-, +), QIII = (-, -), QIV = (+, -). Match the signs of each ordered pair.

Answer: A: Quadrant II, B: Quadrant IV, C: Quadrant III

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 6.4 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

6 +/− ENTER 5

Check the signs of (x, y) on paper first to name each quadrant. On your SC-6, load (-6, 5) with 6 [+/-] [ENTER] 5 so Y holds x = -6 and X holds y = 5.

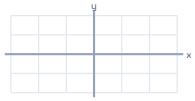
SECTION 6.4 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Name the Quadrant (I, II, III, or IV) that contains the point (7, -4).**

Starter Step 1: Starting at the origin (0, 0), move RIGHT for positive x = +7 and DOWN for negative y = -4 to identify the quadrant.

Way 1 (Paper Work):

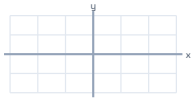


WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Name the Quadrant that contains the point (-9, -2).**

Starter Step 1: Starting at the origin (0, 0), move LEFT for negative x = -9 and DOWN for negative y = -2 to identify the quadrant.

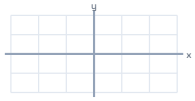
Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Name the Quadrant that contains the point (-5, 11).**

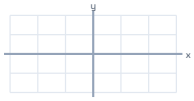
Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Reflect the point (6, 9) across the y-axis. What are the coordinates of the reflected point?**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Reflect the point (6, 9) across the x-axis. What are the coordinates of the reflected point?**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Reflect the point (-3, -7) across the x-axis.**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Reflect the point (-8, 4) across the y-axis, and then reflect the new point across the x-axis.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Which axis does the point (0, -9) lie on, and which axis does the point (-7, 0) lie on?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Starting at (4, -5), you move 7 units left and 8 units up. What are your final coordinates, and which quadrant are you in?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Points A(-4, 2), B(4, 2), and C(4, -5) are three vertices of a rectangle. What are the coordinates of the fourth vertex D, and which quadrant is it in?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 6.4 — Practice B: Modeling Real Life (The Four-Quadrant Coordinate Plane & Reflections)

Textbook Ways: Plot & Count Distance to the Mirror Axis | Coordinate Reflection Sign Rules Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — REFLECTING POINTS ACROSS THE AXES (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: Start at P(-5, 8) in Quadrant II. Reflect P across the x-axis, and then reflect the new point across the y-axis. What are the final coordinates?

WAY 1: Coordinate Sign Rules ((x, -y) Then (-x, y))

1. Across x-axis (keep x, flip y): (-5, 8) → (-5, -8). 2. Across y-axis (flip x, keep y): (-5, -8) → (5, -8) in Quad IV.

WAY 2: Count Equal Steps Across Each Mirror Axis

P is 8 units above the x-axis, so its reflection is 8 units below at (-5, -8); 5 units left of y-axis flips to 5 units right → (5, -8).

Check: (-5, 8) in QII (-, +) flips to QIII (-, -) and then to QIV (+, -).

WATCH OUT: In (x, y), move horizontally (Left/Right) for x FIRST, then vertically (Up/Down) for y. Reflecting across the x-axis flips y!

Answer: (5, -8) in Quadrant IV

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 6.4 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

5 [+/-] [ENTER] 8 [+/-] [⇅] [+/-] [⇅]

Reflect (-5, -8) across the y-axis on paper first (flip x, keep y). On your SC-6, press 5 [+/-] [ENTER] 8 [+/-], then swap to x with [x swap y], flip its sign with [+/-], and swap back to see (5, -8)!

Whenever Mia plotted points like (-4, 3) on a four-quadrant grid, she second-guessed whether to move up-and-down first or left-and-right first, and she tried to memorize rules for reflecting points across the x-axis and y-axis that she immediately mixed up. Then Kai told her how he remembers it on the boat radar: you walk across the deck horizontally (x) before you climb or dive vertically (y). And instead of memorizing reflection rules, folding her graph paper along the horizontal x-axis showed her that a mountain peak at (8, 12) reflects straight down into the lake at (8, -12)—keeping x the same and flipping y!

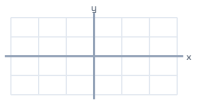
SECTION 6.4 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Treasure Map Quadrants: Identify the quadrant (I, II, III, or IV) for each landmark: Cave (-8, 4), Lighthouse (7, 9), Shipwreck (-5, -6), and Pier (3, -10).**

Starter Step 1: Check the signs of (x, y) for each landmark: QI is (+,+), QII is (-,+), QIII is (-,-), and QIV is (+,-).

Way 1 (Paper Work):

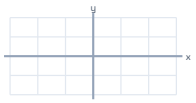


WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

2 **Axis Landmarks: A buoy is at (0, -7) and a dock is at (9, 0). Explain why neither lies in a quadrant, and tell which axis each point lies on.**

Starter Step 1: A point with x = 0 sits directly on the vertical y-axis; a point with y = 0 sits directly on the horizontal x-axis.

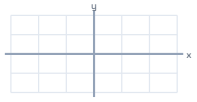
Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

3 **Lake Reflection: A mountain peak is plotted at (8, 12). Its reflection in a lake flips across the x-axis. What are the coordinates of the reflected peak?**

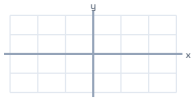
Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

4 **Bird Wing Symmetry: In a biology flight-tunnel diagram, a bird's left wingtip is at (-9, 5) in Quadrant II. Reflect the point across the y-axis to find the coordinates of the right wingtip.**

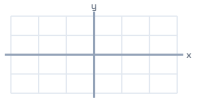
Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

5 **Double Reflection: Start at (-4, -9). Reflect across the x-axis first, and then reflect the new point across the y-axis. What are the final coordinates?**

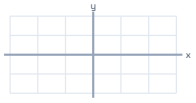
Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

6 **Robot Grid Walk: Starting at the origin (0, 0), a robot moves 6 units left, 8 units down, and then 10 units right. What are its final coordinates and quadrant?**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

7 **Without graphing, identify the quadrant containing P(-2.5, 4.5) and the quadrant containing Q(-1.5, -3.5).**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

8 **Point M(7, -3) is reflected across an axis to M'(-7, -3). Which axis was it reflected across? Explain how you know.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

9 **Square Centered at the Origin: One vertex of a square centered at (0, 0) is (5, 5). What are the coordinates of the other three vertices?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

10 **If x is negative and y is positive, which quadrant contains (x, y), and which quadrant contains (-x, -y)?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 6.5 — Practice A: Distances in the Coordinate Plane

Key Vocabulary: horizontal distance • vertical distance • same quadrant (subtract |x|) • different quadrants (add |x|) Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & VISUAL DIAGRAM

Coordinate Plane Distance: Crosses Axis → ADD |x| | Same Side → SUBTRACT |x|

Different Quadrants (Crosses Axis):

(-6, 4) ←-- y-axis --→ (9, 4)

ADD Distances: |-6| + |9| = 6 + 9 = 15

Coord Diff: |9 - (-6)| = |15| = 15 units

Same Quadrant (Same Side of Axis):

(-12, -3) and (-4, -3) in Quad III

SUBTRACT: |-12| - |-4| = 12 - 4 = 8

Coord Diff: |-4 - (-12)| = |8| = 8 units

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Method 1 (Quadrant Rule): Use absolute values—ADD distances to the axis when points are in DIFFERENT quadrants; SUBTRACT when in the SAME quadrant.

Textbook Method 2 (Coordinate Difference): Subtract the changing coordinates and take the absolute value: $|x_1 - x_2|$ or $|y_1 - y_2|$.

WATCH OUT: If two points are in DIFFERENT quadrants (cross an axis), ADD |x|; if they are in the SAME quadrant, SUBTRACT |x|!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find the distance between (-6, 4) and (9, 4), and between (-12, -3) and (-4, -3).

WAY 1: Add Across Axis / Subtract in Same Quadrant

(-6, 4) to (9, 4) crosses the y-axis → ADD |-6| + |9| = 15. (-12, -3) to (-4, -3) is in QIII → SUBTRACT |-12| - |-4| = 8.

WAY 2: Count Grid Steps / Absolute Difference $|x_1 - x_2|$

From -6 to 0 is 6 steps and 0 to 9 is 9 steps (6 + 9 = 15); from -12 to -4 is 8 steps (|-12 - (-4)| = 8).

Answer: 15 units; and 8 units 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 6.5 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

6 [+/−] [ENTER] 9 [-] [g] [|x|]

Sketch the points on a grid line first to see if they cross an axis. Then check the distance between (-6, 4) and (9, 4) on your SC-6 (or in [f][LINE] mode): Press 6 [+/-] [ENTER] 9 [-] [g][|x|] to confirm 15 units!

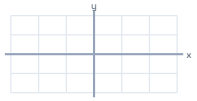
SECTION 6.5 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Find the distance between (-12, 5) and (15, 5).**

Starter Step 1: Since (-12, 5) and (15, 5) are on opposite sides of the y-axis, ADD their absolute distances from 0: |-12| + |15|.

Way 1 (Paper Work):

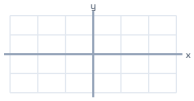


WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Find the vertical distance between (-3, -4) and (-3, -15).**

Starter Step 1: Since both points are in the same quadrant (both y-values are negative), SUBTRACT their absolute distances: |-15| - |-4|.

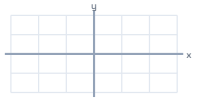
Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Find the vertical distance between (7, 9) and (7, -6).**

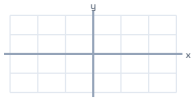
Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Find the horizontal distance between (-14, 2) and (-5, 2).**

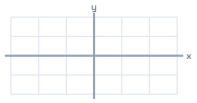
Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Find the distance between (-3.5, 4) and (6.5, 4).**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Find the distance between (2, -4 1/2) and (2, 3 1/2).**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **A horizontal segment starts at (-6, 3) and extends 10 units to the right. What are the coordinates of its right endpoint?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Error Analysis: To find the distance from (-5, 2) to (8, 2), Sam subtracted 8 - 5 = 3. Explain why the distance is 13 units, not 3 units.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Archaeological Survey Grid: A rectangular excavation zone has corners at (-4, 6), (5, 6), (5, -2), and (-4, -2). What are the perimeter and area of the zone?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **City Grid Walk: On a map where 1 grid unit equals 0.5 mile, you walk from (-6, -2) north to (-6, 8), and then east to (4, 8). How many total miles did you walk?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 6.5 — Practice B: Modeling Real Life (Distances in the Coordinate Plane)

Textbook Ways: Add/Subtract Absolute Values to the Axis | Absolute Difference of Coordinates $|x_1 - x_2|$ Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — TOWN PARK BLUEPRINT ON A COORDINATE GRID (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: A rectangular park has vertices at (-4, 6), (5, 6), (5, -2), and (-4, -2). Find the perimeter and area of the park.

WAY 1: Add Distances to the Axes

Width (crosses y-axis): |-4| + |5| = 4 + 5 = 9 units. Height (crosses x-axis): |6| + |-2| = 6 + 2 = 8 units. P = 2(9+8) = 34; A = 9×8 = 72.

WAY 2: Decompose into 4 Quadrant Sub-Rectangles

Sub-areas in the 4 quadrants: QII (4×6=24) + QI (5×6=30) + QIII (4×2=8) + QIV (5×2=10) = 72 square units.

Check: Width = 9, Height = 8 → Perimeter = 9 + 8 + 9 + 8 = 34 units.

WATCH OUT: If two points are in DIFFERENT quadrants (cross an axis), ADD |x|; if they are in the SAME quadrant, SUBTRACT |x|!

Answer: Perimeter = 34 units; Area = 72 square units

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 6.5 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

4 +/− ENTER 5 − g |x| 6 ENTER 2 +/− − f ■

Find the horizontal width |-4 - 5| = 9 and vertical height |6 - (-2)| = 8 on paper first. Check on your SC-6: Press 9 [ENTER] 8 [f][RECT] to verify Area = 72 on X and Perimeter = 34 on Y!

On a harbor coordinate map, Mia needed the length of a suspension bridge stretching from (-8, 5) to (11, 5), and the distance between the school gym at (-14, 6) and the library at (-5, 6). At first she didn't know when to add and when to subtract the absolute values. Plotting both pairs on a simple horizontal line, she saw why you don't need a memorized rule: the bridge crosses right over the y-axis (0), so you add the 8 units on the left plus the 11 units on the right (19 units); the gym and library sit on the same side of 0, so you subtract their distances (14 - 5 = 9 units).

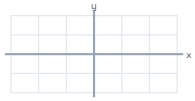
SECTION 6.5 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Suspension Bridge: On a map, the West Tower of a bridge is at (-8, 5) and the East Tower is at (11, 5). How many units long is the bridge?**

Starter Step 1: (-8, 5) and (11, 5) are on opposite sides of the y-axis, so add their distances from the y-axis: |-8| + |11|.

Way 1 (Paper Work):

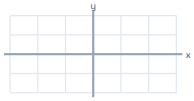


WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Campus Map Walk: On a school map where 1 unit = 10 meters, the gym is at (-14, 6) and the library is at (-5, 6). How many meters apart are they?**

Starter Step 1: (-14, 6) and (-5, 6) are on the same side of the y-axis, so subtract |-14| - |-5| to find the grid units, then multiply by 10 m.

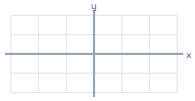
Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Mine Shaft: A vertical elevator runs from a surface tower at (4, 15) straight down to a tunnel at (4, -35). What is the total vertical distance?**

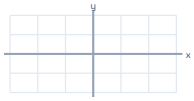
Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Dog Park Perimeter: A rectangular dog park has corners at (-3, 4), (7, 4), (7, -2), and (-3, -2). Find its width, height, and perimeter.**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Dog Park Area: What is the area of the rectangular dog park in Problem 4?**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Subway Line: Stop A is at (-3.5, 2) and Stop C is at (4.25, 2), where each unit is 1 mile. How many miles is it from Stop A to Stop C?**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Find the vertical distance between (2, 3 1/2) and (2, -2 1/4).**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Drone Delivery Route: A drone flies due north from (-6, -2) to (-6, 5), and then turns due east to (4, 5). What is the total distance of its two-leg flight?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Two Possible Locations: Point A is at (3, 4). Point B has the same y-coordinate as Point A and is 8 units away horizontally. What are the two possible ordered pairs for Point B?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Coordinate Triangle Area: A right triangle has vertices at (-4, -1), (6, -1), and (-4, 7). Find its base, height, and area.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

Unit 7: Equations & Inequalities — Mystery Receipts, Roller Coaster Height Bars & Bike Drives

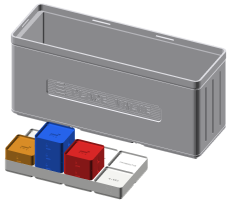
Writing Equations, Solving One-Step Equations, Writing & Graphing Inequalities, and Two-Variable Relationships

In most math worksheets, you are given the starting numbers and asked for the final total. In real life, Maya noticed, it almost always happens backward: you already know the final total on a smudged receipt or the remaining plastic on a 3D-printer spool, and you have to figure out what you started with. When Maya saw 12 robotics shirts billed for $168 total, or Leo saw 185.2 grams left on a spool after a 64.8-gram print, they initially tried random guess-and-check and kept getting frustrated.

Other rules don't even have a single answer. At the amusement park, Leo stood beside a sign reading 'Riders must be at least 54 inches tall,' while a freight elevator plaque warned 'At most 12 passengers.' Leo always mixed up which way the $<$ and $>$ symbols pointed and whether the boundary number itself counted.

Meanwhile, Devon wanted to write a rule connecting how many times he pedals his bike (the input x) to how many times the rear wheel spins (the output y), and see why a bike gear rule always passes through (0, 0) while a store order with a flat $7 shipping fee does not.

When Maya, Leo, and Devon start picturing every equation as a balanced two-pan scale—where doing the inverse operation to both pans unwraps the unknown number—and plugging their solution back into the SC-6 to verify that both sides balance, the guesswork disappears.



Real-World Mathematical Context & Two-Method Mastery • Hands-On Models: Place Value Blocks & Fraction Strips

UNIT 7 LESSONS — LEARN TWO TEXTBOOK WAYS FIRST, THEN CHECK YOUR WORK ON THE SC-6 CALCULATOR

SEC 7.1 (pp.66-67) Writing Equations in One Variable & Testing Solutions

Textbook Ways: Way 1: Substitute Candidates to Test Equality | Way 2: Balance Scale / Mental Unwrapping

Learn by Hand: An equation is a mathematical sentence that uses an equal sign (=) to state that two expressions have the exact same value—just like the two pans of a balanced scale. A value of the variable that makes the equation true is called a solution of the equation.

Check Your Work on the SC-6 Calculator:

4 1 ENTER 4 ENTER 9 × 5 + ≠

Load target 41 [ENTER], evaluate 4 [ENTER] 9 [×] 5 [+], and press [≠]—it flashes NO (zero difference: Left = Right = 41)!

SEC 7.2 (pp.68-69) Solving Equations Using Addition or Subtraction

Textbook Ways: Way 1: Properties of Equality (Inverse Operations) | Way 2: Balance Scale / Part-Part-Whole Bar Model

Learn by Hand: Addition and subtraction are inverse operations—they undo each other. To solve an addition equation like $x + 14 = 39$, use the Subtraction Property of Equality to subtract 14 from BOTH sides so x is isolated on one pan of the balance scale. To solve a subtraction equation like $x - 18 = 45$, use the Addition Property of Equality to add 18 to both sides.

Check Your Work on the SC-6 Calculator:

7 3 ENTER 2 8 - 7 3 ↑↓ 2 8 + ≠

Press 73 [ENTER] 28 [-] to get 45, then load 73 [ENTER] 45 [ENTER] 28 [+] [≠]—it flashes NO to prove 45 + 28 = 73!

SEC 7.3 (pp.70-71) Solving Equations Using Multiplication or Division

Textbook Ways: Way 1: Divide/Multiply by Coefficient | Way 2: Multiply by the Reciprocal | Way 3: Equal-Groups Tape Diagram

Learn by Hand: Multiplication and division are inverse operations. To solve a multiplication equation like $6x = 84$, use the Division Property of Equality to divide both sides by 6. To solve a division equation like $k \div 7 = 14.5$, use the Multiplication Property of Equality to multiply both sides by 7. When the coefficient is a fraction (such as $2/5\ n = 16$), multiply both sides by the reciprocal (5/2).

Check Your Work on the SC-6 Calculator:

1 3 6 ENTER 8 ÷ 1 3 6 ↑↓ 8 × ≠

Press 136 [ENTER] 8 [÷] (17), then test 136 [ENTER] 8 [ENTER] 17 [×] [≠]—it flashes NO to confirm 8(17) = 136!

SEC 7.4 (pp.72-73) Writing and Graphing Inequalities

Textbook Ways: Way 1: Inverse Property of Inequality & Number Line Graph | Way 2: Boundary & Test-Point Substitution

Learn by Hand: An inequality compares two expressions using $<$, $>$, $\leq$ (at most / no more than), or $\geq$ (at least / no less than) and has infinitely many solutions. On a number line graph, use an OPEN circle (o) for $<$ or $>$ (boundary not included) and a SOLID dot (●) for $\leq$ or $\geq$ (boundary included), then shade the arrow in the direction of the true solutions.

Check Your Work on the SC-6 Calculator:

1 2 ENTER 7 + 2 0 f ≥

Solve $x - 7 \geq 12$ on paper first (boundary 19, test x = 20). On your SC-6, press 12 [ENTER] 7 [+] (19) 20 [f][≥]—it flashes YES for $20 \geq 19$ (and use [f][LINE] to see the boundary on the number line)!

SEC 7.5 (pp.74-75) Writing Equations in Two Variables ($y = kx$ and $y = x + b$)

Textbook Ways: Way 1: Test Constant Ratio y/x vs. Constant Difference y - x | Way 2: Input-Output Table & Coordinate Graph

Learn by Hand: In a two-variable equation such as $y = 6x$ or $y = x + 6$, the input variable x is the independent variable (you choose its value freely) and the output variable y is the dependent variable (its value depends on x). In a table of values, if y - x is constant, the rule is additive ($y = x + b$); if $y \div x$ is constant, the rule is multiplicative ($y = kx$).

Check Your Work on the SC-6 Calculator:

1 4 ENTER 2 ÷ 3 5 ENTER 5 ÷ ≠

Press 14 [ENTER] 2 [÷] (7) and 35 [ENTER] 5 [÷] (7), then press [≠] (flashes NO!) to confirm a constant rate $y = 7x$.

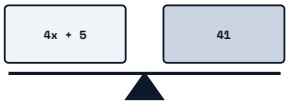
HANDS-ON MODEL: PLACE VALUE BLOCKS

Section 7.1 — Practice A: Writing Equations in One Variable & Testing Solutions

Key Vocabulary: equation • solution of an equation • defining a variable • balance scale • turnaround words ('less than') Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & HANDS-ON MODEL (PLACE VALUE BLOCKS)

Balance Scale Model: $4x + 5 = 41$



At $x = 9$: $4(9) + 5 = 41$ (Balanced!)

Equation Translation Keywords

'is/equals' → = | '12 less than k' → k - 12

Inverse Operations Undo Each Other:

+ = - and × = ÷ (or × b/a)

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept: An equation states that two expressions are equal. To test a value, substitute it for the variable and check if Left Side = Right Side.

Textbook Translation: Translate key words ('is/equals' → =, 'less than' → turnaround subtraction, 'quotient' → ÷) into a one-variable equation.

WATCH OUT: '12 less than k' translates to k - 12 (NOT 12 - k)! Always test your value by substituting it back into the original equation.

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Tell whether $x = 7$ or $x = 9$ is a solution of $4x + 5 = 41$.

WAY 1: Substitute Each Value to Test Equality

Test $x = 7$: $4(7) + 5 = 33 \neq 41$ (not a solution). Test $x = 9$: $4(9) + 5 = 36 + 5 = 41 = 41$ ($x = 9$ is the solution!).

WAY 2: Balance Scale / Mental Math Unwrapping

Ask: 'What number plus 5 equals 41?' → $4x = 36$. Then: '4 times what number equals 36?' → $x = 9$.

Answer: x = 9 is the solution (41 = 41)

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 7.1 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

4 1 ENTER 4 ENTER 9 × 5 + ≠

Test $x = 9$ in $4x + 5 = 41$ on paper first. Then use the SC-6 Equality Test:
Load target 41 [ENTER], evaluate 4 [ENTER] 9 [×] 5 [+], and press [≠]—it flashes NO (zero difference: Left = Right = 41)!

SECTION 7.1 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Tell whether $x = 6$ or $x = 8$ is a solution of $5x - 7 = 33$.**

Starter Step 1: Substitute each candidate value from {6, 8, 10} for x in 5x - 7 until the left side equals 33.

Way 1 (Paper Work):

Sub Value: LHS = _____ = ___

Compare: LHS ___ ?=? RHS ___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

2 **Tell whether $y = 12$ or $y = 15$ is a solution of $y \div 3 + 4 = 9$.**

Starter Step 1: Substitute each candidate value from {12, 15, 18} for y in (y ÷ 3) + 4 until the left side equals 9.

Way 1 (Paper Work):

Sub Value: LHS = _____ = ___

Compare: LHS ___ ?=? RHS ___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

3 **Write the word sentence as an equation and solve: 'The sum of a number x and 14 is 52.'**

Way 1 (Paper Work):

Sub Value: LHS = _____ = ___

Compare: LHS ___ ?=? RHS ___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

4 **Write the word sentence as an equation and solve: '6 times a number k equals 84.'**

Way 1 (Paper Work):

Sub Value: LHS = _____ = ___

Compare: LHS ___ ?=? RHS ___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

5 **Write the word sentence as an equation and solve: '9 less than a number w is 17.'**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

6 **Write the word sentence as an equation and solve: 'The quotient of a number m and 5 is 13.'**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

7 **Four identical pizzas (costing p dollars each) plus a $6 delivery fee total $54. Write an equation to represent the order, and test whether $p = 12$ is the solution.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

8 **Error Analysis: For the sentence '12 less than x is 20', Sam wrote $12 - x = 20$. Explain his error, write the correct equation, and find x.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

9 **Movie Streaming Pass: A streaming service charges a $10 monthly membership plus $3 per movie rental. This month your bill is $28. Write an equation and find how many movies m you rented.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

10 **Restaurant Bill Split: Three friends split a restaurant bill b equally, and each friend pays $14.50. Write an equation and find the total bill b.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

HANDS-ON MODEL: PLACE VALUE BLOCKS

Section 7.1 — Practice B: Modeling Real Life (Writing Equations in One Variable & Testing Solutions)

Textbook Ways: Substitute Candidates to Test Equality | Balance Scale / Mental Unwrapping Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — WRITING & SOLVING A TURNAROUND EQUATION (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: After 8 drones are eliminated in a qualifying round, 25 drones remain in the race. Write and solve an equation to find the starting number of drones n.

WAY 1: Translate Words to an Equation & Use Mental Math

Starting drones (n) minus 8 equals 25: n - 8 = 25. Since 33 - 8 = 25, n = 33 drones.

WAY 2: Part-Part-Whole Bar Model

The total bar (n) splits into 8 eliminated drones and 25 remaining drones → Total n = 25 + 8 = 33 drones.

Watch-Out: '8 less than n' is written n - 8, never 8 - n!

WATCH OUT: '12 less than k' translates to k - 12 (NOT 12 - k)! Always test your value by substituting it back into the original equation.

Answer: n - 8 = 25; n = 33 drones

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 7.1 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

2 5 ENTER 8 + 2 5 ↑↓ 8 - ≠

Write n - 8 = 25 and solve for n = 33 on paper first. Check with the SC-6 Equality Test:
Load 25 [ENTER], evaluate 33 [ENTER] 8 [-], and press [≠]—it flashes NO to confirm both sides equal 25!

Maya used to read an equals sign (=) as 'and the answer is,' which made equations with variables on the left like 15t + 5 = 65 feel backward and confusing. When she looked at her $65 receipt for concert tickets ($15 per ticket t plus a $5 order fee), she realized the equals sign actually works like the center pivot of a balanced two-pan scale: the expression 15t + 5 on the left weighs the exact same as $65 on the right. And to check if t = 4 tickets is right, she didn't have to guess—she could plug 4 in (15 × 4 + 5 = 65) and see both sides balance.

SECTION 7.1 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Tell which value of x from {6, 8, 10} is the solution of 5x - 7 = 33.**

Starter Step 1: Test x = 6, x = 8, and x = 10 in 5x - 7 to see which value makes the equation a true statement.

Way 1 (Paper Work):

Sub Value: LHS = _____ = ___

Compare: LHS ___ ?=? RHS ___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Concert Tickets: Tickets cost $15 each plus a $5 order fee, for a total of $65. Write an equation for t tickets and check whether t = 4 is a solution.**

Starter Step 1: Write the equation (cost of t tickets + $5 fee = $65), then substitute t = 4 to check if both sides match.

Way 1 (Paper Work):

Sub Value: LHS = _____ = ___

Compare: LHS ___ ?=? RHS ___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Write the word sentence '12 less than a number k equals 29' as an equation, and find the value of k.**

Way 1 (Paper Work):

Sub Value: LHS = _____ = ___

Compare: LHS ___ ?=? RHS ___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Write the word sentence 'The quotient of a number n and 6 is 14' as an equation, and find the value of n.**

Way 1 (Paper Work):

Sub Value: LHS = _____ = ___

Compare: LHS ___ ?=? RHS ___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Equilateral Triangle Perimeter: An equilateral triangle has 3 equal sides of length s and a perimeter of 54 cm. Write an equation and find s.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Tell whether m = 12 or m = 15 is a solution of 100 - 4m = 40.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Team Pizza Slices: Dividing p pizza slices equally among 8 players gives each player 3 slices. Write and solve an equation to find p.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Savings Goal: You have $45 saved and add d dollars to reach $120. Write and solve an equation to find d.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Test whether x = 6 is a solution of 3x + 10 = 5x - 2 by substituting x = 6 into both sides.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Sibling Ages: Maya is 13 years old, which is 4 years younger than her brother Leo. Write an equation for Leo's age b and find how old Leo is.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: PLACE VALUE BLOCKS

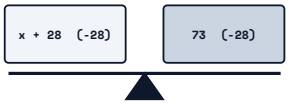
Section 7.2 — Practice A: Solving Equations Using Addition or Subtraction

Key Vocabulary: inverse operations • Addition Property of Equality • Subtraction Property of Equality • isolate the variable

Name: _____ Date: _____ Score: ____ / 100

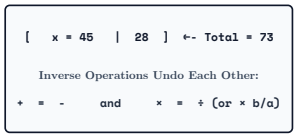
TEXTBOOK CONCEPT & HANDS-ON MODEL (PLACE VALUE BLOCKS)

Balance Scale Model: x + 28 (-28) = 73 (-28)



Remove 28 from BOTH pans → x = 45

Part-Whole Bar Model (x + 28 = 73)



HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept (Inverse Operations): Addition and subtraction are inverse operations. Undo '+ a' by subtracting a from both sides; undo '- a' by adding a.

Textbook Properties of Equality: Subtracting or adding the same number on both sides keeps the equation balanced. Always substitute back to check!

WATCH OUT: Use the INVERSE operation on BOTH sides of the '=' sign (+ undoes -, and - undoes +) to keep the equation balanced!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Solve x + 28 = 73 and check your solution.

WAY 1: Subtraction Property of Equality (Undo + 28)

Subtract 28 from both sides of x + 28 = 73: (x + 28) - 28 = 73 - 28 → x = 45.

WAY 2: Balance Scale / Part-Whole Bar Model

A whole bar of 73 is made of part x and part 28. Removing 28 from both pans leaves x = 73 - 28 = 45. (Check: 45+28 = 73.)

Answer: x = 45 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 7.2 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

7 3 ENTER 2 8 - 7 3 ⇅ 2 8 + ≠

Undo + 28 by subtracting 28 on paper first (x = 45). Check on your SC-6:

Press 73 [ENTER] 28 [-] to get 45, then load 73 [ENTER] 45 [ENTER] 28 [+] [≠]—it flashes NO to prove 45 + 28 = 73!

SECTION 7.2 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 Solve x + 28 = 73 and check your solution.

Starter Step 1: Use the inverse operation of + 28 by subtracting 28 from BOTH sides of the equation.

Way 1 (Paper Work):

Inverse Op Both Sides: _____

Solution = _____ (Check: _____)

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 Solve y - 34 = 58 and check your solution.

Starter Step 1: Use the inverse operation of - 34 by adding 34 to BOTH sides of the equation.

Way 1 (Paper Work):

Inverse Op Both Sides: _____

Solution = _____ (Check: _____)

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 Solve 19 + n = 64.

Way 1 (Paper Work):

Inverse Op Both Sides: _____

Solution = _____ (Check: _____)

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 Solve k - 47 = 29.

Way 1 (Paper Work):

Inverse Op Both Sides: _____

Solution = _____ (Check: _____)

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 Solve w - 19.6 = 34.8.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 Solve p + 8.75 = 25.4.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 Solve m + 3/8 = 5/6.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 Solve a - 1 1/4 = 3 1/2.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 Game Controller Purchase: After spending $18.50 on a custom controller skin, you have $26.75 left in your wallet. Write and solve an equation to find how much money m you had before the purchase.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 Drone Altitude Climb: A quadcopter climbs 42.5 feet to reach a hovering altitude of 110 feet. Write and solve an equation to find its starting altitude h.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: PLACE VALUE BLOCKS

Section 7.2 — Practice B: Modeling Real Life (Solving Equations Using Addition or Subtraction)

Textbook Ways: Properties of Equality (Inverse Operations) | Balance Scale / Part-Part-Whole Bar Model Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — SOLVING DECIMAL & FRACTION EQUATIONS (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: After a 3D print uses 19.6 grams of filament, 34.8 grams remain on the spool. Also solve m + 3/8 = 5/6.

WAY 1: Addition & Subtraction Properties of Equality

1. For w - 19.6 = 34.8, add 19.6 to both sides: w = 34.8 + 19.6 = 54.4 g. 2. For m + 3/8 = 5/6, subtract 3/8: m = 20/24 - 9/24 = 11/24.

WAY 2: Bar Model & Fraction Strip Comparison

1. Whole spool w = used (19.6) + left (34.8) = 54.4 g. 2. On twenty-fourths strips, 5/6 (20/24) is 11/24 longer than 3/8 (9/24).

Check: 54.4 - 19.6 = 34.8 and 11/24 + 9/24 = 20/24 = 5/6.

WATCH OUT: Use the INVERSE operation on BOTH sides of the '=' sign (+ undoes -, and - undoes +) to keep the equation balanced!

Answer: w = 54.4 g; m = 11/24

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 7.2 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

3 4 • 8 ENTER 1 9 • 6 + • 5 • 6 ENTER • 3 • 8 -

Solve both equations by hand first. Check on your SC-6:
Press 34.8 [ENTER] 19.6 [+] (54.4 g) and Two-Dot fractions [.] 5 [.] 6 [ENTER] [.] 3 [.] 8 [-] (11/24), then test each in the original equation with [≠] (flashes NO!).

When Leo saw the equation f - 64.8 = 185.2 for his 3D-printer spool (after a print used 64.8 grams of filament and left 185.2 grams on the scale), his knee-jerk reaction was to subtract 185.2 - 64.8 because he saw a minus sign in the problem. Looking at his answer of 120.4 grams, he stopped—how could the spool start with LESS plastic than was still sitting on the scale?! Picturing the print job like a video playing in reverse, he realized that to undo taking 64.8 grams away, you have to add the 64.8 grams back to both sides (185.2 + 64.8 = 250.0 grams).

SECTION 7.2 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Video Game High Score: After scoring 45 bonus points, your total score is 210. Write and solve an equation to find your score s before the bonus.**

Starter Step 1: Write an equation showing score s plus the 45-point bonus equals 210, then undo the addition.

Way 1 (Paper Work):

Inverse Op Both Sides: _____

Solution = _____ (Check: _____)

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Skateboard Coupon: After using an $18 coupon, a skateboard costs $67. Write and solve an equation to find the regular price p.**

Starter Step 1: Write an equation showing regular price p minus the $18 coupon equals $67, then undo the subtraction.

Way 1 (Paper Work):

Inverse Op Both Sides: _____

Solution = _____ (Check: _____)

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Mountain Summit Climb: From Base Camp, a climber ascends 1,450 feet to reach a 5,200-foot peak. Write and solve an equation to find the Base Camp elevation h.**

Way 1 (Paper Work):

Inverse Op Both Sides: _____

Solution = _____ (Check: _____)

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **3D Filament Remaining: After a print uses 64.8 grams of PLA, 185.2 grams remain on the spool. Write and solve an equation to find how many grams f were on the spool before the print.**

Way 1 (Paper Work):

Inverse Op Both Sides: _____

Solution = _____ (Check: _____)

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Stopwatch Improvement: Leo shaved 0.48 seconds off his previous 100-meter dash time t to run 12.65 seconds. Write and solve an equation to find his previous time t.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Recipe Measuring Cup: You have 3/8 cup of milk and need 7/8 cup in all. Write and solve an equation to find how much more milk m you need.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Trail Hike Remaining: A hiking trail is 4 1/2 miles long, and you have already hiked 2 3/4 miles. Write and solve an equation to find the remaining distance d.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Homing Pigeon Compass Direction: Homing pigeons can sense Earth's magnetic field like a built-in compass! A pigeon turns +38° clockwise so its new compass direction is 125°. Write and solve an equation to find its starting compass direction b.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Submarine Depth Change: A submarine starts at -40 meters and changes elevation by d meters to reach -115 meters. Write and solve an equation for d.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Rectangle Perimeter: A rectangle has a perimeter of 64 cm, and its two widths have a combined length of 20 cm. What is the combined length L of the two longer sides, and what is the length of one longer side?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: FRACTION STRIPS

Section 7.3 — Practice A: Solving Equations Using Multiplication or Division

Key Vocabulary: Multiplication Property of Equality • Division Property of Equality • reciprocal coefficient Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & HANDS-ON MODEL (FRACTION STRIPS)

Balance Scale Model: 8 x-boxes (÷8) = 136 units (÷8)



Divide both sides by 8 → x = 17

Equal-Box Tape Model (8x = 136)

8 boxes = 136 → 1 box = 136 ÷ 8 = 17

Inverse Operations Undo Each Other:

+ = − and × = ÷ (or × b/a)

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept: Multiplication and division are inverse operations. Undo 'a · x' by dividing both sides by a; undo 'x ÷ a' by multiplying both sides by a.

Textbook Fractional Coefficients: To solve (a/b)x = c in one step, multiply both sides by the reciprocal (b/a).

WATCH OUT: To solve (3/4)x = 18, multiply both sides by the RECIPROCAL (4/3); to solve k ÷ 7 = 14.5, multiply both sides by 7!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Solve 8x = 136 and k ÷ 7 = 14.5, and check each solution.

WAY 1: Division & Multiplication Properties of Equality

In 8x = 136, divide both sides by 8 → x = 17. In k ÷ 7 = 14.5, multiply both sides by 7 → k = 101.5.

WAY 2: Equal-Groups Balance Scale / Bar Model (With Blocks)

Split 136 equally across 8 x-boxes (80÷8 + 56÷8 = 10 + 7 = 17); combine 7 equal boxes of 14.5 → 101.5.

Answer: x = 17; k = 101.5 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 7.3 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

1 3 6 ENTER 8 ÷ 1 3 6 ⇅ 8 × ≠

Undo multiplication or division on paper first (x = 17, k = 101.5). Check on your SC-6:
Press 136 [ENTER] 8 [÷] (17), then test 136 [ENTER] 8 [ENTER] 17 [×] [≠]—it flashes NO to confirm 8(17) = 136!

SECTION 7.3 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Solve 8x = 136 and check your solution.**

Starter Step 1: Undo multiplication (8x) by dividing BOTH sides of the equation by 8.

Way 1 (Paper Work):

Inverse Op Both Sides: _____

Solution = _____ (Check: _____)

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Solve 12y = 180 and check your solution.**

Starter Step 1: Undo multiplication (12y) by dividing BOTH sides of the equation by 12.

Way 1 (Paper Work):

Inverse Op Both Sides: _____

Solution = _____ (Check: _____)

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Solve m ÷ 6 = 19 and check your solution.**

Way 1 (Paper Work):

Inverse Op Both Sides: _____

Solution = _____ (Check: _____)

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Solve k ÷ 7 = 14.5.**

Way 1 (Paper Work):

Inverse Op Both Sides: _____

Solution = _____ (Check: _____)

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Solve 4.5w = 63.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Solve 3/4 x = 18.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Solve 2/5 n = 16.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Solve 5/6 p = 2/3.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **3D Tower Print Time: A 3D printer builds a vertical tower at a steady rate of 1.5 cm per minute. Write and solve an equation to find how many minutes t it takes to print a 27 cm tower.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Robotics Club Roster: Exactly 3/5 of the students in the STEM club are 6th graders. If there are 27 sixth graders in the club, write and solve an equation to find the total number of students s in the club.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: FRACTION STRIPS

Section 7.3 — Practice B: Modeling Real Life (Solving Equations Using Multiplication or Division)

Textbook Ways: Divide/Multiply by Coefficient | Multiply by the Reciprocal | Equal-Groups Tape Diagram Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — FRACTIONAL COEFFICIENT EQUATION (3 TEXTBOOK WAYS • ANY 2 = +20 PTS)

Problem: Two-fifths of the songs on a playlist are rock songs. There are 16 rock songs. How many songs are on the playlist in all?

WAY 1: Multiply Both Sides by the Reciprocal (5/2)
Write (2/5)n = 16. Multiply both sides by 5/2: n = 16 × (5/2) = (16 ÷ 2) × 5 = 8 × 5 = 40 songs.

WAY 2: 5-Box Tape Diagram (Find 1/5 First, Then 5/5)
Draw 5 equal fifth-boxes. Since 2 boxes = 16 songs, 1 box = 16 ÷ 2 = 8 songs. All 5 boxes = 5 × 8 = 40 songs.

WAY 3: Convert 2/5 to a Decimal (0.4n = 16)
Write 2/5 as 0.4: 0.4n = 16. Divide both sides by 0.4: n = 16 ÷ 0.4 = 160 ÷ 4 = 40 songs.

WATCH OUT: To solve (3/4)x = 18, multiply both sides by the RECIPROCAL (4/3); to solve k ÷ 7 = 14.5, multiply both sides by 7!

Answer: 40 songs

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 7.3 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[1] [6] [ENTER] [•] [2] [•] [5] [f] [1/x] [×]

Multiply both sides of (2/5)s = 16 by the reciprocal 5/2 on paper first. Check on your SC-6 with Two-Dot fractions: Press 16 [ENTER] [.] 2 [.] 5 [f][1/x] [×] (40), then verify 16 [ENTER] [.] 2 [.] 5 [ENTER] 40 [×] [≠] (flashes NO!).

Maya knew that 12 identical robotics shirts cost $168 (12s = 168), which meant dividing both sides by 12 to find $14 per shirt. But when she hit a fraction equation—a bike on sale for 5/6 of its original price costs $150, or (5/6)p = 150—she froze. Dividing by a fraction inside an equation looked scary until she connected it to what she learned in Unit 2: if 5 fifths of the price is $150, then 1 fifth is $150 ÷ 5 = $30, and all 6 sixths (the full price) is 6 × $30 = $180! Multiplying both sides by the reciprocal (6/5) did both steps at once.

SECTION 7.3 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Robotics Club T-Shirts: A team paid $168 for 12 identical team shirts. Write and solve an equation to find the price s of each shirt.**

Starter Step 1: Write an equation for 12 shirts at s dollars each totaling $168, then divide both sides by 12.

Way 1 (Paper Work):

Inverse Op Both Sides: _____

Solution = _____ (Check: _____)

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Party Pizza Slices: A giant party pizza is cut into equal slices for 9 friends, and each friend gets 3 slices. Write and solve an equation to find the total number of slices p.**

Starter Step 1: Write an equation for p slices divided among 9 friends equaling 3 slices each, then multiply both sides by 9.

Way 1 (Paper Work):

Inverse Op Both Sides: _____

Solution = _____ (Check: _____)

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Train Travel Time: A train travels at 65 miles per hour and covers 325 miles. Write and solve an equation to find the travel time t in hours.**

Way 1 (Paper Work):

Inverse Op Both Sides: _____

Solution = _____ (Check: _____)

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Bulk Almonds: You buy 2.5 pounds of roasted almonds for $16.25. Write and solve an equation to find the price p per pound.**

Way 1 (Paper Work):

Inverse Op Both Sides: _____

Solution = _____ (Check: _____)

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Comic Book Allowance: You spend one-fourth of your savings s on a $9 comic book. Write and solve an equation to find your total savings s.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Water Bottle Survey: Three-fourths of the students in a class brought a water bottle. If 21 students brought a water bottle, how many students c are in the class?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Sale Price Equation: A bike is on sale for 5/6 of its original price p. If the sale price is $150, what was the original price p?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Banner Width: A rectangular banner has an area of 108 square feet and a length of 13.5 feet. Write and solve an equation to find its width w.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Scale Model Rocket: A model rocket is 1/15 the height of the real rocket. If the model is 3.2 feet tall, how tall is the real rocket?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Mystery Number: Forty percent of a number n is 56. Write and solve an equation to find n.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 7.4 — Practice A: Writing and Graphing Inequalities

Key Vocabulary: inequality • solution set • graph of an inequality • open circle (◯ for <, >) • closed circle (● for ≤, ≥) Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & VISUAL DIAGRAM

Graphing Inequalities on a Number Line: Open Circle (◯) vs. Solid Dot (●)

Strict (< or >): OPEN Circle (◯)

19 (Not included)

x > 19 ('more than' / 'fewer than')

Inclusive (≤ or ≥): SOLID Dot (●)

19 (IS a solution!)

x ≥ 19 ('at least' ≥ | 'at most' ≤)

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept: Solve one-step inequalities using the same inverse operations as equations. Graph < or > with an OPEN circle (◯), and ≤ or ≥ with a SOLID dot (●).

Textbook Shading Check: Substitute one test number from your shaded arrow into the original inequality to verify it makes the statement TRUE.

WATCH OUT: Use an OPEN circle (◯) for < or >, and a SOLID dot (●) for ≤ ('at most') or ≥ ('at least')!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Solve x - 7 ≥ 12 and describe its graph on a number line.

WAY 1: Addition Property of Inequality & Boundary Graph

In x - 7 ≥ 12, add 7 to both sides → x ≥ 19. Draw a CLOSED dot (●) at 19 and shade the arrow to the right.

WAY 2: Boundary Equation + Test Value Method

Solve x - 7 = 12 to find the boundary 19 (included because of ≥). Test x = 20: 20 - 7 = 13 ≥ 12 (True → shade right).

Answer: x ≥ 19 (Solid dot at 19, arrow right)

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 7.4 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

1 2 ENTER 7 + 2 0 f ≥

Solve x - 7 ≥ 12 on paper first (boundary 19, test x = 20). On your SC-6, press 12 [ENTER] 7 [+] (19) 20 [f][≥]—it flashes YES for 20 ≥ 19 (and use [f][LINE] to see the boundary on the number line)!

SECTION 7.4 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 Solve x + 8 > 21. Then state whether the boundary point uses an open circle or solid dot, and which way the arrow points.

Starter Step 1: Subtract 8 from both sides to isolate x; remember that > uses an OPEN circle and shades to the RIGHT.

Way 1 (Paper Work):

◯ or ●

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 Solve y - 9 ≤ 15. Then state whether the boundary point uses an open circle or solid dot, and which way the arrow points.

Starter Step 1: Add 9 to both sides to isolate y; remember that ≤ uses a SOLID dot and shades to the LEFT.

Way 1 (Paper Work):

◯ or ●

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 Solve 6n ≥ 78. Then state whether the boundary point uses an open circle or solid dot, and which way the arrow points.

Way 1 (Paper Work):

◯ or ●

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 Solve m ÷ 4 < 9. Then state whether the boundary point uses an open circle or solid dot, and which way the arrow points.

Way 1 (Paper Work):

◯ or ●

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 Write an inequality for the statement: 'You must be at least 48 inches tall (h) to ride the roller coaster.'

Way 1 (Paper Work):

◯ or ●

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 Write an inequality for the statement: 'A freight elevator holds at most 2,000 pounds (w).'

Way 1 (Paper Work):

◯ or ●

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 Which values from the set {4, 5, 6} are solutions of the inequality 4n + 3 > 23?

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 Which values from the set {4, 5, 6} are solutions of the inequality 4x - 3 ≥ 17?

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 Book Fair Budget: You have $50 to spend on paperback books that cost $8 each. Write and solve an inequality to find the maximum number of whole books b you can buy, and state how much money is left over.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 Next-Level Score Target: A player needs at least 500 total points to advance to the next level and currently has 340 points. Write and solve an inequality to find how many more points p the player needs.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 7.4 — Practice B: Modeling Real Life (Writing and Graphing Inequalities)

Textbook Ways: Inverse Property of Inequality & Number Line Graph | Boundary & Test-Point Substitution Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — WEIGHT LIMIT INEQUALITY (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: An adventure pack holds a 15-pound shield plus gold bars weighing 3 pounds each. The pack can hold at most 60 pounds. What is the maximum number of gold bars it can hold?

WAY 1: Write & Solve the Inequality ($3g + 15 \le 60$)

Subtract 15 from both sides: $3g \le 45$. Divide both sides by 3: $g \le 15$ (closed dot at 15, at most 15 bars).

WAY 2: Remaining Capacity Bar Model

Available weight for gold bars = 60 lb - 15 lb = 45 lb. At 3 lb per bar, 45 ÷ 3 = 15 bars maximum.

Check: 15 bars → 3(15)+15 = 60 lb ≤ 60 lb (True); 16 bars → 63 lb (Too heavy!).

WATCH OUT: Use an OPEN circle (◯) for < or >, and a SOLID dot (●) for ≤ ('at most') or ≥ ('at least')!

Answer: g ≤ 15 (at most 15 gold bars)

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 7.4 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

6 0 ENTER 1 5 - 3 ÷ 1 5 g ≤

Solve $3g + 15 \le 60$ on paper first ($g \le 15$). Check the boundary on your SC-6:
Press 60 [ENTER] 15 [-] 3 [÷] (15), then test 60 [ENTER] 3 [ENTER] 15 [×] 15 [+] [g][≤]—it flashes YES (and view on [f][LINE])!

Standing by the roller coaster entrance ('Riders must be at least 54 inches tall') and the freight elevator ('At most 12 passengers'), Leo admitted he always mixed up 'at least' and 'at most' because the word 'least' made him think of the < sign! Then he tested his own height against the 54-inch bar: 'at least 54 inches' means 54 inches is the lowest height allowed, so you must be 54 inches OR TALLER (h ≥ 54). Asking 'Am I allowed to be EQUAL to 54?' also told him when to color in a solid dot (≥ or ≤) versus leaving an open circle (> or <) on the number line.

SECTION 7.4 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Roller Coaster Height Requirement: Riders must be at least 54 inches tall to ride a coaster. Write an inequality for the allowed heights h and describe its number line graph. If Leo is 49 inches tall, how many more inches must he grow?**

Starter Step 1: 'At least 54' means ≥ 54 (solid dot at 54, arrow right); subtract 49 from 54 to find how much Leo must grow.

Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

2 **Elevator Capacity: An elevator holds at most 12 passengers. There are already 5 passengers inside. Write and solve an inequality to find how many more passengers p can board.**

Starter Step 1: 'At most 12' means 5 + p ≤ 12; subtract 5 from both sides to find the maximum number of passengers p.

Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

3 **Free Shipping Threshold: Online orders of $35 or more receive free shipping. You currently have $22.50 in your cart. Write and solve an inequality to find how much more x you must add to get free shipping.**

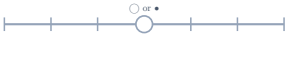
Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

4 **Arcade Budget: You have $28 to spend on VR games that cost $3.50 each. Write and solve an inequality to find the number of games g you can play.**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

5 **Theater Seating: An auditorium needs more than 240 chairs arranged in equal rows of 15 chairs. Write and solve an inequality for the number of rows r. What is the least whole number of rows needed?**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

6 **Drone Battery Warning: A drone uses 25% of its battery during a flight and triggers a low-battery warning because its remaining charge drops below 15%. Write and solve an inequality for the starting battery percent b.**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

7 **Tell which values from {6, 8, 10} are solutions of the inequality $5x - 10 \ge 30$.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

8 **Bake Sale Profit Goal: Your club spends $30 on supplies and sells cupcakes for $2 each. How many cupcakes c must the club sell to make a profit of at least $70?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

9 **School Zone Speed Limit: The speed limit in a school zone is 20 miles per hour. Write an inequality for legal speeds s, give two solutions, and describe the number line graph.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

10 **Cargo Van Weight Limit: A delivery van can carry at most 800 pounds. The driver weighs 170 pounds and each crate weighs 45 pounds. What is the maximum number of crates c the van can carry?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 7.5 — Practice A: Writing Equations in Two Variables ($y = kx$ and $y = x + b$)

Key Vocabulary: independent variable (input x) • dependent variable (output y) • multiplicative ($y = kx$) • additive ($y = x + b$)

Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & VISUAL DIAGRAM

Two-Variable Equations: Independent Input (x) → Dependent Output (y)

Multiplicative: $y = kx$ (e.g., $y = 7x$)	Additive: $y = x + b$ (e.g., $y = x - 4$)
Constant Ratio: $y \div x = 7$	Constant Difference: $y - x = -4$
(1, 7), (2, 14), (3, 21), (4, 28)	(10, 6), (12, 8), (15, 11), (20, 16)
Graph is a line through origin (0, 0)!	Graph crosses y-axis at (0, b)!

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept: In two-variable equations, x is the independent variable (input) and y is the dependent variable (output).

Textbook Rule Comparison: Multiplicative relationships ($y = kx$) have a constant ratio y/x and pass through (0, 0); Additive ($y = x + b$) have a constant difference $y - x$.

WATCH OUT: Check ALL rows of the table! Multiplicative $y = kx$ passes through (0, 0); additive $y = x + b$ has a constant difference $y - x$.

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Table A contains (2, 14), (5, 35), (8, 56). Table B contains (10, 6), (15, 11), (22, 18). Write the equation for each table.

WAY 1: Compare Ratio ($y \div x$) vs. Difference ($y - x$)

Table A ((2,14), (5,35), (8,56)): $14 \div 2 = 35 \div 5 = 56 \div 8 = 7 \rightarrow y = 7x$. Table B ((10,6), (15,11), (22,18)): $y - x = -4 \rightarrow y = x - 4$.

WAY 2: Graph Ordered Pairs & Check the Origin (0, 0)

Plotting Table A gives a line through (0, 0) with steepness 7 ($y = 7x$). Table B shifts every x down by 4 units ($y = x - 4$).

Answer: Table A: $y = 7x$; Table B: $y = x - 4$

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 7.5 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[1] [4] [ENTER] [2] [÷] [3] [5] [ENTER] [5] [÷] [≠]

Test whether $y \div x$ or $y - x$ is constant in each table on paper first. Check on your SC-6:
Press 14 [ENTER] 2 [÷] (7) and 35 [ENTER] 5 [÷] (7), then press [≠] (flashes NO!) to confirm a constant rate $y = 7x$.

SECTION 7.5 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **A table contains the ordered pairs (1, 8), (2, 16), and (3, 24). Write the equation relating x and y, and find y when $x = 10$.**

Starter Step 1: Divide each output y by its input x ($y \div x$) to find the constant multiplicative rate k in $y = kx$.

Way 1 (Paper Work):

x			
y			

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **A table contains the ordered pairs (1, 9), (2, 10), and (5, 13). Write the equation relating x and y, and find y when $x = 20$.**

Starter Step 1: Subtract each input x from its output y ($y - x$) to find the constant amount b added in $y = x + b$.

Way 1 (Paper Work):

x			
y			

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **A table contains the ordered pairs (2, 14), (5, 35), and (8, 56). Write the equation relating x and y, and find y when $x = 12$.**

Way 1 (Paper Work):

x			
y			

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **A table contains the ordered pairs (10, 6), (15, 11), and (22, 18). Write the equation relating x and y, and find y when $x = 30$.**

Way 1 (Paper Work):

x			
y			

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Complete a table of ordered pairs (x, y) for the equation $y = 6x$ using $x = 1, 2, 3$, and 4.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **In the formula $d = 65t$ (representing the distance d in miles a train travels in t hours), which variable is independent and which variable is dependent?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Using the train formula $d = 65t$, how far does the train travel in 4.5 hours?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Does the graph of the multiplicative equation $y = 4x$ pass through the origin (0, 0)? Where does the additive equation $y = x + 4$ cross the y-axis?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Summer Lawn Care: You earn \$12.50 per hour according to $p = 12.50h$. How many hours h must you work to earn \$100?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Climbing Gym Membership: A climbing gym charges a one-time \$25 registration fee plus \$15 per month m. Write a two-variable equation for the total cost c, and find the total cost for 12 months.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 7.5 — Practice B: Modeling Real Life (Writing Equations in Two Variables ($y = kx$ and $y = x + b$))

Textbook Ways: Test Constant Ratio y/x vs. Constant Difference y - x | Input-Output Table & Coordinate Graph Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — TWO-VARIABLE EARNINGS EQUATION (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: You earn $12.50 per hour mowing lawns. Write an equation relating total pay p and hours worked h, and find your pay for 6.5 hours.

WAY 1: Write Multiplicative Equation $p = 12.50h$ & Substitute

Hours h is independent; pay p is dependent: $p = 12.50h$. For $h = 6.5$: $p = 12.50 \times 6.5 = \$81.25$.

WAY 2: Input-Output Ratio Table (6 Hours + 0.5 Hour)

In a table: 1 hr = $12.50, 6 hr = $6 \times \$12.50 = \$75.00$, and 0.5 hr = $6.25. Total = $75.00 + $6.25 = $81.25.

Check: $81.25 ÷ 6.5 hr = $12.50 per hour.

WATCH OUT: Check ALL rows of the table! Multiplicative $y = kx$ passes through (0, 0); additive $y = x + b$ has a constant difference y - x.

Answer: p = 12.50h; $81.25

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 7.5 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

1 2 • 5 0 ENTER 6 • 5 ×

Write $p = 12.50h$ and multiply on paper first. Then check on your SC-6:
Press 12.50 [ENTER] 6.5 [×] to confirm $81.25.

Devon always mixed up which variable was 'independent' and which was 'dependent,' and couldn't tell when a table followed a multiplying rule ($y = kx$) versus an adding rule ($y = x + b$). Looking at his bike, he asked a simple cause-and-effect question: do my feet make the back wheel spin, or does the back wheel force my feet to move? Since his pedal turns (x) come first and cause the wheel turns ($y = 3.5x$), x is independent and y depends on x. And testing $x = 0$ gave him an instant shortcut: 0 pedal turns means 0 wheel turns (0, 0), whereas an order with a $7 shipping fee ($c = x + 7$) never passes through (0, 0)!

SECTION 7.5 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Additive vs. Multiplicative Plans: Plan A adds a $5 fee to a base price x ($y = x + 5$). Plan B charges $8 per month m ($y = 8m$). Tell which equation represents an additive relationship and which represents a multiplicative relationship.**

Starter Step 1: Compare the two equations: adding a fixed constant (+ b) is Additive, while multiplying by a rate (kx) is Multiplicative.

Way 1 (Paper Work):

x			
y			

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **3D Printer Filament Rate: A printer uses 18 grams of PLA per hour h. Write an equation for the total grams g used, identify the independent and dependent variables, and find g for 4.5 hours.**

Starter Step 1: Hours h is the independent input and grams g is the dependent output; multiply the hourly rate by 4.5 hours.

Way 1 (Paper Work):

x			
y			

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Sibling Age Rule: Maya is always 4 years older than her sister Zoe. Write an equation relating Maya's age m and Zoe's age z, and find m when $z = 8$, 12, and 15.**

Way 1 (Paper Work):

x			
y			

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Custom Sticker Sheets: Each printed sheet holds 15 stickers. Write an equation for the total stickers s on p sheets, and find how many sheets are needed for 180 stickers.**

Way 1 (Paper Work):

x			
y			

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Sale Table Rule: Every item on a clearance table is $6 off its original tag price x. Write an equation for the sale price y and find y when $x = \$24$, $35, and $50.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **How High Gear Works on a Bike: Because the front pedal gear is larger than the back wheel gear, 1 pedal turn (x) spins the back wheel 3.5 times. Write an equation for wheel turns y, and find y for 40 pedal turns.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Input-Output Table: An input-output table has pairs (3, 27), (6, 54), (9, 81), and (12, 108). Write an equation relating x and y, and find y when $x = 20$.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Board Game Shipping: Each board game costs $20 plus a flat $7 shipping fee for the order. Write an equation for the total cost c of g games, and find the cost of 6 games.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Which type of relationship always passes through the origin (0, 0) when graphed: a multiplicative relationship ($y = kx$) or an additive relationship ($y = x + b$)? Explain why.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Thunder & Lightning: Sound travels about 0.2 miles per second. Write an equation for the distance d (in miles) sound travels in s seconds. If you hear thunder 15 seconds after seeing lightning, how far away is the storm?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

Unit 8: Surface Area & Volume of 3D Solids — Penguin Huddles, Camping Tents, Aquariums & Ziplines

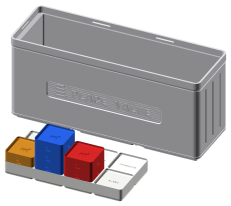
Surface Area of Rectangular & Triangular Prisms, Pyramids, Volume of Prisms, and Circles/Cones/Hypotenuse

Why doesn't an Emperor penguin chick freeze when Antarctic winds hit -40°C? When Zoe read that penguins survive by huddling together to shrink their exposed 'surface area' while keeping their warm 'volume' the same, she admitted she always mixed those two terms up in math class. Liam had the same headache in the workshop: when he sewed nylon for a camping tent or ordered glass for an open-top fish tank, he wasn't sure when to add flat walls and when to multiply length times width times height.

Marcus ran into a surprise of his own at the team pizza night. The pizzeria offered either one 16-inch pizza or two 10-inch pizzas for the same price, and almost everyone at the table assumed two 10-inch pizzas had more food simply because 10 + 10 is 20.

Out on the adventure course, they also needed to find the exact length of a diagonal zipline cable stretched from the top of a 15-meter tower to an anchor 20 meters away on the ground without climbing into the air with a tape measure.

By unfolding cardboard boxes and tents flat onto the table as 2D nets, packing wooden cubes into rectangular trays, and checking prism volumes, circle areas, and right-triangle diagonals on the SC-6, Zoe, Liam, and Marcus master every 3D measurement.



Real-World Mathematical Context & Two-Method Mastery • Hands-On Models: Place Value Blocks & Fraction Strips

UNIT 8 LESSONS — LEARN TWO TEXTBOOK WAYS FIRST, THEN CHECK YOUR WORK ON THE SC-6 CALCULATOR

SEC 8.1 (pp.77-78) Three-Dimensional Figures, Nets & Rectangular Prisms

Textbook Ways: Way 1: Unfold into a 6-Face 2D Net | Way 2: Opposite Face Pairs SA = 2(lw + lh + wh) | Way 3: Lateral Strip Ph + 2B

Learn by Hand: The surface area of a polyhedron is the sum of the areas of all its flat faces, which can be visualized by unfolding the solid into a two-dimensional net. A rectangular prism of length l, width w, and height h has 6 rectangular faces arranged in 3 congruent opposite pairs: SA = 2lw + 2lh + 2wh.

Check Your Work on the SC-6 Calculator:

g CLΣ 8 ENTER 5 f ■ 2 × Σ+ 8 ENTER 3 f ■ 2 × Σ+ 5 ENTER 3 f ■ 2 × Σ+ f Σx

Unfold the 6-face net and find the area of each pair on paper first. On your SC-6, clear stats with [g][CLΣ], add each face pair using [f][RECT] 2 [×] [Σ+], and press [f][Σx] to confirm 158 cm²!

SEC 8.2 (pp.79-80) Surface Areas of Triangular Prisms and Pyramids

Textbook Ways: Way 1: Add All Individual Faces on the 2D Net | Way 2: Base Area + Combined Lateral Area

Learn by Hand: To find the surface area of a triangular prism, unfold its net into 2 congruent triangular bases and 3 rectangular lateral faces, and sum their areas. To find the surface area of a pyramid, add the area of its single polygon base to the areas of its triangular lateral faces (using the slant height of each triangle).

Check Your Work on the SC-6 Calculator:

g CLΣ 1 0 x^2 Σ+ 1 0 ENTER 1 2 g ▲ 4 × Σ+ f Σx

Press [g][CLΣ] 10 [x²] [Σ+] 10 [ENTER] 12 [g][TRI] 4 [×] [Σ+] [f][Σx] to confirm 340 m²!

SEC 8.3 (pp.81-82) Volumes of Rectangular Prisms

Textbook Ways: Way 1: Base Area × Height (V = Bh) | Way 2: Multiply Length × Width × Height (Fractions or Decimals)

Learn by Hand: While Surface Area measures the 2D flat area covering the outside of a solid (in square units), Volume measures the 3D space inside the solid—the number of unit cubes needed to fill it completely (in cubic units). For any rectangular prism, Volume equals the base area B times the height h: V = Bh = l × w × h.

Check Your Work on the SC-6 Calculator:

1 2 ENTER 8 ENTER 5 f ⬢

Type 12 [ENTER] 8 [ENTER] 5 [f][PRISM] to confirm 480 in³ in one step!

SEC 8.4 (pp.83-84) Circles, Cones, and the Pythagorean Theorem

Textbook Ways: Way 1: Exact Form in Terms of π & $a^2 + b^2 = c^2$ | Way 2: Decimal Approximation (π ≈ 3.14) & Scaled Triples

Learn by Hand: In this Grade 7/8 Geometry Bridge: (1) The area of a circle with radius r is $A = \pi r^2$ (where r is half the diameter d). (2) In a right triangle with legs a and b and hypotenuse c, the Pythagorean Theorem states $a^2 + b^2 = c^2$, so $c = \sqrt{(a^2 + b^2)}$. (3) The volume of a cone is one-third the volume of a cylinder with the same base and height: $V = 1/3\pi r^2 h$.

Check Your Work on the SC-6 Calculator:

6 f ● 1 2 ENTER 5 g ⬔

Press 6 [f][CIRC] to verify 113.10 in², and 12 [ENTER] 5 [g][HYP] to verify 13 ft!

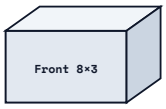
HANDS-ON MODEL: PLACE VALUE BLOCKS

Section 8.1 — Practice A: Three-Dimensional Figures, Nets & Rectangular Prisms

Key Vocabulary: polyhedron ● face ● edge ● vertex ● rectangular prism ● cube ● net ● surface area (SA) Name: _____ Date: _____ Score: ____ / 100

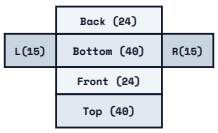
TEXTBOOK CONCEPT & HANDS-ON MODEL (PLACE VALUE BLOCKS)

3D Prism (8×5×3 cm)



SA = 2(40+24+15) = 158

Way 1: Unfolded 6-Face 2D Net



HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Method 1 (2D Net): Unfold the 3D rectangular prism into 6 flat rectangles (Top, Bottom, Front, Back, Left, Right) and add all 6 areas.

Textbook Method 2 (Opposite Face Pairs Formula): Since opposite faces are congruent, use SA = 2lw + 2lh + 2wh = 2(lw + lh + wh).

WATCH OUT: A rectangular prism has 6 faces (3 matching pairs: 2lw + 2lh + 2wh) — don't forget to double all three face pairs!

WORKED EXAMPLE — 3 TEXTBOOK WAYS (ANY 2 WAYS = +20 PTS)

Problem: Find the surface area of a rectangular prism measuring 8 cm long, 5 cm wide, and 3 cm high.

WAY 1: Add All 6 Faces on a 2D Net
Top (40) + Bottom (40) + Front (24) + Back (24) + Left (15) + Right (15) = 80 + 48 + 30 = 158 cm².

WAY 2: Opposite Face Pairs Formula SA = 2(lw + lh + wh)
Sum the 3 distinct face areas and multiply by 2: SA = 2(8·5 + 8·3 + 5·3) = 2(40 + 24 + 15) = 2(79) = 158 cm².

WAY 3: Lateral Perimeter Strip + 2 Bases (Ph + 2B)
4 side walls form a strip of width P = 2(8+5) = 26 cm and height 3 cm (26×3 = 78 cm²), plus 2 bases (80 cm²) = 158 cm².

Answer: 158 cm² 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 8.1 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[g][CLΣ] 8 [ENTER] 5 [f][■] 2 [×][Σ+] 8 [ENTER] 3 [f][■] 2 [×][Σ+] 5 [ENTER] 3 [f][■] 2 [×][Σ+] [f][Σx]

Unfold the 6-face net and find the area of each pair on paper first. On your SC-6, clear stats with [g][CLΣ], add each face pair using [f][RECT] 2 [×] [Σ+], and press [f][Σx] to confirm 158 cm²!

SECTION 8.1 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Find the surface area of a cube with edge length s = 4 cm.**
Starter Step 1: A cube has 6 identical square faces—square the edge length (4^2) to find 1 face, then multiply by 6.

Way 1 (Paper Work):

1 Square Face = s² = _____

Cube SA = 6 × s² = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Find the surface area of a cube with edge length s = 9 in.**
Starter Step 1: A cube has 6 identical square faces—square the edge length (9^2) to find 1 face, then multiply by 6.

Way 1 (Paper Work):

1 Square Face = s² = _____

Cube SA = 6 × s² = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Find the surface area of a rectangular prism measuring 6 cm long, 4 cm wide, and 3 cm high.**

Way 1 (Paper Work):

Faces: l·w=___ l·h=___ w·h=___

SA = 2(___ + ___ + ___) = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Find the surface area of a rectangular box measuring 10 in. long, 5 in. wide, and 2 in. high.**

Way 1 (Paper Work):

Faces: l·w=___ l·h=___ w·h=___

SA = 2(___ + ___ + ___) = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Find the surface area of a rectangular prism with length l = 12 ft, width w = 8 ft, and height h = 5 ft.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Find the surface area of a rectangular prism measuring 5 m long, 4 m wide, and 2.5 m high.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **A cube has a total surface area of 150 cm². What is the area of one square face, and what is the edge length s of the cube?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Error Analysis: For a 5 cm by 4 cm by 3 cm box, Leo added 20 + 15 + 12 = 47 cm². What did he forget to do, and what is the correct surface area?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Open-Top Cold-Storage Battery Cargo Pod: An open-top rectangular battery cargo pod (5 faces, no top lid) is 30 cm long, 20 cm wide, and 15 cm tall. How many square centimeters of insulation paneling are needed to line the bottom and four sides?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Gift Box Wrapping Paper: You are wrapping a gift box that is 15 inches long, 10 inches wide, and 4 inches tall. If you have a sheet of wrapping paper with an area of 550 in.², do you have enough paper to cover the box?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: PLACE VALUE BLOCKS

Section 8.1 — Practice B: Modeling Real Life (Three-Dimensional Figures, Nets & Rectangular Prisms)

Textbook Ways: Unfold into a 6-Face 2D Net | Opposite Face Pairs SA = 2(lw + lh + wh) | Lateral Strip Ph + 2B Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — CUBE & OPEN-TOP STORAGE BIN (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: Find the surface area of a cube with edge length 7 inches, and find the outer surface area of an open-top bin measuring 10 cm by 8 cm by 5 cm.

WAY 1: Add the Exposed Faces on the 2D Net

Cube (6 identical 7×7 squares): 6 × 49 = 294 in^2. Open-Top Bin (5 faces): 1 bottom (10×8 = 80) + 2 fronts (2×50 = 100) + 2 sides (2×40 = 80) = 260 cm^2.

WAY 2: Full 6-Face Formula Minus the Open Top

Full 10×8×5 prism SA = 2(80 + 50 + 40) = 340 cm^2. Subtract the missing 10×8 top lid (80 cm^2): 340 - 80 = 260 cm^2.

Watch-Out: Always check if a real-world box has an open top (5 faces) or all 6 faces!

WATCH OUT: A rectangular prism has 6 faces (3 matching pairs: 2lw + 2lh + 2wh) — don't forget to double all three face pairs!

Answer: Cube = 294 in^2; Open-Top Bin = 260 cm^2

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 8.1 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[7] [x^2] [6] [×]

Calculate $6s^2$ for the cube and the 5 panels of the open-top bin on paper first. Check on your SC-6:
Press 7 [x^2] 6 [×] to confirm 294 in^2 for the cube.

Liam used to plug numbers blindly into the 6-wall box formula SA = 2(lw + lh + wh) without thinking about the actual object—until he ordered too much glass for an open-top aquarium and got the wrong painted area when Zoe glued two 4-centimeter wooden cubes face-to-face to model huddling penguins. When they unfolded a paper model of the fish tank flat on the table, Liam saw why the formula failed him: an open-top tank only has 5 glass panels (no top lid!), and gluing two cubes together hides their 2 touching faces inside, leaving only 10 exposed outer squares instead of 12.

SECTION 8.1 PRACTICE B — MODELING REAL LIFE & APPLICATIONS — 100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Cereal Box Cardboard: A cereal box is a rectangular prism measuring 20 cm long, 8 cm wide, and 30 cm tall. What is its total surface area?**

Starter Step 1: Find the area of the bottom (20×8), front (20×30), and side (8×30) rectangles, add them, and multiply by 2.

Way 1 (Paper Work):

Faces: l·w=___ l·h=___ w·h=___

SA = 2(___ + ___ + ___) = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Foam Die Cube: A giant foam die is a cube with edge length 9 inches. What is the area of one face, and what is the total surface area of all 6 faces?**

Starter Step 1: Multiply 9 in. × 9 in. to find the area of 1 square face, then multiply by 6 faces for the total surface area.

Way 1 (Paper Work):

1 Square Face = s^2 = _____

Cube SA = 6 × s^2 = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Open-Top Glass Aquarium: A rectangular glass fish tank is 24 inches long, 12 inches wide, and 15 inches tall, with an open top (no glass lid). What is the total area of the 5 glass panels?**

Way 1 (Paper Work):

Faces: l·w=___ l·h=___ w·h=___

SA = 2(___ + ___ + ___) = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Wrapping Paper Check: You want to wrap a game console box measuring 15 in by 10 in by 4 in using a sheet of wrapping paper with area 600 in^2. Find the surface area of the box. Do you have enough paper?**

Way 1 (Paper Work):

Faces: l·w=___ l·h=___ w·h=___

SA = 2(___ + ___ + ___) = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Painting a Bedroom: A bedroom is 12 feet long, 10 feet wide, and 8 feet high. You paint the 4 walls and the ceiling, but not the floor. How many square feet do you paint?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Cube Edge from Surface Area: A wooden cube has a total surface area of 150 square centimeters. What is the area of one face, and what is the edge length of the cube?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Storage Trunk: A rectangular storage trunk measures 4 feet by 2 feet by 2 feet. What is its total surface area?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Juice Box Carton: A juice box measures 6 cm by 4 cm by 10 cm. What is its total surface area?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Doubling the Edge of a Cube: Cube A has an edge length of 3 cm and Cube B has an edge length of 6 cm. Find the surface area of each cube. How many times as great is Cube B's surface area?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Two Cubes Glued Together: Two 4 cm by 4 cm by 4 cm cubes are glued together face-to-face to form an 8 cm by 4 cm by 4 cm rectangular prism. What is the surface area of the new prism?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

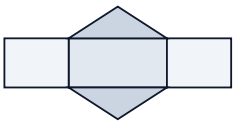
HANDS-ON MODEL: PLACE VALUE BLOCKS

Section 8.2 — Practice A: Surface Areas of Triangular Prisms and Pyramids

Key Vocabulary: triangular prism • pyramid • lateral face • slant height • regular pyramid • 2D net Name: _____ Date: _____ Score: ____ / 100

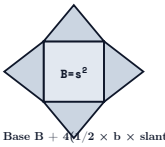
TEXTBOOK CONCEPT & HANDS-ON MODEL (PLACE VALUE BLOCKS)

Triangular Prism Net (5 Faces)



SA = 2 Tri Bases (1/2bh) + 3 Rect Sides

Square Pyramid Net (1 Base + 4 Triangles)



SA = Base B + 4(1/2 × b × slant l)

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Triangular Prisms: Unfold into 2 congruent triangular bases (each 1/2bh) plus 3 rectangular lateral faces, and add all 5 areas.

Textbook Pyramids: Unfold into 1 polygon base B plus triangular lateral faces using the slant height l (each 1/2bl): SA = B + 4(1/2bl).

WATCH OUT: Use the 1/2 on EVERY triangular face (A = 1/2bh or 1/2bl), and count all faces in the net (5 for triangular prism, 5 for square pyramid)!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find the surface area of a square pyramid with a 10 m by 10 m square base and 4 congruent triangular faces each having base 10 m and slant height 12 m.

WAY 1: Add All 5 Faces on the Pyramid Net

Square base = 10 × 10 = 100 m^2. Each triangular face = 1/2 × 10 × 12 = 60 m^2. Total SA = 100 + 60 + 60 + 60 + 60 = 340 m^2.

WAY 2: Base Area + 4 × Lateral Triangle Formula

Since all 4 lateral triangles are congruent: SA = B + 4(1/2bl) = 10^2 + 4(1/2 · 10 · 12) = 100 + 240 = 340 m^2.

Answer: 340 m^2 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 8.2 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[g] [CLΣ] 1 0 [x²] [Σ+] 1 0 [ENTER] 1 2 [g] [▲] 4 [×] [Σ+] [f] [Σx]

Add the square base (10^2) and 4 triangles on paper first. Check on your SC-6:
Press [g][CLΣ] 10 [x^2] [Σ+] 10 [ENTER] 12 [g][TRI] 4 [×] [Σ+] [f][Σx] to confirm 340 m^2!

SECTION 8.2 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 Find the surface area of a square pyramid with a 6 cm by 6 cm base and 4 triangular faces each having base 6 cm and slant height 5 cm.

Starter Step 1: Unfold the pyramid into 1 square base (6 × 6) plus 4 identical triangles (each A = 1/2 × 6 × 5), then add them.

Way 1 (Paper Work):

Base Area(s) B = _____

+ Lateral Faces = _____ → SA=___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 Find the surface area of a square pyramid with an 8 in. by 8 in. base and 4 triangular faces each having base 8 in. and slant height 10 in.

Starter Step 1: Unfold the pyramid into 1 square base (8 × 8) plus 4 identical triangles (each A = 1/2 × 8 × 10), then add them.

Way 1 (Paper Work):

Base Area(s) B = _____

+ Lateral Faces = _____ → SA=___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 Find the surface area of a square pyramid with a 12 ft by 12 ft base and 4 triangular faces each having base 12 ft and slant height 9 ft.

Way 1 (Paper Work):

Base Area(s) B = _____

+ Lateral Faces = _____ → SA=___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 Four-Triangle Pyramid (Tetrahedron): A 4-sided pyramid is made of 4 identical equilateral triangles, each with a base of 6 cm and a height of 5.2 cm. What is the total surface area of all 4 faces?

Way 1 (Paper Work):

Base Area(s) B = _____

+ Lateral Faces = _____ → SA=___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 A triangular prism has 2 right-triangle bases (b = 4 cm, h = 3 cm) and 3 rectangular faces of length 10 cm (widths 3 cm, 4 cm, and 5 cm). What is its surface area?

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 A triangular prism tent has 2 triangular ends (base 8 ft, height 3 ft) and 3 rectangular panels of length 7 ft (widths 5 ft, 5 ft, and 8 ft). What is its total surface area?

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 A regular tetrahedron has 4 congruent triangular faces each with base 8 cm and slant height 6.93 cm. What is its surface area?

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 Compare the nets of a triangular prism and a square pyramid: how many triangular faces and how many rectangular/square faces does each solid have?

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 A-Frame Camping Tent: An A-frame tent is a triangular prism with 2 triangular doors (base 6 ft, height 4 ft), a 6 ft by 8 ft floor, and 2 sloped roof panels (5 ft by 8 ft each). How many square feet of nylon fabric are in the tent?

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 Glass Pyramid Skylight: A rooftop skylight is shaped like a square pyramid with an open bottom (no floor)—only 4 congruent triangular glass panes, each with base 9 ft and slant height 6 ft. What is the total area of the glass panes?

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: PLACE VALUE BLOCKS

Section 8.2 — Practice B: Modeling Real Life (Surface Areas of Triangular Prisms and Pyramids)

Textbook Ways: Add All Individual Faces on the 2D Net | Base Area + Combined Lateral Area Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — SKATEBOARD RAMP (TRIANGULAR PRISM) (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: A skate ramp is a triangular prism with 2 right-triangle ends (base 6 ft, height 8 ft, ramp 10 ft) and a width of 10 ft. Find its total surface area.

WAY 1: Add All 5 Faces on the Unfolded 2D Net

2 Triangles = $2 \times (1/2 \cdot 6 \cdot 8) = 48$ ft^2. 3 Rectangles = $(6\times10) + (8\times10) + (10\times10) = 60 + 80 + 100 = 240$ ft^2. Sum: $48 + 240 = 288$ ft^2.

WAY 2: Combine the 3 Rectangles into One Lateral Strip

Perimeter of triangle base = $6 + 8 + 10 = 24$ ft. Lateral rectangle = $24 \times 10 = 240$ ft^2. Add 2 triangle bases (48 ft^2) → 288 ft^2.

Check: Two 24 ft^2 triangles + 60 + 80 + 100 = 288 ft^2.

WATCH OUT: Use the 1/2 on EVERY triangular face (A = 1/2bh or 1/2bl), and count all faces in the net (5 for triangular prism, 5 for square pyramid)!

Answer: 288 square feet

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 8.2 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

6 ENTER 8 g ▲ 2 × 6 ENTER 8 + 1 0 + 1 0 f ■ +

Add the 2 triangular ends and 3 rectangular sides on paper first. Check on your SC-6:
Press 6 [ENTER] 8 |g|[TRI] 2 [×] (48 ft^2) plus 6 [ENTER] 8 [+] 10 [+] 10 |f|[RECT] (240 ft^2) [+] to confirm 288 ft^2.

When Liam tried to calculate the waterproof fabric needed for a triangular-prism camping tent and the shingles for a square-pyramid gazebo roof, he felt overwhelmed by all the 3D edges and slant heights. He was afraid he would count the wrong number of sides or forget to divide a triangle by 2. Then Marcus suggested mentally 'unzipping' the tent and flattening every wall onto the floor as a 2D net: 1 rectangular floor, 2 rectangular roof slopes, and 2 triangular end doors (5 faces total). Counting and labeling each flat piece in the net before doing any arithmetic kept him from missing a single wall.

SECTION 8.2 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Camping Tent: A tent shaped like a triangular prism has 2 triangular doors (each with base 6 ft and height 4 ft), two 5 ft by 8 ft roof panels, and a 6 ft by 8 ft rectangular floor. What is the total surface area of the tent?**

Starter Step 1: Find the area of the 2 triangular doors (each 1/2 × 6 × 4), the 2 rectangular roof panels (5 × 8), and the floor (6 × 8).

Way 1 (Paper Work):

Base Area(s) B = _____

+ Lateral Faces = _____ → SA=___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

2 **Glass Pyramid Model: A scale model of a square pyramid has a 12 cm by 12 cm square base and 4 congruent triangular faces, each with a slant height of 10 cm. What is its total surface area?**

Starter Step 1: Find the area of the 12 cm by 12 cm square base plus the 4 congruent triangular faces (each A = 1/2 × 12 × 10).

Way 1 (Paper Work):

Base Area(s) B = _____

+ Lateral Faces = _____ → SA=___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

3 **Gazebo Roof Shingles: A gazebo roof is a square pyramid with base edge 8 feet and slant height 6 feet. Find the lateral surface area (the 4 triangular roof faces, not including the open bottom).**

Way 1 (Paper Work):

Base Area(s) B = _____

+ Lateral Faces = _____ → SA=___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

4 **Triangular Chocolate Box: A triangular prism box has 2 equilateral triangle ends (each with area 6.9 cm^2) and 3 identical rectangular sides (each 4 cm by 20 cm). What is the total surface area of the box?**

Way 1 (Paper Work):

Base Area(s) B = _____

+ Lateral Faces = _____ → SA=___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

5 **Wooden Wedge: A right-triangular prism wedge has 2 triangular bases with legs 8 in and 6 in and hypotenuse 10 in. The prism is 5 inches wide. What is its total surface area?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

6 **Four-Sided Game Die: A pyramid-shaped 4-sided die (called a tetrahedron) has 4 identical triangular faces, each with a base of 3 cm and a height of 2.6 cm. What is its total surface area?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

7 **Attic Roof & Gables: An attic prism has 2 triangular gables (each base 16 ft, height 6 ft) and 2 rectangular roof slopes (each 10 ft by 30 ft). What is the combined area of the 2 gables and 2 roof slopes?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

8 **Pyramid Paperweight: A square pyramid has a 6 in by 6 in base and a total surface area of 96 in^2. What is the area of one of its four triangular faces, and what is its slant height?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

9 **Triangular Prism Gift Box: A prism has 2 triangular bases (each with base 12 cm and height 8 cm) and 3 rectangular faces of widths 10 cm, 10 cm, and 12 cm, each 15 cm long. What is its total surface area?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

10 **Comparing Two Pyramids: Square Pyramid A has a 10 cm by 10 cm base and slant height 8 cm. Square Pyramid B has an 8 cm by 8 cm base and slant height 10 cm. Which pyramid has the greater total surface area?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans: _____**

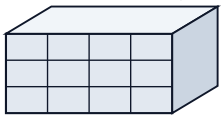
HANDS-ON MODEL: FRACTION STRIPS

Section 8.3 — Practice A: Volumes of Rectangular Prisms

Key Vocabulary: volume (V) • cubic units • base area (B = lw) • V = Bh = lwh • fractional edge length Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & HANDS-ON MODEL (FRACTION STRIPS)

Unit-Cube Layer Packing (V = Bh)



Base B = 12×8 = 96 | × 5 layers = 480 in³

3 Ways to Find Prism Volume:

Way 1 (Layers): V = Base Area B × h

Way 2 (Formula): V = l × w × h

Missing Height: h = V ÷ (l × w)

(Wait: 1/2-in cubes → 8 fit per 1 in³!)

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept: Volume measures the number of unit cubes that fill a 3D solid. Multiply Base Area B = lw by height h: V = Bh = lwh.

Textbook Fractional Edge Lengths: Pack unit-fraction cubes OR multiply fractional dimensions as improper fractions or decimals.

WATCH OUT: Volume is measured in CUBIC units (in.³, cm³). Eight 1/2-inch cubes (2 × 2 × 2 = 8) fit inside 1 cubic inch!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find the volume of a crate measuring 12 in. by 8 in. by 5 in., and an aquarium measuring 3 1/2 ft by 1 1/2 ft by 2 ft.

WAY 1: Layer Stacking V = Bh (With Unit Cubes)

Crate: Base B = 12 × 8 = 96 in²; multiply by 5 layers → 480 in³. Tank (3 1/2 × 1 1/2 × 2): Base B = (7/2)(3/2) = 21/4; × 2 = 10 1/2 ft³.

WAY 2: Multiply Three Dimensions (Improper Fractions or Decimals)

Crate: 12 × (8 × 5) = 12 × 40 = 480 in³. Tank in decimals: 3.5 × 1.5 × 2 = 3.5 × 3 = 10.5 ft³ (10 1/2 ft³).

Answer: Crate = 480 in.³; Aquarium = 10 1/2 ft³

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 8.3 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

1 2 ENTER 8 ENTER 5 f

Multiply length × width × height on paper first. Then check on your SC-6 using the 3-number Prism Volume key: Type 12 [ENTER] 8 [ENTER] 5 [f][PRISM] to confirm 480 in³ in one step!

SECTION 8.3 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Cereal Storage Container: Find the volume of a rectangular plastic container that is 18 cm long, 6 cm wide, and 28 cm tall.**

Starter Step 1: Multiply length × width (18 × 6) to find the Base Area B, then multiply by the height (28 cm).

Way 1 (Paper Work):

Base Area B = l × w = _____

Volume V = B × h = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Find the volume of a cube with edge length s = 5 inches.**

Starter Step 1: For a cube with edge s = 5 in., multiply length × width × height: V = 5 × 5 × 5.

Way 1 (Paper Work):

Cube Edge s = _____

Volume V = s³ = s×s×s = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Camp Lap Pool: A rectangular camp swimming pool is 25 meters long, 10 meters wide, and filled with water 2 meters deep. How many cubic meters of water does it hold?**

Way 1 (Paper Work):

Base Area B = l × w = _____

Volume V = B × h = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **A rectangular prism has a base area of B = 32 m² and a height of h = 7 m. What is its volume?**

Way 1 (Paper Work):

Base Area B = l × w = _____

Volume V = B × h = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Find the volume of a fish tank that is 3 1/2 feet long, 1 1/2 feet wide, and 2 feet tall.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Find the volume of a wooden block measuring 4 1/2 inches long, 3 inches wide, and 2 inches high.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **A rectangular storage box has a volume of 270 in.³, a length of 9 in., and a width of 5 in. What is its height h?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **How many 1/2-inch cubes are needed to completely fill a 1-inch cube (measuring 1 in. by 1 in. by 1 in.)?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Garden Planter Depth: A raised cedar planter has a base measuring 9 inches by 5 inches and holds 360 cubic inches of potting soil. What is the depth h of the soil?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Packing Half-Inch Cubes: A wooden box has a volume of 27 cubic inches. How many 1/2-inch unit cubes can be packed inside the box?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: FRACTION STRIPS

Section 8.3 — Practice B: Modeling Real Life (Volumes of Rectangular Prisms)

Textbook Ways: Base Area × Height (V = Bh) | Multiply Length × Width × Height (Fractions or Decimals) Name: _____ Date: _____ Score: _____ / 100

WORKED EXAMPLE — FINDING MISSING DEPTH FROM VOLUME (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: A cedar planter box is 9 inches long and 5 inches wide, and holds 360 cubic inches of soil. How deep is the soil in the planter box?

WAY 1: Find Base Area B First, Then Divide V ÷ B

Base area B = 9 × 5 = 45 in^2. Substitute into V = Bh: 360 = 45h → h = 360 ÷ 45 = 8 inches deep.

WAY 2: Divide Volume by Length, Then by Width

Divide 360 ÷ 9 in = 40 in^2, then divide 40 ÷ 5 in = 8 inches deep.

Check: 9 in × 5 in × 8 in = 360 in^3.

WATCH OUT: Volume is measured in CUBIC units (in.^{3}, cm^3). Eight 1/2-inch cubes (2 × 2 × 2 = 8) fit inside 1 cubic inch!

Answer: 8 inches deep

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 8.3 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

3 6 0 ENTER 4 5 ÷

Find the base area B = 9 × 5 = 45 in^2 and divide 360 ÷ 45 on paper first. Check on your SC-6: Press 360 [ENTER] 45 [÷] to confirm a depth of 8 inches.

Zoe used to confuse Surface Area (square units, cm^2) with Volume (cubic units, cm^3) until she watched a crow drop pebbles into a rectangular water tube with an 8 cm by 5 cm base. Wrapping paper or glass goes on the flat outside walls (square centimeters), while water and pebbles fill up the 3D space inside (cubic centimeters). Even better, she saw that every rectangular prism fills up like stackable layers: one 1-centimeter-deep layer of water in the tube holds 8 × 5 = 40 cm^3, so when the pebbles push 120 cm^3 of water upward, dividing 120 ÷ 40 shows the water must rise exactly 3 centimeters!

SECTION 8.3 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Swimming Lap Pool: A rectangular pool is 25 meters long, 10 meters wide, and 2 meters deep. What is the volume of the pool?**

Starter Step 1: Multiply length × width × depth (25 m × 10 m × 2 m) to find the volume in cubic meters.

Way 1 (Paper Work):

Base Area B = l × w = _____

Volume V = B × h = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Raised Garden Bed: A cedar garden bed is 8 feet long, 4 feet wide, and 1 1/2 feet deep. How many cubic feet of soil are needed to fill it?**

Starter Step 1: Write 1 1/2 as 1.5 (or 3/2), then multiply length × width × depth (8 × 4 × 1.5).

Way 1 (Paper Work):

Base Area B = l × w = _____

Volume V = B × h = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **3D Printer Build Volume: A 3D printer has a cubic build volume measuring 25 cm on each edge. What is its build volume in cubic centimeters?**

Way 1 (Paper Work):

Base Area B = l × w = _____

Volume V = B × h = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Bento Lunchbox: A rectangular bento box measures 6 1/2 inches long, 4 inches wide, and 2 1/2 inches tall. What is its volume?**

Way 1 (Paper Work):

Base Area B = l × w = _____

Volume V = B × h = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Half-Inch Cube Packing: How many 1/2-inch cubes fit inside a 1-cubic-inch space, and how many 1/2-inch cubes are needed to completely fill a 3 in by 2 in by 2 in box?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Crow Water-Displacement Tank (Aesop's Fable): A crow's rectangular water tube has an 8 cm by 5 cm base. Dropping pebbles into the water displaces 120 cm^3 of volume. By how many centimeters does the water level rise?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Cargo Container: A rectangular cargo box is 5.5 feet long, 4 feet wide, and 3.5 feet tall. What is its volume?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Concrete Patio Slab: A rectangular patio is 12 feet long, 9 feet wide, and 1/3 foot (4 inches) thick. How many cubic feet of concrete are needed?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Acrylic Puzzle Box: A kea parrot's acrylic puzzle box has a rectangular base of 9 inches by 5 inches and a volume of 360 cubic inches. What is the height of the puzzle box?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Aquarium Gallons: A fish tank measures 24 inches long, 12 inches wide, and 16 inches tall. Find its volume in cubic inches, and then find how many gallons of water it holds (1 gallon = 231 in^3).**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 8.4 — Practice A: Circles, Cones, and the Pythagorean Theorem

Key Vocabulary: circle • radius (r) • diameter (d = 2r) • circumference (C = 2πr) • area (A = πr²) • cone (V = 1/3πr²h) • hypotenuse

Name: _____ Date: _____ Score: ____ / 100

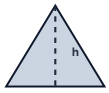
TEXTBOOK CONCEPT & VISUAL DIAGRAM

Circle: A = πr²



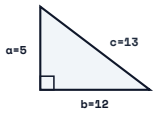
r = d ÷ 2 | 36π ≈ 113.10

Cone: V = 1/3πr²h



Cone = 1/3 of Cylinder!

Right Triangle: a²+b²=c²



c = √(5²+12²) = √169 = 13

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Circle & Cone Formulas: Circle Area A = πr² (if given diameter d, divide by 2 first: r = d/2). Cone Volume V = 1/3Bh = 1/3πr²h.

Textbook Pythagorean Theorem: In a right triangle with legs a and b and hypotenuse c, $a^2 + b^2 = c^2$, so $c = \sqrt{(a^2 + b^2)}$.

WATCH OUT: If given diameter d, divide by 2 FIRST to get radius r = d ÷ 2 before squaring r²! And don't forget the 1/3 for cones!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find the area of a circular pizza with radius r = 6 in. (round to 2 decimal places), and find the diagonal length of a ramp with legs 12 ft and 5 ft.

WAY 1: Formula Evaluation (Exact π & Square Root)

Circle (r = 6): A = π(6²) = 36π ≈ 113.10 in². Right triangle (legs 12, 5): $c^2 = 12^2 + 5^2 = 144 + 25 = 169 \rightarrow c = \sqrt{169} = 13$ ft.

WAY 2: Decimal Estimate (π ≈ 3.14) & Pythagorean Triples

Circle: 36 × 3.14 = 113.04 in² (or 36 × 3.1416 = 113.10 in²). Triangle: recognize the 5-12-13 Pythagorean triple → c = 13 ft.

Answer: Pizza = 113.10 in.²; Ramp diagonal = 13 ft

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 8.4 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

6 f ● 1 2 ENTER 5 g [x²]

Cut the 12-inch diameter in half (r = 6) and calculate πr² and √(12² + 5²) on paper first. Check on your SC-6: Press 6 [f][CIRC] to verify 113.10 in², and 12 [ENTER] 5 [g][HYP] to verify 13 ft!

SECTION 8.4 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Find the area of a circle with radius r = 5 cm (round to the nearest hundredth).**

Starter Step 1: By PEMDAS, square the radius FIRST ($r^2 = 5 \times 5$), and then multiply by π.

Way 1 (Paper Work):

r = _____ | r² = _____

Area A = π × r² = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Garden Sundial: A circular garden sundial has a diameter of 30 cm across the center. Find its area to the nearest hundredth.**

Starter Step 1: Divide the 30 cm diameter by 2 to find the radius r FIRST, then square the radius and multiply by π.

Way 1 (Paper Work):

r = _____ | r² = _____

Area A = π × r² = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Find the hypotenuse c of a right triangle with legs a = 3 cm and b = 4 cm.**

Way 1 (Paper Work):

a² + b² = ___ + ___ = ___

c = √(a² + b²) = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Treehouse Zip-Line Shortcut: A backyard treehouse platform is 6 meters high, and the anchor post is 8 meters away across flat ground. How long is the straight diagonal zip-line cable connecting them?**

Way 1 (Paper Work):

r = _____ | r² = _____

Area A = π × r² = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **A skateboard ramp has a horizontal base of 12 feet and a vertical back height of 5 feet. How long is the sloped riding surface?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **A circular landing pad has a diameter of d = 16 meters. What is its radius r, and what is its area rounded to the nearest hundredth?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Conical Sand Filtration Pile: A conical pile of water-filtration sand at the treatment plant has a base radius of r = 3 meters and a height of h = 2 meters. What is the volume of the sand pile, rounded to the nearest hundredth?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Find the volume of a cone with radius r = 6 inches and height h = 5 inches, rounded to the nearest hundredth.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Cylinder vs. Cone Volume: A cylinder and a cone both have radius r = 3 cm and height h = 10 cm. Find the volume of the cylinder and the volume of the cone, rounded to the nearest hundredth.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Widescreen Monitor Diagonal: Computer monitors are advertised by their diagonal screen length. If a monitor screen is 32 inches wide and 24 inches tall, what is its diagonal measurement?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 8.4 — Practice B: Modeling Real Life (Circles, Cones, and the Pythagorean Theorem)

Textbook Ways: Exact Form in Terms of π & $a^2 + b^2 = c^2$ | Decimal Approximation ($\pi \approx 3.14$) & Scaled Triples Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — VOLUME OF A CONE (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: A waffle cone has a radius of 3 centimeters and a height of 10 centimeters. Find the volume of the cone rounded to the nearest hundredth.

WAY 1: Cone Formula $V = 1/3\pi r^2 h$

Base area $B = \pi(3^2) = 9\pi$ cm^2. Multiply by 1/3 × height: $V = 1/3(9\pi)(10) = 3\pi \times 10 = 30\pi \approx 94.25$ cm^3.

WAY 2: Find Cylinder Volume First, Then Divide by 3

A cylinder with r = 3 and h = 10 has $V = \pi(3^2)(10) = 90\pi \approx 282.74$ cm^3. Divide by 3: 282.74 ÷ 3 ≈ 94.25 cm^3.

Check: 3 cones of 30π cm^3 equal 1 cylinder of 90π cm^3.

WATCH OUT: If given diameter d, divide by 2 FIRST to get radius r = d ÷ 2 before squaring r^2! And don't forget the 1/3 for cones!

Answer: 94.25 cm^3 (exact: 30π cm^3)

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 8.4 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

3 ENTER 1 0 g ▲

Calculate $V = 1/3\pi r^2 h$ on paper first. Then check on your SC-6 using the Cone Volume key:
Type radius 3 [ENTER] height 10 [g][CONE] to confirm 94.25 cm^3!

At the team pizza party, Marcus and his friends were sure that two 10-inch diameter pizzas had more pizza than one 16-inch diameter pizza—until they opened the boxes and saw how huge the 16-inch pizza really was. Working it out on a napkin, Marcus caught the two traps that fool almost everyone with circles: first, a 16-inch pizza has a diameter of 16 inches, so you MUST cut it in half to get the radius (r = 8 inches, not 16!); second, because the area formula squares the radius ($A = \pi r^2$), one 8-inch radius gives $64\pi \approx 201$ in^2, easily beating two 5-inch radius pizzas ($2 \times 25\pi \approx 157$ in^2).

SECTION 8.4 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Trampoline Mat: A circular trampoline has a radius of 7 feet. What is the area of the jumping mat, rounded to the nearest hundredth?**

Starter Step 1: Square the 7-foot radius FIRST (7 × 7), and then multiply by π to find the circular mat's area.

Way 1 (Paper Work):

r = _____ | r^2 = _____

Area $A = \pi \times r^2 \approx$ _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Ferris Wheel Circle: A Ferris wheel has a diameter of 40 meters. What is its radius, and what is the area enclosed by the wheel (to the nearest hundredth)?**

Starter Step 1: Divide the 40-meter diameter by 2 to find the radius r, then square r and multiply by π.

Way 1 (Paper Work):

r = _____ | r^2 = _____

Area $A = \pi \times r^2 \approx$ _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Bicycle Tire Distance: A bicycle wheel has a radius of 14 inches from the center hub to the tire tread. How many inches forward does the bike roll in one full turn of the wheel (its circumference, to the nearest hundredth)?**

Way 1 (Paper Work):

r = _____ | r^2 = _____

Area $A = \pi \times r^2 \approx$ _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Zipline Cable: A zipline tower is 15 meters tall and is anchored to the ground 20 meters away horizontally. How long is the diagonal zipline cable?**

Way 1 (Paper Work):

r = _____ | r^2 = _____

Area $A = \pi \times r^2 \approx$ _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Monitor Diagonal: A rectangular computer monitor screen is 16 inches wide and 12 inches tall. What is its diagonal screen size?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Snow Cone Cup: A paper snow cone cup is shaped like a cone with radius 4 cm and height 9 cm. What is its volume (to the nearest hundredth)?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Highway Safety Cone: A traffic cone has a base diameter of 12 inches and a height of 15 inches. What is its volume (to the nearest hundredth)?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Pizza Size Comparison: Does one 16-inch diameter pizza or two 10-inch diameter pizzas give you more total pizza area? Justify your answer by finding the areas.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Basketball Court Semicircle: The top of a basketball free-throw lane is a semicircle with a radius of 6 feet. What is the area of the semicircle (to the nearest hundredth)?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Ladder Against a Wall: The bottom of a ladder is placed 8 feet from the base of a wall so that the top of the ladder reaches 15 feet up the wall. How long is the ladder?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

Unit 9: Statistical Measures & Data Displays — Basketball Stats, Outlier Prices & Bird Puzzle Trials

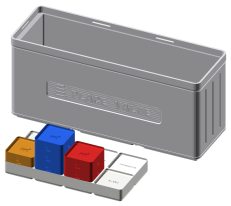
Mean as Fair Share, Median, Mode, Range, Quartiles, Interquartile Range (IQR), Box Plots & MAD

When Leo looked up the prices of five video games—$20, $25, $20, $30, and a rare $105 collector's edition—and calculated the average price as $40, he thought he must have divided wrong. How could the 'average' game be more expensive than four of the five games on the shelf? Meanwhile, on the basketball court, Maya wanted an 85 average across five quizzes after scoring 80, 88, 82, and 84 on the first four, and felt stuck trying to guess what she needed on Quiz 5.

Over in the bird lab, Zoe had an even subtler puzzle. Raven A and Crow B both averaged 9 seconds to unlock a puzzle box across four trials, so on paper the two birds looked identical—even though Raven A finished in a rock-steady 8, 9, 9, and 10 seconds while Crow B swung wildly between 6 and 12 seconds.

And when Maya looked at a box-and-whisker plot of 60 pine trees in the forest, she initially assumed the longest whisker held the most trees—until she learned how quartiles actually split a sorted list into four equal 25% groups.

At first, all the data vocabulary—mean, median, quartiles, IQR, and MAD—sounded like alphabet soup. By working out the data tables by hand first and then entering the data list with `[Σ+]` on the SC-6 to check `[x-bar]`, `[MED]`, `[IQR]`, and `[MAD]`, Maya, Leo, and Zoe learn how to read any data set accurately.



Real-World Mathematical Context & Two-Method Mastery • Hands-On Models: Place Value Blocks & Fraction Strips

UNIT 9 LESSONS — LEARN TWO TEXTBOOK WAYS FIRST, THEN CHECK YOUR WORK ON THE SC-6 CALCULATOR

SEC 9.1 (pp.86-87) Statistical Questions, Dot Plots, and the Mean

Textbook Ways: Way 1: Mean Formula (Sum ÷ Count) | Way 2: Fair-Share Block Leveling | Way 3: Balance Point Deviations

Learn by Hand: The mean (or average, x-bar) of a numerical data set is the sum of the data values divided by the number of data values: x-bar = (Σx) ÷ n. Physically, the mean represents a 'fair share'—if you build cube towers for each data value and move cubes from taller towers to shorter towers until all towers have the exact same height, that common height is the mean!

Check Your Work on the SC-6 Calculator:

[g] [CLΣ] 1 8 [Σ+] 2 4 [Σ+] 1 5 [Σ+] 2 7 [Σ+] 2 1 [Σ+] [g] [x̄]

Add the 5 values and divide by 5 on paper first. On your SC-6, clear stats with [g][CLΣ], enter each value with [Σ+] (or view the dot plot in [FRQ] mode), and press [g][x-bar] to confirm Mean = 21!

SEC 9.2 (pp.88-89) Measures of Center: Median, Mode, and Range

Textbook Ways: Way 1: Ordered List (Least to Greatest) | Way 2: Dot Plot / Frequency Table Analysis

Learn by Hand: When data values are ordered from least to greatest: the Median is the middle value (or the mean of the two middle values when the count n is even); the Mode is the value(s) that occur most often; and the Range is the difference between the greatest and least values (Max - Min). When an extreme outlier is present, the median is usually the best measure of center.

Check Your Work on the SC-6 Calculator:

[g] [CLΣ] 7 [Σ+] 9 [Σ+] 6 [Σ+] 7 [Σ+] 1 0 [Σ+] 8 [Σ+] 9 [Σ+] 7 [Σ+] [f] [MED] [g] [MODE] [RNG]

Press [g][CLΣ], enter each number with [Σ+] (or [FRQ]), and press [f][MED], [g][MODE], and [RNG] to verify Median = 7.5, Mode = 7, and Range = 4!

SEC 9.3 (pp.90-91) Quartiles, Interquartile Range (IQR) & Box Plots

Textbook Ways: Way 1: Split Ordered Data into Lower & Upper Halves | Way 2: Box-and-Whisker Plot & 25% Quartile Sections

Learn by Hand: Quartiles divide an ordered data set into four equal parts (each containing 25% of the data). The median (Q2) divides the data into a lower half and an upper half; the first quartile (Q1) is the median of the lower half, and the third quartile (Q3) is the median of the upper half. The Interquartile Range (IQR = Q3 - Q1) measures the spread of the middle 50% of the data.

Check Your Work on the SC-6 Calculator:

[f] [MIN] [f] [1Q] [f] [MED] [g] [3Q] [g] [MAX] [IQR]

Order the data and split into lower and upper halves on paper first. After entering the list with [Σ+] on your SC-6, press [f][MIN], [f][1Q], [f][MED], [g][3Q], [g][MAX], and [IQR] to check all 5 box-plot values and IQR = 16.5!

SEC 9.4 (pp.92-93) Mean Absolute Deviation (MAD)

Textbook Ways: Way 1: 3-Step MAD Distance Table | Way 2: Number-Line Deviation Balance

Learn by Hand: The Mean Absolute Deviation (MAD) is a measure of variation that tells the average distance between each data value and the mean. To find the MAD in 3 steps: (1) Find the mean x-bar of the data. (2) Find the absolute distance |x - x-bar| of each data value from the mean. (3) Find the mean of those absolute distances!

Check Your Work on the SC-6 Calculator:

[g] [CLΣ] 1 0 [Σ+] 1 2 [Σ+] 1 4 [Σ+] 1 6 [Σ+] 1 8 [Σ+] [g] [x̄] [g] [MAD]

Press [g][CLΣ], enter each value with [Σ+], and press [g][x-bar] (14) and [g][MAD] (2.4)!

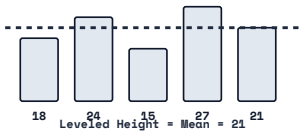
HANDS-ON MODEL: PLACE VALUE BLOCKS

Section 9.1 — Practice A: Statistical Questions, Dot Plots, and the Mean

Key Vocabulary: statistical question • dot plot • peak • cluster • gap • measure of center • mean (fair share) Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & HANDS-ON MODEL (PLACE VALUE BLOCKS)

Way 2: Fair-Share Tower Leveling



Way 1 & 3: Mean Formula & Balance Point

Way 1: Sum ÷ Count = 105 ÷ 5 = 21

Way 3 (Dot Plot Balance at 21):

Left: (21-15) + (21-18) = 6 + 3 = 9

Right: (24-21) + (27-21) = 3 + 6 = 9

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept: A statistical question anticipates variability. The mean (x-bar) is the sum of all data values divided by the number of values (Σx ÷ n).

Textbook Visual Model (Fair-Share / Balance Point): The mean is the leveled height when blocks are shared equally, or the balance point on a dot plot.

WATCH OUT: When finding a missing score for a target mean, multiply Target Mean × Total Count FIRST to find the required sum!

WORKED EXAMPLE — 3 TEXTBOOK WAYS (ANY 2 WAYS = +20 PTS)

Problem: Find the sum, the number of values, and the mean of 18, 24, 15, 27, 21.

WAY 1: Mean Formula (Sum Divided by Number of Values)
Add the 5 values: 18 + 24 + 15 + 27 + 21 = 105. Divide by n = 5 values: Mean = 105 ÷ 5 = 21.

WAY 2: Fair-Share Tower Leveling (With Blocks)
Shift 6 blocks from 27 to 15 (→ 21, 21) and 3 blocks from 24 to 18 (→ 21, 21). All 5 leveled towers equal 21.

WAY 3: Dot Plot Balance Point
On a number line centered at 21, distances left (-6 and -3 = -9) balance distances right (+3 and +6 = +9).

Answer: Sum = 105, Count = 5, Mean = 21

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 9.1 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[g] [CLΣ] 1 8 [Σ+] 2 4 [Σ+] 1 5 [Σ+] 2 7 [Σ+] 2 1 [Σ+] [g] [x̄]

Add the 5 values and divide by 5 on paper first. On your SC-6, clear stats with [g][CLΣ], enter each value with [Σ+] (or view the dot plot in [FRQ] mode), and press [g][x-bar] to confirm Mean = 21!

SECTION 9.1 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Find the sum and the mean of the data set: 2, 4, 6, 8.**
Starter Step 1: Add all 4 data values together to find the total sum, then divide that sum by 4.

Way 1 (Paper Work):

Sum = _____ Count n = ___

Mean = Sum ÷ n = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Find the sum and the mean of the data set: 12, 15, 18, 11, 14.**
Starter Step 1: Add all 5 data values together to find the total sum, then divide that sum by 5.

Way 1 (Paper Work):

Sum = _____ Count n = ___

Mean = Sum ÷ n = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Seed-Bomber Catapult Mean: In three test launches, a seed-bomber catapult flings clay seed pods distances of 150 cm, 162 cm, and 156 cm. What is the mean launch distance?**

Way 1 (Paper Work):

Sum = _____ Count n = ___

Mean = Sum ÷ n = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Find the mean of these four mineral sample masses (in grams): 4.2, 3.8, 4.5, 3.5.**

Way 1 (Paper Work):

Sum = _____ Count n = ___

Mean = Sum ÷ n = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Three measurements are 10, 20, and 25. Find the sum and the mean (rounded to the nearest hundredth).**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **In a dot plot of 6 quiz scores, the score 10 appears 4 times and the score 20 appears 2 times. What is the mean of all 6 scores (rounded to the nearest hundredth)?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Find the mean daily high temperature (°F) for the 6-day period: 68, 72, 75, 71, 64, 70.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **If 5 stacks of blocks have a mean height of 12 blocks, how many total blocks are in all 5 stacks combined?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Target Test Average: You want a mean of 90 across 4 unit tests. Your first three scores are 85, 92, and 88. What score do you need on the fourth test?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Joining the Roster: Six basketball players have a mean score of 15 points per game. A seventh player joins the team and scores 29 points. What is the new mean score for all 7 players?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: PLACE VALUE BLOCKS

Section 9.1 — Practice B: Modeling Real Life (Statistical Questions, Dot Plots, and the Mean)

Textbook Ways: Mean Formula (Sum ÷ Count) | Fair-Share Block Leveling | Balance Point Deviations Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — FINDING A MISSING SCORE FOR A TARGET MEAN (3 TEXTBOOK WAYS • ANY 2 = +20 PTS)

Problem: You want a mean score of 90 across 4 math quizzes. Your first 3 quiz scores are 86, 92, and 88. What score do you need on Quiz 4?

WAY 1: Target Total Sum Minus Current Sum
4 quizzes × 90 mean = 360 total points needed. Current sum = 86 + 92 + 88 = 266. Quiz 4 = 360 - 266 = 94.

WAY 2: Balance Deviations Around 90
Compare each score to 90: 86 is -4, 92 is +2, and 88 is -2 (net = -4). Quiz 4 must be +4 above 90 → 90 + 4 = 94.

WAY 3: Solve the Mean Equation (266 + x) / 4 = 90
Multiply both sides by 4: 266 + x = 360. Subtract 266 from both sides: x = 94.

WATCH OUT: When finding a missing score for a target mean, multiply Target Mean × Total Count FIRST to find the required sum!

Answer: 94

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 9.1 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

4 ENTER 9 0 × 8 6 ENTER 9 2 + 8 8 + -

Multiply 4 tests × 90 = 360 and subtract the first 3 scores on paper first. Check on your SC-6:
Press 4 [ENTER] 90 [×] 86 [ENTER] 92 [+] 88 [+] [-] to confirm 94.

Maya scored 80, 88, 82, and 84 on her first four math quizzes and wanted a mean (average) of 85 across all five quizzes, but she felt stuck trying to guess and check different numbers for Quiz 5. Then she pictured her quiz scores as five towers of blocks that get leveled out to the exact same height of 85. For five towers to each stand 85 blocks tall, there MUST be 5 × 85 = 425 blocks in total! Since her first four quizzes already added up to 80 + 88 + 82 + 84 = 334 points, subtracting 425 - 334 = 91 told her the exact score she needed on Quiz 5 without a single guess.

SECTION 9.1 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Basketball Points Per Game: In 5 games, Maya scored 14, 18, 12, 22, and 19 points. What is the sum of her points, and what is her mean number of points per game?**

Starter Step 1: Add Maya's 5 game scores together to find her total points, then divide the sum by 5 games.

Way 1 (Paper Work):

Sum = _____ Count n = ___

Mean = Sum ÷ n = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **3D Print Batch Weights: Six printed parts weigh 41 g, 43 g, 42 g, 40 g, 44 g, and 42 g. What is the mean weight of the parts?**

Starter Step 1: Add all 6 part weights together, then divide the total weight by 6 parts.

Way 1 (Paper Work):

Sum = _____ Count n = ___

Mean = Sum ÷ n = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Quiz Dot Plot: On a 10-point quiz, three students scored 8, five students scored 9, and two students scored 10. What is the mean score of all 10 students?**

Way 1 (Paper Work):

Sum = _____ Count n = ___

Mean = Sum ÷ n = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Target Test Average: You want a mean score of 85 across 5 tests. Your first 4 test scores are 80, 88, 82, and 84. What score do you need on the fifth test?**

Way 1 (Paper Work):

Sum = _____ Count n = ___

Mean = Sum ÷ n = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Daily Step Counter: Over 4 days you walked 8,200, 9,500, 7,800, and 10,500 steps. What was your mean number of steps per day?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Dropping the Lowest Quiz: A student's 4 quiz scores are 60, 88, 92, and 90. Find the mean of all 4 quizzes, and then find the new mean if the lowest score (60) is dropped.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Fair-Share Leveling: Four stacks of books have 6, 10, 14, and 18 books. If you move books so all four stacks have the same number of books, how many books will be in each stack?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Bluebird Chick Growth: Five nestling bluebirds gained 4.2 g, 5.1 g, 4.8 g, 5.5 g, and 4.4 g this week. What was the mean weight gain?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Explain why 'How many minutes did students in our class read last night?' is a statistical question, while 'How many minutes are in 1 hour?' is not.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Combined Group Mean: In a trivia game, Group A has 4 students with a mean of 20 points, and Group B has 6 students with a mean of 30 points. What is the mean score of all 10 students combined?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 9.2 — Practice A: Measures of Center: Median, Mode, and Range

Key Vocabulary: median (middle value) • mode (most frequent) • range (max - min) • outlier • best measure of center Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & VISUAL DIAGRAM

Measures of Center & Variation: ALWAYS Order Data Least to Greatest First!

6	7	7	7	8	8	9	10

Mode = 7 (Most frequent: 3 times)

Median (Even n=8): (7 + 8) ÷ 2 = 7.5

Range = Max - Min = 10 - 6 = 4 | Use Median when an Outlier skews the Mean!

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Concept: ALWAYS order data from least to greatest first! Median = middle value (or mean of two middle values if n is even); Mode = most frequent; Range = Max - Min.

Textbook Choosing the Best Measure: Use the Median when an extreme outlier skews the Mean; use the Mean when data are symmetric with no outliers.

WATCH OUT: ALWAYS sort the data from least to greatest BEFORE finding the Median! If n is even, average the two middle numbers.

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find the mean, median, mode, and range of the shoe sizes: 7, 9, 6, 7, 10, 8, 9, 7.

WAY 1: Order from Least to Greatest & Cross Off Ends

Ordered (n = 8): 6, 7, 7, [7, 8], 9, 9, 10. Median = (7 + 8) ÷ 2 = 7.5; Mode = 7 (three times); Range = 10 - 6 = 4.

WAY 2: Dot Plot / Frequency Table

Frequencies: 6(1), 7(3), 8(1), 9(2), 10(1). Tallest column = Mode (7); 4th and 5th dots = 7 and 8 → Median 7.5; span = 10 - 6 = 4.

Answer: Mean = 7.875, Median = 7.5, Mode = 7, Range = 4

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 9.2 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[g] [CL Σ] 7 [Σ+] 9 [Σ+] 6 [Σ+] 7 [Σ+] 1 0 [Σ+] 8 [Σ+] 9 [Σ+] 7 [Σ+] [f] [MED] [g] [MODE] [RNG]

Order the numbers from least to greatest on paper first! Then check on your SC-6:

Press [g][CLΣ], enter each number with [Σ+] (or [FRQ]), and press [f][MED], [g][MODE], and [RNG] to verify Median = 7.5, Mode = 7, and Range = 4!

SECTION 9.2 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Shoe Size Survey Mode: A sneaker store records the shoe sizes of 8 customers: 6, 7, 7, 7, 8, 8, 9, 10. What is the mode of the shoe sizes?**

Starter Step 1: Count how many times each shoe size appears—the Mode is the value that repeats MOST often.

Way 1 (Paper Work):

Sorted: [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

Med: _____ Mode: _____ Rng: _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Find the median, mode, and range of the data set: 12, 15, 12, 18, 20, 13.**

Starter Step 1: Sort the 6 numbers from least to greatest FIRST, then average the two middle numbers (3rd and 4th) for the Median.

Way 1 (Paper Work):

Sorted: [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

Med: _____ Mode: _____ Rng: _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Find the mean, median, mode, and range of the data set: 5, 8, 8, 10, 14.**

Way 1 (Paper Work):

Sorted: [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

Med: _____ Mode: _____ Rng: _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **High Temperature Range: Over 7 days, the high temperatures (°F) were 62, 65, 78, 81, 74, 69, and 85. What are the minimum, maximum, and range of the temperatures?**

Way 1 (Paper Work):

Sorted: [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

Med: _____ Mode: _____ Rng: _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Even-Dataset Median: Six daily high temperatures (°F) are ordered from least to greatest: 68, 72, 74, 76, 80, 84. What is the median temperature?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Daily earnings are $15, $18, $15, $20, $17, and $90. Find the mean and median, and state which measure better represents a typical day.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **How do the mean and the median of {10, 12, 14, 16, 18} change if the value 18 is replaced by the outlier 100?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **The minimum score on a science test is 64 and the range of the scores is 31. What was the maximum score on the test?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Find the median and range of the data set {14, 8, 21, 11, 16}.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Neighborhood Home Prices: Five homes on a street sold for (in thousands of dollars): 210, 220, 215, 225, and 980. Find the mean and median sale prices, and explain which measure a home buyer should use.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 9.2 — Practice B: Modeling Real Life (Measures of Center: Median, Mode, and Range)

Textbook Ways: Ordered List (Least to Greatest) | Dot Plot / Frequency Table Analysis Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — CHOOSING THE BEST MEASURE OF CENTER WITH AN OUTLIER (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: Daily dog-walking earnings for 6 days are $15, $18, $15, $20, $17, and $90. Find the mean, median, and mode. Which measure best represents a typical day?

WAY 1: Compare Mean vs. Ordered Median & Mode

Mean = $175 ÷ 6 ≈ $29.17. Ordered: $15, $15, [$17, $18], $20, $90 → Median = $17.50, Mode = $15. Median ($17.50) is best because $90 is an outlier.

WAY 2: Compare With and Without the $90 Outlier

Without $90, the mean is $85 ÷ 5 = $17.00 and median is $17.00. The $90 outlier pulls the mean up by +$12.17 (above 5 of the 6 days!), while the median barely changes.

Rule: When an extreme outlier is present, use the Median instead of the Mean.

WATCH OUT: ALWAYS sort the data from least to greatest BEFORE finding the Median! If n is even, average the two middle numbers.

Answer: Mean = $29.17, Median = $17.50, Mode = $15 (Median is best)

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 9.2 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[g] [CLΣ] [1] [5] [Σ+] [1] [8] [Σ+] [1] [5] [Σ+] [2] [0] [Σ+] [1] [7] [Σ+] [9] [0] [Σ+] [g] [x̄] [f] [MED]

Sort the 6 prices on paper first to see how the $90 outlier pulls the mean up. Enter all 6 prices with [Σ+] on your SC-6 and compare [g][x-bar] ($29.17) against [f][MED] ($17.50)!

When Leo calculated the average price of five video games ($20, $25, $20, $30, and a rare $105 collector's edition) and got $40, he thought he had messed up the division—how could the 'average' price be higher than four of the five games?! He also admitted he sometimes forgot to put numbers in order before picking the middle number (the median). Lining the prices up from least to greatest ($20, $20, $25, $30, $105), he saw why statistics has more than one kind of average: that single $105 outlier yanked the mean up to $40, while the median ($25) sat right in the middle and told the honest truth about what a typical game costs.

SECTION 9.2 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Roller Coaster Speeds: Seven roller coasters have top speeds (in mph) of 55, 62, 48, 70, 62, 50, and 66. Find the median, mode, and range of the speeds.**

Starter Step 1: Sort all 7 coaster speeds from least to greatest FIRST—the middle (4th) value is the Median!

Way 1 (Paper Work):

Sorted: [ ] [ ] [ ] [ ] [ ] [ ] [ ]

Med: _____ Mode: _____ Rng: _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Coding Club Ages: Six club members have ages 11, 13, 12, 14, 12, and 16. Find the median, mode, and range of their ages.**

Starter Step 1: Sort the 6 ages from least to greatest FIRST, then find the average of the two middle ages for the Median.

Way 1 (Paper Work):

Sorted: [ ] [ ] [ ] [ ] [ ] [ ]

Med: _____ Mode: _____ Rng: _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Video Game Prices & Outliers: Five video games cost $20, $25, $20, $30, and $105. Find the mean, median, and mode. Which measure of center best represents the price of a typical game, and why?**

Way 1 (Paper Work):

Sorted: [ ] [ ] [ ] [ ] [ ] [ ] [ ]

Med: _____ Mode: _____ Rng: _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Shoe Size Modes: Find the mode(s) and median of the shoe sizes: 5, 6, 6, 6, 7, 8, 8, 8, 9.**

Way 1 (Paper Work):

Sorted: [ ] [ ] [ ] [ ] [ ] [ ] [ ]

Med: _____ Mode: _____ Rng: _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Sprint Times: Five runners finish a race in 11.2, 11.5, 11.8, 12.1, and 12.4 seconds. Find the median, range, and mode (if any).**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Weekly High Temperatures: This week's daily high temperatures (°F) were 72, 68, 75, 81, 64, 70, and 74. Find the minimum, maximum, range, and median.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Pizza Orders: Daily orders of mushroom pizza over 6 days were 8, 12, 9, 15, 12, and 10. Find the mean, median, mode, and range.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Mystery Data Set: Write a data set of 5 whole numbers that has a median of 10, a mode of 8, and a range of 7.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Bonus Points Effect: Five students scored 70, 75, 80, 85, and 90 on a quiz. If the teacher adds 5 bonus points to every student's score, how do the median and the range change?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Backpack Weights: Eight student backpacks weigh 9, 12, 10, 14, 11, 15, 10, and 13 pounds. Find the mean, median, mode, and range.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 9.3 — Practice A: Quartiles, Interquartile Range (IQR) & Box Plots

Key Vocabulary: quartiles • first quartile (Q_1) • third quartile (Q_3) • interquartile range (IQR = Q_3 - Q_1) • box-and-whisker plot • 5-number summary

Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & VISUAL DIAGRAM

Box-and-Whisker Plot: 5-Number Summary & Interquartile Range (IQR = Q_3 - Q_1)



Middle 50% Box = IQR = Q_3 - Q_1 = 90 - 73.5 = 16.5 | Range = Max - Min = 28

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook 5-Number Summary: Order data to find Min, Q_1 (median of lower half), Median (Q_2), Q_3 (median of upper half), and Max.

Textbook Box-and-Whisker Plot & IQR: The box spans the middle 50% of data from Q_1 to Q_3; Interquartile Range IQR = Q_3 - Q_1. Each section holds 25% of the data.

WATCH OUT: When n is odd, DO NOT include the overall Median in either the lower half or upper half when finding Q_1 and Q_3!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find the 5-Number Summary (Min, Q1, Median, Q3, Max) and IQR of: 68, 72, 75, 78, 82, 85, 88, 92, 96.

WAY 1: Find Medians of the Lower and Upper Halves

Ordered (n=9): 68, 72, 75, 78, [82], 85, 88, 92, 96. Med = 82. Lower half (68, 72, 75, 78) → Q_1 = 73.5. Upper half (85, 88, 92, 96) → Q_3 = 90. IQR = 90 - 73.5 = 16.5.

WAY 2: Read the 5-Number Summary on a Box Plot

Plot whiskers at Min = 68 and Max = 96, box from Q_1 = 73.5 to Q_3 = 90, and line at Med = 82. Box width = 90 - 73.5 = 16.5.

Answer: Min = 68, Q1 = 73.5, Med = 82, Q3 = 90, Max = 96; IQR = 16.5

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 9.3 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

f MIN f 1Q f MED g 3Q g MAX IQR

Order the data and split into lower and upper halves on paper first. After entering the list with [Σ+] on your SC-6, press [f][MIN], [f][1Q], [f][MED], [g][3Q], [g][MAX], and [IQR] to check all 5 box-plot values and IQR = 16.5!

SECTION 9.3 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **For the ordered data set 10, 20, 30, 40, 50, find the 5-Number Summary (Min, Q1, Median, Q3, Max) and the IQR.**

Starter Step 1: Find the middle number (Median) of the sorted list first, then find the median of the lower half (Q_1) and upper half (Q_3).

Way 1 (Paper Work):

Min:___ Q1:___ Med:___ Q3:___ Max:___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **For the ordered data set 4, 6, 8, 10, 12, 14, 16, find Q1, Median, Q3, and the IQR.**

Starter Step 1: The 4th number is the Median; find the middle of the lower 3 numbers for Q_1 and the upper 3 numbers for Q_3, then subtract Q_3 - Q_1.

Way 1 (Paper Work):

Min:___ Q1:___ Med:___ Q3:___ Max:___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **If a box-and-whisker plot has Q1 = 24 and Q3 = 58, what is the Interquartile Range (IQR)?**

Way 1 (Paper Work):

Min:___ Q1:___ Med:___ Q3:___ Max:___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **What percent of the data in a box-and-whisker plot falls above the first quartile (Q1)?**

Way 1 (Paper Work):

Min:___ Q1:___ Med:___ Q3:___ Max:___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Grade 6 Science Scores IQR: Eight ordered science test scores are 65, 72, 75, 80, 84, 88, 92, 96. Find Q1, Median, Q3, and the Interquartile Range (IQR).**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Find Q1, Median, Q3, and the IQR for the 8 ordered wind speeds (mph): 42, 45, 48, 50, 52, 55, 58, 60.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Player A's game scores have an IQR of 4 points, while Player B's scores have an IQR of 19 points. Which player is more consistent? Explain.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Outlier Fence Rule: A data set has Q_1 = 40 and Q_3 = 60. First find the IQR. Then find the upper outlier cutoff by adding 1.5 × IQR to Q_3.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Find Q1, Median, Q3, and the Interquartile Range (IQR) for the data set {12, 15, 18, 22, 25, 28, 34}.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Grade-Level Benchmark Exam: Exactly 200 sixth graders take a benchmark exam whose box plot has Min = 50, Q1 = 70, Median = 80, Q3 = 90, and Max = 100. About how many students scored between 70 and 90?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 9.3 — Practice B: Modeling Real Life (Quartiles, Interquartile Range (IQR) & Box Plots)

Textbook Ways: Split Ordered Data into Lower & Upper Halves | Box-and-Whisker Plot & 25% Quartile Sections Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — INTERPRETING A BOX-AND-WHISKER PLOT (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: A box-and-whisker plot of 40 test scores has Min = 60, Q_1 = 72, Median = 80, Q_3 = 88, and Max = 98. Find the IQR and how many students scored between 72 and 88.

WAY 1: Use Four 25% Quartile Sections

IQR = Q_3 - Q_1 = 88 - 72 = 16 points. The box from Q_1 (72) to Q_3 (88) spans 2 quartile sections = 25% + 25% = 50% of 40 = 20 students.

WAY 2: Divide Total Students into 4 Equal Quarters

40 students ÷ 4 quarters = 10 students in each section (Min–Q_1: 10, Q_1–Med: 10, Med–Q_3: 10, Q_3–Max: 10). Middle two sections = 10 + 10 = 20 students.

Check: 10 + 10 + 10 + 10 = 40 students (100%).

WATCH OUT: When n is odd, DO NOT include the overall Median in either the lower half or upper half when finding Q_1 and Q_3!

Answer: IQR = 16; 20 students (50%)

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 9.3 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

2 5 ENTER 2 ×

Remember each section of a box plot holds 25% of the data (so the middle box holds 50%). Check 50% of 40 students on your SC-6 (40 [ENTER] 50 [%]) to confirm 20 students.

Looking at a box-and-whisker plot of 60 pine trees (Min = 12 ft, Q_1 = 18 ft, Median = 24 ft, Q_3 = 31 ft, Max = 40 ft), Maya initially thought the longest section of the drawing held the most trees. When her coach asked how many trees were taller than 18 feet, she wasn't sure how a box plot could even tell you a headcount. Then she realized how the plot is built: ordering 60 trees and splitting them at the median, Q_1, and Q_3 chops the list into four equal groups of 15 trees (25% in each section), no matter how stretched out or squished a section looks! Since three of the four quarters sit above Q_1 (18 ft), 75% of the 60 trees—45 trees—are taller than 18 feet.

SECTION 9.3 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Archery Scores: Seven ordered archery scores are 40, 45, 50, 55, 60, 65, and 70. Find the 5-number summary (Min, Q_1, Median, Q_3, Max) and the interquartile range (IQR).**

Starter Step 1: Find the middle (4th) score for the Median, then find the middle of the lower 3 scores (Q_1) and upper 3 scores (Q_3).

Way 1 (Paper Work):

Min:___ Q1:___ Med:___ Q3:___ Max:___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Battery Life Test: Eight laptops lasted 6, 7, 8, 9, 11, 12, 14, and 15 hours on a battery test. Find Q_1, the median, Q_3, and the IQR.**

Starter Step 1: Split the 8 sorted hours into the lower 4 and upper 4; average each pair of middle numbers to find Median, Q_1, and Q_3.

Way 1 (Paper Work):

Min:___ Q1:___ Med:___ Q3:___ Max:___

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Robotics Scores: Nine teams scored 30, 15, 45, 25, 35, 20, 50, 40, and 30 points. Order the data and find Q_1, the median, Q_3, and the IQR.**

Way 1 (Paper Work):

Min:___ Q1:___ Med:___ Q3:___ Max:___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Tree Height Box Plot: A box-and-whisker plot of 60 tree heights shows Min = 12 ft, Q_1 = 18 ft, Median = 24 ft, Q_3 = 31 ft, and Max = 40 ft. Find the range and the IQR.**

Way 1 (Paper Work):

Min:___ Q1:___ Med:___ Q3:___ Max:___

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Using the tree height box plot in Problem 4 (60 trees total), what percent of the trees are taller than 18 feet, and how many trees is that?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Using the same 60-tree box plot in Problem 4, how many trees are in the top quartile between 31 feet and 40 feet?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Comparing Consistency: On a math test, Class A has an IQR of 8 points and Class B has an IQR of 19 points. Which class had more consistent scores in the middle 50%?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Outlier Fence Rule: A data set has Q_1 = 20 and Q_3 = 30. An upper outlier is any data value greater than $Q_3 + 1.5 \times$ IQR. What is the upper outlier boundary for this data set?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Push-Up Challenge: Six students completed 18, 22, 25, 29, 34, and 40 push-ups. Find the 5-number summary and the IQR.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Effect of an Outlier on Spread: For the data set {10, 12, 14, 16, 18, 20, 100}, find both the range and the IQR. Why is the IQR a better measure of variation for this data set?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 9.4 — Practice A: Mean Absolute Deviation (MAD)

Key Vocabulary: mean absolute deviation (MAD) • deviation from the mean (|x - x-bar|) • measure of variation • consistency

Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & VISUAL DIAGRAM

Mean Absolute Deviation (MAD): Average Distance of Data Values from the Mean

Data: {10, 12, 14, 16, 18} → Mean = 14



Distances |x - 14|: 4, 2, 0, 2, 4

3-Step MAD Recipe:

1. Find Mean = 70 ÷ 5 = 14
2. Sum |x - 14| = 4+2+0+2+4 = 12
3. MAD = 12 ÷ 5 = 2.4

(Smaller MAD = More Consistent!)

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook 3-Step MAD Algorithm: 1. Find the Mean (x-bar). 2. Find the absolute distance |x - x-bar| of each value from the mean. 3. Average those distances (Sum ÷ n).

Textbook Interpretation: A smaller MAD means the data values are clustered tightly near the mean (more consistent); a larger MAD means more spread out.

WATCH OUT: Distances |x - Mean| are ALWAYS positive (subtract the smaller number from the larger for every data value)!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Find the mean and the Mean Absolute Deviation (MAD) of 10, 12, 14, 16, 18.

WAY 1: 3-Step MAD Distance Table

For {10, 12, 14, 16, 18}: 1. Mean = 70 ÷ 5 = 14. 2. Distances |x - 14|: 4, 2, 0, 2, 4. 3. Average the distances: (4+2+0+2+4) ÷ 5 = 12 ÷ 5 = 2.4.

WAY 2: Number-Line Deviation Balance (Below vs. Above Mean)

Distances below 14 (4 + 2 = 6) equal distances above 14 (2 + 4 = 6). Double one side (2 × 6 = 12) and divide by 5 → MAD = 2.4.

Answer: Mean = 14, MAD = 2.4

1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 9.4 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[g] [CLΣ] 1 0 [Σ+] 1 2 [Σ+] 1 4 [Σ+] 1 6 [Σ+] 1 8 [Σ+] [g] [x̄] [g] [MAD]

Make your 3-step MAD table (mean, absolute distances, average of distances) on paper first. Then check on your SC-6: Press [g][CLΣ], enter each value with [Σ+], and press [g][x-bar] (14) and [g][MAD] (2.4)!

SECTION 9.4 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Seed-Bomber Catapult MAD: For the three catapult launch distances 150 cm, 162 cm, and 156 cm, find the mean and the Mean Absolute Deviation (MAD).**

Starter Step 1: Average the 3 numbers for the Mean → Step 2: Find each number's distance |x - Mean| → Step 3: Average those distances.

Way 1 (Paper Work):

1. Mean = _____
2. |x - Mean|: ____, ____, ____, ____
3. MAD = Sum ÷ n = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Find the mean and the MAD of the data set: 10, 12, 14, 16, 18.**

Starter Step 1: Add the 5 values and divide by 5 for the Mean → Step 2: Find each distance from the Mean → Step 3: Divide the distance sum by 5.

Way 1 (Paper Work):

1. Mean = _____
2. |x - Mean|: ____, ____, ____, ____
3. MAD = Sum ÷ n = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **5-Point Sensor Calibration MAD: Five calibration readings from a pressure sensor are 10, 15, 20, 25, and 30. Find the mean and the Mean Absolute Deviation (MAD).**

Way 1 (Paper Work):

1. Mean = _____
2. |x - Mean|: ____, ____, ____, ____
3. MAD = Sum ÷ n = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **What is the Mean Absolute Deviation (MAD) of five identical values: 7, 7, 7, 7, 7? Explain.**

Way 1 (Paper Work):

1. Mean = _____
2. |x - Mean|: ____, ____, ____, ____
3. MAD = Sum ÷ n = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Find the mean and the MAD of the data set: 20, 24, 28, 32, 36.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Find the mean and the MAD of the 6 sprint times (seconds): 12.4, 13.0, 12.6, 13.2, 12.8, 14.0.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Find the mean and the MAD of the quiz scores: 82, 85, 88, 90, 95.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Team A has a mean score of 50 and a MAD of 2.1. Team B has a mean score of 50 and a MAD of 8.4. Which team's scores are more spread out?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **For the data set {6, 10, 11, 13}, find the mean, list the four absolute deviations |x - x-bar|, and compute the MAD.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Complete Statistical Profile: For the data set {5, 8, 8, 10, 14}, find the Mean, Median, Mode, Range, IQR, and MAD.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 9.4 — Practice B: Modeling Real Life (Mean Absolute Deviation (MAD))

Textbook Ways: 3-Step MAD Distance Table | Number-Line Deviation Balance

Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — COMPARING CONSISTENCY WITH MAD (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: A sprinter's six 100-meter dash times are 12.4, 13.0, 12.6, 13.2, 12.8, and 14.0 seconds. Find the mean and the mean absolute deviation (MAD).

WAY 1: 3-Step MAD Calculation

1. Mean = 78.0 ÷ 6 = 13.0 s. 2. Distances from 13.0: 0.6, 0.0, 0.4, 0.2, 0.2, 1.0. 3. Sum = 2.4 s → MAD = 2.4 ÷ 6 = 0.40 s.

WAY 2: Sum Deviations Below & Above 13.0 s

Below 13.0 s: (13.0-12.4) + (13.0-12.6) + (13.0-12.8) = 0.6 + 0.4 + 0.2 = 1.2 s. Above 13.0 s: 0.2 + 1.0 = 1.2 s. Total = 2.4 ÷ 6 = 0.40 s.

Check: Because total distance below the mean (1.2) equals total distance above (1.2), 13.0 is verified as the exact mean!

WATCH OUT: Distances |x - Mean| are ALWAYS positive (subtract the smaller number from the larger for every data value)!

Answer: Mean = 13.0 s; MAD = 0.40 s

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 9.4 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[g][CL] 12.4 [Σ+] 13.0 [Σ+] 12.6 [Σ+] 13.2 [Σ+] 12.8 [Σ+] 14.0 [Σ+] [g][x̄] [g][MAD]

Calculate the mean and average deviation on paper first. Then enter all 6 sprint times with [Σ+] on your SC-6 and press [g][x-bar] (13.0 s) and [g][MAD] (0.40 s) to verify your table!

In the bird-cognition lab, Raven A solved four puzzle trials in 8, 9, 9, and 10 seconds, while Crow B took 6, 8, 10, and 12 seconds. Both birds had the exact same mean (9 seconds), so Zoe wasn't sure how to prove mathematically that Raven A was much more consistent. When she first heard the name 'Mean Absolute Deviation (MAD),' it sounded terrifying—until she read the three words backward: first find each score's Deviation (distance) from the mean of 9, take its Absolute (positive) value so below-average and above-average distances don't cancel out, and take the Mean (average) of those distances! Raven A's average wobble was just 0.5 seconds, compared to 2.0 seconds for Crow B.

SECTION 9.4 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Nozzle Temperatures: Four temperature readings (°C) are 206, 208, 212, and 214. Find the mean and the mean absolute deviation (MAD).**

Starter Step 1: Find the Mean of the 4 temperatures → Step 2: Find how far each reading is from the Mean → Step 3: Average those 4 distances.

Way 1 (Paper Work):

1. Mean = _____
2. |x - Mean|: ____, ____, ____, ____
3. MAD = Sum ÷ n = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Free-Throw Practice: In 5 practices you made 7, 9, 10, 11, and 13 free throws. Find the mean and the MAD.**

Starter Step 1: Find the Mean of the 5 practices → Step 2: Find each practice's distance from the Mean → Step 3: Average those 5 distances.

Way 1 (Paper Work):

1. Mean = _____
2. |x - Mean|: ____, ____, ____, ____
3. MAD = Sum ÷ n = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Identical Weights: Five brass weights each weigh 50 grams: 50, 50, 50, 50, 50. What is the mean and what is the MAD? Explain why.**

Way 1 (Paper Work):

1. Mean = _____
2. |x - Mean|: ____, ____, ____, ____
3. MAD = Sum ÷ n = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Raven vs. Crow Puzzle Times: Raven A solves 4 puzzle boxes in 8, 9, 9, and 10 seconds (Mean = 9 s, MAD = 0.5 s). Crow B takes 6, 8, 10, and 12 seconds. Find Crow B's mean and MAD. Which bird is more consistent?**

Way 1 (Paper Work):

1. Mean = _____
2. |x - Mean|: ____, ____, ____, ____
3. MAD = Sum ÷ n = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Quiz Scores: Five quiz scores are 76, 82, 85, 88, and 94. Find the mean and the MAD.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Drone Hover Altitude: Six hover altitude readings (in meters) are 15, 16, 14, 15, 17, and 13. Find the mean and the MAD.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Summer High Temperatures: Four daily high temperatures (°F) are 80, 84, 86, and 90. Find the mean, list the four absolute deviations, and find the MAD.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Why Absolute Value Matters: For the data set {10, 15, 20}, the mean is 15. What happens if you add the signed deviations (10 - 15), (15 - 15), and (20 - 15) without taking their absolute values?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Cafe Wait Times: Five customers waited 2, 3, 4, 5, and 11 minutes for their drinks. Find the mean and the MAD.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Full Data Summary: For the 5 scores {12, 15, 15, 18, 20}, find the mean, median, mode, range, IQR, and MAD.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

Unit 10: Grade 7 Bridge — Signed Rationals, 2-Step Equations & Probability — Arctic Weather Swings, Go-Kart Tracks & Token Jars (Grade 7 Bridge)

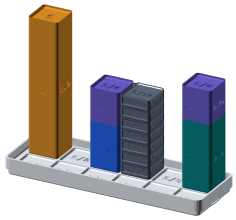
Adding, Subtracting, Multiplying & Dividing Signed Rationals, Two-Step Equations, and Probability/Combinatorics

As the summer wraps up, Kai, Leo, Zoe, and Maya run into challenges where two rules happen at the same time. During a camp trivia night, Kai's team was sitting at -15 points when the judge canceled a -25-point penalty, and nobody at the table could agree whether their score went up to +10 or dropped to -40. At the go-kart track, Leo had $38 to spend when admission cost $8 plus $6 per lap, and he wasn't sure which number to undo first.

In the bird-cognition lab, Zoe watched video from New Zealand showing wild kea parrots peering into clear jars filled with black reward tokens and white non-reward tokens. Before a lid was even opened, the parrots chose the jar with the better fraction of black tokens—while Maya tried to figure out why picking a 1st-2nd-3rd batting order gave six times as many possibilities as picking a 3-person cleanup crew.

Each of these puzzles combines multiple steps—signed numbers with fractions, multiplication with addition in a two-step equation, or counting outcomes where order does or does not matter.

Step by step, the four friends discover that subtracting a negative penalty, unwrapping a two-step equation in reverse order, and comparing permutations (`[nPr]`) with combinations (`[nCr]`) follow clear, logical patterns—and they use their SC-6 calculators to verify every result on their way into seventh-grade math.



Real-World Mathematical Context & Two-Method Mastery • Hands-On Models: Place Value, Fraction & Prime Blocks

UNIT 10 LESSONS — LEARN TWO TEXTBOOK WAYS FIRST, THEN CHECK YOUR WORK ON THE SC-6 CALCULATOR

SEC 10.1 (pp.95-96) Adding and Subtracting Rational Numbers

Textbook Ways: Way 1: Zero-Pair Cancellation | Way 2: Number-Line Jumps | Way 3: Absolute Value Sign Rules & Add-the-Opposite

Learn by Hand: To add two numbers with the SAME sign, add their absolute values and keep the common sign (-6 + (-4) = -10). To add two numbers with DIFFERENT signs, subtract the lesser absolute value from the greater absolute value and keep the sign of the number with the greater absolute value (-8 + 5 = -3). To subtract any signed number, add its opposite: a - b = a + (-b), so 10 - (-4) = 10 + 4 = 14.

Check Your Work on the SC-6 Calculator:

[f] [↔] [8] [+/−] [ENTER] [5] [+] [6] [+/−] [ENTER] [4] [+/−] [+]

Use zero-pair chips or a number line on paper first. On your SC-6, open [f][LINE] Number Line Mode (or [10F] zero-pair frame) and press 8 [+/-] [ENTER] 5 [+] to confirm -3, and 6 [+/-] [ENTER] 4 [+/-] [+] to confirm -10!

SEC 10.2 (pp.97-98) Multiplying and Dividing Rational Numbers

Textbook Ways: Way 1: Same/Different Sign Rules | Way 2: Repeated Addition / Equal Number-Line Jumps

Learn by Hand: When multiplying or dividing two non-zero numbers: if the signs are the SAME (both positive or both negative), the product or quotient is POSITIVE (+); if the signs are DIFFERENT, the product or quotient is NEGATIVE (-). In a product of multiple factors, an EVEN number of negative factors produces a positive result, and an ODD number of negative factors produces a negative result.

Check Your Work on the SC-6 Calculator:

[6] [ENTER] [4] [+/−] [×] [5] [+/−] [x^2]

Press 6 [ENTER] 4 [+/-] [×] (-24) and 5 [+/-] [x^2] (+25).

SEC 10.3 (pp.99-100) Solving Two-Step Equations (ax + b = c)

Textbook Ways: Way 1: Two-Step Inverse Operations | Way 2: Balance Scale / Bar Model Unwrapping

Learn by Hand: A two-step equation contains two operations. To isolate the variable, undo the operations in the REVERSE order of operations: Step 1 undo addition or subtraction first to isolate the variable term (ax), and Step 2 undo multiplication or division second to isolate the variable x.

Check Your Work on the SC-6 Calculator:

[2] [8] [ENTER] [7] [-] [3] [÷] [2] [8] [↑↓] [3] [×] [7] [+] [≠]

Press 28 [ENTER] 7 [-] 3 [÷] (7), then run the Equality Test 28 [ENTER] 3 [ENTER] 7 [×] 7 [+] [≠]—it flashes NO to confirm 3(7) + 7 = 28!

SEC 10.4 (pp.101-102) Theoretical Probability, Combinations & Permutations

Textbook Ways: Way 1: Probability Ratio & Complement Rule | Way 2: Tree Diagram / Outcome List | Way 3: Fundamental Counting Principle

Learn by Hand: Theoretical probability is the ratio of favorable outcomes to total equally likely outcomes: P(event) = Favorable ÷ Total, written as a simplified fraction, decimal, or percent. The probability of an event NOT occurring is the complement: P(not E) = 1 - P(E). For counting arrangements: n! counts ways to order all n items; nPr counts ordered permutations; and nCr counts unordered combinations.

Check Your Work on the SC-6 Calculator:

[•] [4] [•] [1] [0] [g] [SIMP] [5] [$x!$]

Press [.] 4 [.] 10 [g][SIMP] to confirm 2/5, press 5 [x!] to confirm 5! = 120, and explore rolls with [DICE] or [f][1..x]!

HANDS-ON MODEL: PLACE VALUE BLOCKS

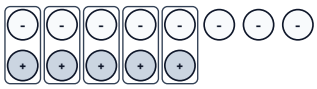
Section 10.1 — Practice A: Adding and Subtracting Rational Numbers

Key Vocabulary: additive inverse • zero pair • Same-Sign Sum Rule • Different-Sign Difference Rule • Add-the-Opposite (a - b = a + (-b))

Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & HANDS-ON MODEL (PLACE VALUE BLOCKS)

Way 2: Zero-Pair Counters (-8 + 5 = -3)



5 Zero Pairs cancel → 3 Red (-) left = -3

Same Signs: Add |x| | Diff Signs: Subtract |x|

Subtracting = Add the Opposite!

a - b = a + (-b)

a - (-b) = a + b

Ex: 10 - (-4) = 10 + 4 = 14

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Addition Rules: Same signs → add absolute values and keep the common sign. Different signs → subtract absolute values and keep the sign of the larger |x|.

Textbook Subtraction Rule (Add the Opposite): To subtract any rational number, add its opposite: a - b = a + (-b), and a - (-b) = a + b.

WATCH OUT: To subtract a signed number, ADD ITS OPPOSITE: a - (-b) = a + b (subtracting a negative moves RIGHT on the number line)!

WORKED EXAMPLE — 3 TEXTBOOK WAYS (ANY 2 WAYS = +20 PTS)

Problem: Evaluate -8 + 5 and -6 + (-4).

WAY 1: Zero-Pair Cancellation Model

In -8 + 5, five (-1, +1) zero pairs cancel to 0, leaving 3 negatives (-3). In -6 + (-4), 6 negatives + 4 negatives = -10.

WAY 2: Horizontal Number-Line Jumps

For -8 + 5, start at -8 and move 5 units right → -3. For -6 + (-4), start at -6 and move 4 units left → -10.

WAY 3: Big Ideas Absolute Value Sign Rules

Different signs (-8 + 5): subtract |8| - |5| = 3, keep sign of -8 → -3. Same signs (-6 + -4): add |6| + |4| = 10, keep '-' → -10.

Answer: -8 + 5 = -3; -6 + (-4) = -10

1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 10.1 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

f [↔] 8 [+/−] ENTER 5 + 6 [+/−] ENTER 4 [+/−] +

Use zero-pair chips or a number line on paper first. On your SC-6, open [f][LINE] Number Line Mode (or [10F] zero-pair frame) and press 8 [+/-] [ENTER] 5 [+] to confirm -3, and 6 [+/-] [ENTER] 4 [+/-] [+] to confirm -10!

SECTION 10.1 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 Evaluate: (a) -15 + 9 and (b) -15 + (-9).

Starter Step 1: In (a) the signs are different, so subtract the absolute values (15 - 9) and keep the sign of -15; in (b) the signs are the same.

Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

2 Evaluate: 14 + (-23).

Starter Step 1: Since 14 and -23 have different signs, subtract 23 - 14 and keep the sign of the number with the larger absolute value (-23).

Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

3 Evaluate: (a) 12 - (-18) and (b) -12 - (-8).

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

4 Evaluate: -9 - 16.

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

5 Ice-Melting Science Lab: Pure water freezes at 0°C. Adding salt lowers the freezing point by 7.8°C, and adding beet extract lowers it by another 6.4°C. What is the new freezing point of the mixture?

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

6 Evaluate: 6.4 - 11.2.

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

7 Evaluate: -7.5 - (-10.2).

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

8 Evaluate: -3/4 - (-5/8).

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

9 Sub-Zero Week Temperature Swing: During a polar vortex week, the lowest recorded temperature was -22°F and the highest recorded temperature was +14°F. How many degrees warmer was the high temperature than the low temperature?

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

10 Evaluate the rational expression: -2 1/2 + 4 1/4 - (-1 1/8).

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

HANDS-ON MODEL: PLACE VALUE BLOCKS

Section 10.1 — Practice B: Modeling Real Life (Adding and Subtracting Rational Numbers)

Textbook Ways: Zero-Pair Cancellation | Number-Line Jumps | Absolute Value Sign Rules & Add-the-Opposite Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — SUBTRACTING SIGNED RATIONAL NUMBERS (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: A mountain sensor rises from -4°C to +10°C. Find the temperature change 10 - (-4), and evaluate -3/4 - (-5/8).

WAY 1: Add the Opposite (a - b = a + (-b))

Rewrite subtraction as adding the opposite: 10 - (-4) = 10 + (+4) = 14°C. For -3/4 - (-5/8): -6/8 + (+5/8) = -1/8.

WAY 2: Number-Line Distance from Start to End

From -4°C up to 0°C is +4°C, and from 0°C to +10°C is +10°C → total rise = 4 + 10 = +14°C. From -5/8 to -6/8 is -1/8.

Rule: Subtracting a negative (- (-b)) is the same as adding a positive (+ b).

WATCH OUT: To subtract a signed number, ADD ITS OPPOSITE: a - (-b) = a + b (subtracting a negative moves RIGHT on the number line)!

Answer: 14°C; and -1/8

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 10.1 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

1 0 ENTER 4 +/− - • 3 • 4 +/− ENTER • 5 • 8 +/− -

Rewrite subtraction as adding the opposite on paper first. Check on your SC-6 (in [f][LINE] mode or with Two-Dot fractions): Press 10 [ENTER] 4 [+/-] [-] (+14°C) and [.] 3 [.] 4 [+/-] [ENTER] [.] 5 [.] 8 [+/-] [-] (-1/8).

During camp trivia night, Kai's team was sitting at -15 points when the host removed a -25-point penalty, writing -15 - (-25) on the board. Seeing two minus signs in a row made Kai's head spin, and half his team guessed their score dropped to -40. Then Kai thought about what 'removing a penalty' actually means in a game: taking away a 25-point loss is the exact same thing as giving the team +25 points! Rewriting - (-25) as + 25 turned the problem into -15 + 25 = +10 points—the same reason the vertical gap between a +4,800 m peak and a -200 m cave is 4,800 - (-200) = 5,000 m.

SECTION 10.1 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Arctic Temperature Swing: At dawn the temperature is -14°F. By noon it warms up by 22°F. What is the temperature at noon?**

Starter Step 1: Write -14 + 22: since the signs are different, find the difference between 22 and 14 and keep the sign of the larger magnitude.

Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

2 **Submarine Double Dive: A research submarine is at -45 meters and dives another 38 meters deeper. What is its new elevation?**

Starter Step 1: Write -45 + (-38): since both changes are negative (diving deeper), add their magnitudes and keep the negative sign.

Way 1 (Paper Work):



WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

3 **Mountain to Valley Elevation Difference: A mountain peak is at +4,800 meters and a valley cave is at -200 meters. How much higher is the peak than the cave?**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

4 **Penalty Reversal: A trivia team has -15 points, and then the host removes a -25-point penalty. Write a subtraction expression and find the team's new score.**

Way 1 (Paper Work):



CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

5 **Account Ledger: Your bank account balance is -$12.50. You deposit $30.00 and then spend $8.75 on lunch. What is your new balance?**

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

6 **Evaluate -5/8 + 3/4.**

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

7 **Evaluate -2/3 - 5/6.**

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

8 **Scuba Dive Log: A scuba diver starts at -18.4 meters, rises 7.9 meters, and then descends 5.5 meters. What is the diver's final elevation?**

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

9 **Cold Front Drop: The temperature drops from +6.5°C in the afternoon to -8.5°C at midnight. What is the signed change in temperature?**

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

10 **Evaluate -14 - (-20) + (-9) - 5.**

Way 1 (Show Paper Steps): Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

HANDS-ON MODEL: FRACTION STRIPS

Section 10.2 — Practice A: Multiplying and Dividing Rational Numbers

Key Vocabulary: multiplicative sign rules • same signs (positive +) • different signs (negative -) • even/odd negative factors

Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & HANDS-ON MODEL (FRACTION STRIPS)

Sign Rules for × and ÷

SAME Signs → POSITIVE (+)

(+) × (+) = + | (-) × (-) = +

DIFFERENT Signs → NEGATIVE (-)

(+) × (-) = - | (-) × (+) = -

Multi-Factor & Power Sign Rule

EVEN # of negatives → Positive (+)

Ex: $(-5)^2 = (-5) \times (-5) = +25$

ODD # of negatives → Negative (-)

Ex: $(-2)^3 = (-2)(-2)(-2) = -8$

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Sign Rules: Same signs → POSITIVE product or quotient (+ · + = +, - · - = +). Different signs → NEGATIVE product or quotient (+ · - = -).

Textbook Multi-Factor Rule: Count the negative factors—an EVEN number of negatives gives a Positive (+); an ODD number of negatives gives a Negative (-).

WATCH OUT: $(-5)^2 = (-5) \times (-5) = +25$ (positive!), whereas -5^2 without parentheses means $-(5 \times 5) = -25$!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Evaluate 6 × (-4), (-6) × (-4), and $(-5)^2$.

WAY 1: Two-Number Sign Rules (Same = +, Different = -)

6 × (-4) has different signs → -24. (-6) × (-4) has the same signs → +24. $(-5)^2 = (-5) \times (-5)$ has same signs → +25.

WAY 2: Repeated Addition & Factor Sign Count

6 jumps of -4 = -24; taking the opposite for (-6)(-4) gives +24. Count negative signs: 1 negative (odd) → -; 2 negatives (even) → +.

Answer: 6×(-4) = -24, (-6)×(-4) = +24, $(-5)^2$ = +25

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 10.2 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

6 [ENTER] 4 [+/-] [×] 5 [+/-] [x^2]

Apply the sign rules (same signs = +, different signs = -) on paper first. Check on your SC-6:

Press 6 [ENTER] 4 [+/-] [×] (-24) and 5 [+/-] [x^2] (+25).

SECTION 10.2 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Evaluate: (a) 7 × (-8) and (b) (-7) × (-8).**

Starter Step 1: Multiply 7 × 8 first; remember that different signs give a NEGATIVE product and same signs give a POSITIVE product.

Way 1 (Paper Work):

Sign Rule: [ ] Same(+) [ ] Diff(-)

Magnitude |a|×|b| = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Evaluate: (a) -45 ÷ 9 and (b) -45 ÷ (-9).**

Starter Step 1: Divide 45 ÷ 9 first; remember that different signs give a NEGATIVE quotient and same signs give a POSITIVE quotient.

Way 1 (Paper Work):

Sign Rule: [ ] Same(+) [ ] Diff(-)

Magnitude |a|×|b| = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Evaluate $(-5)^2$. Explain why the result is positive.**

Way 1 (Paper Work):

Sign Rule: [ ] Same(+) [ ] Diff(-)

Magnitude |a|×|b| = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Evaluate $(-2)^3$. Explain why the cube of a negative number is negative.**

Way 1 (Paper Work):

Sign Rule: [ ] Same(+) [ ] Diff(-)

Magnitude |a|×|b| = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Evaluate: (-3.5) × (-2.4).**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Evaluate: -14.4 ÷ 1.2.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Evaluate: (-2/3) × (9/10).**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Evaluate: -5/6 ÷ (-5/12).**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Evaluate the triple product: (-4) × (-5) × (-6).**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Evaluate using the Order of Operations: (-18) × 4 - (-36) ÷ (-9).**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: FRACTION STRIPS

Section 10.2 — Practice B: Modeling Real Life (Multiplying and Dividing Rational Numbers)

Textbook Ways: Same/Different Sign Rules | Repeated Addition / Equal Number-Line Jumps **Name:** _____ **Date:** _____ **Score:** ____ / 100

WORKED EXAMPLE — DIVIDING SIGNED RATIONAL NUMBERS (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: A research submersible descends to -5/6 kilometer at a rate of -5/12 kilometer per minute. How many minutes did the descent take?

WAY 1: Same-Sign Rule + Multiply by the Reciprocal

Negative ÷ Negative = Positive (+). Multiply by the reciprocal: (-5/6) × (-12/5) = +(5 × 12)/(6 × 5) = +12/6 = 2 minutes.

WAY 2: Common Denominator Division (Twelfths)

Rewrite -5/6 as -10/12 km. Divide numerators: (-10 twelfths) ÷ (-5 twelfths/min) = (-10) ÷ (-5) = +2 minutes.

Check: (2 minutes) × (-5/12 km/min) = -10/12 km = -5/6 km.

WATCH OUT: $(-5)^2 = (-5) \times (-5) = +25$ **(positive!), whereas** -5^2 **without parentheses means** $-(5 \times 5) = -25$**!**

Answer: 2 minutes

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 10.2 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

• 5 • 6 +/− ENTER • 5 • 1 2 +/− ÷

Divide (-5/6) ÷ (-5/12) on paper first (negative ÷ negative = positive). Check on your SC-6 with Two-Dot fractions: Press [.] 5 [.] 6 [+/-] [ENTER] [.] 5 [.] 12 [+/-] [÷] to confirm +2 minutes.

Kai had no trouble seeing why 8 seconds of diving at -15 feet per second lands a submarine at -120 feet (8 × -15 = -120), because 8 groups of a 15-foot drop is a 120-foot drop. What bothered him was why a negative times a negative—like (-4) × (-10)—had to be positive (+40). Worse still, he sometimes mixed up adding two negatives (-4 + -10 = -14) with multiplying two negatives! Thinking of (-4) × (-10) as a referee canceling (-4) penalties of -10 points each made the difference click: combining two debts gives a deeper debt (-14), whereas removing 4 ten-point penalties boosts your score by +40.

SECTION 10.2 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Submarine Descent Rate: A submarine descends at a constant rate of -15 feet per second for 8 seconds. What is its change in elevation?**

Starter Step 1: Multiply the rate (-15 ft/s) by the time (8 s); since the signs are different, check whether the product is positive or negative.

Way 1 (Paper Work):

Sign Rule: [ ] Same(+) [ ] Diff(-)

Magnitude |a|×|b| = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Canceling Game Penalties: In a trivia game, each penalty is -10 points. If the referee cancels (-4) penalties of -10 points each, write a multiplication expression to find your score change and explain why it is positive.**

Starter Step 1: Write (-4) × (-10); apply the Same-Signs Rule for multiplying two negative numbers.

Way 1 (Paper Work):

Sign Rule: [ ] Same(+) [ ] Diff(-)

Magnitude |a|×|b| = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Hourly Temperature Change: Over 6 hours, the temperature changed by a total of -27°F. What was the mean temperature change per hour?**

Way 1 (Paper Work):

Sign Rule: [ ] Same(+) [ ] Diff(-)

Magnitude |a|×|b| = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Evaluate** $(-4)^2$ **and** $(-4)^3$**. Explain why one power is positive and the other is negative.**

Way 1 (Paper Work):

Sign Rule: [ ] Same(+) [ ] Diff(-)

Magnitude |a|×|b| = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Multiply (-3/4) × (-8/9). Write the product in simplest form.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Divide (-7/8) ÷ (7/16). Write the quotient in simplest form.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Evaluate (-2.5) × (-1.6) × (-3).**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Glacier Retreat: A mountain glacier changed in length by -14.4 meters over 4.5 years. What was the average yearly change in length?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Evaluate (-2) × (-3) × (-4) × (-5).**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Evaluate** $(-6)^2 - 4 \times (-5) + (-18) \div (-3)$**.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

HANDS-ON MODEL: PLACE VALUE BLOCKS

Section 10.3 — Practice A: Solving Two-Step Equations (ax + b = c)

Key Vocabulary: two-step equation (ax + b = c) • reverse order of operations • undo addition/subtraction first • undo multiplication/division second

Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & HANDS-ON MODEL (PLACE VALUE BLOCKS)

Two-Step Equation Balance Scale: Solve 3x + 7 = 28 in 2 Steps

Step 1: Undo + or - FIRST

Remove 7 loose cubes from both pans:

3x + 7 - 7 = 28 - 7

3x = 21

Step 2: Undo × or ÷ SECOND

Divide 21 cubes equally into 3 x-boxes:

3x ÷ 3 = 21 ÷ 3

x = 7 (Check: 3(7)+7 = 28!)

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Two-Step Algorithm (ax + b = c): 1. Undo addition or subtraction FIRST to isolate the variable term ax. 2. Undo multiplication or division SECOND to isolate x.

Textbook Verification: Substitute your solution x back into the original two-step equation and follow PEMDAS to confirm Left Side = Right Side.

WATCH OUT: In two-step equations (ax + b = c), undo addition/subtraction FIRST, then undo multiplication/division SECOND!

WORKED EXAMPLE — 2 TEXTBOOK WAYS FOR DOUBLE POINTS (+20)

Problem: Solve 3x + 7 = 28 and check your solution.

WAY 1: Two-Step Inverse Operations (Reverse PEMDAS)

In 3x + 7 = 28: Step 1: Subtract 7 from both sides → 3x = 21. Step 2: Divide both sides by 3 → x = 7.

WAY 2: Balance Scale / Part-Whole Bar Model (With Blocks)

A bar of 28 splits into three x-parts and a 7-part. Take away 7 (leaves 21 for the three x-parts), then divide 21 ÷ 3 = 7.

Answer: x = 7 (Step 1: 3x = 21) 1 Way = +10 | 2 Ways = +20!

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 10.3 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

2 8 ENTER 7 - 3 ÷ 2 8 ⇅ 3 × 7 + ≠

Undo + 7 first and × 3 second on paper (x = 7). Check on your SC-6:

Press 28 [ENTER] 7 [-] 3 [÷] (7), then run the Equality Test 28 [ENTER] 3 [ENTER] 7 [×] 7 [+] [≠]—it flashes NO to confirm 3(7) + 7 = 28!

SECTION 10.3 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 Solve 4x + 9 = 45. Show the result of Step 1 (4x = ?) and the final value of x.

Starter Step 1: Undo + 9 FIRST by subtracting 9 from both sides → Step 2: Undo 4x SECOND by dividing both sides by 4.

Way 1 (Paper Work):

1. Undo +/-: ax = _____
2. Undo ×/÷: x = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

2 Solve 5x - 12 = 38.

Starter Step 1: Undo - 12 FIRST by adding 12 to both sides → Step 2: Undo 5x SECOND by dividing both sides by 5.

Way 1 (Paper Work):

1. Undo +/-: ax = _____
2. Undo ×/÷: x = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

3 Solve 6y + 11 = 59.

Way 1 (Paper Work):

1. Undo +/-: ax = _____
2. Undo ×/÷: x = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

4 Solve 8n - 15 = 49.

Way 1 (Paper Work):

1. Undo +/-: ax = _____
2. Undo ×/÷: x = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

5 Solve m ÷ 4 + 6 = 15.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

6 Solve k ÷ 3 - 5 = 7.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

7 Solve -3x + 14 = 2.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

8 Solve 2.5w + 8 = 23.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

9 Appliance Repair Call: A repair technician charges a $35 service visit fee plus $25 per hour of labor. If the total bill is $160, write and solve a two-step equation to find how many hours h the technician worked.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

10 Skateboard Savings Plan: You already have $45 saved toward a $120 custom skateboard and save $15 each week. Write and solve a two-step equation to find how many weeks w it will take to reach $120.

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) Ans: _____

HANDS-ON MODEL: PLACE VALUE BLOCKS

Section 10.3 — Practice B: Modeling Real Life (Solving Two-Step Equations (ax + b = c))

Textbook Ways: Two-Step Inverse Operations | Balance Scale / Bar Model Unwrapping Name: _____ Date: _____ Score: ____ / 100

WORKED EXAMPLE — TWO-STEP EQUATION WITH A NEGATIVE RATE (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: The temperature is 14°F and drops 3°F each hour. After how many hours will the temperature reach 2°F?

WAY 1: Write & Solve -3x + 14 = 2 Using Inverse Operations

Step 1: Subtract 14 from both sides: -3x = 2 - 14 = -12. Step 2: Divide both sides by -3: x = (-12) ÷ (-3) = 4 hours.

WAY 2: Total Change ÷ Rate of Change (or Countdown Table)

Total drop = 14°F to 2°F = 12°F drop. At 3°F per hour, 12 ÷ 3 = 4 hours (14° → 11° → 8° → 5° → 2°).

Check: -3(4) + 14 = -12 + 14 = 2°F.

WATCH OUT: In two-step equations (ax + b = c), undo addition/subtraction FIRST, then undo multiplication/division SECOND!

Answer: 4 hours

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 10.3 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

2 ENTER 1 4 - 3 +/− ÷ 2 ↑↓ 3 +/− × 1 4 + ≠

Subtract 14 first, then divide by -3 on paper (h = 4). Check on your SC-6:
Press 2 [ENTER] 14 [-] 3 [+/-] [÷] (4), then test 2 [ENTER] 14 [ENTER] 3 [ENTER] 4 [×] [-] [≠]—it flashes NO to confirm 14 - 3(4) = 2!

At the go-kart track, admission was $8 plus $6 per lap r, and Leo had $38 to spend (6r + 8 = 38). At first he tried dividing 38 by 6 right away, got a messy fraction, and felt stuck. Then Maya reminded him of how you get dressed in the morning versus how you undress at night: you put your socks on first and your shoes on second, so to undo it you have to take your shoes off FIRST and your socks off SECOND! In 6r + 8 = 38, r was multiplied by $6 first and $8 was added second—so subtracting the $8 admission fee first (6r = 30) and dividing by $6 second gave a clean 5 laps.

SECTION 10.3 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Go-Kart Track: Admission to a go-kart track is $8 plus $6 for each lap r. If you spend $38 in all, how many laps did you drive?**

Starter Step 1: Write 6r + 8 = 38 → Step 1: Subtract the $8 admission from both sides first → Step 2: Divide by $6 per lap.

Way 1 (Paper Work):

1. Undo +/-: ax = _____
2. Undo ×/÷: x = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **3D Print Shop Quote: A print shop charges a $12 setup fee plus $4 per hour h of printing. If your total bill is $48, how many hours did the print take?**

Starter Step 1: Write 4h + 12 = 48 → Step 1: Subtract the $12 setup fee from both sides first → Step 2: Divide by $4 per hour.

Way 1 (Paper Work):

1. Undo +/-: ax = _____
2. Undo ×/÷: x = _____

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Bicycle Savings Plan: You have $35 saved and save $15 per week w to buy a $140 bicycle. How many weeks will it take to save enough money?**

Way 1 (Paper Work):

1. Undo +/-: ax = _____
2. Undo ×/÷: x = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Bookstore Gift Card: You buy 4 identical hardcover books and use a $10 gift card, paying $46 out of pocket. What was the price b of each book?**

Way 1 (Paper Work):

1. Undo +/-: ax = _____
2. Undo ×/÷: x = _____

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Solve -4x + 18 = -6, and check your solution.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Solve (n ÷ 5) + 9 = 16, and check your solution.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Scuba Tank Pressure: A scuba air tank starts at 3,000 psi and uses 40 psi per minute m until 600 psi remains. How many minutes did the dive last?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Solve (2/3)x - 5 = 11, and check your solution.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Rectangle Perimeter: A rectangle has a length of 18 cm and a perimeter of 56 cm. Write and solve an equation to find its width w.**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Taxi Fare: A taxi ride costs a $3.50 base fare plus $2.25 per mile m. If the total fare is $21.50, how many miles was the ride?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 10.4 — Practice A: Theoretical Probability, Combinations & Permutations

Key Vocabulary: experiment • outcomes • event • sample space • theoretical probability P(E) • complement P(not E) • tree diagram • Fundamental Counting Principle

Name: _____ Date: _____ Score: ____ / 100

TEXTBOOK CONCEPT & VISUAL DIAGRAM

Probability Ratio



P(Blue) = 4/10
= 2/5 = 40%
P(not E) = 1 - P(E)

P(E) = Favorable ÷ Total

Counting: Permutations vs. Combinations

Permutation (nPr) — Order MATTERS!
1st/2nd/3rd of 10: 10 × 9 × 8 = 720

Combination (nCr) — Order Does NOT Matter!
3-person group of 10: 720 ÷ 3! = 120

HOW THE TEXTBOOK & VISUAL MODEL WORK TOGETHER:

Textbook Theoretical Probability: P(event) = Favorable Outcomes ÷ Total Outcomes (from 0 to 1, or 0% to 100%), and P(not E) = 1 - P(E).

Textbook Counting & Permutations vs. Combinations: Fundamental Counting Principle multiplies choices; Permutations nPr (order matters) vs. Combinations nCr (order doesn't matter).

WATCH OUT: Ask 'Does order matter?' — awarding 1st/2nd/3rd medals is a Permutation (nPr); choosing a group/team is a Combination (nCr)!

WORKED EXAMPLE — 3 TEXTBOOK WAYS (ANY 2 WAYS = +20 PTS)

Problem: A 10-sector spinner has 4 Blue, 3 Gold, and 3 Red sectors. Find P(Blue) as a fraction, decimal, and percent, and find how many ways 5 trophies can be arranged in a row (5!).

WAY 1: Theoretical Probability Ratio P(E) = Favorable / Total
Spinner (4 Blue, 3 Gold, 3 Red = 10 sectors): P(Blue) = 4/10 = 2/5 = 40%. P(not Blue) = (3 + 3)/10 = 6/10 = 3/5 = 60%.

WAY 2: Complement Rule P(not E) = 1 - P(E)
Subtract P(Blue) from 1 whole (100%): P(not Blue) = 1 - 2/5 = 3/5 (or 100% - 40% = 60%).

WAY 3: Fundamental Counting Principle (5 Trophies in a Row)
Multiply choices for each spot: 5 choices for 1st × 4 for 2nd × 3 for 3rd × 2 for 4th × 1 for 5th = 120 arrangements.

Answer: P(Blue) = 2/5 = 0.4 = 40%; 5! = 120 ways

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 10.4 PRACTICE A (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

[•] [4] [•] [1] [0] [g] [SIMP] [5] [x!]

Write P(event) = favorable / total and simplify on paper first. Check on your SC-6:
Press [.] 4 [.] 10 [g][SIMP] to confirm 2/5, press 5 [x!] to confirm 5! = 120, and explore rolls with [DICE] or [f][1..x]!

SECTION 10.4 PRACTICE A — KEY IDEAS & MULTI-METHOD PRACTICE

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **Not Rolling a Six: You roll a fair 6-sided die once. What is the probability of rolling a 5, and what is the probability of NOT rolling a 6?**

Starter Step 1: A fair die has 6 equally likely outcomes: count how many outcomes are favorable for P(5) and for P(not 6).

Way 1 (Paper Work):

P(E) = Favorable ___ / Total ___

Simp = ___/___ = _____%

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Complementary Weather Probability: A weather forecast states there is a 0.35 (35%) probability of rain today. What is the probability that it will NOT rain today?**

Starter Step 1: An event and its complement always add to 1 whole (1.00): subtract 0.35 from 1.00 to find P(no rain).

Way 1 (Paper Work):

P(E) = Favorable ___ / Total ___

Simp = ___/___ = _____%

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Double Sixes & High Sums: You roll two fair 6-sided dice (36 equally likely outcomes). What is the probability of rolling double sixes (6, 6), and what is the probability of rolling a sum of 11 or 12?**

Way 1 (Paper Work):

P(E) = Favorable ___ / Total ___

Simp = ___/___ = _____%

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Bookshelf Lineup: In how many different left-to-right orders can 4 different books be arranged on a shelf?**

Way 1 (Paper Work):

P(E) = Favorable ___ / Total ___

Simp = ___/___ = _____%

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Track Finals: Six runners are in the finals of the 100-meter dash. In how many different orders can the 6 runners finish 1st through 6th?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **How many different 3-person project teams can be chosen from 8 students when order does NOT matter?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **How many different ways can 1st, 2nd, and 3rd place ribbons be awarded among 8 science fair finalists when order DOES matter?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Triple Coin Toss: You flip a fair coin 3 times in a row. What is the probability of getting Heads on all 3 flips, written as a fraction and as a percent?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Consecutive Matching Socks: A drawer contains 6 blue socks and 4 gray socks (10 socks total). If you pull out 2 socks in a row without replacement, what is the probability that both socks are blue?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Committee vs. Officers: Compare choosing a 3-person cleanup team from 10 students (where order does not matter) vs. electing a President, Vice President, and Secretary from 10 students (where order matters).**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

TEXTBOOK MODEL: PAPER / GRAPH / FORMULA

Section 10.4 — Practice B: Modeling Real Life (Theoretical Probability, Combinations & Permutations)

Textbook Ways: Probability Ratio & Complement Rule | Tree Diagram / Outcome List | Fundamental Counting Principle Name: _____ Date: _____ Score: _____ / 100

WORKED EXAMPLE — ORDERED ARRANGEMENTS VS. UNORDERED GROUPS (2 TEXTBOOK WAYS = +20 PTS DOUBLE)

Problem: Compare awarding 1st, 2nd, and 3rd place ribbons to 10 runners (order matters) with choosing a 3-person relay crew from 10 runners (order does not matter).

WAY 1: Fundamental Counting Principle (Order Matters)

For 1st, 2nd, and 3rd place ribbons, multiply choices: 10 choices for 1st × 9 for 2nd × 8 for 3rd = 720 ordered ways.

WAY 2: Divide Out Duplicate Orders When Order Does Not Matter

Any 3 chosen runners can be arranged in 3 × 2 × 1 = 6 orders. Divide 720 ordered ways by 6: 720 ÷ 6 = 120 relay crews.

Rule: Order matters → Permutation (720); Order does not matter → Combination (720 ÷ 6 = 120).

WATCH OUT: Ask 'Does order matter?' — awarding 1st/2nd/3rd medals is a Permutation (nPr); choosing a group/team is a Combination (nCr)!

Answer: 720 ordered podiums; 120 unordered 3-runner crews

CHECK YOUR WORK ON THE SC-6 CALCULATOR — SECTION 10.4 PRACTICE B (SOLVE BY HAND FIRST, THEN VERIFY!)

Worked Example SC-6 Check Keystrokes:

1 0 ENTER 3 f $_nC_r$ 1 0 ENTER 3 g $_nP_r$

Decide whether order matters first! Then check on your SC-6:
Press 10 [ENTER] 3 [g][nPr] to confirm 720 ordered podiums (permutations), and 10 [ENTER] 3 [f][nCr] to confirm 120 unordered crews (combinations)!

When Maya's coach needed to pick 3 players from 7 teammates, Maya couldn't understand why picking a 1st, 2nd, and 3rd place batting order had 210 possibilities while picking a 3-person cleanup crew from the same 7 kids had only 35! She wrote out three players—Ali, Ben, and Cam—and suddenly saw why: in a batting order, Ali-Ben-Cam and Ben-Ali-Cam are different lineups (6 different orders for the same 3 kids!), whereas on a cleanup crew, Ali, Ben, and Cam are the exact same group of 3 people no matter whose name is called first. Dividing 210 by those 6 duplicate orders (3! = 6) explained why combinations (35) are six times smaller than permutations (210).

SECTION 10.4 PRACTICE B — MODELING REAL LIFE & APPLICATIONS

100-PT GOAL: Any 5 in 2 Ways (+20) OR All 10 in 1 Way (+10)!

1 **20-Sided Die: You roll a fair 20-sided die numbered 1 to 20. What is the theoretical probability of rolling a multiple of 5? Write your answer as a simplified fraction, a decimal, and a percent.**

Starter Step 1: List the multiples of 5 from 1 to 20 to count favorable outcomes, then divide by the 20 total outcomes and simplify.

Way 1 (Paper Work):

P(E) = Favorable ___ / Total ___

Simp = ___/___ = _____%

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

2 **Complementary Event: The weather forecast says there is a 35% chance of rain tomorrow. What is the probability that it does not rain tomorrow?**

Starter Step 1: Rain and no rain are complementary events that add to 100%: subtract 35% from 100%.

Way 1 (Paper Work):

P(E) = Favorable ___ / Total ___

Simp = ___/___ = _____%

WARM-UP • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

3 **Kea Parrot Token Jar: Researchers show a kea parrot a jar containing 6 black (reward) tokens, 5 green tokens, and 4 white tokens. What is P(black) and what is P(not white) in simplest form?**

Way 1 (Paper Work):

P(E) = Favorable ___ / Total ___

Simp = ___/___ = _____%

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

4 **Custom Skateboard Choices: A skate shop offers 5 deck designs, 4 truck colors, and 6 wheel sets. How many different custom skateboard combinations are possible?**

Way 1 (Paper Work):

P(E) = Favorable ___ / Total ___

Simp = ___/___ = _____%

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

5 **Playlist Shuffle: You put 5 songs on a playlist. In how many different orders can all 5 songs be played without repeating?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

6 **Batting Order: A softball coach chooses 3 players from a group of 7 to bat 1st, 2nd, and 3rd. How many different 3-batter orders are possible?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

7 **Science Lab Trio: A teacher chooses a group of 3 lab partners from 7 students. Since order does not matter, how many different 3-student groups are possible?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

8 **Smoothie Two-Fruit Blend: A smoothie shop has 8 kinds of fruit, and you pick 2 different fruits to blend together. How many 2-fruit combinations can you choose?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CORE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

9 **Expected Outcomes: A game spinner lands on gold 3/8 of the time. If you spin it 160 times, how many times should you expect it to land on gold?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

10 **Coin & Die Compound Event: You flip a fair coin and roll a fair 6-sided die. How many outcomes are in the sample space, and what is the probability of getting Heads and a 6?**

Way 1 (Show Paper Steps):

Way 2 (2nd Method / Check):

CHALLENGE • [ ] +10 (1 Way) [ ] +20 (2 Ways) **Ans:** _____

Appendix A: Middle School Math Formulas & Reference Sheet (Units 1–10)

UNITS 1–2: FACTORS, FRACTIONS & DECIMALS

• Division Check Identity: Dividend = (Divisor × Quotient) + Remainder, with 0 ≤ Remainder < Divisor. Verify on SC-6: [n] [ENTER] [d] [g][÷R].

• GCF & LCM Product Law: For any two positive integers a and b, GCF(a, b) × LCM(a, b) = a × b. Verify on SC-6: [a] [ENTER] [b] [GCF] and [f][LCM].

• Fraction Multiplication & Division: (a/b) × (c/d) = (a·c)/(b·d); (a/b) ÷ (c/d) = (a/b) × (d/c) (Multiply by the Reciprocal!).

UNIT 3: RATIOS, RATES, PERCENTS & MEASUREMENT CONVERSIONS

• Percent Proportion & Equation: Part / Whole = Percent / 100 → Part = (Percent / 100) × Whole. Percent Change = (|New - Old| / Old) × 100%.

• Key Fraction-Decimal-Percent Benchmarks: 1/8 = 0.125 = 12.5% | 1/5 = 0.20 = 20% | 1/4 = 0.25 = 25% | 1/3 ≈ 0.333 = 33.3% | 3/8 = 0.375 = 37.5% | 1/2 = 0.5 = 50% | 3/4 = 0.75 = 75%.

• Customary & Metric Conversions: 1 ft = 12 in | 1 yd = 3 ft | 1 mi = 5,280 ft | 1 in = 2.54 cm | 1 m = 100 cm = 1,000 mm | 1 km = 1,000 m | 1 lb = 16 oz | 1 kg ≈ 2.205 lb | 1 gal = 4 qt = 8 pt = 16 c.

UNITS 4, 6 & 7: ALGEBRAIC EXPRESSIONS, INTEGERS, EQUATIONS & INEQUALITIES

• Distributive, Commutative & Associative Properties: a(b + c) = ab + ac; a + b = b + a; ab = ba; (a + b) + c = a + (b + c).

• Signed Integer Rules: Adding an opposite gives zero: a + (-a) = 0 (1 Zero-Pair). Subtracting a number is adding its opposite: a - b = a + (-b). Product/quotient of same signs is (+); opposite signs is (-).

• Coordinate Plane Distance: Horizontal distance = $|x_2 - x_1|$; Vertical distance = $|y_2 - y_1|$; Reflection across x-axis: (x, y) → (x, -y); across y-axis: (x, y) → (-x, y).

UNITS 5 & 8: 2D POLYGON AREA, ELAPSED TIME & 3D SURFACE AREA / VOLUME

• 2D Polygon Areas: Rectangle A = l·w, P = 2(l+w) | Parallelogram A = b·h | Triangle A = (1/2)·b·h | Trapezoid A = (1/2)·(b_1 + b_2)·h | Circle C = 2·π·r = π·d, A = π·r^2.

• 3D Solid Volumes & Nets: Rectangular Prism V = l·w·h = B·h, SA = 2(lw + lh + wh) | Triangular Prism V = B·h | Pyramid SA = B + Sum(Lateral Triangle Areas) | Cylinder V = π·r^2·h | Cone V = (1/3)·π·r^2·h.

• Base-60 Elapsed Time: 1 hr = 60 min = 3,600 sec. Convert H.MMSS to decimal hours on SC-6 with [f][→HR], and back with [g][→HMS].

UNITS 9 & 10: STATISTICAL MEASURES (MEAN, MEDIAN, IQR, MAD) & PROBABILITY

• Center & Spread: Mean (x-bar) = (Sum of values) / n | Median = middle value of ordered list | Range = Max - Min | IQR = Q_3 - Q_1 | Mean Absolute Deviation (MAD) = (Sum $|x_i$ - x-bar|) / n.

• Theoretical & Compound Probability: P(Event) = (Favorable Outcomes) / (Total Possible Outcomes), 0 ≤ P(E) ≤ 1; P(not E) = 1 - P(E); Independent events: P(A and B) = P(A) × P(B).

Appendix B: SC-6 Keycap Directory & 4-Register Stack Guide

THE 4-FLOOR NUMBER ELEVATOR (REGISTERS T, Z, Y, X) — ZERO PARENTHESES NEEDED

- Floor 1 (Register X — Bottom Line): Shows the number you are typing right now, or the answer to the operation you just pressed.
- Floor 2 (Register Y — Upper Line): Holds your first number after you press [ENTER]. Pressing [+], [-], [×], or [÷] combines Y with X.
- **Floors 3 & 4 (Registers Z & T): Hold extra numbers during multi-step problems. Press [4-LV] to view all 4 floors at once!**

KEYCAP / SHIFT	FUNCTION NAME	EXAMPLE KEYSTROKES	WHAT IT DOES ON THE SC-6
[ENTER]	Stack Lift / Push	1 4 ENTER 6 +	Pushes X up to Y so you can type the 2nd number; [ENTER][+] doubles X.
[g] [÷R]	Quotient & Remainder	2 3 ENTER 4 g ÷R	Divides Y by X, placing whole Quotient in Y (5) and Remainder in X (3).
[g] [FACT]	Prime Factorization	2 4 g 人	Decomposes whole number into prime powers ($2^3 \times 3^1$); [PRIME] tests primality.
[GCF] / [f][LCM]	GCF & LCM	2 4 ENTER 3 6 GCF	Computes Greatest Common Factor (12); press [f][LCM] for Least Common Multiple (72).
[.] n [.] d	Two-Dot Fraction	• 2 • 3 ENTER • 1 • 4 +	Enters proper fraction 2/3 or mixed number w[.]n[.]d; [g][SIMP] simplifies step-by-step.
[FDISP]	Fraction / Dec / %	FDISP	Cycles exact fraction (3/4) = decimal (0.75) = percent (75%) on the LCD.
[g] [a:b]	Ratio Simplifier	1 2 ENTER 8 g a:b	Simplifies ratio Y:X to lowest terms (3:2); press [10F] to step through ratio table.
[≠] / [f][>]	Equality & Compare	4 1 ENTER 4 ENTER 9 × 5 + ≠	Tests Y ≠ X (flashes NO when both sides are equal!). [f][>], [g][<], [f][≥], [g][≤].
[f] [LINE]	Number Line Mode	f ↔ 4 +/− 7 +	Animates signed hops right (+) and left (-) along a visual number line on screen.
[f][RECT] / [g][TRI]	2D Polygon Area	1 2 ENTER 8 f ■	Computes rectangle area (96) & perimeter (40) or triangle area (1/2)·b·h.
[f][PRISM] / [g][CONE]	3D Solid Volume	1 2 ENTER 8 ENTER 5 f ⬢	Computes 3D rectangular prism volume l·w·h (480) or cone volume (1/3)·B·h.
[Σ+] / [g][x-bar] / [IQR]	Statistics Suite	g CLΣ 1 8 Σ+ 2 4 Σ+ g x̄	Accumulates data list; computes Mean [g][x-bar], Median [f][MED], [IQR], and [g][MAD].
[DICE] / [f][nCr]	Probability & Combos	6 ENTER 2 f $_nC_r$	Simulates fair dice/coin trials [DICE] and computes combinations nCr (15) & nPr.
[f] [?] (Tutorial)	Built-In Math Quiz	f ?	Launches interactive step-by-step Tutorial & Quiz Mode for 100-Point Mastery!

Appendix C: Hands-On Classroom Manipulatives & Visual Modeling Guide

MANIPULATIVE MODEL	UNITS	HOW TO MODEL TEXTBOOK CONCEPTS WITH THE BLOCKS
Prime Factor Towers & Venn Tray	**Unit 1**	Stack proportional prime factor blocks (2, 3, 5, 7) to model exponents and place shared prime blocks in the Venn intersection for GCF & LCM.
Place Value Whole-Number Tray	**Unit 1**	Model multi-digit division quotients and remainders across base-10 place columns and hold partial sub-results on stack shelf Y.
Fraction Strips & Comparator	**Unit 2**	Compare fraction strips side-by-side inside the 84 mm twin comparator channel to find common denominators and model fraction division.
Place Value Decimal Tray	**Unit 2**	Trade ones, tenths, hundredths, and thousandths across decimal columns before verifying exact decimal operations on the stack.
Fraction Tape Diagrams & Ratios	**Unit 3**	Use equal-length fraction strips as physical Tape Diagram bars to compare part-to-part vs. part-to-whole ratios, unit rates, and percents.
Place Value & Prime Factor Models	**Unit 4**	Treat variable powers (x^2, x, 1) as distinct place columns that cannot be combined, and use Prime Factor blocks to factor out the GCF.
Fraction & Place Value + Grids	**Unit 5**	Decompose parallelograms, triangles, and trapezoids on grid diagrams, accumulate composite areas with [Σ+], and regroup base-60 time with [CLK].
Place Value, Number Line & Scale	**Units 6, 7**	Model signed opposites and zero pairs along the [f][LINE] number line, and solve one-step equations as balanced scales verified with [≠].
Place Value Unit Cubes & 2D Nets	**Unit 8**	Pack unit blocks into rectangular prisms to see V = l·w·h across 3 stack levels (Z, Y, X) and sum unfolded 2D face pairs with [Σ+].
Place Value Towers & Fraction Ratios	**Units 9, 10**	Level unequal Place Value stacks to find the fair-share Mean, divide ordered data into 4 quarters for IQR, and model probability ratios.

COMPANION RESOURCES & OPEN CURRICULUM LIBRARY

- **Spiral-Bound Lay-Flat Student Workbooks & Unified Teacher's Guide: https://www.lulu.com/spotlight/stackcalc**
- Illustrated Companion Book — From Blocks to the Stack (Amazon): https://link.amazon/B0f5cP0Ur
- Classroom Manipulative Sets & Hardware: https://stackcalcmath.etsy.com | https://stackcalc.io
- Printable Individual Chapter Packets (Units 1–10): https://stackcalc.io/curriculum/big-ideas-middle-school-sc6-workbook/

Appendix D: 100-Point Mastery Certificate & Graph Scratchpad

STACKCALC LEARNING LAB • BIG IDEAS MATH MIDDLE SCHOOL SUPPLEMENT

CERTIFICATE OF TWO-METHOD MATHEMATICAL MASTERY

This certifies that: _____

has successfully completed the 10-Unit Big Ideas Math Middle School Two-Method Block & SC-6 Supplement—solving problems by hand using multiple textbook pathways (+10 Single-Method & +20 Two-Method Mastery), modeling concepts with hands-on manipulatives, and verifying work on the 4-register StackCalc SC-6 calculator.

[ ] Unit 1 [ ] Unit 2 [ ] Unit 3 [ ] Unit 4 [ ] Unit 5

[ ] Unit 6 [ ] Unit 7 [ ] Unit 8 [ ] Unit 9 [ ] Unit 10

Teacher / Parent Coach Signature: _____ **Date:** _____ **100-Pt Towers Built:** _____

EXTRA 5 MM DOT-GRID MATHEMATICAL SCRATCHPAD (TAPE DIAGRAMS, NUMBER LINES, NETS & COORDINATE GRAPHS)

Appendix E: Student Self-Check Answer Key — Units 1 & 2

Solve by hand first and verify on your SC-6 calculator before checking your final answer here. Full worked setups are in the Teacher's Guide.

UNIT 1: NUMERICAL EXPRESSIONS & FACTORS (BIG IDEAS MATH GRADE 6 — CHAPTER 1)

Section 1.1: Whole Number Operations & Division with Remainders

A1: 5 R 1 A2: 9 R 2 A3: 8 R 2 A4: 9 R 4 A5: 12 R 2

A6: 7 full lines, 5 cm left A7: 117 A8: 9 R 3 (Rem < 8) A9: 19 rovers, 2 tires left A10: 8 tents (7 full + 1 for the 29th explorer)

B1: 8 full packs, 5 cells left B2: 11 figures, 6 tickets left B3: 20 canoes (19 full + 1 with 2 campers) B4: 9 full racks, 2 on last B5: 28 knights, 2 m left

B6: 18 full, 19 caches total B7: 187 marbles B8: 8 full trains, 8 on final B9: 9 pizzas (5 slices left) B10: 11 full crates, 17 apples left

Section 1.2: Powers and Exponents

A1: 5 × 5 = 25 A2: 49 (vs. 14) A3: 3 × 3 × 3 = 27 A4: 125 A5: 32

A6: 1,296 A7: 10,000 (4 zeros) and 1 A8: 16 (2×2×2×2 = 16) A9: 96 kN (3 × 32) A10: 8 m^3 per room; 64 m^3 total

B1: 8^2 = 64 pixels B2: 6^3 = 216 blocks B3: 2^6 = 64 bacteria B4: 4 × 4 × 4 × 4 = 256 players B5: 144 + 25 = 169 ft^2

B6: 5^4 = 625 starlings B7: 14 meters B8: 2^5 = 32 > 5^2 = 25 (by 7) B9: 1,024 bytes B10: 27 + 343 = 370 in^3

Section 1.3: Order of Operations

A1: 11 A2: 4 A3: 6 A4: 84 A5: 1

A6: 20 A7: 2 (24 ÷ 12) A8: 24 A9: $92 (60 + 32) A10: $62 (90 - 28)

B1: 4(9 + 5) = $56 B2: 45 + 4(7) = $73 B3: 101 points B4: 10^2 - 6^2 = 64 m^2 B5: $73

B6: 70 points B7: 10 bracelets per box B8: 42 (÷ and × go left to right) B9: 12 muffins per basket B10: 74

Section 1.4: Prime Factorization

A1: $2^2 \times 3$ A2: $2^2 \times 3^2$ A3: 29 is prime; 91 is composite (7 × 13) A4: $3^2 \times 5$ A5: $2^3 \times 3^2$

A6: $2^2 \times 3 \times 7$ A7: $2 \times 3 \times 5 \times 7$ A8: $2^3 \times 3^2 \times 7$ A9: $7 \times 11 \times 13$ A10: $2^4 \times 3^2 \times 5^2$

B1: 1×48, 2×24, 3×16, 4×12, 6×8; $2^4 \times 3$ B2: 2×3^3 B3: 51 is composite (3 × 17) B4: $2^4 \times 5$ B5: 126

B6: $2^2 \times 5^2$ B7: $2^2 \times 5 \times 7$ B8: $2^4 \times 3^2$ B9: Both give $2 \times 3^2 \times 5$ B10: $2 \times 3 \times 5 \times 7 = 210$

Section 1.5: Greatest Common Factor (GCF) & Least Common Multiple (LCM)

A1: GCF = 6, LCM = 36 A2: GCF = 4, LCM = 24 A3: GCF = 5, LCM = 75 A4: GCF = 7, LCM = 70 A5: 24

A6: 28 A7: GCF = 15, LCM = 225 A8: 60 minutes A9: 180 seconds (3 minutes) A10: 12 bags (3 red, 4 blue per bag)

B1: 14 kits (2 bandages, 3 wipes each) B2: 60 years B3: 24 (3 packs of hot dogs, 2 packs of buns) B4: 16 singers per row B5: 10:12 AM (72 minutes later)

B6: 12 inches B7: GCF = 7, LCM = 42 B8: 60 minutes B9: 15 bouquets (3 gold, 5 silver each) B10: GCF=5, LCM=60; 5×60 = 300 = 15×20

UNIT 2: FRACTIONS & DECIMALS (BIG IDEAS MATH GRADE 6 — CHAPTER 2)

Section 2.1: Equivalent Fractions, Mixed Numbers & LCD Addition

A1: 2/3 A2: 8/3 = 2 2/3 trays A3: 13/5 and 2.6 A4: 3 5/6 pizzas (3 whole, 5 sixths) A5: 5/6

A6: 1 3/20 gallons (23/20) A7: 3 5/6 feet A8: 2 7/12 feet A9: 7/8 climbed; 1/8 remains A10: 5 1/4 pints

B1: 5/8 cup B2: 3/4 complete; 1/4 remains B3: 13/8 and 1.625 in B4: 11/24 foot B5: 3 5/6 pizzas (23/6)

B6: 2 3/4 feet (11/4) B7: 3/5 = 0.6 B8: 1/5 liter (0.2 L) B9: 4 5/12 yd used; 7/12 yd left B10: 4 5/8 miles

Section 2.2: Multiplying Fractions and Mixed Numbers

A1: 1/3 A2: 1/2 A3: 1/3 A4: 6 cups A5: 4

A6: 3 A7: 9 cups A8: 3 1/3 (Less than 5 because 2/3 < 1) A9: 14 7/8 miles (119/8) A10: 1/4 of the input speed

B1: 2/5 of the garden B2: 2/5 of a full battery B3: 4 3/8 cups (35/8) B4: 3 in^2 B5: 4 miles

B6: 2/3 yd^2 B7: 10 gallons B8: 1/8 in^3 B9: 8 ft^2 B10: 1/12 of the day (2 hours)

Section 2.3: Dividing Fractions and Mixed Numbers

A1: 2 A2: 4 A3: 2 A4: 8 A5: 1 7/8 (15/8)

A6: 7/8 cup A7: 2 2/3 bike frames A8: 3 A9: 5 full slats (5 7/37 → 3 1/2 in left) A10: 6 glasses

B1: 6 slices B2: 12 pebbles B3: 6 runners B4: 7 segments B5: 40 layers

B6: 6 sections B7: 2 meters B8: 8 cups B9: 5/12 (12/5 × 5/12 = 1) B10: 12 pieces

Section 2.4: Adding, Subtracting, and Multiplying Decimals

A1: 345.67 mm A2: 17.55 A3: 11.35 A4: 2.7 A5: 18.74

A6: 8.1 (Estimate: 4 × 2 = 8) A7: 0.06 (3/50) A8: 35.79 g (35.7864) A9: $6.65 A10: 161 meters

B1: $41.24 B2: $6.35 B3: 0.55 seconds faster B4: $7.80 B5: 1.19 cm^2

B6: $41.40 B7: 30.35 meters B8: $35.75 B9: 2.5 mm gap B10: $35.00 profit

Section 2.5: Dividing Whole Numbers and Decimals

A1: 4 R 7, 4 7/8, and 4.875 A2: 2.3 A3: 45 quarters A4: 8 A5: 130

A6: 6.3 (6 3/10) A7: $20.80 A8: 24 (24 × 0.2 = 4.8) A9: 6 laps A10: $1.25 per bottle; 16 bottles

B1: 4 R 3 = 4 3/8 = 35/8 = 4.375 B2: 16 quarters B3: $8.65 each B4: 72 layers B5: 15 flowers

B6: 6 miles per hour B7: 18 sections B8: 30 cups B9: 7 tickets B10: 6

Appendix E: Student Self-Check Answer Key — Units 3 & 4

Solve by hand first and verify on your SC-6 calculator before checking your final answer here. Full worked setups are in the Teacher's Guide.

UNIT 3: RATIOS, RATES & PERCENTS (BIG IDEAS MATH GRADE 6 — CHAPTERS 3, 4 & 5 (RATIOS, RATES & PERCENTS))

Section 3.1: Ratios and Tape Diagrams

A1: 2 : 3 A2: 3 : 2 A3: 2 : 5 A4: 18 and 24 (Total = 42) A5: 4 L concentrate, 30 L water

A6: 20 dogs A7: 25 seventh graders A8: 30 electric guitars (12 acoustic) A9: 48 gears and 36 gears A10: 77 coins total (49 gold, 28 silver)

B1: Part-to-Part = 3:2; Part-to-Whole = 3:5 B2: 16 strawberries, 12 bananas B3: 6 cups yellow (21 cups total) B4: 35 secret and 25 open B5: 42 gold coins, 18 silver coins

B6: 3 : 1 B7: 4 : 9 (16 out of 36) B8: 16 : 9 B9: 150g almonds, 100g cashews, 250g raisins B10: 3 : 4

Section 3.2: Using Ratio Tables and Double Number Lines

A1: 4:9, 8:18, 12:27, 16:36 A2: 63 A3: 25 A4: 60 pages A5: 48 ounces

A6: $15 A7: 84 beats A8: Ratios scale by multiplication (e.g., ×3 gives 9:15), not addition A9: 20 cups A10: 40 miles

B1: 45 grams B2: 56 scoops B3: $15 for 10 notebooks B4: 112 beats per minute B5: 260 gallons of sap

B6: 25 pounds B7: 15 cups of flour B8: 75 minutes (1 hr 15 min) B9: 8 inches B10: 30 lemons

Section 3.3: Rates, Unit Rates, and Comparing Rates

A1: 65 mph A2: $9 per hour A3: $1.30 per pound A4: 15 grams per hour A5: Supplier B ($0.17/oz vs. $0.18/oz)

A6: 10-pack ($0.82/pen vs. $0.85/pen) A7: 300 miles A8: Pump A (9 gal/min vs. 8 gal/min) A9: 180 mm A10: Car B by 1 mpg (29 mpg vs. 28 mpg)

B1: 150 miles per hour B2: Box B ($0.70/bar vs. $0.75/bar) B3: Spool B ($1.95 vs. $2.20 per 100g) B4: 15 gal/min; 375 gallons B5: B (57 taps/min vs. A's 55 taps/min)

B6: Car A (28 mpg vs. 27 mpg) B7: $9 per dog → 11 dogs B8: 12-pack ($0.42 vs. $0.45 each) B9: 4 inches per hour B10: 4.5 m/s → 90 meters

Section 3.4: Percents, Fractions, and Decimals

A1: 0.6 and 60% A2: 0.36 and 36% A3: 0.45 and 9/20 A4: 12.5% and 1/8 A5: 270 seats

A6: 3,510 grams A7: 84% A8: 1.5 and 1 1/2 A9: 60%, 0.61, 5/8 A10: 1,400 mAh

B1: 0.85 = 85% B2: 0.36 = 36% B3: 20 + 40 = 60 points B4: 104 students B5: 0.15 and 3/20

B6: 45 students total B7: $9.00 B8: 7/8 = 0.875 = 87.5% B9: $480 goal B10: 58% < 3/5 (60%) < 0.62 (62%)

Section 3.5: Solving Percent Problems, Discounts & Conversions

A1: Discount $17; Sale price $68 A2: Tax $10.20; Total $130.20 A3: +400% ((75-15)÷15 = 400%) A4: 25% decrease A5: 25.4 cm

A6: 20 in A7: $39.20 A8: 72 A9: $144 ($200 → $160 → $144) A10: 81.28 cm and 2.27 kg

B1: Discount = $15; Sale Price = $60 B2: Tip = $9.60; Total = $73.60 B3: Tax = $3.60; Total = $51.60 B4: 30% increase B5: 30% decrease

B6: 16.09 km B7: 44.09 lb B8: 3.79 liters; 128 fl oz B9: $72 (10% is taken off $80, not $100) B10: 68°F

UNIT 4: ALGEBRAIC EXPRESSIONS & PROPERTIES (BIG IDEAS MATH GRADE 6 — CHAPTER 3 / CHAPTER 5 (ALGEBRAIC EXPRESSIONS))

Section 4.1: Writing and Evaluating Algebraic Expressions

A1: 33 A2: 54 A3: 13 A4: 47 A5: 27

A6: 55 A7: 8 A8: 15.1 A9: 8 + 4.50h; $26 A10: 150 cm^2

B1: 5g + 6; $26 B2: 50 - 4r; 14 tickets B3: $225 B4: 82 points B5: 144 feet

B6: 96 in^2 B7: 7p + 4d; $41 B8: 78 B9: 111 minutes (1 hr 51 min) B10: 128 - 75 = 53

Section 4.2: Simplifying Expressions by Combining Like Terms

A1: 9x + 13 A2: 5x + 7 A3: 8y + 6 A4: 8a + 13 A5: $6x^2 + 8x + 7$

A6: $4n^2 + 5n + 10$ A7: 5.3x + 2.7 A8: 53 != 80; 5x + 3 is already in simplest form A9: 6x + 16; 76 meters A10: 9x + 8; 53 cm

B1: 9x + 9 B2: 8x + 12 B3: 4x + 14 B4: $6x^2 + 4x + 11$ B5: 13c + 31

B6: 7x + 19 (coeff = 7, constant = 19) B7: (1/2)n + 13 B8: 4y + 3.5 B9: 6m + 9 (both equal 39 at m=5) B10: $3x^2 + 7$

Section 4.3: The Distributive Property

A1: 140 + 21 = 161 A2: 4x + 20 A3: 18x + 24 A4: 16y - 40 A5: 387

A6: 17x + 6 A7: 12n + 13 A8: 5x + 30 (She forgot to multiply 6 by 5) A9: 14x + 15 (Both equal 43 when x = 2) A10: 6(x + 3.50) = 6x + 21; $60 total

B1: 8(50 + 3) = 400 + 24 = $424 B2: 6(10 - 0.10) = $60.00 - $0.60 = $59.40 B3: 7(x + 8) = 7x + 56 B4: 9(2x + 7) = 18x + 63 B5: 4(t - 3) = 4t - 12 ($48 at t = $15)

B6: 14x + 15 B7: 13x + 22 B8: 3x + 5 B9: 2(2x + 5) + 2(x + 3) = 6x + 16 B10: 12(15) = 180 and 120 + 60 = 180

Section 4.4: Factoring Expressions Using the GCF

A1: 4(3 + 5) A2: 4(2x + 3) A3: 5(3y - 5) A4: 12(2x + 3) A5: 16(2n - 3)

A6: 14(3a + 1) A7: 12(3m - 5) A8: 12(2x + 3) (4 and 6 still share a factor of 2) A9: Height = 14 in.; Width = (2x + 3) in. A10: (9x + 12) feet

B1: 8(2x + 3) → 8 bags B2: 7(3x + 5) → length = 3x + 5 m B3: 4(5x + 9) → side = 5x + 9 in B4: 3(6y - 9) → side = 6y - 9 cm B5: 8(5x + 8)

B6: 18(2s + 3) → 18 player kits B7: 6(2x + 3y + 5) B8: 7(5d - 2) → 5d - 2 per row B9: 24x + 36 = 12(2x + 3) B10: 9(3x - 2); at x=2 both equal 36

Appendix E: Student Self-Check Answer Key — Units 5 & 6

Solve by hand first and verify on your SC-6 calculator before checking your final answer here. Full worked setups are in the Teacher's Guide.

UNIT 5: AREAS OF POLYGONS & ELAPSED TIME (BIG IDEAS MATH GRADE 6 — CHAPTER 4 (AREAS OF POLYGONS))

Section 5.1: Areas of Parallelograms and Rectangles

A1: 135 m^2 A2: 36 m^2 A3: 54.4 m^2 A4: 169 cm^2 A5: 26 $in.^2$

A6: 9 cm A7: 18 ft A8: 60 cm^2 (He multiplied by the slanted side 7.5) A9: 666 ft^2 A10: 168 ft^2

B1: 180 ft^2 B2: 60 ft B3: 14.4 ft^2 B4: 15 inches B5: 216 cm^2

B6: 228 ft^2 B7: 2,520 ft^2 B8: 22 in^2 B9: Both equal 144 cm^2 (same base and perpendicular height) B10: 28 ft^2 × $4 = $112.00

Section 5.2: Areas of Triangles

A1: 750 cm^2 A2: 18 ft^2 A3: 88 cm^2 A4: 52.5 m^2 A5: 13 1/2 $in.^2$ (13.5)

A6: 9 cm A7: 12 m A8: 20 cm^2 (He forgot to divide by 2) A9: 210 ft^2 A10: 1,400 cm^2

B1: 150 ft^2 B2: 14 ft^2 (1 side); 28 ft^2 (both) B3: 141 ft^2 B4: 36 inches B5: 6 feet

B6: 10.125 in^2 (10 1/8 in^2) B7: 54 ft^2 B8: 630 cm^2 (315 cm^2 per wing) B9: 13 in^2 B10: 64 ft^2

Section 5.3: Areas of Trapezoids and Composite Figures

A1: 72 cm^2 A2: 135 ft^2 A3: 92 ft^2 A4: 120 $in.^2$ A5: 29.4 m^2

A6: 56 $in.^2$ A7: 10 cm A8: 88 ft^2 A9: 104 cm^2 (80 + 24 = 104) A10: 251 ft^2

B1: 440 in^2 B2: 40 in^2 per side; 160 in^2 total B3: 84 + 50 = 134 ft^2 B4: 42 cm^2 (1 fin); 126 cm^2 (3 fins) B5: 90 + 48 = 138 in^2

B6: 1,152 - 36 = 1,116 in^2 B7: h = 8 ft B8: 280 + 60 = 340 ft^2 B9: 416 - 200 = 216 m^2 B10: 42 cm^2

Section 5.4: Polygons in the Coordinate Plane & Elapsed Time

A1: 8 units A2: 54 square units A3: 10:58 AM A4: 2.5 hours; 4.75 hours A5: 6 hours 50 minutes

A6: 1 hour 45 minutes A7: 2 hours 35 minutes A8: 24 square units A9: 5.5 hours (5 hr 30 min); $101.75 A10: Perimeter = 420 yd; Area = 10,800 yd^2

B1: W=10, H=6, P=32, Area=60 B2: Base=8, Height=6, Area=24 B3: 12:15 PM B4: 3 hours 35 minutes B5: 3 hours 50 minutes

B6: 3.5 hr × $14 = $49.00 B7: 2 hours 45 minutes B8: 4 hours 30 minutes B9: (3, 10); Area = 36 B10: 180 miles

UNIT 6: INTEGERS, RATIONAL NUMBERS & THE COORDINATE PLANE (BIG IDEAS MATH GRADE 6 — CHAPTER 6)

Section 6.1: Positive and Negative Integers & Opposites

A1: -25; Opposite = +25 A2: +42; and +42 A3: 0 m (Sea level) A4: 0 A5: 0 ft

A6: 0 (0 is 0 units from itself) A7: +7°F A8: 0 (Even par) A9: Floor 3 A10: +$18,800

B1: 0 ft (opposites / additive inverses) B2: -282 ft; opposite is +282 ft B3: +19 yards B4: -28°C; opposite is +28°C B5: +$7

B6: -5 (5 under par) B7: -x = +35; -(-x) = -35 B8: +$10 B9: +35 meters B10: +23

Section 6.2: Comparing and Ordering Integers & Rational Numbers

A1: -5°C (-12 < -5) A2: -9 < -2 A3: -15 < 0 A4: -10, -7, -2, 0, 4 A5: -3/4 < -0.6 (-0.6 is greater)

A6: -2 1/2 < -2.3 A7: -5/8 > -7/8 A8: -4.1 is closer to 0 (further to the right) than -4.5 A9: -1 3/4, -1.5, -0.5, 1 1/4 A10: Deepest: -15.2 m; Shallowest: -14 1/2 m

B1: -11°C > -18°C (Tuesday was warmer) B2: Puffin (-45.8 m) is deeper; -45.25 > -45.8 B3: -2/3 > -5/6 (-4/6 > -5/6) B4: -9, -6, -1, +2 B5: -89°C, -39°C, -6°C, 0°C

B6: Sea turtle (-3.1 > -3.25) B7: Stock B (-1.08% > -1.8%) B8: Examples: -3 1/2 and -3.5 B9: -2, -7/4 (-1.75), -1.6, 3/2 (1.5) B10: a is farther from 0 (e.g., -10 < -2)

Section 6.3: Absolute Value and Magnitude

A1: 125 A2: 30 A3: 8.0 A4: 16 A5: |-12| > |9| (12 > 9)

A6: -24 A7: 32 A8: -|-9|, -6, 0, |-3|, |-8| A9: Seagull is closer by 7 feet (25 < 32) A10: Greater debt: -$45 ($45); Greater balance: -$28

B1: |32| + |-14| = 46 meters B2: -$40 is greater balance; Account A owes more ($65) B3: |-18| + |14| = 32 meters B4: 35 and -35 (outside minus applies after | |) B5: -1.5 cm (|-1.5| = 1.5 cm is smallest)

B6: 42 ÷ 6 + 11 = 18 B7: |25| + |-140| = 165 feet B8: 19 and -19 B9: -|-9| (-9), -6, |5| (5), |-8| (8) B10: 4 5/8

Section 6.4: The Four-Quadrant Coordinate Plane & Reflections

A1: Quadrant IV A2: Quadrant III A3: Quadrant II A4: (-6, 9) A5: (6, -9)

A6: (-3, 7) A7: (8, -4) A8: (0, -9) on y-axis; (-7, 0) on x-axis A9: (-3, 3) in Quadrant II A10: D(-4, -5) in Quadrant III

B1: Cave: II, Lighthouse: I, Shipwreck: III, Pier: IV B2: (0, -7) is on y-axis; (9, 0) is on x-axis B3: (8, -12) in Quadrant IV B4: (9, 5) in Quadrant I B5: Step 1: (-4, 9); Final: (4, 9) in Quad I

B6: (4, -8) in Quadrant IV B7: P: Quadrant II; Q: Quadrant III B8: y-axis (x-coordinate changed sign, y stayed -3) B9: (-5, 5), (-5, -5), and (5, -5) B10: (x, y) is in Quad II; (-x, -y) is in Quad IV

Section 6.5: Distances in the Coordinate Plane

A1: 27 units A2: 11 units A3: 15 units A4: 9 units A5: 10 units

A6: 8 units A7: (4, 3) A8: 13 units (They lie on opposite sides of the y-axis: |-5| + |8| = 13) A9: Perimeter = 34 units, Area = 72 sq units A10: 20 grid units = 10 miles

B1: |-8| + |11| = 19 units B2: 9 units = 90 meters B3: |15| + |-35| = 50 units B4: W = 10, H = 6, Perimeter = 32 units B5: 60 square units

B6: 7.75 miles B7: 5 3/4 units B8: 7 + 10 = 17 units B9: (11, 4) and (-5, 4) B10: Base = 10, Height = 8, Area = 40 $units^2$

Appendix E: Student Self-Check Answer Key — Units 7 & 8

Solve by hand first and verify on your SC-6 calculator before checking your final answer here. Full worked setups are in the Teacher's Guide.

UNIT 7: EQUATIONS & INEQUALITIES (BIG IDEAS MATH GRADE 6 — CHAPTER 7)

Section 7.1: Writing Equations in One Variable & Testing Solutions

A1: x = 8 (5(8)-7 = 33) A2: y = 15 (15÷3 + 4 = 9) A3: x + 14 = 52; x = 38 A4: 6k = 84; k = 14 A5: w - 9 = 17; w = 26

A6: m / 5 = 13; m = 65 A7: 4p + 6 = 54; p = $12 A8: x - 12 = 20; x = 32 A9: 10 + 3m = 28; m = 6 movies A10: b / 3 = 14.50; b = $43.50

B1: x = 8 (5·8 - 7 = 33) B2: 15t + 5 = 65; yes, t = 4 is a solution B3: k - 12 = 29; k = 41 B4: n ÷ 6 = 14; n = 84 B5: 3s = 54; s = 18 cm

B6: m = 15 (100 - 60 = 40) B7: p ÷ 8 = 3; p = 24 slices B8: 45 + d = 120; d = $75 B9: Yes (both sides equal 28) B10: b - 4 = 13; b = 17 years old

Section 7.2: Solving Equations Using Addition or Subtraction

A1: x = 45 A2: y = 92 A3: n = 45 A4: k = 76 A5: w = 54.4

A6: p = 16.65 A7: m = 11/24 A8: a = 4 3/4 A9: m - 18.50 = 26.75; m = $45.25 A10: h + 42.5 = 110; h = 67.5 ft

B1: s + 45 = 210; s = 165 points B2: p - 18 = 67; p = $85 B3: h + 1450 = 5200; h = 3,750 feet B4: f - 64.8 = 185.2; f = 250.0 grams B5: t - 0.48 = 12.65; t = 13.13 seconds

B6: m + 3/8 = 7/8; m = 1/2 cup B7: 2 3/4 + d = 4 1/2; d = 1 3/4 miles B8: b + 38 = 125; b = 87° B9: -40 + d = -115; d = -75 meters B10: L = 44 cm; l ength = 22 cm

Section 7.3: Solving Equations Using Multiplication or Division

A1: x = 17 A2: y = 15 A3: m = 114 A4: k = 101.5 A5: w = 14

A6: x = 24 A7: n = 40 A8: p = 4/5 (0.8) A9: 1.5t = 27; t = 18 minutes A10: 3/5 s = 27; s = 45 students

B1: 12s = 168; s = $14 per shirt B2: p ÷ 9 = 3; p = 27 slices B3: 65t = 325; t = 5 hours B4: 2.5p = 16.25; p = $6.50 per pound B5: s ÷ 4 = 9; s = $36

B6: (3/4)c = 21; c = 28 students B7: (5/6)p = 150; p = $180 B8: 13.5w = 108; w = 8 feet B9: h ÷ 15 = 3.2; h = 48 feet B10: 0.40n = 56; n = 140

Section 7.4: Writing and Graphing Inequalities

A1: x > 13 (Open circle at 13, arrow right) A2: y ≤ 24 (Solid dot at 24, arrow left) A3: n ≥ 13 (Solid dot at 13, arrow right) A4: m < 36 (Open circle at 36, arrow left) A5: h ≥ 48

A6: w ≤ 2,000 A7: Only 6 (At n = 5, 23 > 23 is false) A8: 5 and 6 A9: 8b ≤ 50; at most 6 books ($2 left over) A10: p + 340 ≥ 500; p ≥ 160 points

B1: h ≥ 54 (solid dot at 54, shade right); at least 5 in B2: p + 5 ≤ 12 → p ≤ 7 passengers B3: 22.50 + x ≥ 35 → x ≥ $12.50 B4: 3.50g ≤ 28 → g ≤ 8 games B5: 15r > 240 → r > 16 (at least 17 rows)

B6: b - 25 < 15 → b < 40% B7: 8 and 10 B8: 2c - 30 ≥ 70 → c ≥ 50 cupcakes B9: s ≤ 20 (solid dot at 20, shade left down to 0) B10: 45c + 170 ≤ 800 → c ≤ 14 crates

Section 7.5: Writing Equations in Two Variables ($y = kx$ and $y = x + b$)

A1: y = 8x; y = 80 A2: y = x + 8; y = 28 A3: y = 7x; y = 84 A4: y = x - 4; y = 26 A5: (1, 6), (2, 12), (3, 18), (4, 24)

A6: Independent: t; Dependent: d A7: 292.5 miles A8: y = 4x passes through (0, 0); y = x + 4 crosses at (0, 4) A9: 8 hours A10: c = 15m + 25; $205

B1: y = x + 5 is additive; y = 8m is multiplicative B2: g = 18h (h = indep, g = dep); 81 grams B3: m = z + 4; m = 12, 16, 19 B4: s = 15p; p = 12 sheets B5: y = x - 6; $18, $29, and $44

B6: y = 3.5x; 140 wheel turns B7: y = 9x; at x = 20, y = 180 B8: c = 20g + 7; $127 B9: Multiplicative y = kx (when x = 0, y = k·0 = 0) B10: d = 0.2s; 3 miles away

UNIT 8: SURFACE AREA & VOLUME OF 3D SOLIDS (BIG IDEAS MATH GRADE 6 — CHAPTER 8)

Section 8.1: Three-Dimensional Figures, Nets & Rectangular Prisms

A1: 96 cm^2 A2: 486 $in.^2$ A3: 108 cm^2 A4: 160 $in.^2$ A5: 392 ft^2

A6: 85 m^2 A7: One face = 25 cm^2; s = 5 cm A8: 94 cm^2 (He only summed 3 faces instead of doubling for all 6 faces) A9: 2,100 cm^2 A10: 500 $in.^2$ needed; Yes, with 50 $in.^2$ to spare

B1: 2(160 + 600 + 240) = 2,000 cm^2 B2: 1 face = 81 in^2; Total SA = 486 in^2 B3: 288 + 720 + 360 = 1,368 in^2 B4: SA = 500 in^2; yes (100 in^2 to spare) B5: 120 + 192 + 160 = 472 ft^2

B6: 1 face = 25 cm^2; edge = 5 cm B7: 32 + 8 = 40 ft^2 B8: 2(24 + 60 + 40) = 248 cm^2 B9: 54 cm^2 vs. 216 cm^2 (4 times as great) B10: 160 cm^2

Section 8.2: Surface Areas of Triangular Prisms and Pyramids

A1: 96 cm^2 A2: 224 $in.^2$ A3: 360 ft^2 A4: 62.4 cm^2 A5: 132 cm^2

A6: 150 ft^2 A7: 110.88 cm^2 A8: Triangular prism: 2 triangles + 3 rectangles; Square pyramid: 1 square + 4 triangles A9: 152 ft^2 A10: 108 ft^2

B1: 24 + 80 + 48 = 152 ft^2 B2: 144 + 240 = 384 cm^2 B3: 96 ft^2 B4: 13.8 + 240 = 253.8 cm^2 B5: 48 + 120 = 168 in^2

B6: 15.6 cm^2 B7: 96 + 600 = 696 ft^2 B8: 15 in^2 per face; slant height = 5 in B9: 96 + 480 = 576 cm^2 B10: Pyramid A (260 cm^2 > 224 cm^2)

Section 8.3: Volumes of Rectangular Prisms

A1: 3,024 cm^3 A2: 125 $in.^3$ A3: 500 m^3 A4: 224 m^3 A5: 10 1/2 ft^3 (10.5)

A6: 27 $in.^3$ A7: 6 in. A8: 8 half-inch cubes A9: 8 inches A10: 216 half-inch cubes

B1: 500 m^3 B2: 48 ft^3 B3: 15,625 cm^3 B4: 65 in^3 B5: 8 per in^3; 96 half-inch cubes

B6: 3 cm rise B7: 77 ft^3 B8: 36 ft^3 B9: 8 inches B10: 4,608 in^3 ≈ 19.95 (about 20) gallons

Section 8.4: Circles, Cones, and the Pythagorean Theorem

A1: 78.54 cm^2 A2: 706.86 cm^2 A3: 5 cm A4: 10 meters A5: 13 ft

A6: r = 8 m; Area = 201.06 m^2 A7: 18.85 m^3 A8: 188.50 $in.^3$ A9: Cylinder = 282.74 cm^3; Cone = 94.25 cm^3 A10: 40 inches

B1: 153.94 ft^2 B2: r = 20 m; Area ≈ 1,256.64 m^2 B3: 87.96 inches B4: 25 meters B5: 20 inches

B6: 150.80 cm^3 (48π) B7: 565.49 in^3 (180π) B8: One 16-in pizza (201.06 in^2 > 157.08 in^2) B9: 56.55 ft^2 B10: 17 feet

Appendix E: Student Self-Check Answer Key — Units 9 & 10

Solve by hand first and verify on your SC-6 calculator before checking your final answer here. Full worked setups are in the Teacher's Guide.

UNIT 9: STATISTICAL MEASURES & DATA DISPLAYS (BIG IDEAS MATH GRADE 6 — CHAPTERS 9 & 10 (STATISTICAL MEASURES AND DATA DISPLAYS))

Section 9.1: Statistical Questions, Dot Plots, and the Mean

A1: Sum = 20, Mean = 5 A2: Sum = 70, Mean = 14 A3: 156 cm A4: Mean = 4.0 g A5: Sum = 55, Mean = 18.33

A6: 13.33 (80 ÷ 6) A7: 70°F A8: 60 total blocks A9: 95 A10: 17 points

B1: Sum = 85; Mean = 17 points B2: 42 grams B3: 89 ÷ 10 = 8.9 B4: 91 B5: Total = 36,000; Mean = 9,000 steps

B6: 82.5 → 90.0 (+7.5 points) B7: 12 books per stack B8: 4.8 g B9: Reading time varies from student to student B10: 260 ÷ 10 = 26 points

Section 9.2: Measures of Center: Median, Mode, and Range

A1: Mode = 7 A2: Median = 14, Mode = 12, Range = 8 A3: Mean = 9, Median = 8, Mode = 8, Range = 9 A4: Min = 62°F, Max = 85°F, Range = 23°F A5: 75°F

A6: Mean = $29.17, Median = $17.50 (Median is best) A7: Mean increases from 14 to 30.8; Median stays 14 A8: 95 A9: Median = 14, Range = 13 A10: Mean = $370k, Median = $220k (Use Median due to the $980k outlier)

B1: Median = 62, Mode = 62, Range = 22 B2: Median = 12.5, Mode = 12, Range = 5 B3: Mean=$40, Med=$25, Mode=$20; Median (outlier $105 skews mean) B4: Modes = 6 and 8 (bimodal); Median = 7 B5: Median = 11.8 s, Range = 1.2 s, No mode

B6: Min = 64, Max = 81, Range = 17°F, Median = 72°F B7: Mean = 11, Median = 11, Mode = 12, Range = 7 B8: Example: 8, 8, 10, 12, 15 B9: Median increases by 5 (to 85); Range stays 20 B10: Mean = 11.75, Median = 11.5, Mode = 10, Range = 6

Section 9.3: Quartiles, Interquartile Range (IQR) & Box Plots

A1: Min = 10, Q1 = 15, Med = 30, Q3 = 45, Max = 50, IQR = 30 A2: Q1 = 6, Med = 10, Q3 = 14, IQR = 8 A3: 34 A4: 75% A5: Q1 = 73.5, Med = 82, Q3 = 90, IQR = 16.5

A6: Q1 = 46.5, Med = 51, Q3 = 56.5, IQR = 10 A7: Player A (smaller IQR means the middle 50% of scores are closer together) A8: 90 (Any score > 90 is an outlier) A9: Q1 = 15, Med = 22, Q3 = 28, IQR = 13 A10: About 100 students (50% of 200)

B1: Min=40, Q_1=45, Med=55, Q_3=65, Max=70; IQR = 20 B2: Q_1 = 7.5, Med = 10, Q_3 = 13, IQR = 5.5 B3: Q_1 = 22.5, Med = 30, Q_3 = 42.5, IQR = 20 B4: Range = 28 ft; IQR = 13 ft B5: 75% = 45 trees

B6: 25% of 60 = 15 trees B7: Class A (smaller IQR means less spread in the middle 50%) B8: 30 + 15 = 45 B9: Min=18, Q_1=22, Med=27, Q_3=34, Max=40; IQR = 12 B10: Range = 90, IQR = 8 (IQR is not distorted by the outlier 100)

Section 9.4: Mean Absolute Deviation (MAD)

A1: Mean = 156 cm, MAD = 4.0 cm A2: Mean = 14, MAD = 2.4 A3: Mean = 20, MAD = 6.0 A4: Mean = 7, MAD = 0 (Zero spread) A5: Mean = 28, MAD = 4.8

A6: Mean = 13.0 s, MAD = 0.40 s A7: Mean = 88, MAD = 3.6 A8: Team B (a larger MAD indicates greater variation from the mean) A9: Mean = 10; deviations: 4, 0, 1, 3; MAD = 2.0 A10: Mean = 9, Med = 8, Mode = 8, Range = 9, IQR = 5.5, MAD = 2.4

B1: Mean = 210°C; MAD = 3.0°C B2: Mean = 10; MAD = 1.6 shots B3: Mean = 50 g, MAD = 0 g (no deviation from the mean) B4: Mean = 9 s, MAD = 2.0 s; Raven A is more consistent B5: Mean = 85; MAD = 4.8

B6: Mean = 15 m, MAD = 1.0 m B7: Mean = 85°F; deviations 5, 1, 1, 5; MAD = 3.0°F B8: The signed deviations (-5 + 0 + 5) always add to 0 B9: Mean = 5 min; MAD = 2.4 min B10: Mean=16, Med=15, Mode=15, Rng=8, IQR=5.5, MAD=2.4

UNIT 10: GRADE 7 BRIDGE — SIGNED RATIONALS, 2-STEP EQUATIONS & PROBABILITY (BIG IDEAS MATH GRADE 7 BRIDGE — CHAPTERS 1–3 & CHAPTER 7)

Section 10.1: Adding and Subtracting Rational Numbers

A1: (a) -6; (b) -24 A2: -9 A3: (a) 30; (b) -4 A4: -25 A5: -14.2°C

A6: -4.8 A7: +2.7 A8: -1/8 A9: 36°F A10: 2 7/8 (2.875)

B1: +8°F B2: -83 meters B3: 4,800 - (-200) = 5,000 meters B4: -15 - (-25) = +10 points B5: +$8.75

B6: 1/8 B7: -9/6 = -1 1/2 B8: -16.0 meters B9: -15.0°C B10: -8

Section 10.2: Multiplying and Dividing Rational Numbers

A1: (a) -56; (b) +56 A2: (a) -5; (b) +5 A3: +25 (2 negative factors = positive) A4: -8 (3 negative factors = odd count = negative) A5: +8.4

A6: -12 A7: -3/5 (-0.6) A8: +2 A9: -120 A10: -76

B1: -120 feet B2: +40 points (same signs give a positive product) B3: -4.5°F per hour B4: $(-4)^2$ = +16; $(-4)^3$ = -64 B5: +2/3

B6: -2 B7: -12 B8: -3.2 meters per year B9: +120 B10: 36 - (-20) + 6 = 62

Section 10.3: Solving Two-Step Equations (ax + b = c)

A1: 4x = 36; x = 9 A2: 5x = 50; x = 10 A3: 6y = 48; y = 8 A4: 8n = 64; n = 8 A5: m / 4 = 9; m = 36

A6: k / 3 = 12; k = 36 A7: -3x = -12; x = 4 A8: 2.5w = 15; w = 6 A9: 25h + 35 = 160; h = 5 hours A10: 15w + 45 = 120; w = 5 weeks

B1: 6r + 8 = 38 → r = 5 laps B2: 4h + 12 = 48 → h = 9 hours B3: 15w + 35 = 140 → w = 7 weeks B4: 4b - 10 = 46 → b = $14 per book B5: x = 6 (-4·6 + 18 = -6)

B6: n = 35 B7: 3000 - 40m = 600 → m = 60 minutes B8: x = 24 B9: 2w + 36 = 56 → w = 10 cm B10: 2.25m + 3.50 = 21.50 → m = 8 miles

Section 10.4: Theoretical Probability, Combinations & Permutations

A1: P(5) = 1/6; P(not 6) = 5/6 A2: 0.65 (65% or 13/20) A3: P(double 6s) = 1/36; P(sum 11 or 12) = 3/36 = 1/12 A4: 24 ways A5: 720 ways

A6: 56 teams A7: 336 podium arrangements A8: 1/8 (12.5%) A9: 1/3 (6/10 × 5/9 = 30/90) A10: 120 combinations vs. 720 permutations (differ by 3! = 6)

B1: 4/20 = 1/5 = 0.20 = 20% B2: 65% (13/20 = 0.65) B3: P(black) = 2/5; P(not white) = 11/15 B4: 120 custom skateboards B5: 5! = 120 orders

B6: 210 batting orders (permutations) B7: 35 lab groups (combinations) B8: 28 fruit combinations B9: 60 times B10: 12 outcomes; P(H and 6) = 1/12